Routledge Advances in Game Studies

INDEPENDENT VIDEOGAMES

CULTURES, NETWORKS, TECHNIQUES AND POLITICS

Edited by
Paolo Ruffino

ROUTLEDGE

Independent Videogames

Independent Videogames investigates the social and cultural implications of contemporary forms of independent videogame development. Through a series of case studies and theoretical investigations, it evaluates the significance of such a multi-faceted phenomenon within videogame and digital cultures.

A diverse team of scholars highlights the specificities of independence within the industry and the culture of digital gaming through case studies and theoretical questions. The chapters focus on labor, gender, distribution models and technologies of production to map the current state of research on independent game development. The authors also identify how the boundaries of independence are becoming opaque in the contemporary game industry – often at the cost of the claims of autonomy, freedom, and emancipation that underlie the indie scene. The book ultimately imagines new and better narratives for a less exploitative and more inclusive videogame industry.

Systematically mapping the current directions of a phenomenon that is becoming increasingly difficult to define and limit, this book will be a crucial resource for scholars and students of game studies, media history, media industries and independent gaming.

Dr. Paolo Ruffino is a Lecturer in Communication and Media at the University of Liverpool. He is the author of *Future Gaming: Creative Interventions in Video Game Culture* (Goldsmiths and MIT Press, 2018). He has co-curated, with Marco Benoît Carbone, a special issue of *Games and Culture* in 2017 on the work of Roger Caillois, and of *GAME The Italian Journal of Game Studies* on videogame subcultures in 2014. His research focuses on independent videogame development, the automation of play, and contemporary practices and technologies of gamification and quantification of the self. He is chair of DiGRA Italia and a member of British DiGRA.

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1 After independence

Paolo Ruffino

Studying independent videogames

What are independent videogames? A brief history of an unresolved question

This book collects research produced around independent videogames and their makers. The authors of this volume have been investigating how independence has framed discourses surrounding videogame production in North and Central America, the United Kingdom, Europe, and Australia over a period of more than a decade. The duration and geographical extension of this research are significant. It is symptomatic of the pervasiveness of the notion of independence in videogame culture. As this book will attempt to demonstrate, independence can be identified across a broad range of practices and contexts, and in very distant geographical areas, such that it can no longer be ignored when trying to understand how games are made and played. This collection argues that at the root of the notion of independence there is a struggle to find alternative modalities to make, play, and distribute videogames, and that these modalities are now part and parcel of the global game industry. At the same time, it also shows that studying the videogame industry and culture requires an understanding of other contexts of creative production, such as cinema, music, and fashion, where the notion of independence has been used and debated for much longer. Yet this does not imply that independence can be easily understood, or that it has the same meanings across such a broad spectrum of human activities. Independence is a common thread across (game-)making practices in different areas of the globe, but it also generates productive contradictions at a local level.

In 2012, Bart Simon from Concordia University edited an issue on independent games in *Loading... The Journal of the Canadian Game Studies Association*. Published on January 1, 2013, the issue exemplifies the initial curiosity on the part of the research community. Prior to the issue's publication, it seemed relatively easy to identify in the discourses produced by players and developers a general idea of what made a videogame "independent".

However, on closer inspection, a more precise and unequivocal definition appeared much more difficult to determine. The general understanding was that independent videogames were made by individuals or small companies, with limited budgets, and distributed privately or via online platforms bypassing traditional publishers. But it was difficult to determine whether these new modalities of production could be defined as being more or less independent of those relations of power that characterized previous forms of game development, and whether they offered a real form of creative and financial emancipation. Moreover, the independent scene had distinctive aesthetic traits, which defined it in terms of audiences, but which were not necessarily related to the claims of economic and technical independence from the systems of production and distribution of the mainstream industry. It soon became clear that seeking a fixed definition of independence was a pointless task, as it changed across the various actors involved and their temporary and fluctuating interpretations of their own work. Garda and Grabarczyk (2016) made this problem more prominent when they argued that there is a difference between being independent in terms of creativity, distribution, or financing methods, and that, in each case, these three areas do not necessarily overlap or co-exist. Their analysis calls into question the various degrees, types, and modalities of independence that co-exist within game culture. As Simon asked in the introduction to the issue of *Loading…*: "Are we talking about a social movement, an art movement, a cultural scene, a fad, an ethics, a value orientation, a social identity, an assertion of authority, a cultural politics, an accident, a new form of capitalism…?" (2013, 1).

In 2002, 10 years before the publication of the journal issue, Eric Zimmermann was already writing about the apparently irresolvable problem of defining independence by articulating a series of opposing arguments. Published in the catalog of the exhibition *Game On*, his piece posed the existential question: 'do independent games exist?' (Zimmermann, 2002). The answers were set out against a list of cultural, economic, and technological factors, which demonstrated how contradictory trends were framing the new contexts of game-making, with no ultimate resolution across such a spectrum. Independence appeared as being defined through an intractable tension between opposing trends. A similar belief can be grasped from the words of game developer Molleindustria, who interpreted independence as a "degree of compromise with the capital" (Molleindustria, 2012). In his view, rather than being a defined status, independence is negotiated with and through various cultural, aesthetic, technological, and economic factors. Temporary positions of independence can be obtained when making decisions regarding how videogames are made and how they are published, according to Molleindustria. It matters for whom, and with whom, developers make videogames, and how they consider their inevitable relations of "dependence" as an opening toward new spaces of creative and economic autonomy.

Yet independence cannot be disentangled from its historical and geographical specificity. Questions over its meaning and potential cannot ignore the fact that the concept has emerged in videogame culture at a specific time and place. The argument that self-funded and self-distributed productions, relatively independent of a publisher, existed before in the history of the medium reinforces the importance of understanding why certain modalities of labor only came to be known as "independent" in the early 2000s. In light of a saturated market and the difficulty in effectively defining independence, by 2014, articles on gaming websites were already claiming that "indie is over", and that the movement had lost its momentum. The online distribution platforms were overcrowded with titles made by developers who aspired to work on their own games and make a career as independents, in what came to be known as the "indiepocalypse" (see the chapter by Nadav Lipkin in this collection). Many people thought that the promises of creative and financial independence had already gone, leaving a hyper-competitive market saturated with low-budget titles. The problem that was emerging at the time is still clearly visible in 2020.

Bennett Foddy, in a talk at IndieCade East 2014, argued that such an understanding was based on a false historical perception of the medium that had brought many North-American developers and commentators to identify low-budget productions and the absence of a publisher as entirely new phenomena. Similar productions, as Foddy argues, existed before the "indie" tag became popular. In parts of the world such as the UK and Europe, the videogame industry in the 1980s and 1990s was characterized by small studios and titles produced independently of a publisher, and by the wide distribution of programmable home computers, which encouraged game-makers and hobbyists. Similarly, the early years of the medium consist of experimental work largely produced outside of an identifiable industrial complex. "In the beginning", Foddy argues, "everything looked like an indie game" (IndieCade, 2014a). Independence was the name given to an approach to game production that was not entirely new, but re-branded as new at that particular time and in the particular context of North-American game development.

Anna Anthropy developed a similar line of argument at another talk at IndieCade 2014 (IndieCade, 2014b). The queer and trans game design scene existed long before the indie tag became popular, and it provided a context for all those who felt marginalized by the industry. Before the indie movement became popularized, games were already being made by "freaks, normal, amateurs, artists, dreamers, drop-outs, queers, housewives, and people like you" (Anthropy, 2012). Software such as Twine provided a free and accessible tool for many game-makers who desired to produce biographical and intimate accounts of their life experiences (see the chapter by Bonnie Ruberg in this collection). "Trans women invented indie games", Anthropy declared provocatively at IndieCade, showing how less visible communities of developers proliferated prior to the 2000s, offering welcoming contexts of mutual assistance. Furthermore, Heikki Tyni and Olli Sotamaa also noted how the LAN and demo party scene subculture in Finland in the early 1990s has framed and shaped the local game industry (Tyni and Sotamaa, 2014). Stephanie Boluk and Patrick Lemieux also argue that non-Western cultures of modding and hacking, popular in China and Japan, and the many construction kits and level editors produced in the 1980s and 1990s are regularly ignored when historicizing independent videogames. They conclude that "ironically, the emergence of the term *indie* game as a label and genre in the late 2000s signals the moment independent game development became dependent" (Boluk and Lemieux, 2017, 34).

However, while a broader perspective on the different ways in which videogames have been made and played can certainly reveal something new around the geographical and historical specificity of independent game development, it still does not tell us why the notion of "independence" within videogame culture came about in a specific time and place. As Laine Nooney argues, responding directly to Foddy's intervention, there is no "true" history of indie development. Many early forms of experimentation that might resemble contemporary indie games did not have, in their own time, any major industrial complex to be independent *from*. Moreover, the indie label has been used in the past to distinguish hardware or software producers from others and to claim a degree of artistic merit, in ways that are not immediately comparable to the values and meanings associated with the same word at the beginning of the 21st century. In other words, a historical analysis cannot overlook the contemporary fascination with being "indie" that brings together game developers to festival and workshops, that creates forms of aggregation and the sharing of knowledge via social media, that encourages many to experiment with game development and take creative risks, and that equally generates promises, anxieties, and new forms of social exclusion. As Nooney concludes:

> "Indie" is a referent for a curious balancing act between art and business, commerce and creativity, one made possible by very specific technological affordances, the economic operations of late capitalism, varied aesthetic imperatives about the contours of "personal expression," a kind of neoliberal hucksterism around the power of individual creativity, and the endlessly scalable conditions which drive our creative desires.
>
> (Nooney, 2014)

When trying to understand the meaning of independence and its historical specificity, we should identify the various discursive performances and practices that are organized around this notion in all their complexity and contradiction. The academic research around independent videogames, to which this book is dedicated, has been trying to provide a structure for such an investigation.

While presenting a panel on independent games at the DiGRA 2013 conference, Felan Parker argued that, to say it with Bruno Latour, academic researchers should "follow the actors" who mobilize the concept of independence in their discourses and practices (Latour 2005, 12; Parker, 2013). Parker claims that early studies on the history of independence had already revealed a much more complex reality where the definition of independent, mainstream, "formal" or triple-A varies depending on where and when the text was written. Martin and Deuze (2009), for instance, have observed that the appeal to authenticity and self-expression in the videogame industry emerges predominantly (but certainly not for the first time) around the early 2000s, when similar tendencies were visible in the marketing and promotion of digital and creative products. The newness of independence must also be understood, according to the authors, alongside the desire to market these new products as such.

Following Parker's suggestion, the chapters in this book are organized around specific analytical contexts. In so doing, the collection creates a strategic mapping of the notion of independence, exploring how historically and geographically situated actors are currently defining the new meanings associated with the concept. Such a perspective, which gives the same importance to both global and local dynamics, echoes the analysis of Aphra Kerr in her book *Global Games: Production, Circulation and Policy in the Networked Era*. Kerr argues that understanding local contexts and interactions is the only viable strategy for comprehending the trans-national networks of production, consumption, and distribution of the videogame industry (Kerr 2017, 201). In a similar vein, the Creative Territories research project, which looked at hubs of game development in the United Kingdom and The Netherlands, concludes that "video game makers [...] live in interconnected, overlapping, local, regional, national and global 'places'" (Creative Territories 2015, 4). This collection takes as its point of departure the realization that the temporary and strategic definitions of independence given in specific geographical contexts are fundamental to understanding the global contemporary videogame industry.

While trying to compare this editorial project with previous publications and interventions in the same area, we should also acknowledge that independent videogames in 2020 are an even more intricate topic of research. At the time when Parker was writing, independence appeared as a series of contradictory tendencies, as Zimmermann had already noticed in 2002, but it was still identifiable in the discourses produced by game developers, in the marketing of new products, and in the distribution strategies of new platforms. As Jennifer Whitson has pointed out, in the early 2010s game developers started to find "escape routes" from consoles and move toward mobile and social platforms (Whitson, 2013). This process has broadened the contexts in which independents can work and distribute their games, incentivizing the involvement of game-makers in multiple projects and platforms. For us, writing in 2020, the knots of independence are even more difficult to disentangle. They have become blurred, almost transparent, in the broader discourses around game production. As recent publications and industry reports have revealed, the practices that came to be defined around this notion are now the standard condition for those who are involved in videogame production. No longer an exceptional or alternative choice, independence, in all its myriad meanings, is now considered the most common type of employment in the videogame industry, or at least a required stage in the career of a game worker. "Following the actors" might be even more difficult in this scenario.

An industry of indies

The yearly Game Developers Conference publishes a periodic survey on the "state of the game industry". The 2020 report, based on 4,000 respondents, revealed that 36% of developers worked on the marketing of their own game and without a publisher (GDC, 2020, 14). The same survey concluded that 51% of respondents financed their project through the company's existing funds, and 31% used personal finances (GDC, 2020, 14–15). The results are consistent with previous surveys. In 2013, GDC surveyed 2,500 North-American game developers and revealed that 53% of them identified as "indie" (GDC, 2013). A subsequent survey by IGDA concluded that 48% of US game developers self-identified as independent (IGDA, 2014). In the United Kingdom, an even more prominent trend is visible. In 2013, the trade association TIGA estimated that "83% of all studios that started up in 2011 and 2012 were independent (as opposed to publisher owned)" (TIGA, 2013). In 2017, up to 95% of the workforce in the UK was employed in small or micro companies and could be defined as being "independent" of publishers and producers (UKIE, 2017). In Australia, Apperley and Golding argue that independent companies now form the "backbone" of game development (Apperley and Golding, 2015, 61). As noted by Brendan Keogh, independence is the only available solution in a context, such as the Australian game industry, where the "mainstream" has never been significantly present (2019a). Similar trends are visible in other regions that are not mapped with the same specificity and attention as the major areas of game production. In Italy, the almost complete absence of medium to large digital entertainment companies has led to the establishment of local contexts of aggregation for individuals and small companies (rarely formed by more than two people) that constitute, *de facto*, the only visible aspect of the national videogame industry (Ruffino, 2020). The Latin-American game development scene, as discussed by Guevara-Villalobos in this book, is mostly formed by studios that are using the possibilities offered by digital distribution and mobile markets to regain creative and economic independence after long periods of dependence on global networks of outsourcing.

Independence is no longer a marginal or alternative mode of production (if it ever was); instead, it is the most common type of organization within the videogame industry. It appears that almost every game developer is now partially or temporarily "indie" within their career, and the trend is expected to grow, consistent with recent developments in the cinema, music, and fashion industries (Hesmondhalgh, 2013; McRobbie, 2016). Moreover, it is becoming harder to detach the analysis of the mainstream videogame industry from the presence of smaller, local, grassroots forms of production, and the emergence of "informal" types of labor and collaboration. As Keogh notes, "contemporary videogame development is defined by a complex network of actors and practices that is not captured by the straightforward developer-publisher-retailer model through which videogame development and distribution have been traditionally understood" (2019b, 15). Keogh goes on to observe that, following a period of "formalisation" in the 1990s and early 2000s (a phenomenon also evidenced by Jørgensen et al. (2017) in relation to the Scandinavian region), game production is transitioning toward a period of "intense in/formalisation" (2019b, 16). Videogames are now frequently made "mundanely, precariously, and across intensifying networks of professional and amateur labour" and by "localised studios and individuals creating and distributing videogame works in a much vaster range of socio-economic contexts" (2019b, 16).

This is what John Vanderhoef, who also contributes to this collection, has defined as an "industry of indies" (2016). Vanderhoef notes that the boundaries separating the amateur from the professional are becoming more porous, and the two approaches are often co-existent within the same product. Such a co-existence creates its own contradictions. Small independent companies often adopt the same managerial approach of the larger companies, appropriating a similar neoliberal ethos of competition and self-determination. On the other hand, the desire for financial autonomy on the part of those seeking creative independence is forced to compromise in the face of platforms of distribution and crowdfunding that are often shaped by the same neoliberal ideology that independents might have been originally trying to challenge (Tyni, 2020). In 2005, Julian Kücklich's earlier analysis of modding, a much more specific phenomenon of game labor, already identified how the amateurial and professional mindsets are entangled in and influenced by neoliberal capitalism in the framing of digital labor (Kücklich, 2005). Independent game development has undergone a similar transformation.

Furthermore, indies and larger companies frequently co-habit in the same geographical areas. In Montreal, Ubisoft established its headquarters in 1997 giving visibility to the city as one of the most important global hubs of production and bolstering a lively indie scene. However, as Jason Della Rocca argued in 2013, Ubisoft found in Montreal a prolific substrate of developers who did not necessarily identify as indie and might not have been internationally visible, but who provided the required skills and manpower to establish a larger company (Della Rocca, 2013). As already noted at IndieCade East 2014 by Bennet Foddy in his historical reconstruction, in an "industry of indies", the boundaries around the independent territory are less clearly delineated. Instead, we find continuous relations of *dependence*, which involve methods of management and use of resources, geographical proximity, as well as ethical approaches to labor. These relations give rise to mutual economic benefits and intellectual influences, as well as forms of co-optation and exploitation.

The second wave of independence

The in/formalization of game development, to use Keogh's terminology, and the merging of professional and hobbyist practices do not necessarily bring about a more stable model for those people involved in it. On the contrary, many of the contributors to this book share the vision that (independent) game development is now more risky than ever and unlikely to provide sufficient economic stability for creative experimentation. Jennifer R. Whitson argues that the videogame industry has adopted, citing Boltanski and Chiapello (1999), the "new spirit of capitalism": the initial enthusiasm surrounding access to tools of production and distribution has ultimately made it acceptable to overload single individuals with excessive risks and responsibilities, it has exacerbated pre-existing gendered barriers to inclusivity and ultimately limited the opportunities for creative expression (Whitson, 2019).

One of the supporting arguments of this volume is that the emergence of the notion of independence within videogame culture has worked as a discursive justification for the introduction and legitimization of modalities of labor that typically characterize the creative industries (Leadbeater and Oakley, 1999; Kerr, 2006, 43–74; Hesmondhalgh, 2013, 358–362). In the current historical period, the game industry is experiencing what Angela McRobbie identified, in relation to the creative industries of the late 1990s, as the rise of a second wave of independence. Such a second wave, as McRobbie noted, marks (perhaps "ironically", as she put it) "the decline of 'the indies'[,] the rise of the creative sub-contractor and the downgrading of creativity" (2002, 519).

The upshot of this process can be seen clearly in the work of those who have been paying close attention to the dynamics within development studios and at promotional and networking events. Casey O'Donnell has been arguing throughout his research that "work/play" and "passion" shape a semantic field that justifies exploitative regimes of labor and systematically denies any chance for creative expression and self-realization (O'Donnell, 2009, 2014). As highlighted by Aleena Chia in her survey of North-American indies and hobbyists, the incentive to "do what you love" prompts many to adopt a moral stance, where direct financial remuneration is no longer expected but is still required, particularly when compared to the amount of time and effort needed to release a marketable product (Chia, 2019).

The ethos of independence has also jettisoned professional figures, such as the producer, which sustained the process of intermediation with audiences and external collaborators. The duties of this specific profile have been re-distributed to developers, increasing their workload and required levels of expertise, in the name of a more direct and intimate contact with the audience. Whitson et al. (2018) conclude their research on the effects of the "missing producer" by arguing that indie developers are now called on to mimic the practices of larger studios, despite the significantly reduced number of people involved in a project, while facing economic conditions that are hardly sustainable. The goal then becomes to "keep on keeping on", as one of their interviewees put it. Keogh reaches a similar conclusion while investigating Australian independent developers, noting how "survival" becomes the chief barometer in measuring the success of an indie company, rather than economic profitability (2019a).

Survival often means being visible and accepted in the circuits of cultural intermediation that connect developers with consumers and other developers, which can frame new collaborative and support networks. The work of cultural intermediaries who organize festivals, workshops, coworking spaces, and who generate grants and funding opportunities is becoming a defining characteristic of the "second wave" of independence. Significantly, IndieCade, a festival, co-founded and co-organized by Celia Pearce, who contributes to this volume with an account of her experience, has fueled the perception and ethos of "indie" games more than any videogame product or individual developer. As Parker et al. (2018) argue, while investigating the role of the itinerant Indie Megabooth showcase, similar events act as cultural intermediaries and shape the attitudes of consumption of videogame products. Matthew Perks and colleagues conclude that the fundamental role of intermediaries is often overlooked when analyzing the new dynamics of game production, but this feature is now becoming pivotal in the survival strategies of the new indies (Perks et al., 2019). In a context in which indie and triple-A productions are becoming harder to distinguish in their strategies and tools of development, cultural intermediators come to play a defining role in framing the new cultures of production.

Indie game studies after independence

In 2013, at a time when the notion of independence was at the peak of its popularity among game developers and consumers, Felan Parker made a case for "indie game studies": a research project that should put at its center the complexity of the new modes of production of videogames (Parker, 2013, 8). Inspired by Parker's call, this volume seeks to take his perspective even further, and to collect research conducted at a time when independence, though still used as a signifier, has become almost synonymous with game-making practices *tout court*. Indie is something that most game developers have been, are, or intend to be. It is discussed and defined in a variety of places, such as hubs, festivals, and coworking spaces. It is brought up as a tag in crowdfunding campaigns and distribution channels. It is situated in time and space. It is strategic. The question is how we, within game studies (very broadly intended), might still find ways to track independence, to disentangle it and connect it with other *things*, and reveal the relations of power that it otherwise renders opaque.

Since 2013, many researchers have been involved in various projects and publications around this area of investigation. It would be impossible to list them all here, but it is worth mentioning some of the works carried out by authors included in the present collection, which have largely framed the approach of this editorial project. The Indie Interfaces project, involving Canadian scholars from Concordia University, the University of Toronto, and the University of Waterloo, has been looking at the various types of intermediation that take place across the production, distribution, and reception of indie games. The group has been looking at how festivals, incubators, platform holders, and other types of social and cultural "interfaces" frame the meanings of independence and what comes to be counted as part of it. The Refiguring Innovation in Games (ReFiG) project, supported by the Social Sciences and Humanities Research Council of Canada since 2015 and involving a large number of international scholars, has focused more closely on issues of inclusivity and gender. While not exclusively invested in independent game-making, ReFiG has shed light on the barriers that prevent access to game development and has offered numerous contexts for the sharing of knowledge and skills. The Creative Territories project, supported by the Arts and Humanities Research Council in the UK and initiated in 2014, has posed fundamental questions regarding the strategies that can help to ensure that independence becomes more than a "bubble" and is transformed into a viable concept for economically sustainable and culturally relevant projects. Creative Territories has also paid close attention to how independence has changed the technological conditions for the creation and distribution of games and what kind of possibilities have been opened, and closed, in this new landscape.

These three major research projects have largely framed the rationale behind this volume and its keywords. In what follows, I will introduce the individual contributions and situate them within the broader vision of this publication.

Introduction to independent videogames: cultures, networks, techniques, and politics

Cultures

First, independence is a **cultural** phenomenon: a concept that has changed the cultural value of videogames and the meanings associated with being part of the contexts of production and consumption. It has brought attention to the dynamics of inclusivity within the game industry, the relations of work and care that game workers establish with each other, and the passion put into the process of crafting a game. It has also raised fundamental questions regarding the aesthetics of games and the presence of niche markets, and how new technologies have changed how games are made and how they look (see Juul, 2019). Game culture "after independence" appears much more varied and complex.

The collection starts with a contribution by Aphra Kerr, whose work on the global and local trends of the videogame industry has greatly inspired the research of many of the authors in this collection. Chapter 2 looks at the results of a series of initiatives for inclusivity in game development implemented between 2016 and 2019. The chapter challenges the common assumption, maintained in discourses surrounding independence, that new development tools and digital channels of distribution have democratized game production and made it more accessible. These visions do not match with the reality of the contexts of game development, where the majority of participants remain largely white, male, and from the same socio-economic background as those who typically have the widest access to the job market. Game jams are taken by Kerr as a privileged point of access to understand and tackle this contradiction. Game jams are contexts that attract participants with varied game-making skills, for a limited amount of time, with the objective of making a playable game. These events are experimental and inclusive in scope, although they rarely succeed in attracting participants who are not already immersed in the culture of game production. The main referent for local game jams is the Global Game Jam initiative, which sets rules of conduct and good practices for inclusivity and to incentivize broader participation. However, the global initiative must be analyzed in its local instantiations, Kerr argues, where the guidelines are put into practice and confronted with the gendered assumptions and expectations of those who take part. The chapter draws on the experience gathered while organizing three jams in Dublin, Limerick, and Cork in Ireland, followed by six female-friendly game-making workshops. It reflects on the technical and spatial assemblages that make similar events more or less welcoming for participants who are typically left at the margins of production cultures.

Bonnie Ruberg (Chapter 3) looks at the meaning and ethos of independence in videogame culture and argues that the "indie imaginary" is often limiting and hostile to those who do not already conform to the requirements of the game industry. The best-known indie games are often "about" videogames: they reflect their structures, logics, and aesthetics. Developers' narratives about "going indie" are often predicated on the assumption that one can afford to take risks in one's career, knowing that returning to a steady job, if necessary, will be relatively easy: an assumption that hides the gendered and racial inequalities of labor in the game industry. Ruberg observes that alternative ways of thinking about videogame design are more likely to be found in what they call the "queer games avant-garde". These are games made by, for, and often about queer and trans players. In these texts, we find not only new aesthetics and alternative narratives, but also different ways of thinking about relations established while making and playing videogames. Queer indie games are challenging the very notion of independence, finding and bringing to the fore the relations of interdependence that connect various human beings in the production of a videogame. Ruberg's work is based on a corpus of interviews with queer and trans indie game-makers, collected while working on Ruberg's most recent monograph (Ruberg, 2020). Ruberg draws a picture of an artistic avant-garde, independent *of independence*, interested in drawing from other forms of expression rather than looking back at the traditions of videogame culture. It is a movement that attempts to queer the styles and textual strategies that make videogame culture often exclusionary and self-referential.

Independence is often predicated in oppositional terms, and the difference with the triple-A context has been marked through technical and aesthetic strategies. Paweł Grabarczyk's Chapter 4 looks at a very specific aspect of contemporary indie games: their use of virtual reality technologies. As Grabarczyk argues, the various techniques adopted in the early experimental products have yet to produce a shared "VR aesthetic". For these reasons, independent developers have found in virtual reality (VR) a technological device and a niche market where new titles have no defined "opposition". Grabarczyk identifies the strategies adopted by indie developers when approaching VR productions, and how they have found their own ways of claiming a form of artistic and creative independence, despite the absence of a set of practices to be independent of. Thus, indie VR games are caught in a peculiar moment: they reveal the drive toward niche markets and unexploited audiences that many indies must pursue in order to survive economically, but they also open up a new territory of creative experimentation in which to investigate how new trends and standards might emerge. VR games are also seen as an outgrowth of a hyper-competitive approach that tends to saturate the market as soon as new niches are identified. Grabarczyk concludes by observing that the current status of VR games is likely to expand toward independent games that resemble triple-A productions and eventually establish an aesthetic that new projects will imitate.

Networks

Second, this book explores the **networks** of professional contacts and exchange of knowledge that are made visible while identifying with the notion of independence, as well as the practices and actors that are made opaque as part of the same gesture. Independence has been presented in its earliest accounts as a hyper-individualized practice, where self-made individuals could allegedly reach global audiences, bypassing publishers, producers, and marketing teams, thanks to the new tools of funding, production, and distribution. We know from many of the studies already cited in this introduction that independence has actually made intermediaries, gatekeepers, and trans-local hubs of production more important, even if they might appear less prominent. This section looks at the methodologies required to study networks that bring together developers at hubs, incubators, and new workplaces. It explores some less visible connections, such as those between indies and academia. These networks are one of the most significant aspects of game production in the wake of the first "wave" of independence.

Pierson Browne and Jennifer R. Whitson (Chapter 5) start with the assumption that networking is essential for game development and use Social Network Analysis as a theoretical framework to examine the difference between distributed and local networks, how they are constituted, and who benefits the most from them. The advantages of a large network of contacts are largely dependent on the geographical location of the participants. Having regular access to a local community of producers facilitates serendipitous connections to contacts, contracts, and opportunities through what Browne and Whitson refer to as "leaky pipes". For those who live in areas that are not strongly tied to the same circuits, and in cities that do not constitute a hub of game-making, game festivals and events become crucial sites for conducting more structured and intense networking, utilizing the temporary proximity with professionals who might generate funding opportunities and enhance visibility. But the benefits of these events are more limited, acting as short-term "fire hoses" rather than "leaky pipes". In the second half of their chapter, Browne and Whitson use statistical data analysis from 123 surveys carried out with participants at PAX Prime 2015, Pax East 2016, and the Game Developers Conference 2016 and supplement their data with semi-structured interviews to conclude that geographically situated hubs of production facilitate the constitution of networks which, in turn, effectively make the career of an independent developer more likely to become sustainable in the long run. While the common conception of independent developers suggests that individuals are responsible for their own careers, a more attentive look reveals that the connections developers establish while outside of their own studio (or bedroom) could be far more vital than their personal creativity. However, access to those networks is often unequal, as it largely depends on geographical distance, economic resources required for traveling, the cost of attendance, and social capital.

Celia Pearce's chapter (Chapter 6) explores how academic institutions have facilitated the establishment of new videogames companies. Through three case studies, Pearce unveils a fundamental, and rarely acknowledged, network of exchanges of knowledge, funding opportunities, and contexts of visibility, which are provided by university institutions and academic incubators. These connections are often based on unpaid and voluntary labor, and they certainly complicate the romantic image of the indie developer as a self-made artist. Academia has played a major role in the rise of the indies, Pearce argues, and not only in the more predictable sense that many developers have graduated from the new game design degrees that have spawned across the globe. Academia has offered a slew of volunteers for testing and experimenting, festivals and occasions of visibility, incubators, and spaces in which to grow a potential business. The network has both cultural and economic implications. The impact of academia is visible in the three case studies analyzed by Pearce. First, the festival IndieCade, which has significantly shaped indie culture and aesthetics through a community of producers in the USA and Europe, began thanks to the fundamental contribution of academic scholars and students. The Copenhagen Game Collective, one of the most celebrated groups of independent game-makers, was born from the classes and game jams of the IT University (ITU) in Copenhagen, Denmark. The final case, Oculus, explores how the now-famous VR company leveraged an IndieCade sponsorship which was forged in its academic roots at University of Southern California. Pearce looks at these stories from her own personal perspective as an academic at Northeastern University who has been involved for a considerable time and as a founding member in the curation of the IndieCade festival and in the establishment of indie culture. Her chapter demonstrates how cultural and economic exchanges are often made invisible, particularly when a stronger narrative of self-determination is made more prominent. Independent game development is deep-rooted in institutions that have provided the cultural, technical, and economic conditions for the emergence of a new mode of game-making.

Techniques

Third, games "after independence" are novel **technical** objects that require new methods of production and distribution, and revised methodologies of research. They are born in a new technical landscape and serve to frame the technologies that are used in the broader digital and creative industries. One of the major examples is the reach of Unity, one of the software tools most commonly used by the major game companies. It is now used by triple-A productions as well as game-makers, hobbyists, and students. As Benjamin Nicoll and Brendan Keogh point out, Unity is more than just a tool, as it frames an entire "circuit of cultural software" that influences the types of skills needed to access game-making practices, the methodologies of production, and the workflow of a new project (Nicoll and Keogh, 2019). Middleware (computer software used to connect games and applications with the operating system) is a cultural intermediator (Charrieras, 2011).

Patrick Crogan's Chapter 7 is both a philosophical inquiry and the result of a continued research project on the new social and technological conditions for making and playing videogames. Crogan looks at the broader digital technocultural transformations of the creative industries through the work of Bernard Stiegler. Through Stiegler, Crogan interprets the wide adoption of software tools and channels of distribution as a *pharmakon*: both medicine and poison, they enable new cycles of creativity and at the same time inhibit the technical exploration of alternative modes of production. Within game development, as in many other forms of production in neoliberal political economies, the drive toward "creative disruption" is captured by standardized techniques of production, distribution, and consumption, and the cycles of permanent innovation make political and cultural reflection harder to achieve. Modalities of production and new techniques go hand in hand, and as game development becomes allegedly more accessible and democratized, it also drives new technical solutions for its own capitalistic capture. Crogan investigates such a condition and argues that, through the notion of independence, human creativity could be de-routed from its economistic understanding and re-evaluated for its cultural and critical potential. Ultimately, Crogan suggests three possible paths for the future of independent game-making. First, game design should be become more aware of its reflective and critical potential (as also noted by authors such as Ian Bogost 2007, 2011). Second, the ready-made and automated elements of the tools of production facilitate development while also limiting the explorative possibilities of game design. Their effect must be understood and negotiated. Finally, the ways in which players are engaged must be re-calibrated. The use of freemium models of distribution and the co-optation of crowdfunding by capitalistic systems of exploitation are depriving contemporary videogames of the creative aspects of shareware and peer-to-peer cultures. These models must be re-thought in light of their original ethos.

The new strategies of game development have also given rise to novel solutions in both production and publishing. Independent videogame development has long been seen as the next stage after a publisher-centric model, where a small number of publishing houses could exercise control over the creation and marketing of a game. As discussed above, the labor once under the control of publishers has not disappeared, and it is now, in most cases, the direct responsibility of the independent developer. Felan Parker (Chapter 8) investigates the case of Annapurna Interactive, a Californian company that has re-established the role once played by publishing houses while adapting it to the new conditions of production. Annapurna Interactive is part of Annapurna Pictures, a film company situated as part of the rise of "indiewood" cinema productions: films that, while indie in their aesthetic, are targeted at a mainstream audience and produced with a substantial budget (Tzioumakis, 2012). Annapurna Interactive has been responsible for critically acclaimed, small or medium team productions such as *What Remains of Edith Finch*, *Florence*, *Gorogoa*, and *Donut County*. As Parker argues, the success of Annapurna Interactive is significant of broader changes in the creative and cultural industries, where new actors are re-interpreting the role of old intermediaries, accommodating the needs of the new independents while identifying new audiences and consolidating their position of power and prestige. Parker looks at the history of Annapurna, their aesthetic and brand image, their relationship with developers, and how they are changing the ecosystem of indie production.

Casey O'Donnell's Chapter 9 takes a different route. His intervention analyzes the different methodologies and implications of studying game development studios. O'Donnell provides a fundamental account of the various methodologies of academic research on game production, while also exploring the specificity of the practices that are being studied by academic scholars. Game development, he argues, is a human practice where culture, technology, politics, and economics are all co-present and equally important. Researching game-makers involves all these aspects, regardless of the size of the studio. Indie studios can be even harder to analyze, as indies often operate in coworking spaces, from their own flats, or while on the move in temporary locations. They work in solitude and in collaboration with others who are distantly located or temporarily joined in shared spaces. They work on multiple projects at the same time. A large number of game workers are "below the line". For instance, testers (see Ozimek, 2019), translators, or those employed at platforms such as Kickstarter, are rarely considered as what "counts" as part of game development. However, they are essential in the production process. The study of these forms of labor can shed light on areas of production that are otherwise invisible. The researcher is confronted with difficult decisions about how to delimit the practices to analyze and how to collect "data" from conversations and exchanges that are often informal. O'Donnell explores the broader heritage of studio studies in relation to production studies, anthropology, and sociology. He concludes by providing some useful guidelines for researchers who wish to understand the complexities of independent game-making. Staying in close contact with those who work with(in) videogames is fundamental in the attempt to reach a better understanding of the dynamics involved.

Politics

Politics is the fourth keyword of this collection. Attending to the new techniques and contexts of production has raised awareness of the human relations that are inevitably part of game-making. Game developers do not work in isolation. The fourth section of this collection entertains the idea that the future of game labor will be determined by understanding how indies-after-independence can make sense of the co-presence of others in their daily jobs. The two contributions collected in this section approach the same issue from distinct perspectives.

Jamie Woodcock (Chapter 10) considers the theoretical and practical challenges of collectively organizing independent developers. Woodcock takes as his point of departure the growing interest in unionizing that has gained momentum around the global organization Game Workers Unite. Woodcock, who has been closely following and personally involved in the establishment of the group in its early stages, addresses some of the key questions that have emerged in relation to labor unions in the videogame industry. The chapter's initial concern consists in individuating the fundamental literature that can help us understand the dynamics of competition and co-operation in (indie) studios. Complementing Casey O'Donnell's chapter, Woodcock looks at the specificity of studying studios of game development and explores the specificity of independent developers' workplaces. Indies have a much looser and more flexible relation with their "employers": sometimes, they manage their own work, or they might be responding to crowdfunding campaigns, or they could be working with temporary subcontractors. It is often a combination of different types of labor. In all these circumstances, there are connections with platforms and other workers, who are involved through formal and informal relations of labor. While the immediate impression would be that indie labor is "unorganizable" in a collective effort, Woodcock finds similarities with other sectors of the gig economy. He identifies a comparable example in the 24-hour strike carried out by food delivery drivers in Greece in April 2019. The Driver Workers' Informal Assembly organized an effective disruption of service by organizing a large number of dispersed, isolated workers. The couriers organized for better pay and pension conditions, and the strike raised awareness of the exploitative conditions that drivers have to endure. Videogames are never made in solitude, the author argues, and this means that struggles for better conditions of work can be organized collectively. Identifying shared strategies across the gig economy, even in sectors and geographical areas that might appear disconnected from videogame development, can shed light on the possibilities of labor organizing in the new contexts of production.

One of the factors that has largely contributed to the climate of increased competitiveness, precarity, and anxiety among independent developers is the saturation of the market. We frequently hear that too many games are now being produced, sometimes more than what consumers can play or even notice. Nadav Lipkin's Chapter 11 addresses the insecurity that game workers must tolerate when facing a surfeit of games on the market. The saturation of the offer is an oft-debated issue within the independent game development scene. It has been said for some time now that an "indiepocalypse", that is, the collapse of the current conditions of production, should be hitting indie developers anytime. The risk is particularly high and causes anxiety for those who are constantly struggling to make a living out of their independent labor. The indiepocalypse might not have materialized yet, but Lipkin argues that the threat of it already produces responses among indie developers. Game development is imbued with an ethos of labor that encourages and celebrates risk-taking. Access to means of production and distribution is, for instance, rarely discussed in relation to the higher levels of competitiveness that it naturally brings with it. Lipkin looks at the economic, social, and technical reasons that have brought conditions of hyper-competitiveness to game development in the West. The current conditions are also responsible for raising high barriers to inclusivity since only a very few can afford to take such risks. In conclusion, Lipkin argues that such conditions can only be changed by a form of collective organizing. The unionization discussed previously by Woodcock is seen as a possible response to the threat of an "indiepocalypse".

Local indie game studies

The fifth and final section of the book collects research focused on regional or geographically embedded case studies. The various contributions bring perspectives from very distant areas, some larger (such as Guevara-Villalobos' analysis of Latin America) and some much more defined (as in Vanderhoef's participatory observation of the games expos at the Los Angeles Convention Center). Each piece highlights different interpretations and approaches to game development being mobilized by the notion of independence.

As discussed above, independent videogames and videogame production can only be understood if the two dimensions – the global and the local – are taken into account in their mutual dependence. The first chapter in this section is a perfect example of how this dynamic comes into play. Orlando Guevara-Villalobos (Chapter 12) looks at Latin America and the recent history of game development in the area. Guevara-Villalobos argues that in order to understand the current state of independent production in Latin America, we must look at the multi-scalar networks that studios rely on and reinforce through their practice, and that put them in relation with each other as much as with international actors. Latin America, despite not being at the center of the history of the videogame industry that we typically receive, has been an important context of outsourcing. A lively grassroots scene in the early 1980s has provided the required skills and manpower to the global videogame industry while building up the knowledge and experience required for the establishment of a local industry. With the advent of digital distribution and mobile markets, Latin America has seen the emergence of a scene that can be defined as independent. However, the denationalized spaces of digital production that existed before these technological changes must now find new ways to engage with local public institutions in order to sustain their needs. For Guevara-Villalobos, economic support for local digital economies and appropriate infrastructure are two fundamental demands that could support the emerging industry in Latin America and provide the foundation for an independent game industry. Thus, compared to the global North, the meaning of independence in this area is much more closely entangled with the intervention of economic and state institutions.

Brendan Keogh's Chapter 13 examines independent game development in Melbourne, Australia. The chapter presents the findings and conclusions of a series of interviews with 40 videogame makers based in Melbourne. Keogh concludes that game-making practices in the Australian city shape, what Pierre Bourdieu calls, a "cultural field": a broad range of practices, aesthetics, and value regimes composed by actors coming from (sometimes radically) different economic, social, and technical backgrounds. In recent years, the area surrounding Melbourne has seen an increase in the number of people involved in game-making practices. Keogh notes that this is the result of public interventions oriented toward the cultural and creative industries. In the absence of an established videogame industry, the Australian videogame sector comprises a large number of actors who are, in one way or another, independent, that is, working on a variety of small-scale projects and without an identifiable employer. The interests and ambitions of these projects are extremely varied and include actors who are attempting to establish a sustainable career and others who do not consider game-making as the primary source of their subsistence. Drawing on the responses gathered through his research, Keogh argues that Melbourne includes different *scenes*, organized around locations, communities, or the use of the same software and production tools. His analysis sheds light on what I have previously defined as the "second wave" of independence (which, in contexts such as Australia, is *de facto* the only wave): in this particular historical moment, videogame production is a heterogeneous practice comprised varied values and interests, which are often invisible as they do not translate into internationally renowned products, but nonetheless shape a social, economic, and cultural sector. Being "indie", in this scenario, is more than just being alternative or oppositional to the dominant modalities of production; instead, being indie becomes a signifier for a broad spectrum of identities. It also becomes the inevitable status quo for many who attempt to start a career in game development.

Olli Sotamaa's Chapter 14 provides an analytical comparison of two case studies that perfectly exemplify the heterogeneous approaches to game production that are often labeled as independent. Sotamaa looks at two "indie" games produced in Finland: *Nex Machina* (2017) by Housemarque and *Baba is You* (2019) by Arvi Teikari. Housemarque is one of the earliest game studios in Finland. Established in 1995 from the merging of Bloodhouse and Terramarque, Housemarque has been responsible for games such as *Resogun* (2013) and *Matterfall* (2017), released on Sony's PlayStation consoles. On the other hand, Arvi Teikari, also known as "Hempuli", is known for his small-scale games, freely available on his website (hempuli.com), and for his participation in the Independent Games Festival 2018 with *Baba is You*, where he won the awards for "Best Student Game" and "Excellence in Design". The game was commercially released in 2019 on PC and Nintendo Switch. While both representing commercially successful and internationally visible examples, the two case studies are radically different in scale. Sotamaa observes how Housemarque and Hempuli deploy different strategies of engaging with the local development scene and international collaborators, their consumers, publishers, local institutions, and their funding opportunities, and how they have different understandings of the needs and pressures of a constantly mutating context of production. The two developers face similar challenges in terms of economic survival, which are also shared by the other local developers, but they each establish radically divergent and unique strategies and identities in relation to their work. Both case studies are the result of a concentration of digital and creative companies in the Finnish capital, Helsinki, which has fostered the growth of a local hub of videogame production. On Sotamaa's account, while the international distribution of both products might make local relations seem less prominent, they have been made possible, thanks to the presence of game jams and access to private and public funding opportunities in Helsinki.

Independent videogames have been marking their own independence through a variety of strategies, which involve their aesthetics, funding, methods of distribution, narratives, and themes. Festivals and companies might also mark their distance and separation from the mainstream and triple-A industry geographically and spatially. John Vanderhoef's work (Chapter 15) provides a closer look at IndiE3, an event for independent developers in Los Angeles. Started in 2014, IndiE3 takes place adjacent to the E3, the more famous game industry expo which gathers the major publishers and producers of the global industry. In recent years, E3 has lost its original appeal and has been abandoned by some of the most important publishers, but it remains one of the most important annual meetings for representatives of the game industry. IndiE3 emerged as an alternative for those who did not feel represented by the grandiose style of E3 stands and press conferences. Vanderhoef discusses his personal engagement with IndiE3 and analyzes the strategies adopted by some of its most popular stands, namely those of Devolver Digital and Good Shepherd. IndiE3, Vanderhoef concludes, marks its "independence" aesthetically, discursively, and spatially by positioning itself in opposition to the E3. However, it also takes advantage of its spatial and temporal proximity to the E3, while replicating and even exacerbating its toxic masculinity and strategies of gatekeeping that E3 and videogame culture are often associated with. Independence does not necessarily imply being progressive and inclusive, and Vanderhoef's analysis of IndiE3 provides a geographically situated case in point. The author concludes by questioning whether claims of independence should not be brought forward along with a sense of responsibility and self-reflectivity. While the triple-A industry is often challenged for its gendered and exclusionary dynamics, its parodies should not be immune to the same kind of scrutiny. The case of IndiE3 analyzed by Vanderhoef shows that the absence of critical discourse might equally be shared by events and projects that claim to be alternative and oppositional.

Alison Harvey's Chapter 16 provides a close look at how higher education students re-interpret independence as part of their career narratives.

Harvey's analysis is based on a piece of qualitative research, inclusive of 49 interviews and direct observation, conducted in five higher education institutions in the United Kingdom. The geographical context is significant. First, the UK is in the process of leaving the European Union, which is likely to affect the freedom of movement of workers, and which is indirectly influencing the aspirations of British and foreign students in this country. Second, in the UK, the relation between industrial sectors and higher education is the subject of a plethora of documents and reports produced by policymakers and stakeholders. These draw a picture of the imagined student of a game design degree and in turn shape the students' expectations of what kind of position they should occupy by the end of their studies. Education ends up being structured around the norms and values of the triple-A industry. Independence, Harvey notes, emerges as a desired aspiration and end-goal in the interviews, particularly when students refer to the kind of diverse and creative game design they wish to see in the sector and when imagining their own future careers. However, while identifying the structural and systemic issues of the industry that thwart their hopes of independence, students have an individualized imagination of their careers where independence is only seen as either a privilege or a strategic move while building a portfolio. Harvey concludes that collective action, such as the union organizing described by Woodcock in the Politics section of this collection, could provide the right opportunity to make students' dreams of independence more viable. In the meantime, the "impossibility of indie" directly affects the possibilities of a more diverse, innovative, and experimental game sector in the UK, achieving the opposite result of what policymaking institutions would have hoped.

Conclusion

This edited collection should prove valuable for anyone who is willing to understand the new dynamics of production of independent videogames and their surrounding culture. However, it also has many limitations. There are broad areas of investigation which have not been touched on at all here or mentioned only tangentially or in passing. For instance, the types and attitudes of consumption of indie games, their aesthetic, and their textual strategies have attracted the attention of international scholars in the past one or two decades. There are only minimal traces of those areas of investigation in this volume. This collection is certainly not exhaustive, and the shifting contexts of production will likely make some details outdated fairly quickly. However, it is hoped that the directions of research, questions, methods, and concerns outlined in this collection will endure and provide inspiration for further research. The main concern of this editorial project has been to highlight the importance of understanding how games are produced. Without paying attention to the material and discursive elements involved in the production of videogames, it becomes difficult to understand their cultural, creative, economic, and political potential. Independent games have made this fundamental concern more prominent and almost unavoidable for *game studies* – intended in the broad sense of the study of (video)games and the various ways in which humans and nonhumans become involved in them.

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Part I

Cultures

2 Decoding and recoding game jams and independent game-making spaces for diversity and inclusion

Aphra Kerr

The growth of independent games production is often associated with a discourse that anyone can now make games because of the widespread availability of free game-making tools and new distribution channels. This discourse parallels a mainstream discourse that coding is a core skill in contemporary economies. These discourses emanate from Information and Communications Technology (ICT) companies, but also from European governments which invest significant efforts in promoting science, technology, engineering, and maths (STEM) as an occupation and educational choice. The efforts aimed at attracting children and teenagers into STEM often use hackathons and game jams. Despite these discourses and promotional efforts, the numbers of women and minorities in technology-based industries have stabilized and are going down in many countries, including Ireland. This is regardless of the increase in female participation in the workforce overall. Clearly, technology cultures, education, and occupations, of which games is a part, have a persistent issue with attracting and keeping women and other minorities. This is the wider context that shapes any initiative aimed at encouraging greater participation by women and others into independent and amateur games education and production.

This chapter presents the findings of a collaborative research project exploring diversity and inclusion in game jams and game design workshops conducted in different cities in Ireland between 2016 and 2019. It focuses on game jams as a site of significant "game labor politics" (de Peuter and Young, 2019) – particularly in relation to inclusivity, gender, and approaches to making games. While the Global Game Jam (GGJ) provides the dominant template for a game jam, local instances of game jams are shaped by both global and local formations. Many of the first academic papers on game jams presented them as a positive and inclusive space for collaborative cultural production, innovation, and networking. Game jams arrived in Ireland over the past decade alongside free to download game-making software and a proliferation of local participation in online game-making groups. And yet my observations at local game-making events and game networking events jarred with the dominant discourses of widening participation and inclusion. The game-making events I attended in Ireland seemed to be populated by the same white, male, young faces that I had interviewed a decade earlier in professional companies. If the tools and distribution channels are democratizing and game jams are a social space where people of all skill levels can freely come together to collaboratively learn and create, why are the demographics of those attending these events so similar to the demographic profile of the mainstream games industry? And how independent are they really from the global structures of the games industry?

This chapter first reports on research conducted at three adult game jams which took place in three different cities in Ireland. These events were informal, organized by volunteers and usually attended voluntarily. The game jams were free to attend and open to everyone over 18 years. The chapter discusses who participated and their motivations for participating. It outlines how the design of these events can implicitly and explicitly exclude women and others and, in turn, may reproduce gendered representation in the wider game-making community and industry. The chapter then draws upon our experiences of organizing six adult female-friendly game-making workshops in two different cities in Ireland. It reflects on strategies to overcome recruitment and participation barriers and our reasons for organizing female friendly, rather than women-only events. It proposes that the techno-spatial assemblages created in informal learning spaces can lead to the production of particular code/spaces (Kitchin and Dodge, 2005) which can limit the effectiveness of diversity interventions. Finally, the chapter reflects on how informal educational activities are shaped by the international political economy of the games industry.

The findings on both the game jams and workshops are based on surveys of attendees, observations, informal interviews, and an analysis of relevant online resources. Much of the academic literature on game jams is positive about their role in game development and innovation. However, relatively little of this literature attends to issues of non-participation or the demographics of those who attend. Similarly, the literature on independent games production and the rise of free to download game production tools focuses on the democratizing potential of these tools, rather than empirically mapping if their availability really is democratizing, and for whom. The chapter concludes by asking the reader to consider if some informal events, like game jams, are in fact reproducing both local, and distant, gendered, raced, and classed production cultures.

From game development to game-making cultures in Ireland

Game studies has a strong legacy of work focused on both professional and semi-professional industry and production studies (Dovey and Kennedy, 2006; Kerr, 2006; Dyer-Witheford and De Peuter, 2009). In the last decade, new game production logics have broadened beyond the processes, roles, and distribution channels of established triple A and online PC games

(Kerr, 2017). This includes the development of new distribution channels for game-makers, ranging from the Android and Apple application stores to specialist websites like itch.io. While independent game start-ups are often supported by local and national industry programs, they co-exist with more informal creative collectives and communities of practice (Kerr, 2017: 159–164). These range from city-based clusters of game developers in the UK (Guevara-Villalobos, 2013) and Australia (Keogh, 2019a) to dispersed individual modders (Jarrett, 2019). Independent game developers range from DIY "craft gamemakers" (Westecott, 2012), to amateur "everyday developers" (Vanderhoef, 2016: 35), to "everyday gamemakers" which includes professional, amateur, and player game-making (Young, 2018).

Despite the emergence of a discourse of independent game developers, many independent game developers replicate the processes and game genres of established commercial studios (Ruffino, 2013), and demographic statistics on professional employment in the games industry in North America and Europe finds that it remains dominated by young, white, heterosexual men. The data on the employment and pay differentials in core development roles in the professional industry in the US, Canada, the UK, and Ireland remains stubbornly gendered and raced, particularly for programming and engineering roles. Surveys by the International Game Developers Association (IGDA) of full-time and freelance game developers over the past decade give us some insights into diversity and inclusion issues. The latest survey (n = 996 respondents) found that while workers and companies in the industry recognize the importance of diversity in relation to representation in the workforce (81%) and content (85%), respondents felt that there was unequal treatment of workers in the industry, and a sizeable minority reported experiencing or witnessing unequal treatment (IGDA, 2018). These responses came from respondents who are themselves not very diverse – male (74%), white/Caucasian/European (68%), and heterosexual (81%). Many respondents did not know if their company had a diversity or formal complaints policy. We continue to see barriers and backlash against opening up game development to a wider range of perspectives and games, as #gamergate demonstrated (Shaw, 2012; Mortensen, 2018).

A government commissioned report published in 2017 in Ireland estimated that the audio-visual sector supported almost 17,000 full-time worker equivalents and just over 10,500 directly (Olsberg SPI and Nordicity, 2017). They found that just over 2000 people were employed in development and publishing of video games – but that the vast majority were employed in distribution-type activities including customer support and localization. This largely confirms previous research which tracked the undulating growth of professional employment in game development, and the rapid growth of below-the-line game support and distribution functions in Ireland (Kerr and Cawley, 2012; McCormick, 2012). Throughout the 2000s, the bulk of employment growth in Ireland was in support, rather than development roles. Consultancy surveys rarely give detail on the demographics of developers, but Kerr and Cawley (2012) note that the demographics of development companies in Ireland largely mirror the findings of the IGDA survey in terms of gender, race, ethnicity, and age. Over 90% were aged under 35 years and the respondents employed less than 13% women, falling to 7% when one looked at core development roles. The majority of the game companies who responded were less than five years old. In 2011, the Irish government launched an action plan for the Irish games industry with a focus on incubators, early stage investment, and education (Forfás, 2011). A local trade association 'Games Ireland' was established by indigenous and multinational companies, and for three years, it ran an annual 'Games Ireland Gathering' (GIG) conference. By 2011, there were almost 30 graduate game development and game technology courses in universities and colleges around Ireland.

As Keogh (2019b) has pointed out, formal surveys and action plans largely focus on established companies, multinationals, and the talent pipeline. As in other countries, the decade, since 2010, has seen the graduation of game-making students in Ireland and the growth of non-professional game-making. Participant observation and direct engagement in the local game-making scene by the author in Ireland would suggest that there are many more non-professional "everyday gamemakers" than those formally employed in games. Some are very early stage and not formally registered as a company. Some are surviving on social welfare. Some are working full time in banking, retail, or social media companies. Some are still students. These game-makers are an important part of the local games community. Game developers, in general, and game programmers, in particular, move between the large and small companies, contracting for technology companies to pay the rent and working independently to develop their own game projects. Sometimes, the same people are founders of multiple companies. The large and small companies combined with freelance and part-time game-makers have given rise to a vibrant local game production culture – although perhaps national, rather than local, is a better term to use. While many of the larger professional companies are located in the cities of Dublin or Galway, other game-makers are spread around the country enabled by the spread of broadband and seeking lower living costs.

Independent game developers are often self-employed or working in micro companies of less than five employees. They are distributed throughout the island of Ireland, and they rely on non-formalized forms of face-to-face and digital social networking to meet. Most cannot afford to pay formal membership fees for national trade or professional associations. As in other creative industries, annual face-to-face gatherings can provide appropriate levels of "buzz" and networking opportunities to generate new employment prospects, new projects, or move old ones along (van Egeraat et al., 2013). However, what has been striking are the number of ephemeral face-to-face and online activities that have emerged including a range of festivals, game jams, and online groups (Shepherd and Kerr, 2014). These activities are all the more interesting, given that the formal professional associations, including the Irish chapter of the IGDA and the trade association Games Ireland, are currently inactive. In their place, a local not-for-profit association focused on game-makers has emerged. Imirt – which is the Irish word for play – is the Irish Game Makers Association, and it specifically positions itself to represent developers and makers of both analog and digital games, in line with the broader game-makers term used by Young (2018). It charges a small membership fee and has an elected board.

Local game jams – open to everyone?

The growth of temporary social game-making events like game jams are not unique to games, and we can find such activities in music, arts, and crafts. Mark Banks (2007) argues that non-market clusters in music, art, and design signal a "re-moralization" of social and economic activities and provide support against the increasingly individualized risks associated with commercial cultural production. It would be a mistake however to think that all bottom-up informal social making events are opposed to the market, and, in our research, we noted that independently run game jams can be shaped by the (sometimes competing) interests of sponsors, venues, software, and the expectations of the attendees. Indeed, attendees at informal game-making events are also subject to the unrelenting pressure in contemporary cultural economies for students to find internships, for freelance workers to develop extensive portfolios, and for everyone to continuously upskill in the latest tools.

Early game studies scholars highlighted the benefits of teaching people how to make games (Kafai, 2006). Today, there is considerable literature on game jams within game studies and the education field. The best-known contemporary game jam emerged from within the professional games industry. The annual GGJ was established in 2008 by the IGDA. The GGJ website describes the event as a "hackathon focused on game development", where game development is "condensed into a 48-hour development cycle" (GGJ, 2019). The website also notes that, in 2018, the annual event took place in "803 sites in 108 countries and with 42,800 registered participants". These produced and uploaded 8,606 games. One academic defines game jams as "accelerated and constrained" forms of collaborative game-making (Kultima, 2015). Attendees must design a game in a pre-defined length of time on a theme that is announced at the start of the day. Research on game jams would suggest that they are a useful way to motivate people to learn content, technical, and collaborative skills (Kultima, 2015, Locke et al., 2015). This work also found that there are many different organizers of game jams, including the games and technology industry, universities, and schools. A number of Irish universities and companies host GGJ events each year, but it appears that most are run by volunteers.

Rather than focus on the GGJ events, our research focused on an Irish-based independent not-for-profit organization called GameCraft, which runs four to eight game jam events each year. GameCraft was established in 2012 and uses the tagline "Connect, Create, Collaborate". Its website states that "GameCraft is a games jam event designed around building the gaming community. We aim to create events which allow game-makers to meet, share ideas, have fun, compete for prizes and most importantly make games!" (Gamecraft, 2019). It was established by two female programmers and is still predominantly run by one of them. A number of developers, makers, and academics serve on the board. GameCraft has been invited to organize events in London, New York, Paris, and Vienna, but, over the past six years, most of their events have taken place in Ireland. We focused on GameCraft, because it differs from GGJ in a number of important ways. First, GameCraft events are shorter at 10–12 hours. Second, the event has a prominent code of conduct. Third, it encourages and provides materials for non-digital game-making and crafting. In all other respects, it appears to conform to the standard template – sign up is online, a theme is announced at the start of the day, people can work in teams or alone, and there is an end of the day play session, voting, and prizes. The website states that no prior experience is required to attend.

The author has attended a number of GameCrafts since 2013, but this chapter draws specifically on the findings of survey and observational research conducted in three different cities in Ireland between 2016 and 2019. Of the total attendees at the three events, just over half, or 53 in total, completed the survey. Our results largely confirmed our observations. Most respondents at the events identified as male (77% in Dublin, 93% in Limerick, 80% in Cork), white (92% in Dublin, 93% in Limerick, 85% in Cork), and straight (77% in Dublin, 93% in Limerick, 85% in Cork). Respondents in Dublin were fairly evenly distributed between different age groups, while over 85% of attendees in Limerick and Cork were aged from 18 to 24 years. These findings mirror the findings of studies of those who attend the GGJ. A 2013 survey of GGJ participants found that participants were 86% male, 56.5% were aged 21–29 years, and 60% had a college or degree-level qualification (Fowler et al., 2013).

Many of the attendees of these game jams were studying or had a college or degree qualification. Most of the students were studying programming or game technology. Of those that were not studying, most were working at least part time in the IT industry. This "open to everyone" event, which had sought to attract individuals with no game design experience, attracted mostly males who were already studying games or working in the IT industry. A majority of attendees at both events were programmers (85% in Dublin, 57% in Limerick) or studying game development and game technology (81% in Cork). When asked about their motivations to attend, most respondents said that they were trying to improve game-making skills (92% in Dublin, 93% in Limerick, 69% in Cork) and to meet others in the

Irish games making community (92% in Dublin, 64% in Limerick, 31% in Cork). Of note is that some of the students attending the game jam in Cork obtained formal course credits for attending. The fact that these events were held on university campuses, and some students had to attend for course credit, further reinforced the number of technical students attending and the blurred lines between formal and informal education in this context.

Of interest in terms of inclusion is that word of mouth, course lecturers, and game-specific online social media groups emerged as equally important in terms of recruiting people to attend these events, and many people arrived with friends and pre-formed teams. It appears that 'insiders' who are within existing social networks were first, hearing about the events, and second, felt secure in attending because they knew other people who would be attending. It was also striking that many already had some level of technical and game-making skills, and some had released their own social or educational games. While beginners were welcome to attend, few attendees were absolute beginners. Further, people had to bring their own equipment – and clearly this might also had constituted a barrier to attendance for some. These findings mirror those of recent studies which explore why people do not attend game jams. Key barriers include the timing of events (weekends/length), the location, the cost, and a fear of turning up and not being skilled enough to participate (Preston et al., 2012; Meriläinen and Aurava, 2018; Meriläinen, 2019). However, these barriers also reflect research on barriers to adult education more generally. Boeren (2011) for example, found significant gender and class barriers to participation in lifelong learning in their analysis of the European Adult Education Survey. They distinguish between situational (work/life balance), institutional (timing, location), and dispositional (self-confidence) barriers to life-long learning and help us to situate game jams within wider educational research. In their analysis, cost and caring responsibilities were the most significant barriers to participation in lifelong learning for women.

Existing game jam research rarely comments on the extent to which these events rely upon volunteer labor. GameCraft relied upon significant unpaid and largely unseen labor by one key female volunteer, with support from mostly male full-time academics and industry representatives. The main organizer has an MA-level qualification in programming and is highly experienced at organizing IT and games events. She takes care of the technical, catering, and communicative structure of the events and takes care of the set up and well-being of attendees on the day. She is also the person people report any misconduct to. This role involves a large degree of face-to-face affective labor (Kennedy, 2018). More free labor and technical knowledge are embodied in the free and open source software used to advertize the event, run the sign-up process, and post the games after the event. Further, the attendees can be conceptualized as aspirational labor in terms of their temporal investment in self-training and social networking. The organization of these events, and their overall contribution to the "local ludic economy" (Kennedy, 2018), was completely reliant on the voluntarily given, and freely taken, labor of the main organizer.

Our surveys, observations, and interactions with attendees at these game jams helped us to understand the demographics and motivations of participants and the invisible labor involved in staging a games jam. These findings suggest that the communication channels and messages employed by GameCraft were very successful at attracting young males who specialize in programming and are already interested in games, but were not so successful in reaching outside gaming communities and pre-existing social networks. Attendees were those with the requisite social, economic, and gaming capital and were insiders in the local game-making culture and word of mouth social networks. They were also those who were able to take advantage of the social networking opportunities that such events offer – becoming familiar with local formal educational institutions, companies/sponsors, and other game-makers. Local games or IT companies sponsored the prizes and sometimes promoted internships and job opportunities at the events.

The findings from these game jams point to a stark-gendered division of labor with the female, organizational and communicative work largely unseen and unvalued, and the largely male programming and creative outputs at the events celebrated and often visible afterward online. Even though these events were shorter than the GGJ, they were still attracting a very narrow demographic, while the focus on making things work within a very short time frame places a priority on programming skills over other types of game-making skills. The intense day or weekend-long game jam format reproduces the intense, iterative, and agile production models adopted in the software industry, but is also reflective of wider patterns of project work in the creative industries. Based on this evidence, we would have to state that informal independent game jams are not necessarily an inclusive social space for learning, and the events can be heavily structured by local gender, race, and class structures. Further, if game jams are positioned as an entry point into the games or IT industry pipeline, they may be simply reproducing the problematic demographic structures already identified in the wider games industry.

Recoding inclusive game-making events

Respondents to our surveys were aware that GameCrafts were not very diverse – they noted that women, people over 30 years of age, and "non-Irish" people were missing. They were at best ambivalent about how to address this, and many did not want their game jams specifically addressing diversity issues. However, our research project was informed by a commitment to engaged research and the feminist perspectives of academics and activists participating in the Refiguring Innovation in Games (ReFiG) research project. This five-year project aimed to intervene in game-making cultures, education, and industry to improve diversity. As a research partner, we endeavored to diversify the game-making community in Ireland through the development of more inclusive informal game-making events.

Many of our ReFiG partners had experience running single and mixed-sex after-school clubs, women-only incubation centers, and game jams for women and LGBTQ participants (Fisher and Harvey, 2013; Harvey and Fisher, 2014; Kennedy, 2018). Our local collaborators in Ireland had experience running 'female-friendly' programming workshops. These local events encouraged female attendees to bring along male allies, and those who identified as queer, non-binary, or trans. Our focus on equity of opportunity, and interventions for social change, was at times starkly different from the increasingly corporatized international approach to diversity and women in technology events. We also wanted to be independent of international and local technology diversity programs. Experience has shown that events focused on equity and social change are not always welcome. Some local event organizers have been criticized for organizing events solely for women and discriminating against others. Some local game-making themes have been viewed as too political. For example, a local independent game jam that focused on the 'repeal' abortion referendum in 2018 in Ireland had problems finding a venue to host them.

Following Boeren (2011), we can divide the key barriers to participation in local game jams into three main types: institutional (timing, length, location, venue), situational (life stage, caring responsibilities, income, access to technology), and dispositional (self-confidence, knowledge of games, knowledge of local games culture, peer network). We felt that we could address the first two sets of barriers in the organization of our events and the third through the content of our events. In the summer of 2016, we ran three beginner and female-friendly workshops for adults on successive weekends in Dublin city, and we repeated them in Galway city in 2018. We had 30 attendees in Dublin and 33 in Galway. The half-day workshops focused on game design, narrative design, and game coding, and many attended all three. Across the workshops, tutors introduced a number of software and game engine tools that independent game developers use including Twine, Fungus (a Unity plugin), and the industry standard tool, Unity. Elsewhere, we detail how we addressed the structural barriers to participation in game jams (Kerr and Savage, 2020), and we have produced a free to download manual and roadmap (Kerr et al., 2020). Key to attracting a more diverse set of participants were: learning from diversity research conducted elsewhere; advertizing outside online game channels; partnering with local diversity champions; monitoring signups for diversity; and changing the when, where, and how of the event format. Key to successful learning outcomes were carefully considering pedagogical and tutoring strategies, having low technology learning supports, pacing, and questioning the tools and examples we used to teach. We successfully managed to diversify participants in our workshops and attracted a majority of female and older attendees from a variety of creative backgrounds, from jewelry making to graphic designers. And feedback from exit surveys was overwhelmingly positive. One person stated that they "loved the day, very warm and welcoming". Another "excellent hands on course. Well done to all". Collaboration was encouraged, competition was not. There were no prizes, deadlines, or ticking clocks. The tutors, organizers, and research assistants were paid for their time and expertise.

Our workshop goal was to work with local organizers, tutors, and diverse participants to create inclusive game-making spaces. Our interactive fiction and board game design workshops appeared to largely achieve these aims. However, we continued to encounter significant techno-spatial barriers when we attempted to run our game coding workshops. In an attempt to overcome the taken for granted requirement that learners "bring your own computer", we partnered with a local innovation hub in one instance to use their training center. However, the center had its rows of computers bolted to the tables – quite literally, the spatial layout presupposed an individualized learning experience with an expert tutor guiding the process. The training computers were too slow to adequately run the educational version of the game engine Unity – revealing another taken for granted presumption of the game engine developers. In addition, the organizational labor required to support the coding workshops was significantly greater than the other workshops. This involved downloading and pre-installing the software, setting up individual student accounts, and having extra support tutors in the rooms. Yet again, we were struck by the dissonance between the free game-making tools aimed at democratizing game development and the invisible labor and barriers to actually making something with these tools. This free software only served to assimilate the tutors and the participants into the wider rationalistic, generic, and data-driven economies of the professional game-making software industry. Unity was developed in 2005 as a tool to create a first-person shooter game and then became a separate business product for the games studio. The company aimed to democratize game development and provides a range of pre-designed kits, assets, and drag and drop as well as coding functionality. While educational licenses are free, the software charges larger development companies, a subscription fee based on game revenues. It has become the *de facto* tool for independent mobile, virtual reality, and PC game development. The Unity game engine now supports the development of content for 25 different platforms and is being used to develop content in other audio-visual industries and related fields.

The increasing dominance of game software tools like Unity in formal education and the games industry demands more analysis based on our experiences with informal learning workshops (Nicoll and Keogh, 2019). Our techno-spatial learning environments can be conceptualized as code/spaces in the sense that Kitchin and Dodge (2005, 2011) developed the concept. For them, code/spaces are assemblages created by the interaction of computer code and spaces of deployment. In our learning environments, this assemblage includes coded objects, infrastructures, and processes. While game engines might make game development more efficient, this efficiency comes at the cost of foreclosing some pedagogical approaches, design options, and ways of working. Users trade development time and design choice for precoded sets of assets, assumptions, and relationships. Engines are dominated by mathematical logic (Freedman, 2018) and exert "authorial power" over the process of games development, as Malazita (2018) has argued. In our workshops, a significant temporal dissonance emerged when we tried to introduce novice adult game-makers to Unity. For beginners, the pace of the session was too fast. Feedback forms noted that the interface was not intuitive. Attendees wanted paper handouts. Peer learning and collaborative supports were restricted by the layout of the room. Many of these attendees had significant everyday computer experience and were working or studying in university. These potential game-makers were struggling to conform to the prescription of what constitutes an everyday developer.

In the final analysis, this chapter would like to suggest that both informal and formal learning environments need to reconsider the empowering and democratizing discourse associated with game engines and tools (Keogh, 2019b). While clearly these tools are used by an increasing number of independent game-making companies and individuals to create games, they impose significant technical and creative limitations on game-makers and presuppose certain gaming, technical, and design capabilities. They also require users to establish accounts and, by so doing, incorporate users into a transnational economy of datified learning. Research and interventions for inclusion need to consider how game engines transduce and, in some cases, conflict with feminist and critical pedagogical objectives and creative autonomy. Using game engines developed by the global games industry in highly structured local teaching spaces imposes significant technical and social constraints on creativity and innovation and may further contribute to the continued lack of diversity and inclusion in everyday game-making.

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3 Queering indie

How LGBTQ experiences challenge dominant narratives of independent games

Bonnie Ruberg

This article argues that the notion of "indie" in indie games is a specific and limited cultural construct – one that can be made visible and radically destabilized when considered alongside the experiences of LGBTQ (queer) independent game-makers. Rather than being a distinct category of videogames or game production, indie is best understood as an imaginary: a set of dominant narratives about what independent games are, who develops them and how, and what makes these games meaningful. Here, I draw from a series of interviews with contributors to what I refer to as the "queer games avant-garde", a vast and broad contemporary network of queer and transgender independent game developers, to demonstrate how these dominant narratives fail to account for significant portions of indie game-making. The experiences of the queer and trans game-makers interviewed also reveal how queer indie game-making, as a broader artistic and subcultural practice, resists and reimagines the problematic ideological underpinnings of "indie". It brings into question the very identity of indie games, challenging notions of authenticity, diversity, and even independence. It offers alternative, queerer, and more socially just visions of what independent games might be, while also provocatively suggesting that the true work of "indie" may be to abandon independence or destroy videogames entirely.

The interviews I draw from here appear as primary source material in my book *The Queer Games Avant-Garde: How LGBTQ Game Makers Are Reimagining the Medium of Video Games* (2020), which contains roughly two dozen interview-based profiles of queer and trans independent game developers. Most of these interviews were conducted in 2017; some were conducted in 2018. As an interviewer, my central goal for these conversations was to move beyond typical narratives of diversity and inclusion and explore the politics, aesthetics, and personal stakes of making queer games. There are many possible ways of studying queer indie games, such as by analyzing these games themselves (e.g. Pow, 2018), situating them within historical contexts (e.g. Harvat, 2018), or reflecting on queer gaming communities (e.g. Pozo et al., 2017). For this work, I chose to focus on how queer and transgender game developers describe their own work, placing value on the insights of marginalized creators themselves. The developers interviewed for *The Queer Games Avant-Garde* represent only a fraction of the hundreds of queer and trans creators currently making digital and analog games. Queer game-making is often misunderstood as niche, but it is in fact the vanguard of independent game development today.

To demonstrate how the experiences of queer and trans indie game-makers challenge the dominant narratives of indie games, I first articulate the characteristics of what I call the "indie imaginary". For this, I turn to Jesper Juul's 2019 book *Handmade Pixels: Independent Video Games and the Quest for Authenticity*, as well as Stephanie Boluk and Patrick Lemieux's 2017 book *Metagaming*. A number of the terms I use in this article are, by nature, difficult to define. I talk more at length below about the definition of independent games. Though "independent games" and "indie games" are often used interchangeably, here, I most frequently use "indie games" or more simply "indie". First and foremost, I am concerned with this constructed idea of *indie-ness*: its contours, its biases, and the ways it can be shifted. In my experience, the shorthand "indie" – punchy, cool, with the air of insider tech speak – captures this sense of independent games not as a stable constellation of works or creators but as an ethos. I also frequently use the term "queer games". By this, I mean digital and analog games, commonly scrappy and experimental, made by (and often *about* and *for*) queer and transgender people. As a category, queer games encompass a huge spectrum of works. As of this writing, there are more than 1,000 games marked with the "LBGTQ" tag on the online distribution platform Itch.io alone. The creators whose interviews I draw from here are likewise very different; each speaks from their own unique experiences of intersectional identity. I am not suggesting that queer games should be considered a unified category or that queer game-makers should be seen as one specific community. However, as these interviews show, there are common threads that cross these interviews, which powerfully resist the stories that are typically told about independent games.

Imagining indie

What makes a game "indie"? I ask this question not because there is one clear or correct way to define indie games but precisely because – as we will see, and as scholars like Maria B. Garda and Pawel Grabarczyk (2016) have argued – the very definition of indie is shifting, uncertain, often contradictory, and fundamentally an imagined *construct* rather than a fixed category. Identifying the characteristics of this construct is important for understanding how queer indie games and their creators do not fit these dominant ideas of what makes a game or a game developer "indie".

One way to define indie is through indie games themselves. When the term "indie" first became a buzzword in the games industry and sectors of gaming culture in the mid-2000s, it was used to describe a specific paradigm of videogame distribution, namely developers releasing and distributing videogames "independently" from secondary game publishers. Yet, as Stephanie Boluk and Patrick Lemieux explain, somewhere around the turn of the decade, with the release of games like *Braid* (2008) and *Super Meat Boy* (2010), "the term indie game ceased to function solely as a label for a particular mode of independent production or digital distribution and became the common designator for a genre of videogames with a shared history and common aesthetic" (2017, 28). Today, the criteria for what counts as an indie game are increasingly messy and multiple. To some, an indie game might be one developed by a small team (though many so-called indie games are made by increasingly large teams), or one distributed using seemingly independent platforms (though major corporate players like Nintendo and PlayStation now distribute independent games), or one that appears in indie game venues like IndieCade (though IndieCade now partners to showcase games in decidedly non-indie spaces like the Electronic Entertainment Expo). As Juul writes, "Independent game and indie game are labels for which there are no agreed-upon definitions, yet they play a large role in game development and consumption" (2019, 7). Thus, the concept of indie games remains compelling even as the definition of "indie" is itself messy.

Another possible way to define the "indie" in indie games is through the ethos of newness. At their core, writes Juul, independent games are those which "contain a fundamental newness: they are about playing in new ways, solving new problems, solving old problems for new reasons, being free to ignore something we used to have to do, or framing video games in a new way" (6). Juul's description of this "fundamental newness" is in keeping with the larger practices, found in both game industry space and popular reporting on games, of associating indie games with innovation. Most commonly, this means mechanical innovation: new spins on game mechanics taken from classics of the medium. The widely popular indie game *Undertale* (2015), for example, received accolades for reimagining the fighting mechanics typically found in role-playing games. However, upon closer inspection, it becomes clear that many games celebrated for their newness are, in other ways, far from progressive or are stripped of their progressive implications by their reception (Ruberg, 2018). Using *Braid* as an example, Boluk and Lemieux describe how the game participates in long histories of "obsessive, masculine desire", which manifest in "the ludic challenges of retro videogames, the courtly love quest for an unattainable princess, and the scientific models of technological progress that drive the military industrial complex" (2017, 28). That is not to say that all indie games fit this description; many truly experimental, radical, and joyfully strange works are being produced under the banner of "indie". Yet, as I have argued elsewhere, "the best-known indie designers are still white, cisgender men, and the most popular and widely celebrated indie games rarely challenge the status quo of limited diverse representation" (Ruberg, 2019b, 87).

In truth, indie games are so hard to define because "indie" is not one clear category or concrete community. Instead, it is an imaginary. Indie is best understood as a specific set of ideas or dominant narratives about independent videogames, the people who make them, how those people do their work, and what makes these games meaningful. Common cultural reference points for this indie imaginary, like *Indie Game: The Movie* (2012), present a very specific vision of indie game-making – one in which indie developers are lone white men (Boluk and Lemieux, 2017, 33) and the trajectory of developing indie games is overcoming obstacles to attain financial success (Juul, 2019, 12). The indie imaginary also tells certain stories about how one becomes an independent developer. At video game industry events, it is common to hear developers talk about "going indie". In this context, "going indie" is understood to mean quitting one's job in the mainstream videogame industry to develop games independently. Among game developers, it is often seen as a bold and enviable, if potentially foolhardy, move: to quit the grind of crunching on a AAA or mobile game for a huge company, take a gamble, and commit full-time to that passion project that has always been on the back burner. Yet only certain people have the privilege of fitting this narrative. Before leaving the mainstream game industry, one must first be hired in the mainstream game industry, which notoriously discriminates against women, people of color, LGBTQ people, etc. One must also have enough financial stability and money saved up to step away from a steady job. Thus, the dominant narrative of how a developer *becomes* indie feeds into the imagined notion of who *is* indie: straight, cisgender, white men who can take a gamble and making something "new". This, as Sarah Sharma writes, is the privilege of "exit": being able to walk away from a problematic situation or system (Sharma, 2017).

Ironically, at the same time that the indie imaginary presents a vision of indies as straight white men (often singular, self-fashioned "auteurs"), indie games are commonly imagined as being more diverse and inclusive than other types of videogames. Commonly heard wisdom posits that, because indie games are made on smaller budgets than large-scale, AAA titles, they can be less traditional in their content, creating opportunities for exploring diverse representation that mainstream development studios might avoid for fear of pushback from their core player-bases (Fron et al., 2007). To an extent, this imagined dynamic is accurate. Yet, as the experiences of the queer indie game-makers discussed below highlight, the reality of diversity and inclusion in indie games is far more complicated. This is one of the key ways that queer indie games and the experiences of their creators challenge the indie imaginary, bringing into focus the very fact that "indie" as we know it *is* an imaginary: a limited and limiting construct that privileges some experiences while leaving many others out.

Complicating narratives of diversity and inclusion

Perhaps, the most obvious way in which the work of the queer games avant-garde challenges the dominant narratives of indie games is by showing that, contrary to the indie imaginary, many different types of people make videogames. In a way, this is a realization of the future that Anna Anthropy called for nearly a decade ago in her influential 2012 book, *Rise of the Video Game Zinesters*, where she writes, “What I want from videogames is a plurality of voices. I want games to come from a wider set of experiences and present a wider set of perspectives” (8). Indeed, as a game-maker, Llaura McGee states in the interview I conducted with her, “What’s more important than the fact that I’m making a game about being queer or trans… is the fact that, as a trans, queer person, I’m making games at all” (70). As the contributors to the queer games avant-garde evidence, there are many more types of people developing indie videogames than those who fit the standard vision of the indie imaginary.

At the same time, these queer indies themselves issue reminders to be wary of narratives of diversity and inclusion. Such narratives are often self-congratulatory and un-self-critical, and they threaten to integrate “diverse” people into an already problematic and exploitative games industry (Ruberg and Phillips, 2018). Indeed, as game-maker Kara Stone describes in her interview, sexist attitudes associated with broader gaming cultures still regularly rear their heads in community spaces associated with indie games. Stone tells this story:

> I showed my game *Cyber Sex Adventure* at IndieCade a few years ago. Because it’s about sex and I was standing right there, apparently people felt like I was inviting them to talk to me about sexuality and comment on my appearance. When I’m physically present in games exhibition spaces, I also get grilled a lot about my expertise—like, “Did you program this? Do you know a coding language?” Because I’m femme, people doubt that I know what I’m doing.
>
> (129)

Experiences like Stone’s prompt us to reconsider whether the world of “indie”, as it is commonly imagined, is indeed more inclusive than the games industry more broadly. Stories of this sort highlight how discrimination and marginalization still regularly manifest even at events like IndieCade, which has a stated goal of “helping to create a public perception of games as rich, diverse, artistic, and culturally significant” (https://www.indiecade.com/about/).

Even as the queer and trans indie game-makers interviewed for this project seem to represent the diversification of “indie”, they describe intentionally complicating notions of diverse representation through their games.

Both Aeevee Bee and Mattie Brice reflect on the need for queer and trans people to tell their own stories because they are aware of the complexities and frictions around queer identity. "Representation can't just be a list of identity categories", states Bee, who cautions against the umbrella of queerness as an "amorphous blob". "It's not really representation unless you're creating complexity… Our experiences are specific to our lives. Focusing only on identity, especially identity without experience, reduces everyone to an abstraction" (56, 57). Says Brice, reflecting on her time with other queer game-makers in the San Francisco Bay Area:

> There are a lot of assumptions that, because people are queer, we should automatically get along. In my experience, that wasn't the case. In hindsight that's obvious, but when you're a member of a minority, you feel like you need to gather people around you who are like you.
>
> (140)

These statements reflect a productive tension within the work of the queer games avant-garde, which both values queer community and seeks to deconstruct the tokenizing, flattening visions of LGBTQ experience that are associated with simple pushes for "representation".

Indeed, the experiences of queer and trans indie game-makers encourage us to move away from neoliberal, corporate interest in *diversity*, and toward a more nuanced consideration of *difference* in independent games. This difference manifests not just as the difference between indies and AAA developers – that is, the core difference through which "indie-ness" is commonly defined – but also as a difference within the landscape of indie game development. Says McGee, pointing out the striking socioeconomic disparities between different types of indie game developers:

> Indie games are this weird space where there are marginalized people right next to people who are super well off… I wish there was more of a culture of financially successful [indies] giving back, because the people who are doing the really experimental stuff are definitely not making a living wage.
>
> (72)

Another difference that the experiences of these queer and trans game-makers bring to the surface is the different ways in which developers *become* indie. To fit the narrative of "going indie", a developer must already work in the game industry and then, at least temporarily, sever themselves from it. By contrast, Stone describes a very different path into indie games:

> I thought I was going to keep making experimental video art, but I randomly went to a talk by someone from [the Toronto-based organization] Dames Making Games. I was like, "Oh my god, women can

> make video games?" I had never heard of indie game designers…. After the talk, I rushed over and asked how I could get involved. They were hosting a ten-week workshop and I signed up. I've been following the current of video games ever since.
>
> (128)

Relatedly, many of the game-makers interviewed for this project report working not *independent from* but rather *entangled with* the mainstream game industry. Mo Cohen, for example, explains that they fund their indie game-making by working a day job as a programmer. Seanna Musgrave is both an experimental VR designer and a VR developer for larger studios that do not fall under the indie umbrella. This suggests that the "independent" in independent games may itself be more an imagined construct than a lived reality. It also indicates that only certain people may have the privilege of so-called independence.

Videogames that don't care about videogames

In addition to demonstrating the limited and problematic nature of the indie imaginary, the work of the queer games avant-garde also brings into question the fundamental ideologies that underlie both the dominant narratives of indie games and the ways that scholars have, to date, identified what makes these games meaningful. Here and in the sections that follow, I turn again to Juul's *Handmade Pixels*, which helpfully articulates key features of the indie ideology. While there are many elements of Juul's text that can be fittingly applied to the queer games avant-garde, I focus here on the characteristics of indie games, as Juul articulates them, that do not fit queer indie games, and, moreover, those that queer indie games actively resist and often destabilize.

"Independent games are often *about games* … commenting on game conventions, highlighting or subverting them, sometimes to make cultural or political statements", writes Juul (19). Certainly, it is true that many of the best-known indie games, including Anna Anthropy's own *Dys4ia*, are meta-reflections on (or explicitly play with) the established tropes of videogames. However, as a broader constellation of work, the games created by the queer games avant-garde offer an important counterpoint to this characterization of indie games. For the most part, these queer games are not "games about games". More striking still, the ways that the queer and trans developers interviewed for this project describe their work show us a version of indie game-making that is intentionally and radically *uncaring* toward videogames.

This quality of the queer games avant-garde – that is precisely *not* about games – manifests in these interviews in two primary ways. First, the game-makers I spoke with rarely pointed toward other videogames as the inspirations for their work. Instead, they situated themselves within different artistic genealogies. Game-makers Mattie Brice and Kara Stone both point to feminist performance art as a key influence. Even Dietrich Squinkifer, who readily admits that their *Dominique Pamplemousse* games are inspired by the point-and-click adventure games they played in their youth, describes their work as most closely aligned with the theater of Bertolt Brecht. In fact, far from making games about games, some interviewees stressed the importance of paying *less* attention to games. As advice for aspiring game developers, Llaura McGee says, "Often, when people are making video games, they come up with a cool mechanic, but it doesn't go anywhere. Games could learn a lot from looking at things that aren't games" (71).

A second way that this turn *away* from games manifests is as an emphasis among queer indie developers on challenging (or simply not caring about) what counts as a game. As a growing number of game studies scholars have argued, ongoing debates about what constitutes a "real" game are fueled by the same discriminatory attitudes that characterize toxic gamer culture (Chess, 2017; Kagen, 2017; Consalvo and Paul, 2019). While some experimental indie developers argue for their work to be included in the games umbrella, the queer and trans game-makers interviewed here repeatedly disavowed an interest in the label of "games". Andi McClure states, "Personally, I'm not interested in games themselves. I'm interested in the space of what is possible when you have these mechanical systems encoded in software.... What does it matter if it's a game?" (79). Relatedly, Squinkifer describes making games that are not intended for gamers:

> I like to say that my work is loved by tenured professors and disliked by 'gamers'.... People who are interested in the weird, experimental possibilities of interactive media like my work. That's in part because it doesn't require as much mechanical skill as a lot of video games do. It's important for game controllers to be intuitive for the average person, especially if we want to reach other queer people or trans people or people of color.
>
> (40)

This quote from Squinkifer reflects a version of indie game development that is not invested in games or gamers. At the same time, it is a subtle indictment of more mainstream independent videogames like *Cuphead* (2019), which is in fact designed to be difficult and thereby exclusionary, hailing gamers while refusing entry to other players.

Yet, many of the contributors to the queer games avant-garde go beyond not caring about games; they actively seek to "destroy" the medium as we know it today. Juul describes how *Indie Game: The Movie* and related popular press reporting presented an image of notorious indie developer Jonathan Blow as "out to save video games" (12). What exactly he aims to save videogames from is less clear; in any case, this depiction of Blow casts indie developers as ones who are committed to protecting the medium they love and steering it away from artistic demise. By contrast, a number of the interviewees I spoke with expressed their desire to destroy videogames. In these conversations, destruction emerged as a powerful symbol for deconstructing the oppressive status quo and opening up new, queerer possibilities for the medium. Squinkifer describes how their aptly titled game *Quing's Quest VII: The Death of Video Games* was created as part of a game jam "for people who wanted to ruin video games". Says Squinkifer, "It's a dark topic, but it's also fun and frivolous… [It's] about resistance through this destructive sense of joy" (37). In a particularly compelling quote – my favorite of the entire book project – McClure reflects on the early days of her game-making practice:

> Traditional gamers thought indie would destroy the medium, which didn't happen obviously—but, for this period, I was like, "I can destroy something? Great, I have this awesome destructive power! I'm going to be your super villain. I'm going to make something so avant-garde it will actually destroy the medium and there will be nothing left.
>
> (79)

Statements like these suggest a vision of independent game-making that looks very different than the one presented by the dominant narrative of the indie imaginary. They paint a picture of the queer games avant-garde as a site of joyful, transgressive ambivalence, using the medium of videogames itself to unapologetically and even gleefully dismantle the medium of videogames. These are not games about games. If anything, they are games *against* games: games against the very idea that new work should be loyal to the creative and cultural norms of the work that came before it.

From independence to (inter)dependence

It may seem self-evident that "independence" is important to indie games. In *Handmade Pixels*, Juul explores this notion of independence, breaking it out into different forms across different historical moments. In Juul's taxonomy, games may be financially independent, aesthetically independent, or culturally independent (12–14). In each case, "independent" can be translated as meaning "different from the mainstream" – whether that be mainstream modes of production, mainstream visual styles, or the types of representational content found in mainstream videogames. Yet, in this formatulion, a set of core concerns remain un-interrogated: Why value independence above all? When we valorize independence, what people and forms of work are devalued or pushed out of view? If we want to reimagine "indie" in ways that are truly more inclusive, what alternatives might we find to the "independent" in independent games?

In contrast to the work that Juul describes, the queer games avant-garde, taken broadly, rejects these notions of independence. In place of independence, the queer and trans game-makers I interviewed focused on its opposite: they stressed the importance of what we might call, in contrast to independence, *dependence* or *interdependence* – the ways that individuals (especially marginalized individuals) help one another and join together to form systems of support and shared ways of making meaning. In our conversations, these queer and trans game-makers did not discuss breaking from tradition or setting themselves apart from the mainstream – the defining acts of independence, in Juul's description. Nor did not they represent themselves as singular creators operating, in that sense, independently. Instead, what emerged across our discussions of their work, their creative processes, and their interpersonal connections was a sense that queer indie game was made possible by the way that marginalized people depend on and support one another.

Collaboration was a recurring topic in these interviews. Many of the queer and trans game-makers I spoke with emphasized that their work was the result of group effort. In response to a question about whether so-called accessible development platforms now make it possible for a single individual to develop a videogame, Mo Cohen replies, "To anyone who thinks that, I would say, 'That is ridiculous'. It's true that a lot of game development tools are free these days, but no one has all the skills to make a game". Cohen also highlights the importance of compensation for collaborative labor. "Let's say I want to hire an artist or a musician. I have to pay them. Their time is worth money" (213). Themes of interdependence likewise appeared in the ways that some of the queer indies I interviewed described their creative practices. Andi McClure explains:

> I view the process of creation as a collaboration between you and your medium, or, if you're working with computers, a collaboration between you and a machine.... You have something you want to do in the creative process, but also your tools have something they want... I get my best work out of letting the machine drive and seeing what it can do. (75)

Here, McClure argues that the human game-maker is not independent from the computational tools she uses to develop her games. In this vision of indie game development, the interplays between designer and design tool become productive sites of negotiation, exploration, and mutual respect between two entities with their own desires.

Resisting the lure of so-called independence is also key to the ways that many of the game-makers interviewed discussed queer community. Aevee Bee describes seeing growing numbers of queer and transgender attendees at the annual Game Developers Conference in San Francisco. For Bee, community growth of this sort is important precisely because it undercuts the expectation that marginalized game-makers should have to stand out (or stand alone) to succeed. The size and variety of this group, says Bee:

> helps reduce this competitive urge that is the most poisonous and deadly thing for the queer community: the Highlander syndrome. 'There can be only one person who will get the money to survive. There can be only one person who will get the fame'.... That attitude can do real damage. (58)

Statements like Bee's reflect how many queer indies disavow the competitive spirit that goes hand in hand with valorizing independence, even as these same queer indies face striking financial precarity: working multiple jobs, or experiencing homeliness while developing their indie games (Ruberg, 2019a). In this sense, refusing to participate in the "Highlander syndrome" is a political stance; it suggests that some queer and trans indie game-makers may be willing to sacrifice "getting the money to survive" in order to foster a more mutually supportive community. At the same time, Bee's comment gestures toward the fact that the very notion that "only one person [can] get the fame" is itself flawed and harmful, especially when applied to marginalized people. It incorrectly suggests that queer game-making is too small a realm for multiple developers to succeed and worrisomely encourages the tokenizing of one individual as "the" queer indie game-maker.

This turn away from independence and toward dependence can also be seen in the very games created by these queer indies. Many of these works foreground intimate experiences of co-creation and interdependence between games and players. In her game *Dysforgiveness*, Seanna Musgrave prompts players to interact through embodied interdependence; players must swap body part cards, which are velcroed to their clothing. Musgrave explains, "There's this thing between trans feminine and trans masculine people that I've always thought was sweet where they talk about trading body parts – like 'Here you can have these. I don't want these anymore'" (149). Work such as this literalizes the turn away from independence and toward interdependence, prompting players to turn to one another to reach the physical forms that feel most true to themselves. *Dysforgiveness* and other works by queer indies use play itself to highlight how, for marginalized people, art and self-actualization are made possible not by independence but by interpersonal connection and dynamics of mutual care. At the same time, the task of swapping body parts in *Dysforgivesness* is challenging and awkward, reminding us that offering this kind of mutual support is rarely easy or without its own complications.

Against authenticity

In Juul's articulation, authenticity is key to understanding independent games: what they strive for, what makes them meaningful, and how they are understood in their wider cultural reception. We might distill from

Juul's writing in *Handmade Pixels* that indie games are, in some fundamental way, defined by their quest for authenticity – or, at least, their affiliation with such a quest. Writes Juul, "Independent games are continually presented as the *authentic* alternative to mainstream games—and to previous independent games… [Indie] developers continue to strive to make new, *truly* authentic video games" (6) (emphasis in the original). This may well be true in many sectors of independent game development, as well as in the indie imaginary. However, in the work of the queer games avant-garde, we see a form of indie game-making that interrogates, resists, and condemns the very notion of authenticity.

Queer cultures and queer theory have long challenged "authenticity" and "realness" as culturally constructed categories closely tied to sexuality and gender. For queer subjects, the term authentic may bring to mind societal expectations for what it means to be an "authentic" man or an "authentic" woman – concepts that typically conflate biological sex with gender and thereby promote discrimination against transgender and non-binary people. In queer subcultures of ballroom and drag, authenticity and realness have been reappropriated and transformed, playing up the constructedness and performativity of supposedly authentic identities.

It is fitting then that many of the queer and trans game-makers interviewed for this project describe playing with or rejecting the expectation that the games they make, as marginalized people, should be "authentic". For contributors to the queer games avant-garde, calls for authenticity typically come from outside. Straight, cisgender players and news outlets reporting on how videogames are becoming more "diverse" impose worrisome narratives onto queer indie games. They present these games as confessional, autobiography "empathy machines" that are supposed to give more privileged players access to the real, authentic experiences of marginalized people (Pozo, 2018). By contrast, in our conversations, these game-makers condemned the empathy narrative and described deliberately striving to complicate what it might mean to translate one's own "authentic" experiences into a game. In discussing her game, *We Know the Devil* (2015), Bee states:

> I'm wary of confessional work… *We Know the Devil* is super autobiographical, but it's also about me unpacking that autobiographical history and finding alternative avenues out of it. A confessional story about my own childhood would be this static product… Instead, the game offers alternatives. It says, 'Here's a thing that brings you in and allows you to both experience queer suffering but also to escape from it.' In making a game like that, you can control and process your personal truth instead of just reproducing it'.
>
> (55)

This quote from Bee offers alternatives to dominant narratives of authenticity in indie games by highlighting the importance of allowing players – especially fellow queer and trans people – to play through other versions of history. Rather than replicating what "really" happened, work like Bee's eschews authenticity in favor of something far more powerful: "finding alternatives", in Bee's words, and creating new possibilities.

If we understand authenticity as a performance, then we can see the desire to be deemed authentic found in some (more straight, more cisgender, more masculine) sectors of independent game development as a kind of performance anxiety. Paradoxically, in the version of indie game-making that Juul describes, *being* authentic is one in the same with *being read as* authentic by someone from the outside. Yet, many of the game-makers I interviewed described themselves as being uninterested in how their work would be received or perceived by people whose life experiences are not already reflected in their games. For instance, Brice states, "More than trying to social change, I make art to affirm myself and others who might be experiencing similar things". She continues, "Think about who makes our media and who our media is made for…. The whole world is already directed at white, straight, hetero, cisgender men. The act of being like, 'Who else can this be for?'… that's really powerful" (132). This quote draws out the connection between questions of authenticity and questions of audience. Rather than pursuing authenticity, Brice characterizes her art as affirmational. Authenticity is what is demanded of marginalized creators by those who do not hold similarly marginalized subject positions. Affirmation, by contrast, makes no claim to representing what is authentic or real. It offers support to others without making marginalized experiences consumable, or playable, by those who expecting to encounter the authentic queer experience.

Queering indie?

Looking to the experiences of queer and trans indie game-makers prompts us to question and reimagine the types of games – and the types of people – we imagine when we talk about "indie". Precisely because many of the contributors to the queer games avant-garde do not fit the dominant narratives about indie games and their creators, their stories bring into relief the constructedness of indie as an imagined category, as well as the limitations and biases inherent in that imaginary. To draw attention to these stories is to argue that "indie", as it is commonly conceived, does not sufficiently account for the work of game creators at the cultural margins. Yet, in another sense, queer and trans developers are not actually at the margins of indie games, though the indie imaginary would make it seem so. They are, as I argue in *The Queer Games Avant-Garde*, actively leading changes to the medium today. What does this mean for the indie imaginary? It suggests that this imaginary is not simply overlooking the experiences or works of queer and trans creators. Instead, it seems that the indie imaginary as it currently exists is actively working to dictate and delimit who belongs in the realm of "indie" and who should be understood as existing at the margins.

Attending to how the dominant narrative of indie games can be challenged through the experiences of LGBTQ people is one way of "queering" indie. When deployed as a verb, "queer" becomes an action (ideally done by and in the service of queer people): a way to disrupt or reframe a notion, system, or media object once tied to oppressive norms of sexuality and gender. To queer indie is to reveal how indie has excluded queer subjects and thereby to argue for a new vision of indie that includes and values them. To queer indie is also to reveal how indie has been queer all along, even as it attempts to distance itself from queerness. At the same time, to queer indie is to destabilize the very ideology that underlies indie-ness: to question authenticity, independence, and a nostalgic longing to develop games about games. In place of these things, a queered form of indie might value interdependence, affirmation, difference, and the joys of destroying what has come before. As game studies scholars, we must recognize the crucial, ongoing contributions of LGBTQ people to the broad constellation of work typically referred to as independent games. Yet also, in keeping with the spirit of queerness itself, we must interrogate and destabilize the very notion of "indie", bringing to the surface the cracks in the dominant narratives of independent games and seeking out other models for imagining and celebrating videogames developed outside the mainstream.

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4 Virtually indie

On the characteristics of independent game development for virtual reality headsets

Paweł Grabarczyk

Indies and VR – worlds apart?

On the face of it, virtual reality games (VR games for short) and independent games should not result in much of an overlap. VR games embrace the technology that powers them and celebrate technological progress. The discussions surrounding VR games can be seen as prime examples of techno fetishism where different resolutions, degrees of field of view, and advanced techniques such as eye-tracking are the main point of focus.[1] The narrative surrounding independent games could not be further from that. As pointed out by Jesper Juul (2014), independent games are often consciously opposed to the technological side of the medium and sometimes make an active stand against it. The look and feel of independent games can often be interpreted as a conscious disregard or critique of the technological fixation of the gaming industry. This characteristic attitude is a part of 'independent spirit' – the intellectual backbone of independent game production and one of the main reasons why contemporary-independent games differ from analogous phenomena known from the past. As pointed out by Orlando Guevara-Villalobos (2015), the independent spirit can be best explained as a specific attitude to labor, characterized by cooperative sociality, freedom (which often leads to unconstrained experimentation), and authorship.[2] Apart from the obvious economic context, freedom can be also understood as the ability to break out from the common perception of how a videogame should look like. As argued by Chris DeLeon at GDC 2010, cited by Guevara-Villalobos:

> We are in great haste to construct a higher fidelity technology; but it may be we have nothing important to communicate that requires higher fidelity… as if the main object were to talk elaborately, and not to talk sensibly. [N]o videogame ever stood the lower in my estimation for having low fidelity graphics, yet I am sure that there is greater anxiety commonly to have fashionable visuals, or at least high definition and 3-D graphics, than to have sound meaning.
>
> (DeLeon in Guevara-Villalobos, 2015)

VR games are very different in this respect as they demand a certain minimal level of graphical fidelity and technical prowess in order to be playable. This is especially true of the minimally acceptable resolution and especially framerate which has to be higher than the framerate of non-VR games. It seems that, in contrast to independent games, the focus on technological advancement is inherent to VR development.

Additionally, many independent games directly or indirectly challenge the contemporary and technologically advanced imagery by embracing the past via retro-aesthetics. As pointed by Jesper Juul (2019), this choice is not coincidental as it can be seen as a display of anti-modernism of independent games. This resulted in a visible trend of retro games, such as *Oniken* (JoyMasher, 2012) or *VVVVVV* (Cavanagh, 2010), which can sometimes mimic the aesthetics of past games down to the choice of the color palette. What is more, many games using retro aesthetics deliberately do not try to mimic the most advanced games from the past. For example, even if *VVVVVV* was released on Commodore 64 in the 1980s, it would have probably not been considered to be graphically impressive. The usage of pixel graphics is thus both an aesthetic choice and a critique of contemporary labor practices of big budget games.

Basing on this observation, Jesper Juul (2004) introduced the notion of the ‘independent style’ which refers to the practice of imitating older technologies (such as pixel graphics). As pointed out by Juul, this decision is typically a necessity in disguise as the economic realities of independent development dictate technological choices. The novelty of the independent style is that this necessity has been at some point consciously embraced by independent developers and turned into a positive narrative.

Needless to say, not every independent game created after 2005 resorts to retro-aesthetics. Still, the number of technologically advanced independent games is fairly low. Moreover, the existing examples, such as *The Vanishing of Ethan Carter* (The Astronauts, 2014) or *Hellblade: Senua’s Sacrifice* (Ninja Theory, 2017) are fairly specific, as both games have been created by veteran AAA developers experienced with big budget projects. Some independent developers may even be critical of independent style – for example, Tale of Tales’ *Sunset* (Tale of Tales, 2015) ridicules the independent style by introducing a special pixelated mode called ‘indie mode’. As acknowledged by Michaël Samyn in an interview (Juul, 2015), the company never wanted to be associated with lo-fi games, but this kind of attitude is an exception. Interestingly, all of the three abovementioned developers turned toward VR development.

VR landscape contrasts with this sharply as games do not attempt to visually refashion the medium’s past, and there are practically no games which could be described as VR-retro. One interesting exception can be found in *Timescope* (Somasoft, 2018), a freeware Gear VR experience that gives the users the chance to look at classic daguerrotypes (which can be seen as the early attempt at VR due to the similarity of the devices used to watch them). Still, most VR games that make use of retro aesthetics (such as *Pixel Ripped*, *1989* (Arvore, 2018)) bank on the nostalgia for traditional platforms (in this specific case, the handheld console Gameboy) and not VR nostalgia. The most probable reason for this is rather trivial – there is basically no VR nostalgia to tap into, as the earlier VR generations were too niche, to develop a recognizable aesthetic. Their user base was also too fragmented to establish a common past that new games can refer to or rework. The few players who tried VR in the 1990s may remember it differently, because they probably encountered completely different exotic devices or prototypes.

What is more, even if independent games do not refer to a particular era of past gaming aesthetics, they very often embrace the independent style in a general sense and relate to unspecified past technology via a combination of pixel art, cartoon graphics, and even non-digital aesthetic styles, such as crayon drawings or paintings. This artistic choice does not translate well to VR games because of the technological demands of the platform. For example, even though creating games played on a 2D plane or top-down games in VR is possible,[3] the experience will end up being too similar to that of playing these games on a big screen.

The difference in how VR and independent games relate to technology goes beyond the representational side. Even a cursory look at the hardware requirements for games classified as 'indie' on the Steam platform shows that the majority of them do not require a lot of computing power. What is more important is that these modest demands are not just coincidental, as they add to the narrative presenting independent games as more egalitarian, strengthening the image of independent games as the opposition of so-called 'AAA' games, which typically aim to push the hardware progress. This difference in attitude can be best seen in the case of the narrative surrounding the changes of console generations. A very common criticism geared toward games released for the new generation of machines is that they do not utilize the new hardware in any meaningful way, that they look and feel 'last gen'. In a way, the new wave of games has to somehow make an excuse for their existence by utilizing the additional power granted by the hardware.[4] It can be argued that the rise of popularity of independent games on consoles disrupted this cycle. For example, PlayStation 4's library contains a sizable collection of independent games that could easily be released on PlayStation 3 (or even previous consoles), and yet, this point is almost never raised in their reviews. Contrary to this, the demand for the 'excuse for the existence' is very common in VR games reviews which can often declare the games as "possible on non-VR setups".[5] The prototypical 'indie game' and the prototypical 'VR game' seem to be perceived by their consumers as sitting on the opposite side of a spectrum with regards to the use of technological innovation.

And yet, as can be seen on every popular storefront which sells VR games, a considerable number of current VR games come from independent developers. For example, at the time of writing, independent VR games constitute 67% of all VR games available for the Steam platform. This percentage does not fall far from the overall percentage of games tagged as independent on Steam (74%).[6] Furthermore, a significant number of the best received and most commercially successful games released for VR platforms come from independent developers. Examples of such games are *Beat Saber* (Beat Games, 2018) (the first VR game to break one million sales), *SuperHot* (Superhot Team, 2016), *Arizona Sunshine* (Vertigo Games, 2016), *Moss* (Polyarc, 2018), or *Job Simulator* (Owlchemy Labs, 2016). On the list of 12 best-selling Steam VR games from 2018, only two games are not classified as 'indie'.

The discrepancy between the image of a prototypical 'indie game' captured by the concepts of 'indie spirit' and 'independent style' and the demands of VR games provokes two questions: (1) what are the reasons for such a big popularity of VR amongst independent developers? (2) Are indie VR games different enough from their traditional counterparts, so that they demand new academic descriptions and conceptualizations – is there such a thing as 'indie VR spirit' or 'VR-independent style'?

Some conceptual distinctions

Before we proceed to answer both questions, let me pause to address a potential terminological issue and say something more about the way terms 'independent', 'indie', and 'VR'" are understood in this paper.

The way I will use the words 'independent' and 'indie' will be based on the model presented in Garda and Grabarczyk (2016). According to the model, the term 'independent game' can be understood in at least three senses: first, it may refer to the financial independence of the developer; second, it may refer to the independence of the developer from a publisher (the game is self-published); and finally, it may refer to the creative independence of the author. In practice, most existing independent games combine more than one of these types of independence, but the point is that they are conceptually separate. The first two types of independence are fairly self-explanatory. The third one refers to the artistic autonomy of the creators – to the fact that developers may try to implement their vision without conforming to any external forces or suggestions – be it from the employer, publisher, or, most importantly, the audience. Different authors writing about independent games may focus on or prioritize one of these senses of independence.[7] It can be thus said that the notion of independence remains to be ambiguous, and it is always important to specify the type of independence we have in mind.

For the purpose of this chapter, I will be interested only in the first sense of independence, that is financial independence.

As pointed out in Garda and Grabarczyk (2016), whenever games that are independent in any of the abovementioned senses share the same economic and cultural context, they may start to share some of their internal characteristics. For example, many independent games may use pixel graphics, because it happens to be cheaper in production, and they may be distributed digitally because it solves the problem of manufacturing discs and logistics. These internal properties of games start to function as indie 'markers' – visible traits which help the users to classify games as independent. The obvious upside of the combination of these markers is that they enable the players to recognize independent games without investigating the actual production process of the games (which is not something the user can simply see).[8] The downside of these traits is that they are completely contingent and can change whenever the economic or technological context changes (for example, whenever a new, cheaper, method of creating graphics is invented). For this reason, they can never be treated as very stable predictors of whether a given game should be classified as independent. As suggested by Garda and Grabarczyk (2016), it is useful to reserve the word 'indie' to refer to this collection of visible traits. Most of the aspects that constitute the indie spirit or the independent style belong to this category. Indie markers do not have to be visual – traits such as small development team, digital distribution, unconventional topics, or small hardware requirements may also contribute to the recognition of game's independence.

As for the way VR is understood in this chapter, the reader should simply assume that VR games refer to any of the games produced for the current batch of consumer headsets, such as Oculus Rift, Vive, PSVR, Windows Mixed Reality, Oculus Quest, Oculus Go, or even Gear VR. Grouping all of these headsets together may seem risky as the differences between them could, in principle, affect their relation to the independent market. The listed headsets depend on different large corporations (Sony, Microsoft, Valve, Samsung, and Facebook), and they vary vastly in terms of technological complexity. What is more, they cover three different game market sectors: the console, PC, and mobile development. Still, since the aim of this paper is to paint a broader picture of the relation of independent and VR production, I decided to not differentiate between them.

Why are there so many indie VR games?

The reasons for the popularity of VR development amongst independent developers have been divided into three groups of factors. In the first section, I analyze some of the practical reasons which, although probably not decisive, definitely helped to ease the tension between indie and VR. In the second section, I go back to the notions of independent style and independent spirit and reexamine their impact on VR. In the third section, I look at the most important reason for the popularity of indie VR that is unique to the current VR market conditions – the fact that they have not been established as an opposite of so-called AAA games.

Practical reasons

The most direct practical reason for the popularity of VR technology amongst VR developers comes down to the availability of development tools. Unity – the most popular game engine used in independent games development[9] – added VR support in version 5.1 in June 2015, just in time for the release of the first wave of consumer headsets. The ease of exporting the project to VR and the support for many different headsets made the engine the preferred choice of both: independent developers and VR developers, creating a natural overlap (Bonfiglio, 2018). This proved especially true in the case of the less demanding headsets, such as Gear VR (Matney, 2017). The availability of good introductory literature distributed via channels related to independent games development[10] could have only boosted this process. What is more, the consumer headsets started to be widely available in education institutions, especially those which teach game development (on the relation between independent games and educational institutions, see the work of Celia Pearce in this same volume).

The second practical reason relates to the changes and demands of the current market for independent games. The early successes of independent game developers such as Jonathan Blow as well as the possibility of self-publishing both on consoles and Steam resulted in a dramatic increase in the number of games produced.[11] Recent years proved to be more and more challenging for independent developers as it became increasingly difficult for their games to stand out or to be profitable (Fenlon, 2018). It is obvious that game magazines and websites cannot review all releases, but the number of games devoid of any media coverage is bigger than ever. Moreover, games promoted by Independent Games Festival are often already well covered in the press. New methods of promotion, such as YouTube channels, Steam Curators, or Twitch, help to alleviate the situation to a certain extent, but some of the recent trends show that they may no longer be as useful as they were earlier.[12] The situation looks even worse once we take the existence of even less publicized distribution channels, such as itch.io into account.

On the surface, the situation of VR games may look even grimmer. Shortly after the release of the three main VR headsets, games released for these systems found coverage in the specialized press. For example, all of the launch PSVR games and many of the games released in the Oculus Store (such as *EVE Valkyrie* (CCP Games, 2016)) have been reviewed on popular mainstream gaming websites. In 2019, this is true only of one of the headsets (PSVR) and applies only to very few game releases. This lack of coverage extends even to announcements and plain information on the release schedule. One prominent example of this can be found in the Wikipedia entry for yearly game releases for all current platforms.[13] In 2016 and 2017 games released for Oculus Rift, Vive and PSVR have been listed in their respective entries as all of these three headsets have been treated as separate platforms. This changed in 2018 when the headsets started to be treated rather as accessories than platforms and the games produced for them started to be listed simply as PC or PS4 games. On the surface, this change in classification may look purely cosmetic, but the result of it is that only a fraction of releases available for all of the three headsets have been listed. The reason for it is that games that could have been considered as 'important enough' for the VR platform end up being ignored once they are considered in the broader pool of releases. What is especially interesting is that games released for the newly launched Oculus Quest have been completely ignored on the list. In this respect, VR games started to be treated similarly to mobile titles which, after a short wave of initial interest, stopped being covered by the mainstream gaming press.

And yet, paradoxically, the visibility of VR games improved due to these exact reasons. VR games became an established niche, and their users learned that they have to search for the information about them in places dedicated only to VR. Exiled from the mainstream press, VR gaming attracted enough interest to generate its own microcosm of media coverage, consisting of specialized websites, such as Upload VR, Road to VR or VR focus, specialized YouTube channels,[14] and even an aggregation site similar to Metacritic but focusing only on VR games for all platforms (VRGamecritic). What is more, since the number of all VR releases (independent productions included) is significantly lower than the number of independent game releases, it is actually possible to cover most of them. This enables the aforementioned sites and channels to maintain a healthy balance – there are enough new releases and announcements for them to publish new content often but not so much so they lose their status of places for 'all VR news'. It is then much easier for independent VR games to stand amongst the crowd, to be noticed by the VR press, and to fight for the attention of the customer. In addition to that, VR games can also typically be sold for a higher price. This aspect is especially relevant for the independent games market, as customers and developers can sometimes drastically undervalue independent games (Consalvo & Paul, 2018).

Style and spirit

As mentioned in the beginning, two key aspects for the understanding of the phenomenon of contemporary-independent games, that is the independent style and the indie spirit, seem to be incompatible with the development demands and, especially, the narrative surrounding VR games. Still, as I am going to show in this section, the relation between both of these conceptual categories and VR is much more nuanced.

Even if most of the aspects of the spirit of independent labor make it a hard fit for the technologically oriented VR development, at least one aspect stands out as fairly easy to reconcile with the demands of this new medium. Independent games are often heralded as innovative, fresh, and experimentative. There are no doubts that employing this form of experimentative spirit in a risk-averse working environment is difficult. Many game formulas do not change for years, and the investors may be reluctant to back up radically different gameplay ideas. Even most of the successful crowdfunding campaigns capitalize on nostalgia and reviving old ideas and genres, rather than experimentation. This is at least the perception of the industry many independent developers have (Guevara-Villalobos, 2015). Things are very different in the VR space as there are no proven formulas to adhere to. More importantly, consumers expect innovation and surprise as many of the existing solutions are not considered satisfactory. A conservative attitude which is often seen in AAA development is possible only because the consumers accepted the existing design solutions as optimal or final.

For example, there is still no consensus on the way VR games should implement locomotion. The biggest problem related to the movement in VR relates to simulation sickness which may occur to the player. The alternative solution for free movement can be provided by room-scale VR – simulations in which the movement of the player in the physical space matches the movement in virtual space. The problem with this solution is that it constrains the developer (only fairly small, horizontal spaces can be simulated this way) and demands the user to dedicate a big space for VR use. Some of the early popular approaches, such as teleportation (in which the players point at places to which they wish to be instantly transported) and snap-turning (movement in which the players can move freely within the z-axis and turn their head, but the turning of the body is restricted to predefined jumps of 30 or more degrees) have been lately superseded by independent developers' innovative solutions. In *Budget Cuts* (Neat Corporation, 2018), the players traverse the space by opening and physically entering portals, similar to Valve's *Portal* (Valve, 2007) which combines teleportation and walking. In *Spring Vector* (Survios, 2017), the players embody runners by moving their hands in such a way so they mimic the movement of the hands of the runner (while still standing) in order to trick the brain into feeling the momentum. This experimental method of movement proved successful enough for another independent developer to create a model that enables this type of locomotion in many existing VR games.

There is also no consensus on which existing genres can be successfully adapted to the new medium. Strategy games (for example, *Brass Tactics* (Hidden Path Entertainment, 2018)) and rhythm games (for example, *Thumper* (Drool, 2016) and *Beat Saber* (Beat Games, 2018)) have found their place in the VR space in a quite natural way, but other genres still await reinvention. This creates a possibility for independent developers to try their hand at genres which are normally completely dominated by big companies – *Orbus VR* (Orbus Online, 2018) (an MMO RPG) and *Pavlov* (Vankrupt Games, 2017) (a competitive PVP online shooter) can be seen as good examples of this. It is also very unclear how many completely new genres will VR inspire in the future. It seems that, at this point, the only well-established genre in the VR space is the so-called 'wave shooter' (*Robo Recall* (Epic Games, 2017) and *Space Pirate Trainer* (I-Illusions, 2017)), but a plethora of the existing titles defy our current genre categorizations.

The third example of basic game functionality that demands reconceptualization in VR is the implementation of menus and inventory systems. Displaying standard interface elements in VR creates a surreal effect of menus floating in the air which may break immersion. This forces developers to think of completely new ways of solving old problems. Many of them opt for embodied interfaces (it is, for example, fairly typical for shooters to demand a physical change of weapons and reloading), but the variety of competing solutions show that VR is still far off from established interface conventions.

The reason I am pointing out these examples is that they show that the VR audience expects innovation on a fundamental level, such as genres or basic mechanics (traversal, inventory manipulation). This makes VR a fertile ground for unconstrained experimentation which may be very attractive for independent creators.

The experimental, conceptual nature of many independent projects translates to a relatively short length of many independent games. For example, *The Graveyard* (Tale of Tales, 2008) or *The Passage* (Rohrer, 2007) can be easily finished in less than 15 minutes. This prompted a wave of criticism from players accustomed to longer experiences.[15] In contrast, short experiences in VR are more of a norm than an exception. VR technology is still very often experienced during presentations and exhibitions (Golding, 2019) or in VR arcades. All of these environments dictate short play sessions. What is more, many players are not comfortable with long play sessions in VR because of disorientation or nausea. This makes VR a much more welcoming space for short experimental projects.

The evolution of aesthetics of independent games is the main reason why the incompatibility of the independent style and VR did not stop the development of independent VR games. As predicted in the model mentioned in the section above, what has been identified as an independent style is an amalgamation of different traits resulting from the technological, economic, and social circumstances of the games' production. Since all of these circumstances change over time, so is the independent style. Independent games evolve fast and make our categorizations obsolete (Simon, 2013). It seems that pixel graphics which was dominant in early years of contemporary-independent games started to be slowly matched by low polygonal untextured graphics (so-called low-poly style). Games such as *Surgeon Simulator* (Bossa Studios, 2013), *The Witness* (Thekla, 2016), *Firewatch* (Campo Santo, 2016), or *Quadrilateral Cowboy* (Blando Games, 2016) can serve as good examples of this graphical style. In contrast to pixel graphics, low-poly style can be used in VR games in a very easy way and is actually helpful as it translates well to the resolutions of contemporary headsets, making games sharper and clean. This resulted in a fairly high adoption of this style amongst VR developers – *Job Simulator*, *SuperHot VR*, *Accounting* (Crows Crows Crows, 2016), or *Orbus VR* can serve as good examples. It is also worth adding that the adoption of low-poly style has been pivotal for the success of less powerful headsets, such as Gear VR or Oculus Quest. One important difference between the use of the low-polygonal graphics and the use of pixel graphics is that the former is not typically combined with a low-tech approach described by Juul (2014). Low-polygonal VR games do not present themselves as less technologically advanced or content-oriented. They instead swap one type of technological prowess for another. Simplified graphics allows the developers to create much bigger open worlds or increase draw distances (Orbus VR) or replace the focus on traditional aesthetics for a more kinesthetic approach, presenting the player with a multitude of affordances (*Job Simulator*) or natural body movement (*SuperHOT VR*). For this reason, it was possible for these games to be marketed as showpieces of new technology – something that would have not been possible for games created in the independent style in Juul's sense of the term.

Indies in a vacuum

As we have seen in the previous section, the specificity of VR games makes it hard for some of the concepts used in the description of independent games to be applied to VR. The upshot of the discussion was that even if they can be used, they have to be employed with caution. The last reason for the popularity of VR amongst independent developers that I wish to look at in this chapter goes even further into this direction because it points to a fundamental difference between VR and non-VR development. A difference that makes any analogies between regular and VR-independent games is rather risky. As we are going to see below, this particular aspect of VR games makes them a good case study of what happens to independent development once it gets removed from its original economic and cultural context.

As I have already mentioned, independent development is typically positioned as an alternative to the mainstream (Newman, 2011). It can be seen as a reaction to the dominant market conditions (Pedercini, 2012). Independent production functions as the other side of the coin for the so-called AAA games scene.[16]

The term AAA game is vague and its meaning is disputed,[17] but the most popular understanding of it is that it refers to a game with a large budget (Lipkin, 2013) which is, of course, relative to the historical production conditions. As pointed out by Juul (2014), one of the motivations behind the independent style is the ability to produce games with a small budget that are free of negative connotations of being made with restricted economic resources. This was crucial for the independent developers as it allowed them to create their identity against both AAA productions and "budget games" (Juul, 2014). The key to understanding the current VR market is that this constitutive opposition of indie *versus* AAA no longer applies. To put it bluntly, the AAA sector of the VR market is practically non-existent. It can be argued that the only existing VR game which that can be classified as a big budget production is *Resident Evil 7* produced by Capcom in 2017. All of the other big-budgeted productions, such as *Skyrim VR* (Bethesda Game Studios, 2017), *Fallout 4*, or *Borderlands 2 VR* (Gearbox Software, 2018) are simply inexpensive VR ports of existing big-budgeted games. The other closest analogy to AAA VR can be found in VR modes added to some of the big budget releases available for PSVR. Problem is that, even all of these VR experiences tie with big budget productions, the VR portion of these games appears to be very insignificant. They function either as VR commercials for their respective games (*Call Of Duty: Infinite Warfare* (Infinity Ward, 2016) and *Kingdom Hearts III* (Square Enix, 2019)) or simplified modes, which give the users only a taste of possible AAA VR games (*Tekken 7* (Bandai Namco Studios, 2017), *Star Wars Battlefront* (Dice, 2016), *Ace Combat 7* (Bandai Namco Studios, 2019), and *Gran Turismo Sport* (Polyphony Digital, 2017)). This puts independent VR games in a unique position – they are not perceived as an opposition to the mainstream, because there is no VR mainstream. Moreover, it is perfectly possible for them to become the new mainstream as they are the pioneers of the new technology. Independent VR games did not have to position themselves against larger productions, because videogame companies decided not to make significant investments in the VR market and produced only smaller-scale games, such as *Marvel Powers United VR* (Sanzaru Games, 2018), *Lone Echo* (Ready At Dawn, 2017), or *Until Dawn: Rush of Blood* (Supermassive Games, 2016). This enabled independent games to set the horizon of expectations for the users. They did not need to recontextualize the necessities of independent game production positing it as a deliberate choice of independent style, because their games did not contrast with the juggernaut of corporate multi-million projects.

Moreover, the lack of big-budget VR games resulted in a void in the market as the audience demanded for "bigger" VR games. For this reason, presenting games as "small", "simple", or "humble" may have been detrimental to their sales as the difference in scope between VR games produced by independent developers and corporations was at this point still negligible.

This difference in attitude makes VR-independent games similar to the PC shareware market from the early 1990s. Even though on the surface games such as *Commander Keen* (Ideas from the Deep, 1990) could be perceived as similar to the current indie games, the technological and cultural context of their origin, as well as the attitude of their creators was very different. As proven by the later transformative success of *Wolfenstein 3D* (id Software, 1992) and *Doom* (id Software, 1993), the main actors of the shareware scene were interested in pushing the technological boundaries of PC gaming, more than contemporary low-level independent game production. In hindsight, this attitude can be even attributed to early products such as the aforementioned *Commander Keen*, which was an attempt at creating a console-style experience on PC. Born amidst early attempts such as *Catacombs 3D* (id Software, 1991), the FPS became the new mainstream. Judging by the list of VR bestsellers cited above, something similar has happened to VR games as well – independent VR games proved to be successful enough to function simply as 'VR games', they became the reference point of how a VR game looks like, and they have set the horizon of expectations for the new technology.[18] What is interesting is that this process may have been, at least partially, the result of a conscious strategy of claiming new technology before it normalizes. This can be seen in Robert Yang's blogpost in which he raises this point in a very evocative manner:

> Imagine video games except AAA titles barely exist, and thus no one can pointlessly compare you to them… and that's the current state of VR. If we get in early enough, we can define the general public's first significant impressions of VR, and influence how people value VR experiences. We need to develop the theory, the language, and the touchstones that others will have to adopt in order to seem fluent -- we need to be the new normal here, and we could possibly do it, because no one else has defined the norms yet.
>
> (Yang, 2016)

Conclusion: change of landscape?

Despite the differences in the narrative surrounding VR and independent game development exemplified by the phenomena of independent style and indie spirit, the independent VR market is thriving.[19] As argued in this paper, the reasons for it are numerous. The demand for innovation and experimentation in the VR space makes the technology fitting for some of the elements of the independent spirit as VR creators are not expected to follow any proven formulas. The idea of 'chasing your own vision' does not function as a romanticized ideal – it becomes a necessity when there are no trends to chase. In addition to that, a combination of purely practical reasons made VR development even more attractive for the developers. The availability of tools such as Unity, better visibility in the market, and higher potential of income contributed to the popularity of the new platforms. Last but not least, the fundamental difference in market dynamics and the lack of an established AAA VR market disrupted the well-known context of the duality between mainstream and indie development. Since the production and work conditions of independent VR development were not an exception but rather the norm, there was simply no need of creating the narrative to explain them and give them identity.

Needless to say, the main reason for this characteristic of VR development is that VR technology is still a nascent platform. It is difficult to predict its further development and trajectory. It seems that even though its reception was significantly less enthusiastic than some of its proponents predicted (Wordsworth, 2013), it is not a failure either. In this respect, the current wave of VR headsets differs from all of the earlier attempts at introducing the technology to the mass market (for example, the short-lived popularity of VFX-1 helmets in the late 1990s). Even if VR helmets will not replace traditional gaming, contrary to earlier experimentations, it is now likely that they will continue being used by game consumers. The most visible confirmation is that all of the three main companies promoting VR helmets (Valve, Facebook, and Sony) announced or even introduced new iterations of the hardware. This growth of VR market may result in a change in economic relations and can lead to the establishment of the duality between AAA and independent development in the future. This scenario is likely as some of the early signs of this trajectory can already be seen. For example, the marketing efforts related to several games produced lately by Sony for the PSVR platform (such as *Farpoint* (Impulse Gear, 2017), *Astro Bot Rescue Mission* (SIE Japan Studio, 2018) or *Blood and Truth* (SIE London Studio, 2019)) as well as early games available of the platform (such as *Robinson The Journey* (Crytek, 2016) or *Battlezone* (Rebellion Developments, 2016)) position these games as premium VR products. All of these games have been reviewed by the mainstream gaming press and sold as physical releases for a higher price than most of the digital-only VR games. The increase in computing power of the most popular current VR solution – PSVR (which has been confirmed to be compatible with the upcoming PlayStation 5) will inevitably increase the cost of realistically looking games which may broaden the divide between 'premium' and 'low-poly' VR games, establishing a new VR-independent style. It thus seems reasonable to predict that all of these factors combined may change the landscape of VR development and contribute to the rise of the indie-AAA duality in the VR space.

Notes

1 This can be seen on popular websites, such as roadtovr.com and uploadvr.com which choose to devote much space to discuss even minor hardware novelties, despite the growing number of unreviewed software.
2 Many ideas comprising the 'independent spirit' can be traced back to the Scratchware Manifesto (Designer X, 2000) and has been elaborated on in the following works (Zimmerman, 2002; Jahn-Sudmann, 2008; Lipkin, 2013; Guevara-Villalobos, 2015).
3 *The Lost Bear* (Fabrik Games, 2017) can be a good example of this.
4 See Leadbetter (2013) as an example of this. The following quote nicely encapsulates the attitude I am referring to: "Speaking from an entirely personal perspective, we've not yet sampled the entire launch line-up, but if someone gave us a PlayStation 4 along with *Killzone Shadow Fall*, *Battlefield 4* and *Resogun*, we'd be absolutely thrilled: each of these games offers up an immense,

high-quality experience that is simply unparalleled on current-gen console – and that's what it's all about".

5 See Gamespot's *SuperHyperCube* (Kokoromi, 2016) review for an example of this (Brown, 2016).
6 The percentages were calculated by comparing the number of games using "independent" and "vr" tags on the Steam storefront.
7 For example, Anthropy (2012) clearly favors the third sense of independence.
8 This has been also observed in Lipkin (2013).
9 This can be for example seen on itch.io statistics site: https://itch.io/game-development/engines/most-projects
10 For example, the Humble Bundle bundles.
11 See https://www.statista.com/statistics/552623/number-games-released-steam/
12 As reported by Davies (2019), the number of games covered by top 10 influencers dropped from 300 different games in 2016 to 28 games in 2018.
13 The relevant pages are: https://en.wikipedia.org/wiki/2016_in_video_gaming, https://en.wikipedia.org/wiki/2017_in_video_gaming and https://en.wikipedia.org/wiki/2018_in_video_gaming
14 Such as PSVR without parole (https://www.youtube.com/channel/UCnsAq8DF3xfWjC9sm_r_h2w).
15 Plenty of examples of such opinions can be found in the review section of *The Graveyard*'s Steam page: https://store.steampowered.com/app/27020/The_Graveyard/.
16 Not every author agrees to this dichotomy. See Martin and Deuze (2009) and Parker (2013) for examples of criticism.
17 Which should not come as a surprise because it depends on what has been defined as indie (Kogel, 2012).
18 As seen in McCrea (2013), a similar effect could sometimes be achieved on a local scale.
19 As shown in the recent GDC survey (GDC, 2020), even though the percentage of developers interested in VR is fairly small, it is at the same time stable. The same percentage of developers who currently work on a VR project (15%) plan their new project to be in VR. Moreover, the number of developers interested in VR in the future is 27%. And 82% of surveyed developers were independent in financial sense.

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Part II

Networks

5 Network or die? What social network analysis can tell us about indie game development

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> You can look at your network of people like a computer network. In a computer network, you want maximum connectivity between computers. Ideally, every computer is directly connected to every other computer. That way, if one link gets disconnected, it's only two hops to bridge that gap. You should strive to make your personal network equally connected. A connected network is a strong network.
>
> (Kazemi, 2005)[1]

For aspiring developers, advice like that above flows through game forums, websites, and conferences. In our own work, we duplicate these flows, pointing toward the 'boys clubs', insiders, gatekeepers, and fan communities that developers need to connect to in order to sustain game careers (Whitson et al., 2018). Even for "informal" developers – artists, makers, hobbyists, and others who may not aspire to commercial game-making – these networks, particularly local ones, are acknowledged as important (Keogh, 2019). For the most part, building one's network is spoken about as a somewhat vague yet unqualified good: the bigger and more robust network you have, the better off you are in terms of both creative inspiration and economic sustainability. For developers who do not live in cities with pre-existing developer communities, events such as *PAX* and the *Game Developers Conference* (GDC) become ideal sites to connect with other developers, fans, streamers, mainstream press, publishers, funders, and others in a compressed timeline.

Elsewhere, we show how these events "intermediate" indie games culture, noting the financial cost and physical exhaustion behind this affective, relational labor (Parker et al., 2018). In this chapter, we use both qualitative and quantitative data from developer interviews and surveys gathered through the Indie MEGABOOTH (IMB) at PAX East, PAX West, and GDC in 2015–2016 to examine networking practices.[2] We first outline theories of Social Network Analysis (SNA), defining what we mean by networking, highlighting how these theories can be applied to small-scale game development, and summarizing important takeaways. In the second half of the chapter, we use survey data to discuss the differences in both the economic stability and networking practices that are linked to one's geographic location. While quantitative analysis is somewhat novel for indie game studies, this empirical data helps us better understand how both creative ideas and funding opportunities circulate within indie networks, as well as the limitations of events and online communities as stand-ins for long-term face-to-face communities of practice.

SNA and developer networks

Introduction to social network analysis: background and context

Being connected to others ostensibly opens doors, reinforcing the belief that "it's not what you know, but who you know". The focus on strategically building one's personal and professional networks – and the slippage between the two – ties in with what Andreas Wittel (2001) termed "network sociality" in response to the explosion of more ephemeral social ties due to digital communications. For Wittel, social relationships are increasingly commodified and leveraged as a business resource, particularly in new media work. What is new here, for Wittel, is the explicit recognition of social relations as a form of capital, and how these practices are not restricted to boardrooms, but also permeate entry-level work. Wittel points to a blurring of work and play that, today, is exemplified by gaming event floors and monthly meetups:

> On the one hand the commodification of social relationships (doing a pitch, getting funds, finding work) is highly obvious, on the other, it is important to hide this commodification by creating a frame (music, alcohol, etc.) that makes people comfortable, that suggests a somehow 'authentic' interest in meeting people.
>
> (Wittel, 2001, 56)

With flexible and intrinsically insecure work, networks provide security: "The bigger the networks are, the better for everyone who participates in them" (Wittel, 2001, 57). Linking this to game-making: in an ecosystem with readily accessible development tools and online distribution platforms, there is no shortage of games. Even well-crafted games fail in this crowded market, disappearing under the hundreds of other launches that month. Thus, studios continually build and leverage their respective networks to signal-boost their game's launch and reduce the inherent risk.

We argue that SNA can add more nuance to this "bigger is better" understanding of networks and community-building. SNA emerged as a recognizable field in the 1970s as a product of Harrison White's emphasis on the structures within which social actors operate (Freeman, 2004). The

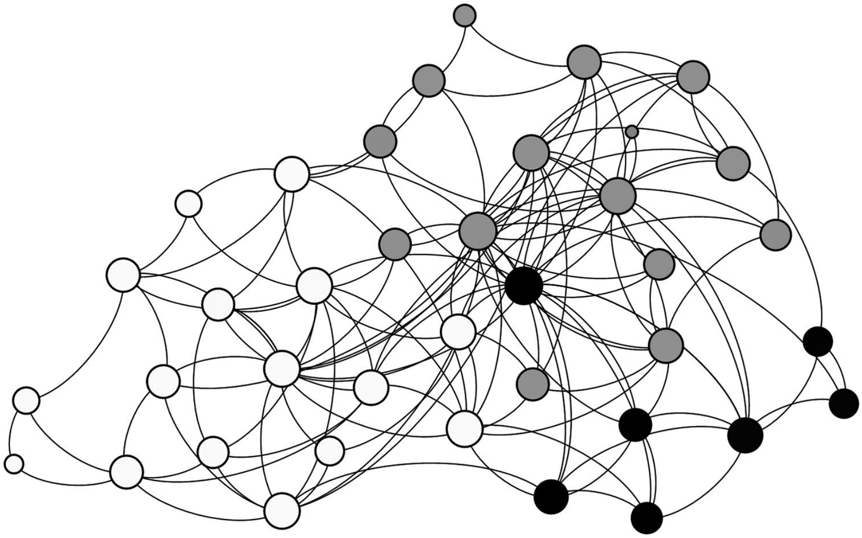

Figure 5.1 Indie interfaces SNA.

structure of a network (i.e. its location, who is in it, how they each connect, what, and how they share) is equally as important (if not more so) as a network's size in determining how useful it is for those who are a part of it (Borgatti et al., 2009). SNA emphasizes the use of empirical data, graphical imagery, and computational models to examine this social structure. Social Network Analysts make extensive use of network diagrams such as the one above.

The above-anonymized diagram depicts the connections that attendees at our *Indie Interfaces 2017* event drew between each other. Each node represents an individual, and the ties connecting them indicate a relationship. Highly connected people demonstrate their centrality visually: their nodes are commonly depicted as larger. In this diagram, most attendees were well-connected to each other. The different groupings depicted in grayscale in Figure 5.1 were algorithmically calculated but generally coincide with Canadian scholars (gray), international scholars (black), and those from the game industry (white). Oftentimes, the networks under examination are larger, multifaceted, and encompass several distinct groups with varying degrees of overlap.

Weak ties, brokers, and structural holes

The popular conception of networks is heavily influenced by Granovetter's (1973) work on the "Strength of Weak Ties". Granovetter broke with conventional wisdom by claiming that the most valuable connections in one's personal network are not necessarily the strongest ones (such as those between family members, close friends, etc.). Instead, useful information is more likely to arrive via weak ties, such as those between acquaintances. This is because, according to Granovetter, everyone in a tight-knit community already has access to most of the same information. In contrast, the information that individuals glean from weak ties – such as new ideas or word of job openings – is unlikely to be common knowledge.

Burt (2004) expands on Granovetter's work, showing how weak ties can be thought of as bridges spanning "structural holes" that separate tight-knit communities. A structural hole, in Burt's view, is a collection of potential ties between nodes which, if they existed, would interconnect two separate groups in a network. Because these ties do not exist, their absence appears as a hole in the structure of the network. Gesturing toward their powerful position vis-à-vis their networks, Burt uses the term "broker" to refer to those whose personal connections bridge structural holes. These brokers, according to Burt, enjoy privileged, early access to novel information flowing through the network.[3] For Burt, brokerage is both a position and an act that can take many forms. Brokerage can involve transmitting information about groups' strengths and difficulties or spreading knowledge of a particularly effective technique or practice between groups. Brokerage can also involve synthesizing the wider body of knowledge and practice to which brokers are exposed to something innovative. Burt puts it thus: "people who stand near the holes in a social structure are at higher risk of having good ideas" (2004, 349–350).

Opportunities for brokerage are ripe in any arena of human endeavor where socially disconnected groups harbor expertise and information that might be mutually beneficial if shared. For example, artists and tools engineers may struggle to solve a shared problem but are unaware of each other's progress due to the conceptual gulf separating their two groups (Whitson, 2018). In such cases, both groups would benefit from the addition of an individual capable of understanding both groups' struggles in their own terms – hence the Technical Artist's recent rise to prominence: a role combining art and engineering proficiencies.

Propinquity and spillover effects

In their study of innovation and group membership in the Boston Biotech sector, Owen-Smith and Powell (2004) demonstrate that ties between network actors are not perfect conduits. Rather than simply and strictly transmitting from point A to point B, the pipes that carry information and resources are "leaky". When fungible materials flow through a network, they slosh over the boundaries of the ties, making them available for others to utilize without needing to rely on brokers. An actor's capacity to take advantage of a network's "spillover effect" is conditional on belonging to the network in question and propinquity, referring to the geographic collocation of the network's members. These spillover effects occur incidentally within collocated networks – at parties, industry meet-ups, coworking spaces, and watercoolers, or talking shop over beers with friends. In such networks, membership alone is linked to success and innovative ideas (in Owen-Smith and Powell's case, patents).

As noted by a Toronto developer, local networks are where one learns how to survive as an indie:

> … before I started working alone, I got together with a bunch of people who had been running studios anywhere from, like, half a year to ten years, and just asking them questions. And I wouldn't have been able to do that – or *they wouldn't even have given me the time of day if I hadn't been involved and we hadn't known each other for three years at this point*. So, I don't know how anyone could do it without building some sort of circle… it's a circle of acquaintances or peers but not necessarily friends. But it's really important, at least to me …
>
> (Male solo developer from Canada, PAX Prime 2015. emphasis added)

Belonging to and being a recognized member of a collocated network is enough to reap benefits from the information flowing within it. In contrast, if a network is geographically distant, belonging to the network provides few, if any, spillover benefits. It is far more important, in such cases, to occupy a strong, central location in the network; in such cases, one is conceivably benefitting from brokerage advantages.

One developer and IMB team member from the Midwest United States reflects upon how attending events is a stand-in for belonging to a local community, both for themselves and other developers:

> The team I was thinking of is from [another Midwest state]. They're in the same situation that we are in…and there's not a big game dev scene there. [This event] is kind of their link to the outside world beyond the TIGSource forums[4] or whatever their internet portal you have to go on.
>
> (Male developer and IMB team member from USA, PAX Prime 2015)

When asked why he volunteers at events, he describes the benefits that accrue to his brokerage position as:

> making and maintaining connections, particularly given that it's kind of my lifeline to the outside world a little bit. Because you can only interact with people so much via, you know, Twitter and Email and that kind of thing. You know you see people at conferences and that's nice, but in general [by organizing] I'm getting to work directly with people on something of this magnitude whether they're the other people on the team,

> or they're the exhibitors, the volunteers. You kind of form that greater connection. If you're here in Seattle, a lot of people have that all the time.

In this case, online communities and physical events are limited stand-ins for local face-to-face communities, such as those in Seattle. By becoming a broker via his IMB organizing, he feels more able to form meaningful connections to, and help, others. Advice, leads, collaboration opportunities, feedback, and potential sources of publicity are shared in both locally clustered and distributed communities, but in distributed communities such as PAX and TIGSource, benefits are preferentially distributed toward those seen as important or central and thus actively singled out and included in interactions. Online networks are especially prone to "network effects", wherein popularity and obscurity are both self-reinforcing, which results in the vast majority of the available resources and attention being diverted away from the less-known toward the better-known (Uzzi, 1996; Browne, 2015).

Bigger isn't always better

Returning to our opening quote, Darius Kazemi explicitly references Granovetter's argument about the strength of weak ties and encourages developers to take it to heart.

> When I try to illustrate weak ties, I often use the example where there are ten people you know, but not very well at all. But you know that one of them is a guy who does research on serious games. And you know that another one is thinking of getting into that sector. You introduce them, and all of a sudden you have their gratitude. And you did nothing but send them both a brief email.
>
> (Kazemi, 2006)

Not only, he argues, will an abundance of weak ties provide you with better chances of receiving useful information or resources, they also allow you to act as a broker. However, there is a significant workload associated with becoming a broker and maintaining one's position (Whitson et al., 2018).

Podolny (2001) argues that the logic of brokerage encourages the single-minded pursuit of network expansion. Such advice, he concludes, ignores the "negative consequences that derive from this expansion" (2001, 35). *Who* you are connected to impacts your reputation. For example, if a broker's network is populated by gambling/casino management, their reputation within indie scenes may be suspect. Having too many disparate and superficial ties can also be seen in a negative light – in terms of being 'all talk' and attempting to be too many things. In both cases, the consequences are perceptual: they are predicated on others' perceptions of the actor's position in the network. In this way, Podolny claims, ties between actors act both as pipelines (conduits of resources and information) *and* as prisms (lenses which color others' views of the actors involved).

Stovel and Shaw (2012) argue that expanding one's network can cause others to regard an actor with suspicion and hostility. Actors who have managed to secure a brokerage position in a network can benefit from it, but the very act of doing so is likely to undermine this position. Brokers who use their brokerage position for personal gains are viewed as corrupt, and if brokers "fail to effectively manage these gains, they risk undermining the very relationships that keep them at the center of potential transactions and interactions" (Stovel and Shaw, 2012, 154). When brokers are seen as extracting more than they contribute to their network, this damages their standing. This is less of a problem for individuals who work in a larger studio and/or are otherwise financially stable – they can better afford to be selfless with the benefits of their network position and have less to lose if their position in a network is negatively impacted. As we detail elsewhere, cultural intermediaries such as event and coworking space organizers are paradigm brokers and continually walk this fine line as they navigate their own precarious positions (Perks et al., 2019). For example, a co-working space organizer that asks their network to promote their game, contribute free services, and/or accepts 'too many' donations from large corporations could be seen as exploiting their position, undermining relationships built upon mutual goodwill and support.

Additionally, networking and becoming a broker to many diverse groups – which, for the purpose of this chapter, we refer to as becoming a "super-broker" – can result in spreading oneself too thinly. Clement et al.'s (2018) study of French game production companies shows how super-brokers have both positive and negative impacts upon those in their networks. They gather information from different communities, but this also leads to a multitude of collaboration opportunities: the super-broker becomes so invested in other projects and communities, they have little time to dedicate to their own. A studio with a super-broker member risks that broker diverting their labor and resources away from the team's efforts. This leads to a dilemma for small teams who must network to be successful. They require brokers, but cannot support or sustain a dedicated member of their team to do so. Thus, all members wear multiple hats and take on double shifts (Whitson et al., 2018).

The benefits and drawbacks of being connected to a super-broker depend on ones' role (Clement et al., 2018). Creativity-focused roles, such as designers and writers, are likely to benefit from the information spillover that comes from a relationship with a super-broker. Efficiency-focused roles, such as producers and programmers, are likely to suffer from a super-broker's ability to rapidly and selectively extend or withhold information and resources without being affected by the consequential suffering of their contacts. In other words, if a studio is attempting to finish a game and/or is undergoing a crisis that requires a team's full attention, having a super-broker on the team can be a detriment, as their time and energy are commonly diverted elsewhere. For developers contracting on several side-projects, this becomes a common occurrence.

Finally, SNA lends some general insight into exhibiting games at events such as PAX and GDC. While the end-stage of production is the ideal time to promote a game, it is also the most resource-intensive stage, and developers have few remaining resources (both time and financial) to exhibit on show floors. Events may also generate new creative directions and dead ends that divert attention. Pragmatically, there is no easy formula in terms of balancing the need to network with shipping a game, but it does help explain why indies face crunch when, unlike AAA studios, they have more autonomy in setting their release dates: everything needs to be done in the final months of production.

To summarize this section, SNA theory tells us that networking is important and takes different forms in terms of helping developers access material resources and support, as well as generating and refining game ideas. There is a difference between how ideas and resources flow between local networks and distributed networks that are online or temporally restricted (such as events): benefits flow more freely to all members of local networks due to "leaky pipes". In distributed networks, benefits generally flow to brokers who connect different communities and those with already established reputations. While "bigger is commonly better", the perceived pressure to connect with many different communities to help market one's game comes with considerable risks in terms of the work needed to establish and maintain a positive reputation as a broker, and the associated dangers of being perceived as selfish or having a tainted network reputation, and spreading oneself too thin across different work roles and communities.

In the second half of this chapter, we take a closer look at local versus distributed networks. We begin with a discussion of hubs, pointing to key literature on local game-making communities. We then test networking theories empirically, using quantitative and qualitative analysis to demonstrate how geographical location can impact one's trajectory as a developer. We then highlight the strengths and weaknesses of ephemeral events that attempt to distill the benefits of watercooler conversations and other leaky pipes into the span of a few days.

Using SNA to understand local and event-based networks

We start this section with a testable hypothesis generated from SNA theories: developers that have access to local face-to-face development communities exhibit key differences from developers that lack access. Before moving to a discussion of our data, it is important to first situate development hubs as local networks.

Defining "hub"

Florida's influential (2002) work on the creative class, Leadbeater and Oakley's *The Independents* (1999), and McRobbie's more recent (2016) critique all foreground the impact of local communities on cultural production writ large. In game studies, *The Good Hubbing Guide* (Crogan, 2015) emphasizes the importance of "creative territories", detailing strategies for setting up and maintaining local communities of practice that are economically stable in the long-run (Figure 5.2).

For Crogan, a hub is a specific venue/location/organization, usually a co-working space or similar. These physical spaces facilitate the exchange of expertise, allow for spontaneous encounters, and help sustain what is colloquially referred to as "critical mass". Well-established hubs also maintain ties to nearby education institutions, community groups, and representatives from municipal and regional governments, which provide resources

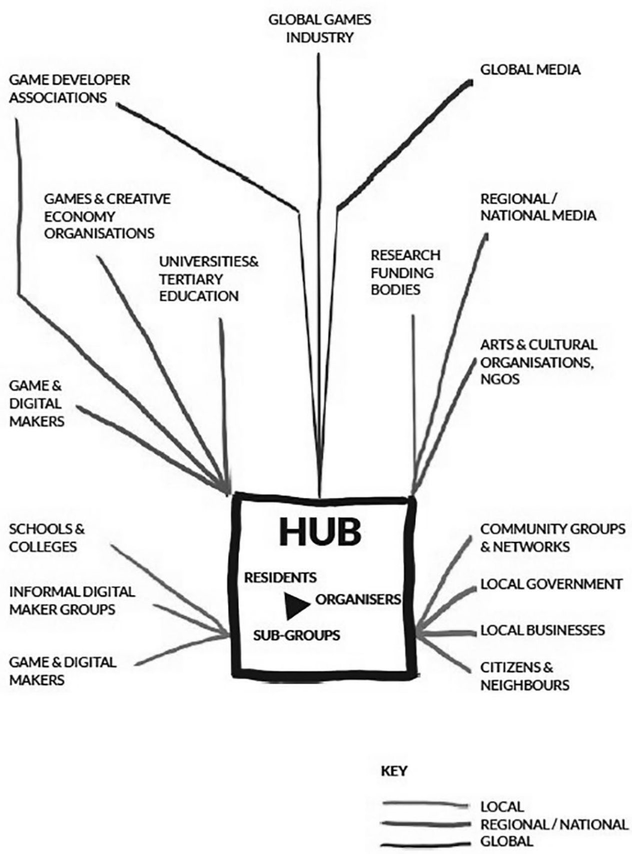

Figure 5.2 Hub diagram. Image credit: Crogan (2015, 5).

and promote interchange. These dynamics are also captured by Guevara-Villalobos' (2011) work on communities of practice.

In contrast, Canadian game studies researchers use "hub" in a different way, discussing whole cities as national hubs (Gouglas et al., 2010; Parker and Jenson, 2017). For Parker and Jenson (2017), the benefits of hubs and the local network effects they generate can be visualized as operating on a spectrum: cities like Toronto, Montreal, and Vancouver exhibit a high concentration of developers and indie activity, whereas locations such as Ottawa or Calgary are more marginal in that they struggle to maintain critical mass, even though they have local community organizations and spaces that could, by Crogan's definition, be defined as hubs. At the far end of the spectrum are geographically isolated developers with access to online communities only. These online networks provide "some access to community, knowledge, resources, and transnational markets"; however, they are a far cry from local scenes which "stabilize and formalize the diverse social/economic networks of actors and activities that encompass indie cultural production" (Parker and Jenson, 2017, 872). This difference is articulated by one interviewee who describes the local Boston community (the site of PAX East, 2015):

> [There's] a group called the Indie Game Collective and they're just a bunch of indie developers that kind of work together. They work in the same [spot] and they throw ideas off each other. It was slightly curated for different companies that brought different things to the table and they just share a lot of ideas and can help out one another. I think it's – you can go alone but it's better to have like kind of that support system around you.
>
> (Non-binary developer from the United States, PAX Prime 2015)

In short, developers in hub cities can find everything they need locally, whereas isolated developers must 'go it alone' and work much harder to find and secure resources and connections.

For the purposes of this chapter, we used a developer's city/town to operationalize whether they were from a hub or not. We formulated a number of criteria that, to us, characterize sites of local leaky pipe exchanges. These are sites where serendipitous meetings and conversations generate flows of contacts, resources, and ideas. These criteria include:

- The existence of production-oriented expertise and resources (for example, studio space, skills/knowledge-base, investors/funding contacts). In this sense, hubs are generally sites with established studios and/or a number of released games;
- The existence of an accessible community of practice that promotes the frequent free exchange of ideas (for example, regular game development meetups, and/or support organizations such as co-working spaces, which promote flows beyond individual studio walls);

- Internal recognition as a developer hub in terms of a collective identity. A sense of 'we're-all-in-this-together' defines the open flow of resources and support within indie communities of practice;
- The existence of public-facing events, such as showcases, that reflect an ability to interface with other institutional structures and scenes;
- External recognition as a developer hub, particularly by cultural intermediaries and state actors. This is connected to economic and legal support which affords access to greater resources.

We then forwarded our criteria to game academics and industry experts with a list of locations and asked them to pinpoint hub cities. In line with Parker and Jenson (2017), there were a few marginal locations that met only some of the criteria. We sorted and coded the data according to whether developers came from a clearly defined hub or elsewhere. It is worth noting that the above criteria also characterize the networking and exchange of cultural, social, and economic capital that takes place at events such as GDC and PAX. The difference is a temporal one: developers in hub cities have access to these leaky pipes at all times, whereas, at events, developers only have access to time-limited resources that act more like firehoses.

Data and analysis

Our data span three different events: PAX Prime 2015 (now known as PAX West), PAX East 2016, and the Game Developers' Conference (GDC) 2016. Along with gathering 123 surveys, we conducted semi-structured interviews and participant observation. The surveys collected demographic and economic information from game studios, including where studios were located, their size, team composition, and launch histories, as well as how they supported their team and funded their activities. Data from the surveys were shared directly with developer communities (Whitson, 2015; Parker, 2015). The interviews additionally asked interviewees about their personal goals, their support systems, and the perceived health of independent production scenes. The data was then cleaned, coded, and collated.[5] We were interested in two variables: whether developers came from a hub city (79 respondents) or not (28 respondents)[6] and whether they considered networking opportunities as a motivation for attending said event. To examine the relationship between these two binary variables, we used Chi-square tests of independence.

We found a number of statistically significant results that support some of the theories of networking detailed above. Not surprisingly, 86.2% of respondents from non-hub cities viewed networking as an important motivation for exhibiting, whereas only 61.5% of respondents from hub cities said the same.[7] Developers from hub cities were also more likely to have launched games before. Only six non-hub developers had launched games before compared to 39 hub developers. We also examined funding sources. Developers that expressly went to events to network were more likely to be interested in connecting with publishers/investors and were more likely to not have had previous contact with publishers/investors. Not surprisingly, this could be interpreted as meaning that developers who do not have regular access to publishers/investors in their home locations go to events expressly to connect with them.

We also wanted to know if developers had differential access to funding depending upon where they came from, including locational differences in whether they relied on past savings, partners/family (i.e. were 'kept devs'), day jobs, and government funding. Statistically speaking, there was no correlation between whether one was a full-time indie and whether they were from a hub or not. There was very little difference in the *number* of funding sources: developers from hubs, on average, reported 2.77 funding sources, whereas non-hub developers reported 2.71, meaning that regardless of where one was from, indie development is funded piecemeal, through multiple sources. But there was a difference in the *forms* of funding (Figure 5.3).

As demonstrated in the graphs below, respondents from geographical hubs were more likely to have game-related day jobs and to use this to

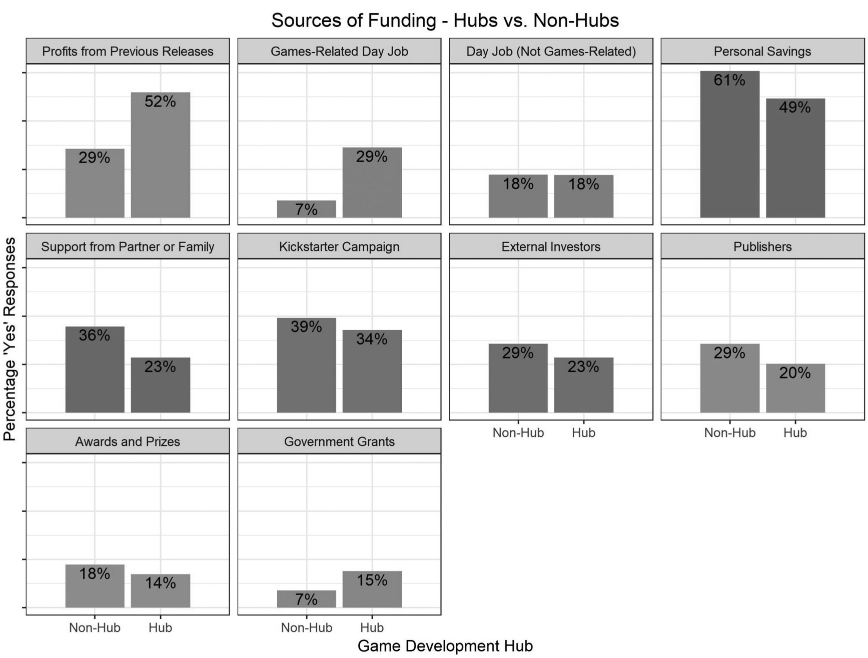

Figure 5.3 Funding graph.

fund their indie work.[8] This is a sensible (if somewhat obvious) result, as those from hubs are likely to have more opportunities to work with game development studios and thus gain experience. Those interested in working in the game industry also likely gravitate toward such hubs. Second, developers from hubs were more likely to fund their games using profits from previous releases. This result corroborates that of the relationship between the 'Hub City' and 'First Launch' variables: if developers have not launched a game yet, they will not have any profits to put toward funding game development; game developers from hubs are more likely to have released a game before *and* to be funding their current game with profits from a previous release. In short, what this means is that developers from hub cities, as a whole, were more experienced, more able to focus on game-making (without needing to rely on non-game related employment), and more targeted in terms of their event activities.

Our qualitative data support these findings. Building what are seen as useful connections to others is simply easier in hubs, with many serendipitous opportunities, such as those described by one participant from Seattle:

> [I]f you are making your own game or you want to break into the games industry, all you have to do is go to a local party. We have like four or five indie game parties a month at this point. So, you're going to show up there and there will be recruiters from Microsoft or ArenaNet or from whoever, you know. ID and Xbox might show up and take an interest in your game, other developers might want to partner with you. If you need people on your team you can always find people at these parties, so… yeah. The success of your game definitely depends on the size and success of your community as well.
>
> (Male developer and IMB team member from USA, PAX Prime 2015)

However, qualitative data adds more nuance to this discussion, as Jorge Blacio, a developer from South American demonstrated. His country lacked a hub, and he and his team were acutely aware of the lack of community, support organizations, funding, and both informal and formal education opportunities. However, he portrayed isolation as having both good and bad elements:

> That's one of the bad things. You do not get [work] experience in your country. And you have to – if you're really really starting out and you don't have any money to do it, how would you be able to get these experiences? But the good thing is that, since you're isolated from the rest of the scene, you're not like tainted through the mainstream stuff that's going on. And you can come up with really unique stuff. Like when

> we came out with our game people see and say, "How the hell did you come up with a flying door?".
>
> (Jorge Blacio, Game Producer at Freaky Creations, PAX Prime 2015)

In this sense, by not belonging to a hub, Blacio's game stood out within a crowded ecosystem on a creative level because it was not shaped by common tastes and influences. While a deeper examination of creativity and how it links to where a developer is from is beyond the scope of this chapter, we think dissenting views such as this are important to include, as they add context that is missing with quantitative data alone.

We would like to also note a significant finding which warrants further investigation: we found a negative correlation between the desire to 'network' at events and the desire to 'engage with fans'. In contrast, there was no relationship between the desire to 'network' and the desire to 'playtest'. Potentially, this could support Wittel's theory of network sociality, in terms of seeing networking as something instrumentally undertaken to access high(er) status connections, resources, and/or services, such as player feedback on how to improve one's game. However, there may be alternate explanations. For example, those interested in networking are generally less connected and experienced as developers and thus may not feel that they have an established fan base with which to engage. This finding is worth exploring further, as we often take for granted that a desire for audience engagement is a fundamental principle of being independent in the first place.

Concluding thoughts

Game developer networks provide resources, mutual support, and privileged access to other individuals and organizations such as publishers, press, and cultural intermediaries. These communities allow developers access to uncommon information, such as feedback on ideas and prototypes and other sources of inspiration, thus spurring innovation. Creativity is not the purview of the individual – it lives within networks, particularly the boundaries separating groups from one another (Uzzi and Spiro, 2005; de Vaan et al., 2015).

This chapter used SNA theory to deepen the oftentimes vague discussions about networking in game development. It questions whether time and resources networking are always beneficially spent, looking at how benefits are differentially distributed according to one's location, role, and stage of development. As we have demonstrated, there is a difference between local communities of practice – hubs – which have continued access to leaky resource pipelines, and other, more isolated developers.

The quantitative analysis supports the theory that there are distinct benefits of belonging to geographic hubs. These benefits can only be partly replicated by temporally limited networks such as events. Developers from hubs are generally more experienced in terms of previous launches and more able to focus on game-making without needing to rely on non-game-related employment. Qualitatively, they are able to access advice and assistance that would not otherwise be available. Temporary developer communities form around events such as GDC and PAX. These events attempt to act as 'watercoolers' for developers; however, these networks are more ephemeral and take much more effort to build and maintain.

While the quantitative analysis can help us understand more about how and why developers network, as well as provide empirical evidence as to the benefits of being able to access local game development hubs, it is worth noting the limitations of this analysis, both in terms of sample size and because we coded for hubs/non-hub ourselves rather than asking developers to self-identify. Our focus on funding and economic stability is both a function of data that is amenable to statistical analysis and our own interest in understanding and generating research that could assist in reducing financial precarity in cultural work. However, as Keogh (2019) points out, it is a mistake to assume that all game-makers aspire to commercialization and careers.

While it seems common sense to claim that geographical hubs are important to cultural production, we think our mixed-method analysis *empirically* – rather than anecdotally – demonstrates that there are distinct differences between those that can easily access resources and those that cannot. This leads to two key takeaways. First, whether one can have a long-term career making games, at an indie scale, is strongly shaped by where one lives. Advantages accrue to those in hubs. In a crowded and competitive ecosystem, developers in hubs have stronger and more easily accessed support networks and thus more assistance 'surviving' than those 'going it alone', particularly in terms of access to funding and skills-development. Not all developers outside of hubs can afford to travel and showcase their games, thus building important connections to others, leading to further structural differences in terms of who is able to sustain a career in games.

This points to our second takeaway. While the existence of hubs makes cultural production more sustainable in these geographical areas, hubs themselves are precarious and typically rely on state funding and support (McRobbie, 2016). Accordingly, state-level funding initiatives are an important area of study when examining how to foster stable, long-term creative communities. If the physical manifestations of hubs such as meet-ups, co-working spaces, and local events disappear, or key studios relocate or withdraw, then these communities of practice have a harder time cohering and sustaining themselves. In these cases, the collapse of a studio or workspace could lead to a cascading effect throughout the local community. Accordingly, as a unit of analysis, we would like to further emphasize the need to study hubs rather than focus on individual games or studios.

Acknowledgments

We would like to thank Felan Parker, Bart Simon, our industry collaborators, Kelly Wallick, Christopher Floyd, and Jason Della Rocca, and most importantly, the many developers who shared their expertise with us.

Notes

1 Darius Kazemi is a developer, artist, and co-founder of *Feel Train*, a creative technology cooperative. This quote is from his article, 'Effective Networking in the Games Industry'.
2 Interview and survey participants were recruited through the Indie MEGABOOTH (IMB), an organization that collectively purchases floor space at large gaming events and conventions, allowing small developers to pool resources and occupy the same space as large multinationals. Quotes and context were approved by interviewees in the writing stage of this chapter. Final quotes are anonymous, unless otherwise requested by interviewees.
3 In Figure 5.1, nodes are all generally well-connected through multiple routes, meaning that there are no distinct communities, and thus no brokers are needed to bridge the structural holes between separate communities.
4 TIGSource is a public online forum for independent game developers to discuss, create, collaborate, and share.
5 The process for re-coding one question bears mentioning: in the GDC and PAX 2016 surveys, respondents were asked to rank their reasons for applying to the IMB exhibit using a five-point Likert Scale. The PAX 2015 data, however, posed the question as a true/false proposition which asked respondents to check all the appropriate boxes. As such, the decision was made to recode the Likert responses from the 2016 surveys as binary true–false responses, where anything listed as a 1 (Most Important), 2, and 3 was considered 'important', whereas anything listed as a 4 or 5 (Least Important) was considered 'not important'.
6 Sixteen survey respondents only identified their country/state. They were removed from this analysis.
7 Because of the way the Likert scale for the 'Networking' variable was interpreted and coded, with a random distribution of responses, we would expect to see 66.7% of respondents indicate that networking was an important motivation.
8 For the funding graph, the only differences that are statistically significant are in the Games-Related Day Job ($p < 0.05$) and Profits from Previous Releases ($p < 0.1$). The graphs showing that non-hub respondents were more likely to rely on personal savings and their partner/family were not statistically significant but indicate an interesting avenue for investigation.

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6 Strange bedfellows

Indie games and academia

Celia Pearce

The indie ecosystem

The definition of 'indie games' is a constantly morphing social construct. Initially, in the early aughts, they were defined as games made without the financial support of a major publisher, typically members of the Entertainment Software Association, which represents the mainstream game industry. Even with this seemingly clear-cut definition, the term often became murky and subjective, as well as exclusionary. Indeed, a variety of game types that met these criteria were often excluded, such as serious games, artgames, games created in an academic context, Twine games, casual, and even mobile games. This had the perhaps unintended consequence of marginalizing people of color and what I call "people of gender". Since then the definition has evolved, and even individual games can change status over time. Furthermore, new publishing and funding models have shifted the role of publishers. In practice, 'indie' may now be more of an ideological stance, as well as an aesthetic conceit (Juul, 2014; 2019).

In my prior work on online play communities (Pearce, 2009), I used the term "ecosystem" to describe the environment in which a specific set of emergent behaviors takes place. In trying to form a holistic view of the current situation with indie games, I was able to identify 10 factors of the indie ecosystem, which are involved in a complex web of interdependencies and feedback loops. While a more detailed description is provided in my book *IndieCade: A History* (2020), the summary below provides an overview and context for the case studies that follow. (Specific factors are indicated via italics.)

The question of games as a legitimate form of cultural production has long been controversial. Since the late 1990s, the notion of *Games as Art* – as expressive form rather than commodity – has helped to elevate their cultural status. However, it took nearly a decade for shifts in *Public Policy and Perception* to follow suit. In the United States, it was not until 2011 that videogames were deemed Free Speech by the Supreme Court, and the same year, they were added to as a funding category by the National Endowment for the Arts. *Academia* (the third factor) had a major hand in both developments. Many of the early artgame practitioners emerged from art schools; and influential curators, art historians, and critics responsible for their appearance in museums, galleries, and alternative exhibition spaces were often academics (Clarke and Mitchell, 2007; Parker, 2013; Schrank, 2014; Sharp, 2015). Academics also played a major role in shifting the public perception of games beyond moral panic, both tangibly, in contributing to legal amicus briefs, and intangibly, through working to advance public discourse. This also led to new approaches to *Game Journalism and Criticism* beyond merely product reviews. Indeed, the term "game journalism" has become a contested site in light of controversies surrounding its so-called "ethics" (Colbert and Hoskinson, 2014; Consalvo, 2009; Nieborg et al., 2009); however, the growth of indie games could not have taken place without a substantive contribution from game writers and critics, who often skirt around the intersections of academia, public intellectualism, and journalism.

From the economic side, three significant factors contributed to the rise of indie games. *New Tools* lowered barriers of entry to game development. Adobe Flash, plus fan and artist modding practices (Clarke and Mitchell, 2007; Pearce, 2006; Schrank, 2014; Sharp, 2015), eventually led to the development of stand-alone 3D game engines such as Torque, Unity, and Unreal; and later, RPG Maker, Gamemaker, and Twine further broadened access to game creation. Alongside the development of tools has been two major shifts in the economics of game development. The first Among these is what I call '*Techtonic*' *Shifts (Changing Platforms)*. By far, the biggest game changer here has been the rise of smartphones over consoles as the primary gaming platform (Wijman, 2018), further broadening the gaming demographic. Additionally, the resurgence of virtual reality (VR), captured here in the third case study, has also had a major impact in the platform arena. These shifts have also reflected changes in *IndieNomics (Publishing & Funding Models)*. This includes the advent of early self-publishing portals like Manifesto, Steam, and, more recently, itch.io, the App Store, and crowdfunding sites like Kickstarter and IndieGogo, as well as more traditionally structured venture capital groups such as the Indie Game Fund. At the same time, the mainstream industry found itself in the increasingly unsustainable situation of rising production costs, a self-inflicted wound caused in part by the battle for higher graphics quality, without a requisite increase in retail price (Rivera, 2019). Indie games became a low-cost alternative to populating downloadable console catalogs. At the same time, producers of indie content in other media – such as Franco-German TV Station ARTE France and US indie film studio Annapurna – have gotten into the act. In addition, there has been a growing trend of independent studio acquisition.

The two final factors of the indie ecosystem revolve in large part around community-building. The growth of *Creator Communities* has built a culture of "autodidactic communalism" (Pearce, 1997). This includes short-term provisional communities, such as game jams and incubators (Conditt and Tremp, 2018; *Global Game Jam®*, 2014), as well as indie collectives, which are common in the USA, and hubs, which are more pervasive in the UK and Europe (Crogan, 2015). These provide developers with a cushion against the economic precarity of the indie lifestyle (Keogh, 2015; O'Donnell, 2014; Parker, 2013; Parker et al., 2017; Whitson, 2013) as well as a level of nimbleness and scalability.

At the intersection of all these various components of the indie ecosystem are *Festivals, Exhibitions & Awards*, the primary focus of my larger monograph *IndieCade: A History - The Interdependence of Independents* (Pearce, 2020, as well as the first case study in this chapter). These function as cultural intermediaries (Bourdieu, 1984; Parker et al., 2017), as fulcrums for the other components in the indie ecosystem – bringing together game-makers, audiences, publishers, tech companies, academia, the press, etc. From industry-embedded exhibitions, such as the Game Developers Conference's Independent Game Festival (IGF) and the Indie Megabooth, to stand-alone festivals such as IndieCade and A MAZE., to game art parties such as Wild Rumpus and "artcades" such as BabyCastles, to more formal artworld venues such as galleries and museums, they serve as tastemakers, raising the bar for quality and innovation, exposing a broader public to new works, and serving a market-testing function for publishers, who often serve as sponsors.

Case studies

Case study #1: IndieCade

IndieCade is a stand-alone festival of independent games that takes place annually in Los Angeles every Fall and also hosts other events and showcases, including the IndieCade Showcase @ E3 (the giant commercial game expo), IndieCade East, and IndieCade Europe (Pearce, 2020). It functions as a hub and cultural intermediary among the other nine elements of the indie ecosystem, as well as a community of play and practice where developers come together to network, share, playtest and celebrate one another's work. It is not an exaggeration to say that IndieCade would be impossible without the support of academics. As co-founder of IndieCade, and an academic myself, I have a unique insight into its complex interdependencies and entanglements with academia.

Two of IndieCade's co-founders – Stephanie Barish and myself – initially met while working at the University of Southern California (USC). Sam Roberts, our third co-founder, is working there at this writing. After leaving USC, I co-curated ALT+CTRL (not to be confused with the Game Developers Conference showcase focusing on alternative controllers, alt.ctrl.GDC) at UC Irvine's Beall Center for Art & Technology, a juried follow-on to a prior exhibition by my co-curators, Antoinette Lafarge and Robert Nideffer. In addition to being located at a university, ALT+CTRL brought together works by a number of artgame practitioners who were coming out of art schools and universities. Among them were founders of L.A.'s C-Level, the first artgame collective in the US, founded by graduates from California Institute of the Arts, whose work represents a particular aesthetic "school" of critical art practice, which interrogates game culture in a variety of ways. Other featured artists included Molleindustria and Tale of Tales, both of which went on to become household names in the indie scene.

Shortly after ALT+CTRL, I co-founded Ludica with Tracy Fullerton, Janine Fron, and Jacquelyn Ford Morie, all of whom were employed at USC at the time. Ludica produced a series of academic workshops and papers that critiqued aspects of what we came to call the 'Hegemony of Play' (Fron et al., 2007), the ways in which the mainstream game industry controlled the discourse about the definition of games, play, and fun. Concurrently, I published a paper on the use of games in the 20th-century avant-garde which traced the roots of artgames back to movements such as dada and Fluxus (Pearce, 2006). These papers formed the basis of some of IndieCade's underlying philosophies. They were also part of a larger movement of feminist game criticism that was taking place within academic circles at Digital Games Research Association (DiGRA) and Foundations of Digital Games, as well as game journals such as *Journal of Game Studies, Games & Culture,* and *Well Played*, and the feminist game journal *Loading....*

2006 was a pivotal year in terms of indies and academia. Two student "indie blockbusters" hit the festival scene: *Narbacular Drop*, later renamed and published by Valve as *Portal*, and *Cloud*, the game that led to the formation of thatgamecompany. Both took awards at the Slamdance Guerilla Games Competition, as well as becoming IGF finalists and later going on to win mainstream game awards. They created a new aspiration for students entering academic game programs.

The following year saw a controversy engulf and ultimately destroy Slamdance's brief foray into games. Colloquially known as 'Slamgate', it involved the banning by the larger the organization of finalist *Super Columbine Massacre*. This led to a widespread boycott of the showcase by universities and colleges, including USC. It also spawned widespread criticism and discussion among academics – also intersecting with Game Criticism – about the double-standard applied to film and the need to build game literacy. In the midst of this maelstrom, Slamdance's game showcase director Sam Roberts joined the team of IndieCade.

The first IndieCade event took place in the summer of 2007, ironically hosted at E3 by the Entertainment Software Association, the very organization to which indie festivals define themselves in opposition. The ESA relationship had arisen as a direct result of the organization's prior support of academic conferences and research, including an ongoing relationship with Stephanie, and a conference that I organized at USC on the future of interactive entertainment (*Entertainment in the Interactive Age*, 2001).

From the beginning, IndieCade chose not to segregate student games, as well as taking the somewhat unprecedented step of showing work by their professors. One of the first examples was *The Night Journey*, which IndieCade showed in prototype form at its first showcase. The game, created by video artist Bill Viola in collaboration with Tracy Fullerton at USC's Game Innovation Lab (which had also funded *Cloud*), had been controversially rejected by the IGF, but won an award at IndieCade 2008. *The Night Journey* epitomizes the "USC school" of design, characterized as "un-games", "anti-games", or "zen games". Academic games also intersect with other previously excluded genres such as serious and art- and alternative controller games.

IndieCade, like many other venues, is highly dependent on unpaid labor (Parker et al., 2017), most of which comes out of academia, from program chairs, to jurors, to student volunteers. While this can be viewed as exploitive (rightly so), the reality is much more nuanced. First, as has been pointed out earlier, game academia has a major hand in bringing cultural cachet to videogames, both in terms of the Public Perception and Policy dimension of the ecosystem, and in terms of promoting Games as Art. Being involved in events like IndieCade provides a platform for academics to influence the larger direction that indie games are taking. Second, participation in events like IndieCade has a marketing function for academic institutions. They were among the first to sponsor, and exposure of student and faculty work can influence rankings, reputation, publication deals, and the formation of indie studios. Third, in addition to serving as a career onramp for students, we've seen a growing fluidity in careers across academia and industry, sometimes integrating both. Fourth, academia operates on a currency system orthogonal to the capitalist monetary measurement of labor. For tenure track faculty, community engagement counts as service, which impacts tenurability. For students, volunteering serves a networking function, leading to internships, employment, collaborations, and new studios. Some schools also offer special rewards for participating in IndieCade, such as USC's badge system and student travel grants for festival acceptance. The use of student volunteers is not unique to IndieCade or even to games. Across disciplines, they are a mainstay of academic conferences, festivals, and even for-profit industry events such as GDC and PAX (Penny Arcade Expo).

Aside from the "karma" factor of contributing to IndieCade as a community of play and practice, there are also tangible outcomes, most notably industry exposure. Numerous academics and their students have received funding and publishing deals due to participation in IndieCade, through both direct introductions and incidental encounters. The festival's marketing function allows sponsors to get a sense of the potential audience for a game and also provides press exposure for game-makers (intersecting with the Game Criticism component of the ecosystem), which is one of the few ways indies can afford to market their games. Many indie developers also take advantage of IndieCade as a platform for promoting Kickstarters or other crowdsource funding, intersecting with the IndieNomics aspect of the ecosystem. Therefore, the unpaid labor calculation has for some resulted in monetary benefits, which are not well-measured or documented.

In part because of this synergistic relationship with academia, the IndieCade scheduling cycle has always been tied to the academic calendar. The Spring submission deadline coincides with the point where both students and faculty are typically finishing up projects, particularly thesis projects, which often win awards. The E3 showcase happens in early summer, a time when students and academics have more flexibility in terms of time and travel. Due to the festival's timing in mid-Fall, the bulk of the labor that goes into jurying, programming, and conference planning takes place in the academic off-season of summer.

Finally, as alluded to above, a number of key schools also support IndieCade with in-kind and financial sponsorship. At the first IndieCade in Bellevue, Washington, computers and student volunteers were provided by nearby Digipen. Schools sponsor tables, show demos, host sessions and parties, and support IndieCade by providing space: IndieCade 2016 was hosted at USC, the 2018 and 2019 Festivals were at Santa Monica City College, and IndieCade Europe launched 2016 at the École nationale supérieure d'arts et métiers.

Aside from those already mentioned, other programs that have been involved in IndieCade, whether through volunteer labor, sponsorship, or exhibition include: the NYU Game Center and Parson The New School, which were instrumental in launching IndieCade East; Georgia Tech and Savannah College of Art & Design; Carnegie Mellon's Entertainment Technology Center; UC Santa Cruz, UCLA and UC Irvine, and the IT University of Copenhagen, which is heavily implicated in our second case study.

Case study #2: Copenhagen game collective

This case study will focus on the synergy between Academia and Creative Communities, how they interact with each other directly, and how they connect to and through other ecosystem factors. The academic center of gravity of this narrative is the IT University of Copenhagen. Copenhagen ITU (as it's also known) was founded in 1999, one of the first universities in Europe to offer a games degree. In 2003, its Centre for Computer Games Research spun out of the Department of Digital Aesthetics and Communication. One of ITU's distinguishing features was the merger of theory and practice, an idea which was fairly new at the time. Started by humanists and social scientists within a technical university, ITU's games program was highly influential in the fledgling academic study of games. Co-founding faculty Espen Aarseth and Susanna Tosca were also involved in launching the *Journal of Game Studies* in 2001, one of the first academic publications in the field (Erard, 2004), and DiGRA, the Digital Games Research Association in 2003. This made ITU a magnet for the brightest thinkers in game studies, spawning well-known game scholars such as Jesper Juul and Gonzalo Frasca.

In 2008, a group of Ph.D. students from Copenhagen ITU were accepted into the first IndieCade festival in Bellevue, Washington with an unusual project entitled *Dark Room Sex Game*. This Wiimote-based, audio-only local multiplayer erotic rhythm game was described thusly by its creators:

> Our idea was that a sex game would in fact be even more erotic if one stripped away all visuals and forced players to use their imaginations. Our hope was to use a combination of humor and embarrassment to make players uncomfortable, yet strangely satisfied. One could view the game as a commentary on the dull fixation on visuals in the spheres of sex and also videogames.
>
> (*Copenhagen Game Collective*, n.d.)

Dark Room Sex Game earned the IndieCade 2008 Most Fun Game Award precisely, according to award jurors, because it managed to blend fun with embarrassment in a satisfying way. The game was presented by three ITU Ph.D. students: Lau Korsgaard, Dajana Dimovska, and Douglas Wilson. There, they met another Danish student, Nils Deneken, who won an award for *Rückblende* (or *Flashback*), his final project for his design degree at Essen University in Germany. Recognizing that there was another Dane at the Festival, the ITU students reached out to Deneken, who had already founded Die Gute Fabrik in Copenhagen.

Together, they formed the Copenhagen Game Collective (CGC) in 2008. CGC now functions as a nonprofit, but began its life as a creative collaborative. The founders described it in a 2014 interview as "…a loose network of creative individuals…that likes to run events, exhibitions and throw parties around experimental independent games". (Barron, 2014) In part due to being forged in a humanistically focused academic program, CGC's distinctively Nordic style – what might be termed the "Copenhagen School" – is known for irreverent cultural commentary through highly social "digital folk games" for adults that combine technology with physical play.

Around the time that CGC was formed, Wilson officially joined Die Gute Fabrik as a partner, along with ITU grad Christoffer Holmgård. The studio has a synergistic relationship to CGC, serving in some ways as its commercial counterpart. An early example is *B.U.T.T.O.N. (Brutally Unfair Tactics Totally O.K. Now)*, which demonstrates CGC members' signature tactic of subverting the intended use of game controllers to promote physical, social play. It is reminiscent of the children's game "Mother May I?", but with Xbox controllers. *B.U.T.T.O.N.* appeared at IndieCade's E3 Showcase in 2010 and was awardee. It became the first CGC game to be published on a console; development is credited to CGC, Die Gute Fabrik, and NapNok (formerly KnapNok) Games, all affiliated with CGC. *B.U.T.T.O.N.* is also a case of what would traditionally have been viewed as an artgame transitioning into a commercial one. Since then, members and affiliated studios have published numerous commercial games, including Bernhard Schulenburg's 2020 IndieCade Finalist *Where is My Heart?*, NapNok's *Affordable Space Adventures* (which grew out of an IndieCade game jam project) on Nintendo WiiU, *Spin the Bottle: Bumpie's Party* on Wii, and Doug Wilson's *Johann Sebastian Joust*, which won awards at both IndieCade and GDC and later became the foundation for the Sony PlayStation bundle *Sports Friends*.

The latter illustrates more broadly some of the synergies between academia and Creative Communities. Of the four games included in *Sports Friends*, three were designed by academic research/practitioners, including Ramiro Corbetta's *Hokra*, and *Super Pole Riders* by Bennet Foddy, an NYU faculty member. The bundling strategy allowed for smaller games to be packed together thematically; but equally importantly, it was among the first titles published on a platform whose developers included academics. These three developers are exemplary of a kind hybrid research/practitioner who both creates games and publishes academic writing, often about or related to their practice, a profile that many CGC members embody.

Most of the groups and individuals mentioned here (including ITU) have a relationship with the "Spilhuset" or "Play House" in Copenhagen Central City. The Spilhuset comes out of the traditional Danish "Culture House", a kind of combination co-working space and community/cultural center. A converted apartment building, the Spilhuset serves as a central hub for a number of indie studios in Copenhagen. In addition to those mentioned earlier, its resident studios have included Playdead (creators of *Limbo* and *Inside*), Robocat (later Northplay), GameSwing, and PressPlay (a spin-off of Flashbulb Games), as well as a host of freelancers and solo developers.

ITU was also instrumental in the launch of the Nordic Game Jam, a locus of the Danish indie scene and an example of a Provisional Creator Community. The first Nordic Game Jam, held in 2006 at ITU, hosted around 30 participants, working in one room in Flash. Copenhagen-based Unity showed up with an early Mac-only version of its game engine, pitching it as a replacement for Flash and Gamemaker and bringing a message of developer emancipation. Here, we see the intersection of Tools, Academia, and Creative Communities. The next section will also explore those relationships in terms of "Techtonics" and changing platforms. CGC itself has hosted events such as the Yonderplay Showcase at the Nordic Games Conference, w00t, a stand-alone festival of physical play in Copenhagen, and Hebocon a Denmark-wide series of robot tournaments. Having a non-profit collective also gives members (including commercial game affiliates) an umbrella under which to experiment with artgames and public projects which might not have immediate commercial potential. It also allows the group to be eligible for public funding.

These ongoing connections to academia are crucial in three ways: first, from an economic perspective, indie studios have notoriously unstable revenue which makes it difficult to sustain a full-time staff. Academia enables these studios to be sustainable by supplementing the incomes of key team members. Second, having team members in Academia means creators can keep one foot in research and artistic practice, which feeds back into their studio's design practice. For instance, Die Gute Fabrik's current project, *Mutazione*, is highly informed by Holmgård's psychology and AI background, and Doug Wilson's has published peer-review articles about Die Gute Fabrik's games (Wilson, 2010). It is also important to note that for full-time academics, having a best-selling commercial game is of less consequence in terms of academic currency than having a festival award-winning game. Third, having members in academia maintains a connection to the pipeline that produces up-and-coming developers, which can be beneficial in terms of cultivating new talent and keeping the community vibrant and relevant.

CGC is not the only Creative Community with ties to academia; in fact, this scenario is quite common. L.A.'s indie game collective Glitch City, for instance, whose initial headquarters was walking distance to IndieCade, boasts a high percentage of graduates from USC as well as UCLA. In some cases, as with the NYU Game Center Incubator, the Creative Community is actually housed within the University itself.

Case study #3: Oculus

The story of Oculus touches many other aspects of the indie ecosystem and provides perhaps the best illustration among the three studies of the complex web of interdependencies between its factors, as well as the sometimes hidden or underreported influences of academia. Putting aside the more recent controversies surrounding him, early journalistic accounts of Oculus typically focused on its founder, inventor Palmer Luckey. As a teenager, he began modding obsolete vintage VR headsets before trying to build his own at 16 (Kumparak, 2014). Given the technology's history within academic research labs, at least a few of the VR headsets he tinkered with were, in all likelihood, either created or used in academic settings. He took advantage of the fact that these once-high tech systems could be gotten for a song in order to better understand earlier generations of devices.

Before famously dropping out of Cal State Long Beach, Luckey got a job at University of Southern California's Institute for Creative Technologies, an academic R&D lab bridging military technologies and entertainment. There, he found a mentor in Mark Bolas, a "bona fide 20th Century VR pioneer" (Crecente, 2016), who had been instrumental in so-called "first wave VR" in the 1990s, the very generation of vintage gear that Luckey had cut his teeth on. Bolas had studied under NASA VR pioneer Scott Fisher, who was the Chair of the USC's Interactive Media Division at the time both Bolas and Luckey worked there.

While working at USC, Luckey was exposed to a community of content creators from the School of Cinematic Arts' Interactive Media Division, as well as faculty and students in the IMD including Tracy Fullerton, Sam

Roberts, VR art pioneer Jacquelyn Ford Morie, and Laird Malamed, an MIT grad and game industry veteran who would become Oculus' COO. Among them was Nonny de la Peña, in whose Immersive Journalism Lab Luckey – initially a journalism major – also worked. De la Peña would eventually integrate the Oculus Rift into her own work. Thus, the academic environment also exposed Luckey to collaborators and content creators who became key components of Oculus' business strategy.

Oculus' intersection with the IndieNomics factor of the ecosystem is well-reported, having raised $2,437,429 million on Kickstarter in August 2012, nearly 10 times its initial ask of $250,000. As it turns out, a significant number of supporters were academics and indies. Importantly, the first iteration of the Oculus Rift headset was not targeted to consumers:

> Instead, this first release was meant for developers and early adopters to get their feet wet with VR. To cleanse the palate of any bad taste left by the headsets of yesteryear. To get people building things in VR. (Kumparak, 2014)

After its Kickstarter success, Oculus made a series of strategic moves aimed at amplifying the early adoption of the Oculus DK1 (DevKit 1) among content creators, including academics, students, and indies. Learning from the mistakes of its forebears, the company understood that content was going to make or break its platform (Lien, 2013). To fortify this strategy, Oculus launched three initiatives to ensure its success.

First, the company started sending free Devkits to university labs. This often-overlooked part of the story meant that dozens of DK1s were sitting around university labs waiting to be experimented with by curious academic researchers and students. Oculus knew: (1) they'd be willing to work with a prototype; (2) they'd make innovative work, unencumbered by investor return considerations; (3) students would learn how to use it and then take those skills into industry. One example occurred in my own lab at Georgia Tech, the Emergent Game Group {egg}. Over the summer, a team of students decided to port our underwater MMOG *Mermaids* to Oculus Rift. Members of that team subsequently spawned two indie studios that went on to receive some notoriety for their VR work: Turbo Button, which created *SMS Racing* and *Floor Plan*, was also involved in IndieCade's VR Jam with Oculus, and The Stork Burnt Down, creators of *Home Improvisation*, which became a Sony VR Launch title, were all shown at IndieCade.

The second strategic move was to forge partnerships with toolmakers. In a 2013 Polygon interview, Luckey explained:

> There's never been a virtual reality headset with a really good set of tools that makes it easy for game developers to work with it… We're built into Unity and Unreal Engine, which a lot of games are made in. If I'm a developer, it's very easy for me to make my game support VR.

> It's not a huge research project. It's something you can have one guy working on, and get it at least sort of working in a couple of days and you can really flesh it out in just a few months.
>
> (Lien, 2013)

This intersection with the Tools factor made the system more accessible to developers of all sizes, but particularly academic researchers, students, and indies, which brings us to the company's third strategic move.

In 2013, Oculus began a long-standing partnership with IndieCade. Forged through its relationships at USC – particularly COO Laird Malamed, who was also a professor there – Oculus derived three benefits from working with IndieCade: first, association with IndieCade earned Oculus "indie cred", putting the company in the midst of a high concentration of a major target developer audience and rewarding the indie community that had supported it on Kickstarter. Second, the company had figured out early in the game that the best way to sell the headset was to let people try it. Some of the first public demos of DK1 content were commissioned by IndieCade and shown at its 2013 Electronic Entertainment Expo (E3) showcase, also creating an opportunity for the company to network with indie devs (Futter, 2013; Hilliard, 2013) as well as mainstream publishers. Third, the partnership produced tangible economic outcomes – the funding of innovative content for the platform. The initiative launched with the Oculus/IndieCade VR "Slow Jam", a two-week distributed game jam timed to coincide with the IndieCade Festival taking place in October. The rationale for the jam arose from Oculus' commitment to indies:

> For the past year, the creators of the Oculus Rift have maintained that the best, most innovative experiences for their virtual reality headset will come from small, independent developers. They're the ones willing to take risks, they say, the game-makers unburdened from traditional thinking.
>
> "The VR rulebook is being written right now," Nate Mitchell, VP of product at Oculus, told Polygon in an interview. "And there's a very high probability that the killer apps [for Oculus Rift] don't come from AAA developers, but from indies just trying new things. It's pretty much a level playing field in terms of game design."
>
> "We're big believers that great software drives hardware adoption," Mitchell said. "Helping developers build great games and content makes the content that much stronger. Without great games, the Rift is just a heavy pair of ski goggles, right? By working with developers to do a game jam, giving them the opportunity to innovate, it gave them an opportunity to try new things"
>
> (McWhertor, 2013)

Because the DK1 was in limited circulation, there were only two ways to participate: (1) be a Kickstarter supporter with your own headset; (2) go to a designated IndieCade "playtest hub", typically either indie collectives or – you guessed it – academic labs that already had Oculus Rift DK1's on hand. Any hub without one could get a loaner system through IndieCade (Gilbert, 2013). There were two $10K grand prizes. One went to *Ciess* by E. McNeil, which went on to be published as *Darknet* (James, 2014); the other was *Virtual Internet Hacker* by Lau Korsgaard of CGC and NapNok (featured in Case Study #1). Second place slots were shared by Brian Schrank, one of my former Georgia Tech Ph.D. students, with his students at DePaul University, and Robert Yang, an indie developer known for his queer-themed games, who also taught at NYU and Parsons in New York City.

The widespread circulation of Oculus headsets to universities also meant that they would be available for other Game Jams. The following year, three Carleton University alumni – Ben Kane, Brian Fetter, and Allen Pestaluky – formed a team at the 2014 Global Game Jam (GGJ) at Algonquin College in Ottawa to produce what was arguably the first indie VR blockbuster: *Keep Talking And Nobody Explodes*. The game went on to win the 2014 IndieCade Media Choice Award and became something of a cult classic.

Between the GGJ and IndieCade 2014, Oculus announced that it was being acquired by Facebook for a reported $2 billion dollars and later updated to $3 billion. It would be two years before the consumer headset was released. In the meantime, indie developers, students, and academics were making dozens of Oculus games, showing them at festivals, and publishing them on *Steam*. And this is at the point where the only people who could play Steam-published Oculus games were people who had access to DevKits (the DK2 was released in 2014), many of which were in academic labs. Oculus went on to fund a number of VR titles, many through its partnership with IndieCade.

Conclusion

In this chapter, I have provided an overview of the "indie game ecosystem" and used case studies of three of its components to illustrate the complex and often invisible entanglements between academia and indie games. What these narratives tell is that, far from being "academic" in the pejorative sense, academia has played a vital role in both the cultural and economic rise of indie games. Some of its contributions have been more obvious and better documented – such as success of student games in the indie marketplace – while others have gone largely unreported, such as academia's role in legitimizing games as a form of cultural production, supporting the development of new technologies in varying ways, or producing different "schools" – aesthetics and philosophies of game design. We also explored academia's economic impact, whether providing a financial cushion for indie developers and studios, or serving as a labor pool for festivals in exchange for tangible and intangible currencies such as program visibility, publication of student and faculty games, and tenure service points.

More broadly, this analysis underscores the need for more research into understanding the complex dynamics which have led to independent games going from a small niche to a dominant sector of the video game industry.

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Part III
Techniques

7 The conditions of videogame production

The nature and stakes of creative freedom in Stiegler's philosophy of technicity[1]

Patrick Crogan

In 2014, the National Endowment for Science, Technology and the Arts (NESTA), a UK creative industry and innovation charity, released the report "A Map of the UK Games Industry" (Bakshi and Mateos-Garcia, 2014). This influential report, one of several which has contributed to the increased influence of NESTA in public policy and funding initiatives concerning the creative economy, claimed that games production has been significantly understated in estimates of the UK's "creative" and "cultural industries".[2] The report is based on an innovative "big data" methodology that the authors argue has picked up more accurate and up to date information about the dynamic state of the games sector. By "scraping" data from "product directories, wikis and games reviews sites", the report found that many smaller and medium enterprises remain un-reported via more conventional industry measurement techniques and survey instruments (Bakshi and Mateos-Garcia, 2014). The report included a map of the UK that identified various regions of the country, where a previously invisible sector of smaller, less-established games producers were present and displayed a dynamic and lively emergence.

In a blogpost accompanying the announcement of the report, co-author Juan Mateos-Garcia summarizes the key findings which confirm widely perceived trends in the global game industry over the last decade such as the expansion of production due to growth in the mobile and tablet app and software markets (above all Apple's iPhone and iPad), the lowering of capital and cost barriers for entry to these, and other platforms encouraging smaller developer enterprises. The report presents what it argues is a more effective means of representing the UK games industry's changing profile as part of the dynamic shifts in the global digital media landscape. It showcases some useful new means for tracking and representing the shifting state of games production and, in particular, in registering the emergence of "indie" game producers and the progress of smaller and startup firms from small to small-to-medium enterprises. It also provides further confirmation of the prominent place of games production as part of the shifting profile of the media industries in the UK. Its articulation of the value of these tracking and profiling methods is resolutely located in existing creative economy terms – the new picture of games production revises estimates of the economic (monetary) worth of the industry and is analyzed in terms of how it tallies with established accounts of creative clustering and IT infrastructure, education, and training provision. The primarily economic orientation of this account of the structural and contextual factors enabling the production of creative forms has little to say about their nature or value other than in monetary or employment terms, as gross domestic product, as enhancement to a "Creative UK" brand, and so on.

It is arguably unreasonable to insist that an organization with NESTA's history and mission statement should provide a meta-critical analysis of its influential contributions to public debate about and policy-making advocacy for the creative industries. One might suggest, however, that this report's lack of a substantial engagement in other ways of considering or advocating the potential value of innovation in game or other creative production seems to represent an evolution from NESTA's historical commitment to help with the "creation of products or services with commercial, *cultural or social* potential" (NESTA, 2017, my italics). That job falls increasingly to those critics, proponents, and producers seeking ways to look past the blindspots of creative industry modes of interpreting the contemporary state of media and cultural production. In this regard, it should also be noted, however, in keeping with the general drive toward the instrumentalization of university labor (both in teaching and research) in the UK and elsewhere in the globalization of what is seen as the tertiary education "market", that NESTA has attained a very influential position in setting the agenda of UK national funding of scholarly research in videogames and other digital media. For instance, its 2013 "Next Gen" report (NESTA, 2013) set the terms of the first ever round of the UK's Arts and Humanities Research Council Funding dedicated to videogames in 2013. Next, Gen had the goal of supporting "better research-oriented university/industry collaborations with a view to accelerating innovation and maintaining competitive advantage in the videogames industry" (Livingstone and Hope, 2013). It was explicitly cited as the rationale for the AHRC's Video Games Research Networking Scheme call for proposals in guidance to applicants document that has since been taken down from the AHRC's website.

Along with Jonathan Dovey and other colleagues at my university's Digital Cultures Research Centre, and with Helen Kennedy at Brighton University, Joost Raessens and Stefan Werning at Utrecht University, principals of game production hubs in Bristol, Leamington Spa, Utrecht, and other researchers and game developers, I led the Creative Territories project investigating the cultural dimensions and implications of the current developments in the games industry in the UK and as part of global trends (Crogan, 2015). I begin with this UK instance of efforts to map and measure the current trends in national computer games production to indicate the context for exploring the nature and significance of games as bearers of potential value beyond a strictly economic conception of value. This was a key aim of

Creative Territories. For one thing, this mode of state-funded institutional framing of computer games as an industrial and economic object of interest does not readily accommodate the kinds of approach and engagement with computer games that go with the term philosophy – however diverse the schools, methods, and allegiances that could be gathered together under that banner.[3] The philosophical questioning of contemporary phenomena is increasingly difficult to maintain today where university funding is increasingly distributed and evaluated according to principles of immediately tangible real-world applicability and utility. The criteria arise from instrumental, pragmatic, and commercially focused prerogatives which presume to know what is most pertinent today and measured by techniques of data analysis and variants of the marketing survey and testimonial narrative.

In addition, the purportedly more "democratized" conditions (Garcia, 2014) of game-making that NESTA's mapping of the industry evidence are important not only for narrowly philosophical but also for all critical interrogations of computer games as an already significant and yet still emerging form of cultural production, expression, and experience. The question of the conditions – technological, economic, environmental, and cultural – of making and acting in the digital technocultural milieu is of primary importance for Bernard Stiegler's philosophical activist project, a project which seeks to combine and cross-fertilize with other critical and practical interventions in digital culture. Through a multi-level strategy, Stiegler's project seeks both to understand and to intervene in the way that digital media innovations are adopted and become established practices and routines. This goal is pursued through various activities – research, policy advocacy, labs and prototyping, partnerships, and outreach – which are conducted both by the independent association Ars Industrialis he co-founded and as the Pompidou Centre's Institut de Recherche et d'Innovation that he directs. Stiegler and his various collaborators develop alternative perspectives and alternative experimental models for fostering a less narrow and more comprehensively valuable digitally mediated culture and society.

In this text, I would like to offer some thoughts about the ongoing becoming of computer games inspired by Stiegler's project. While Stiegler himself has had little to say about them, the changing conditions of game production today are another important instance of the ongoing digital transformation of human cultural becoming with which he is chiefly concerned. They bear the potential of digital and networked making and sharing to reshape the established modes of commercial industrial production. This potential is for the emergence of a more participatory or "contributory" model of the collective production of forms of expression and experience, one which revises existing means of valuing games and supporting their development, just as much as it is for forms that accelerate the techno-capitalist globalization of industrial mediation and exacerbate its injurious effects on cultures, communities, and economies – even in those places (such as the UK) seeking to optimize the opportunities arising from this dynamic.

This is what Stiegler calls the "pharmacological" character of technological development understood as an integral part of human becoming – its potential as both poison and cure to human existence. Stiegler's philosophy is complex and wide-ranging in its ambition to engage with contemporary technoculture. I will walk a fine line in this paper between explaining his key concepts as they pertain to the subject of this study and providing an adequately comprehensive overview of his philosophy. Technology is always a *pharmakon* in Stiegler's account of the constitutive role that it plays as prosthetic supplement to human being (Stiegler, 2013). A core claim of his philosophy concerns what he describes as the irreducible technicity of the human: there is no human being without technical supplementation. The *pharmakon* – which means both medicine and poison – of technical forms is not optional; it is a constitutive part of the human as "technical life". For what distinguishes human being from the other members of the animal kingdom is the technical character of human evolution (Stiegler, 1998). Like all species, the human being is an evolving becoming rather than a permanence, but its becoming is inextricably bound up and composed with technical becoming.

This construction of the current technocultural context and the critical framework for my inquiry may appear perhaps too weighty to lay at the door of computer games production or play. Can videogames, which are ostensibly just about play, free time, what one does during one's time off from the serious business of living, be so significant? One response is to recall that the same kind of question has been asked of movies, radio, and television, other industrially produced entertainment forms that have played their part in (re)shaping lived experience across the world. For his part, Stiegler approaches these industrial media forms as providing entertainment, information, and communication. They not only organize leisure-time (and therefore also work and rest), but also affect the means of knowledge production and dissemination and, consequently, the shape and character of collective experience (Stiegler, 2011). Because of their global industrial materialization of commodities, experiences, and services which serve to rhythm existence, videogames are far from superfluous, inconsequential, or tangential to the serious questioning of the contemporary state of things. They are an important contributor to the wider dimension of digital technocultural transformation.

For the same reasons, the pharmacological possibilities offered by changes in the production, distribution, and playing of videogames should be understood and assessed through a situated philosophical–critical engagement in the contemporary technocultural milieu. I will propose that creative cultural production is vitally important today and that consequently, questions about the nature and conditions of game production and the place of games as cultural expression and reflection, as modes of critical dialog and collective action, should be posed to the possible future of independent game making and playing. Consequently, in what follows, I will examine the claims made for the claimed democratization of game development in recent years. The task will be to extract the curative potential from the predominantly commercial exploitation of industrial technological innovation in which such claims often become merely rationalizations for continuing capitalist logics of creative destruction. This therapeutic potential resides in such things as the possibilities for the cultural and aesthetic diversification of videogame renderings of human experience, the opening up of game development as a more sustainable business for smaller enterprises, and enriching forms of participation and collective engagement in mediated sociality beyond platform capitalist parameters.

Individuation and the conditions of production

To consider further the pharmacological character of videogame production as it is currently developing, let me return for a moment to NESTA's presentation of a more timely mapping of videogame production. The results portray an expanded and more dynamic sector, with a significant increase in smaller enterprises. Our research in the Creative Territories project involved a partnership with the Bristol Games Hub, an independently established and managed collocation of game-makers including indie firms and individual freelancers. It bore out the general picture of a recovering and re-forming industry offering new opportunities for developers, programmers, graphics/animation designers, and other game-makers. It also confirmed, in broad terms, the notion of a "democratizing" aspect in this expansion because of the changed technological and commercial conditions of production and publishing. On the production side, the emergence of widely available game engines – above all the cross-platform *Unity* system (Unity Technologies, from 2005), but also YoYo Games' *Game Maker: Studio* (from 1999) – has provided low-cost access to suites of production tools and software and to developer and community-resourced asset libraries. The emergence of the new publishing and distribution opportunities associated with mobile and tablet platforms – with Apple most prominent for now – and the "return" of pc-gaming via the growth of Valve's online Steam distribution site have significantly expanded the demand for games. One could add here developments in game bundling and other models of distribution, and the access to capital via crowd-sourcing, as some of the principal elements of the changed conditions supporting the expansion of possibilities for games-makers to start their own enterprise and retain significant control of the results of their creative labor (Crogan, 2015).

The cultural and social implications of the expansion of markets and the rise in numbers of makers of videogames are not in view in this report. The so-called "democratizing" of access to the means of production wears this positive political label in a familiar neoliberal reflex identifying enhanced market competition with increased democratic freedom – optimizing freedom to enter the market is cast as a fundamental aim of democracy. I will leave aside a detailed critique of the contradictions and ironies of this prevailing economic discourse, especially as it issues here from state and state-supported agencies supporting economic policy goals to support the growth of the national UK industry in a global marketplace. The task of evaluating the potential value of the widening of access to market for game-makers remains for those who would offer something other than this narrow valorization of the contemporary situation. This is a large task, already tackled by many researchers on game making and games as cultural forms. I offer here some observations on how to address these questions of value and potential from the perspective of Stiegler's project. This address must begin not only with the classic political economic issue of ownership of the means of production (and hence the Marxist issue of access to and control of these means); or with its elaboration in formulations of social or cultural capital (Bourdieu, 1984) nor with their appearance in the guise of the figure of the "creative class" (Florida, 2002), but with a better grasp of the conditions of production.[4]

The conditions of production involve individuals, collectives organized in different configurations, and technical objects and systems. The classic Marxist position accounts for these in terms of the political economic relations concerning the contest over the ownership of the means of production For Stiegler, because of the irreducible technicity of human being, the relations of production affect and condition this struggle and the cultural political coordinates of existence. In Stiegler's account, technics pervade the history and the question of human becoming. Technology is the inescapable supplement, the third element individuating itself in relation to human becoming (Stiegler, 1998).[5] This means that the becoming of technics is related to the becoming of human individuation, but it is not the same thing nor is it merely a subordinated aspect of human becoming. Stiegler has a neat formula for this notion of a complex folding together of the human and the technical: technics is the "and" between "psychic [individual] and collective individuation" (Stiegler, 2014, 48). Technics mediates between individual and collective. In turn, it becomes in relation to both individual and collective. Through the collective's synchronization of practices, operations, implementations of investment in material production, pursuit of research agendas, and so forth, technics develops along certain lines. At the same time, individuals, through their adoption of technical forms and through their own particular, idiosyncratic employment of them, produce the potential for their unexpected, unprogrammed deployment and innovation and, consequently, for the appearance of new programs and new materializations. I will return to this notion of idiosyncratic use below in relation to the idea of the program and the unprogrammed.

This means that technics are the means of stabilization (or at least metastabilization) of the relations that condition individual and collective identities and the modality for their becoming other. Technics as artefactual realization of experience, expression, and interpretation of existence enable the transmission of ways of life to those who come after. The most decisive feature of Stiegler's theory of technics is that technical artifacts are interpreted as an exteriorized form of memory. And Stiegler insists on a "strong" reading of technical memory as more than memory-support or archive: technics *are* memory inasmuch as they are co-constitutive of the human capacity to think, reflect, and remember. As the now-prevailing neuroscientific theory of neuroplasticity confirms (Malabou, 2008), technocultural forms condition interior, organically based memory.[6] This exterior memory is what enables ethnocultural becoming to dominate genetic, species-based evolution in human individuation in what Stiegler terms "epiphylogenesis", a singular form of becoming based on the technically enabled capacity to retain and pass on individually lived experiences. Conversely, this also means that technological change can destabilize cultural transmission and put into question the very sense of existence. This is the other side of its pharmacological potential.

The individuation of creativity (or capture and democratization)

This is why technics or rather, "technicity" – the relational dynamics composing human and technical becoming – is so critical in understanding the conditions of production. For instance, Celia Pearce was correct in many ways when in her presentation for the Creative Territories project symposium, "Mapping the Collective", she called 2014 a "watershed moment" in videogames as the industry moves away from a predominantly "film studio" model toward a "variegated industry of diverse studios and creators" making games for different platforms and for an "increasingly heterogeneous audience" (Pearce, 2014). The promise of what lies beyond this watershed moment, however, must be properly grasped and articulated if it is to be realized. In the remaining pages, I will try to sketch out the terms of such a critical articulation of the changed technological and technocultural conditions of videogame production, distribution, and reception by parsing them in terms of the general account of human technicity in Stiegler's work I have briefly elaborated here.

As I just said, Pearce is right to identify a shift in the industrial organization of production as crucial to a transformation of videogames as cultural form. For Stiegler, a cultural politics today must prioritize this task of reformulating industry – and therefore the economy – because the industrial production of mediating forms of experience is central to the maintenance (or alteration) of technocultural norms. And today, these norms have largely become toxic, poisonous, not only for the natural environment that is incorporated within the human world (and that the prevailing scientific opinion warns is well on the way toward crisis conditions in the wake of industrialization). For Stiegler, the prevailing capitalist dynamics organizing industry and the production of cultural experience are also toxic to the environment of psychic and collective individuation (Stiegler, 2010, 43).[7] Stiegler's diagnosis of this "ill-being" of human individuation (as he names it in *Technics and Time 3*, Stiegler, 2011) shares features with other critiques of consumerism of the globalizing, hegemonic regulation of attitudes, aspirations, and desires coordinated with the production of goods and services, perpetuating its mode of permanent innovation, upgrade, and so forth, and addressing the consumer as one who is increasingly dependent on product and service provision to guide her on how to live and what she should know (Stiegler, 2014). Stiegler argues that this has become the default mode of adoption of technological becoming in the industrial era, and that it has reached a crisis in our current period, the signs of which are readable across economic, social, and global geopolitical spheres as much as in the ecological domain. This crisis is one in which a sustainable metastabilization of becoming is increasingly difficult to achieve. The permanent race between systems, platforms, devices for making and sharing things and experiences, for storing and accessing knowledge and memories, the race to gain control and monopolize and monetize the investments of new ventures, and to capture the imaginations and the attention of consumers, this race tends to "short-circuit" the composition of psychic and collective human individuation, leaving less and less time or space for critical or cultural reflection and appraisal of the other possibilities and implications of technics as medium for human individuation. It is precisely this questioning that tends to be excluded in this short-circuiting of longer circuits of individuation that pass through technical intermediaries – from books and films to videogames – and allow for the formulation of hypotheses, theories, principals, and programs for shaping their adoption.

Consequently, if the industrial context of computer game production is shifting today, then, it needs to be considered whether this is part of the ongoing overturning of established technocultural models described above. In some ways, it resembles recent upheavals in modes of production and distribution based on a lowering of technological and capital-cost entry barriers and the opening up of online distribution avenues such as have been witnessed in, for example, the music industry and video production. In this regard, what is happening at this "watershed moment" for videogames can be understood as a further iteration of the destabilization of broadcast media industrial structures and part of a wider evolution of globalizing, online, networked mediation.

Identifying the new conditions for videogame-making should also consider technical tendencies with wider implications such as the increase in storage capacity, processor size, and efficiency, the provision of low or no-cost cloud data storage and the lowering of costs, and increased production and supply of portable devices and smart phones. These have each played a part in encouraging the growth of smaller videogame-makers willing to venture out on their own, while also proving crucial to the diversifying of platforms for videogames, from mobile and device releases, to Steam and pc bundling, to social media casual games, and to the expansion of gaming media culture via online video streams, "Let's play", E-sports, and so on. Considering all these factors together, it is possible to see the changes in videogame production and mediation as a particular configuration of a digital technical tendency the speed and scope of which has led to a regime of "permanent innovation" that has tended to outstrip the ability of cultural and political spheres to reflect on and inflect its course.[8] My point here is that the changes in the conditions of the industrial production of games do possess potentials for diversification and democratization but that these emerge as part of a more general dynamic of industrial and technocultural transformation, one which we have been subject to for some time, and one whose wider impacts and implications remain to be comprehensively reviewed.

To venture a comparison, the mixed blessing of iTunes or Spotify as an alternative to the old "record label" arrangement for musicians (and not just for the tiny number of major music celebrities) may offer something in the way of a caution for game creators seeking to establish themselves on the iOS platform and elsewhere. Musicians pay the largest price for the solution to the piracy challenge for the music business, with profound ramifications in the longer term for the future of musical creativity as the archive of recordings ages. In a similar way, the acceptance of risk by the indie game developer launching their game is inversely matched by the exploitation of game content as part of massive catalogs of free or low-cost media experiences available to the platform's user base. The trends indicate the repetition of a familiar pattern inasmuch that, relative to the expanding number of new game experiences available, fewer and fewer games are making a viable amount of money for their makers on the Apple store (Thoa, 2012; Garth, 2013; Rawlings, 2013).

In their default, systematic implementation of technical developments toward commercial ends, these trends led by the competition between platforms for users, for user retention, and for user attention in the form of content, reputational, and review metadata and other forms of monetizable data tend to dominate the conditions of the creation of videogames. This impacts upon the promise of creative diversity that Pearce identifies in the current situation. It is right to identify the increased opportunities for smaller developers to enter the market with their own IP as a significant change. It promises to allow much game creativity to avoid what Nick Dyer-Witherford and Grieg de Peuter characterized as the capture function of global "game-capital" by major publisher corporations (de Peuter and Dyer-Witherford, 2005). A greater number of entrants have the chance to retain their intellectual property, but they enter into a globalized commercial market, still accessed via major transnational corporations providing the material mediation of their "immaterial" creativity and providing a principal avenue of their eventual (re)capture by global corporate capital in the manner of *Minecraft* or of the *Oculus Rift* 3D system. The possibilities for a "counter-mobilisation" of independent, democratized, creative "immaterial labour", however, rest on more than this temporary opening up of the market (de Peuter and Dyer-Witherford, 2005). Here, the question concerning the conditions of creativity within the conditions of production assumes its full significance.

A few years ago, developer Alistair Doulin coined the term "mindie" in posing the question to indie developers about their relation to the mainstream game industry:[9]

> Most indie developers I know don't have the luxury of making games just for the love of it. At the end of the day, we all need money to live. Mindie developers are mindful of making a profit from their games but not at the expense of the game itself. The current landscape of game distribution is the perfect time for this lofty ideal. Digital distribution, micro-transactions and huge user-bases on platforms like Facebook mean that we no longer have to make a boxed product that sinks or swims based on the first few months of sales. This gives Mindie developers the freedom to flex our creative muscles and then find a sales model that fits our product. There are many options to take opening up many more possibilities for game design and creativity.
>
> (Doulin, 2011)

"Are you mindie?" was a question about how the indie ethos sat with the default mainstream trajectory of game development as a viable business, requiring significant capital investment, commercial relationships with distribution platforms (if not with publisher-investors). The question is extremely pertinent, but the "gap" between indie and mainstream cultures of creative production needs a better critical articulation to propose an alternative to Doulin's suggested solution of a bridging maneuver allowing the indie to pass to and fro between the commercial mainstream and creative freedom.[10] I would argue that such a course of action is better understood and adopted as a strategic move, necessary perhaps but not sufficient, for it does not address (except to accommodate) the systematic issue of the mainstream mobilization of experience and expression for commercial ends or the systemic possibilities of game-making technics to reinvent the potentialities of digital mediation for human individuation.[11]

And this is where and why the situation of game-makers today speaks so cogently to crucial issues for our collective, globalizing technocultural development. They encounter the pharmacological character of the opening up of opportunities to generate a livelihood out of their interest in and love for game-generated experience, expression, and community. Their creativity is key to opening and holding open the window of opportunity for cultural reflection on and access to those systemic possibilities of game-making as part of the digital mediation of human becoming. This holding open can create an avenue for intervening in the systematic capture of the productivity of human individuation by the default techno-logics dominating technical transformation today. For it enables the time and space for the longer and less-directed circuits of individuation to form that are required to suspend the default mode of adoption of technological advance.

As explained above, the dynamic of psychic and collective individuation is powered by individual responses to the collective synchronization of modes and models of living. Creativity is an emergent potentiality of this responsiveness. The individual reacts to the existing cultural prescriptions (realized and accessed in the form of transindividual artifacts, rites, and routines) and repeats them more *or less* faithfully. This is characterized by Stiegler as the relationality of the idiomatic and the idiosyncratic. The idiosyncratic repetition of the practices established in the cultural idiom is the mode of their transformation. The idiosyncratic is peculiar, unique, singular, but makes sense and makes a difference only in and as a relation to the regular, the typical, and the idiomatic. The idiom is itself the result of the selective synthesis of the vast accumulation of idiosyncratic responses amounting to the metastable form of a synchronizing program. A pro-gram is a "writing forward".[12] As such, it is the condition of production of the idiosyncratic as much as the surface registering its appearance and filtering its effect (Stiegler, 2009, 69ff).

From this perspective then, the creativity of game creators can be understood as the more *or less* faithful repetition of the established cultural modes for "doing things with videogames" (as Ian Bogost, 2011 has it) via the adoption of those digital (and other) technologies which form part of those technical systems that mediate human becoming. The stakes of sustaining and fostering this creativity become clearer from this point of view. Beyond considerations of expanding the opportunity for personal freedom of expression (and protecting against the appropriation of its market value), or even of promoting diversity of experience as a general cultural benefit, creativity is crucial to the maintenance of the dynamic of technical and human individuation.

All of these aspects of creativity – which is usually understood as an innate quality of psychic individuals – assume their full significance in this compositional view of human-technical individuation. Creativity can enable the time and space for the longer circuits of individuation to form that are so crucial to a properly "human" inflection of our technical becoming. Creativity realizes technical forms that mediate between individual and collective identity; it concretizes in space the temporal dynamic of individuation. The artifact spatializes time, providing the material substrate of individuation. Individual experience amounts to the re-temporalization of the technical artifact. Playing a videogame, watching a film, listening to music, reading a book: this is how the individual (re)animates the dynamic of their own psychic development and becomes part of the collective iteration of culture.

This is why making games, films, writing, recording music, graphics, and so on are critical for composing the different dynamics of our becoming.

Videogames are especially important today as so-called "native" digital forms of mediation familiar to succeeding generations. Our digital technical development continues to overturn and reframe the conditions of mediation of cultural and individual experience, both globally and locally. The speed of transmission, the incorporation of older media forms and of new spheres of experience and sociality not subject to industrial mediation before, and in other ways that have been analyzed in terms of real-time, ubiquitous and pervasive media, "smart" networked devices, and so on, the mediation of existence has been transformed in a sweeping and systemic fashion.

Videogame production can play a crucial role in responding to the challenge of forging these longer circuits of the reinvention of human becoming that move us beyond the default settings of the "digital revolution". Game creatives can open up new adoptions of a digital technicity to which we are tending to become adapted rather than adopting it in a creative and inventive mode. They can make a major contribution to composing human autonomy with the extraordinary power and reach of digital programming. Cultural renewal is the form of human individuation that composes itself through reanimating the artifacts that have been left to us by those that are dead or, at least, those who are no longer who they were. It is plain to see that this task of transformation is urgent today and that it needs to go far beyond the venture capitalist model of Silicon Valley's version of "creative destruction". Given the pervasive, cross-generational currency of videogames, game making can make a critical contribution to this task.

Conclusion

I will conclude by noting three specific aspects of this challenge and this possibility as they relate specifically to the current state of games production:

First, the game design process is a key battleground for this struggle: Ian Bogost showed that as software, games are a procedural form able to build in a critical or polemic "procedural rhetoric" that the player encounters through their interactions (Bogost, 2007). In this regard, Mary Flanagan's work (2009) on thinking values in game creation, along with her proposals in *Critical Play* about revising the design cycle to incorporate a reflective-critical phase of iteration, indicates an important path toward the much-needed "longer circuits" of cultural programming.

Related to this is the need to evaluate all the standard and pre-created elements that come with the game design engines and other making tools that are now more widely available to independent developers. At the Creative Territories event, "Mapping the Collective", Stefan Werning identified some of these operating a "built-in bias" in the current opening up of access to game creation packages such as *Game Maker: Studio* and *Unity*. These include: automatic design processes in game engines and other software, ready-made game assets and free or low-cost assets in online marketplaces, option pre-sets (defaults), tutorials, and demos. Werning pointed out that these bring benefits in ease and availability and allow less experienced and smaller developer teams to explore design concepts and make expressive games without being hampered by technical challenges and tasks. But they can also limit the scope and potential exploration and realization of the enormous technical possibilities of these software systems. Their pharmacological character must be grasped and negotiated collectively if indie game expression is to go beyond a collection of shining examples or a few mega-hits.

Finally, the question of the creative agency of the player is crucial here. This is in itself an enormous topic – that of the current and future collectives that game-makers develop in relation to as well as altering through their work. In relation to the preceding discussion of the capture and exploitation of the consumer in the prevailing commercial logics of industrial mediation, it is clear that the "freemium" model of monetization from online software distribution is a topic of central importance.[13] The freemium model of app distribution that so many indie game-makers rely on when publishing on the new platforms offers significant benefits in terms of the retention of intellectual property, but the pathway to commercial viability is pressuring game design in new ways as they take on the responsibility to generate a revenue stream from a "culture of free".

Freemium could be approached as a "perversion" of the shareware distribution practices prevalent in the earlier years of online culture. In a similar vein, crowd-funding has funded a lot of games and other startup ventures in recent years, involving user communities as investors and often including them in more or less significant ways in the production process. Crowd-funding is showing signs of being reabsorbed in more familiar routines of global capitalization and value extraction, where it is deployed in carefully managed marketing strategies masquerading as "grassroot community building", and in which an increasingly slick and sophisticated crowd-funding pitch and "perk" design is taking a depressingly familiar appearance, namely, as precisely what crowd-funding as a peer-to-peer exchange is supposed to be different from.

In this, as in other aspects of digital online culture, games have been a leader. But to take advantage of this avant-garde position, videogame creativity has to lead a trend in what is possible via online digital networked culture that becomes more than a temporary fashion in monetizing attention or experience. This is a challenge that is wider in scope than videogames and the opportunities that are emerging today for game-makers and their collectives, but they are no small part of the possibilities for renewing the technocultural program today.

Notes

1 This is an extended and revised version of the keynote delivered under the same title at Philosophy of Computer Games 8: Freedom in Play, Istanbul Bilgi University, 13–15 November 2014, http://gamephilosophy2014.org/. My thanks

to the conference chair, Tonguc Ibrahim Sezen, the program committee chair, Rune Klevjer, the Games and Philosophy association and all the kind helpers at the host institution, Istanbul Bilgi University. Part of this research was supported by the UK Arts and Humanities Research Council through the Video Games Research Network scheme in the form of the "Creative Territories" project I led (2013–2015).

2 NESTA is a UK national lottery-funded agency. A proportion of gambling revenue collected from the state's gaming enterprises is channeled to various organisations with public service aspirations.

3 Readers can recall that this paper was prepared in response to an invitation to speak at the 8th Philosophy of Computer Games conference.

4 It should be noted that Florida revisited and to some extent, recanted his claims for the emergence of a new digitally savvy "creative class" of (what Bourdieu would call heavily culturally and socially capitalised) citizens as the driving force of urban renewal in the new century (Florida, 2012). The damage had largely been done, however, by this influential thesis which dovetailed neatly with neoliberal property speculation and complicit urban redevelopment practices prevalent at the turn of the century, and still largely so today.

5 Stiegler speaks about technics rather than technology to draw a distinction between modern industrial technology as particular, singular development and understanding (logos) of technics, and technics as the "exteriorisation" of organic, biological functions in forms, practices and artefacts characteristic of human individuation. Technics includes what is usually understood as technology. In approaching technics as exteriorisation Stiegler mobilises the account of "hominization" given by influential evolutionary anthropologist André Leroi-Gourhan (Stiegler, 1998).

6 Readers familiar with Andy Clark and David Chalmers' work on extended cognition will see the parallels with their influential "extended mind" thesis in the philosophy of mind (Chalmers and Clark, 1998).

7 In *Taking Care* Stiegler proposes that for the issue of the pollution of psychic and collective individuation by commercial mediation, we need something comparable to the public acknowledgment of the environmental dangers of global warming and the effort to develop public awareness and debate, pedagogical policies, and so forth to promote a substantial and widely negotiated and supported response to the challenge that our governments in their default mode of operation seem unable to sustain.

8 "Permanent innovation" is Stiegler's description in David Barison and Daniel Ross' *The Ister* (2004) of the characteristic of industrial modernisation, one which intensifies and "strikes us with an unprecedented force" in the contemporary digital era.

9 I learnt of Doulin's blogpost through Neils Hoogendoorn's presentation "Who's Indie Now?" at the first Creative Territories workshop in Bristol in May 2015.

10 As Stiegler reflects (Stiegler, 2011, 1), this problem and this "solution" were articulated by Hitchcock – albeit in a less sanguine mode – when reflecting on his situation within Hollywood, as one required to manage and deploy industrial capital investment made for commercial ends while seeking to realize his own artistic program.

11 In the years since Doulin's hopeful articulation of the Mindie situation, NESTA's mapping of a new videogame industrial landscape and my address to the 2014 Philosophy of Computer Games conference, it has become increasingly apparent that the freedom Doulin sees in the creative independence of indie developers to "find a sales model that suits our product" has been significantly curtailed. On the contrary, as David Nieborg has argued in his analysis of the

game app business (Nieborg, 2015), it is the power of digital platforms over distribution and marketing that is shaping the very design processes of developers looking to produce commercially viable games for these platforms.

12 Stiegler develops this notion of the cultural program from André Leroi-Gourhan's account of the development of the ethnocultural program as the means of human evolution in and through the differentiation of forms and modes of the technical exteriorization of functions. See the first chapter of Stiegler (1998) and Leroi-Gourhan (1993).

13 This has received increasing attention since this talk was delivered in 2014. See Nieborg (2015) and Crogan (2018).

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8 Boutique indie

Annapurna interactive and contemporary independent game development

Felan Parker

The Game Awards are one of the most prominent awards shows in mainstream gaming culture, celebrating fan-beloved and commercially successful games each December. In 2018, the Mobile Game category included *PlayerUnknown's Battlegrounds* and *Fortnite: Battle Royale*, two wildly popular multiplayer action games, but neither won. Seemingly against all odds and much to the consternation of some gamers, they lost to *Florence*, a quiet, emotionally resonant game about a romantic relationship, published by a relatively new but already influential actor in the field of independent games, Annapurna Interactive (Patange, 2018). This apparent upset is symptomatic of broader changes in the industry and culture of digital games, in which indie games, once considered a marginal phenomenon, "not only compete with established 'AAA' hegemons but also converge with the latter in hybrid configurations and reciprocal dependencies" (de Peuter and Young, 2019: 2).

Annapurna Interactive is a branch of the independent film company Annapurna Pictures, dedicated to financing and publishing "personal, emotional, and original games that push the boundaries of interactive content and encourage artists to bring new visions to the medium" (Annapurna Pictures, 2016). Although the entry of a Hollywood company into the game industry is nothing new (Brookey, 2010), there is something unique about Annapurna's approach. Rather than expanding on existing franchises and intellectual properties as is the norm, they publish original games by independent developers in an eclectic variety of genres. Through their well-financed operations, meticulous curatorial and promotional strategies, and distinctive branding, I contend that Annapurna Interactive is fostering a conception of games as a legitimate cultural form that challenges conventional cultural categories and imagined audiences. At the same time, they are deeply embedded in the cultural and economic processes that structure indie game development and the game industry at large.

To develop these arguments, I adopt a media industry studies approach based on a critical discourse analysis of popular press and industry-oriented coverage in both film and game venues, promotional paratexts, and popular/critical reception. My analysis is supplemented by in-depth, semi-structured interviews with two game developers who have partnered with Annapurna: Erik is an experienced developer whose game has been released, and emerging developer Todd's game is currently in development. To maintain confidentiality, interviewees are referred to by pseudonyms. After a brief overview of the current state of indie game publishing and indie film production, I will examine Annapurna Interactive in three aspects: first, their operations and game developer partnerships; second, their curatorial style and brand persona; and finally, their imagined audience and position in the wider game industry ecosystem.

Indie games and Indiewood

The game industry has traditionally been dominated by a small number of powerful publishers and hardware manufacturers (Kerr, 2016: 44). The popularization and discursive construction of 'indie games' in the mid-2000s can be seen in part as a narrative of emancipation from the publisher-centric model and its perceived creative limitations, barriers to entry, and high costs to developers (Ruffino, 2013). However, this narrative belies the increasing power of digital distribution platforms over all modes of game development, many of which are still controlled by the same handful of major corporations, as well as the "second shift" of work traditionally performed by publishers now demanded of already overworked developers (Tyni, 2017; Whitson et al., 2018). Cultural and economic intermediaries of various kinds continue to be crucially important in indie game development – including, ironically, a new breed of indie publishers.

John Vanderhoef has documented the rise of "a new generation of born-digital, indie friendly publishers" like Devolver Digital, Raw Fury, and Good Shepherd Entertainment, as well as successful indie developers like Chucklefish and Double Fine who extend publishing services to other developers (2016: 22, 2020a, 2020b). These indie publishing endeavors offer similar forms of financial and promotional support to traditional publishers, but with more equitable deals and fewer strings attached, notably granting developers greater creative autonomy and allowing them to retain intellectual property rights (Vanderhoef, 2016: 198). Indie publishers, alongside indie-oriented divisions of larger publishers and platforms, now play a central role in commercial indie game development and are seen as highly desirable, even essential, to drive discoverability and capture an audience in an oversaturated attention economy (Kuchera, 2014; Sinclair, 2014; Wiltshire, 2017). Brendan Keogh conceptualizes these ebbs and flows of industry power as successive waves of formalization, informalization, and re-formalization, as new modes of independent game production open up new creative and/or economic possibility spaces which are then formalized and incorporated back into the established game industry (Keogh, 2019a).

American indie film is a key antecedent to the idea of indie games. Annapurna Pictures is an independent film production, financing, and distribution company founded in 2011 by Megan Ellison, the daughter of billionaire

Oracle co-founder Larry Ellison (Lyons, 2016: 54). Thanks to significant financial backing and close partnerships with powerful Hollywood institutions, Annapurna has built a reputation for "sophisticated, high-quality films" by well-established auteur directors that "might otherwise be deemed risky by contemporary Hollywood studios" (Annapurna Pictures quoted in Lyons, 2016: 54). Ellison has also been an outspoken advocate for filmmakers marginalized due to their gender, race, sexual orientation, and/or politics (Sims, 2019). Annapurna Pictures' catalogue includes Paul Thomas Anderson's *The Master* (2012), Kathryn Bigelow's *Zero Dark Thirty* (2012), Spike Jonze's *Her* (2013), the Coen Brothers' *The Ballad of Buster Scruggs* (2018), Barry Jenkins' *If Beale Street Could Talk* (2018), Boots Riley's *Sorry to Bother You* (2018), and Olivia Wilde's *Booksmart* (2019). As this laundry list of high-profile filmmakers and critically acclaimed, award-courting films indicates, Annapurna Pictures markets itself as a "home for auteurs and edgy fare" (Siegel, 2019) and occupies the cultural-economic middle ground between popular blockbusters and esoteric art films.

Annapurna Pictures can be situated as part of the "Indiewood" phenomenon of pseudo-independent film production and distribution presaged in the late 1980s by ambitious indie studios like Miramax and popularized via the Sundance Film Festival (Perren, 2012). This phenomenon encompasses not only financially independent productions but also "boutique" divisions of major studios specializing in indie-style mid-budget features (Tzioumakis, 2012: 7). As Alisa Perren argues, the popularization of indie film can be situated as part of a larger corporate restructuring of media industries in the 1990s that emphasized the production of niche media products for narrow targeted demographics as a complement (rather than alternative) to big-budget blockbusters targeting a mass audience (Perren, 2012: 6). Indiewood films attempt to marry the edgy, hip, or "smart" aesthetics and themes associated with low-budget indie movies with the bigger budgets and aggressive promotion necessary to reach a wider audience (Sconce, 2002).

The incorporation of independent film into the logics of global entertainment conglomerates parallels the mainstreaming and formalization of indie games described above, with indie aesthetics and modes of production now occupying a prominent, rather than marginal or oppositional position, in both industries. Annapurna Pictures has had considerable critical and commercial success with their high-end Indiewood model (Lyons, 2016: 57–58), making Ellison a powerful presence in Hollywood and allowing the company to expand their operations beyond film into other cultural industries including television, theatre, and, most recently, games.

Operations and developer partnerships

Annapurna spent years behind the scenes building a team, courting indie developers, and laying groundwork before officially announcing the Interactive branch and their initial slate of games at the end of 2016 (Cieply and

Barnes, 2011; Favis, 2018; McWhertor, 2016; Spangler, 2016). Annapurna staffed their new initiative with two executives from their film division, producer Neale Hemrajani and technologist James Masi, and hired a team of seasoned game industry veterans. The majority were poached from various branches of Sony, including Deborah Mars, James Masi, Hector Sanchez, Nathan Gary, and Jeff Legaspi, all of whom had worked closely with indie developers to bring their titles to PlayStation (Favis, 2018: 17; Siegel, 2016; Spangler, 2016), helping Sony cultivate a reputation for “helping creative, inspirational games come to market” by securing platform-exclusive indie content (Vanderhoef, 2016: 189, 192). Jenova Chen, the creative director of thatgamecompany, a previously Sony-backed indie studio responsible for the critically acclaimed artgames *Flower* and *Journey*, is also credited as an advisor and “scout” for Annapurna Interactive (Martens, 2019).

Working closely with the larger company, Annapurna’s Interactive division was well-positioned to leverage established networks and personal relationships with high-profile indie developers (Favis, 2018: 16–17). The familiar faces on the team also acted as assurance to prospective partners that this as of yet unannounced new publisher was serious about supporting indies. According to Erik, “They have a lot of experience working with indies and understanding what indies want, and aspire to, and maybe what some of their weakness are and how they can help out”, which he says was an important factor for his team. All of this is very much in keeping with Annapurna Pictures’ film production strategy of tapping established talent and industry networks to mitigate the potential risk of edgier or less marketable projects.

In addition to poaching staff, Annapurna Interactive was also able to take on promising projects that had previously been in production with Sony. Their first published game was *What Remains of Edith Finch* (2017), a narrative exploration game made up of magical-realist vignettes. *Edith Finch* began as a Sony-backed project developed by Los Angeles-based studio Giant Sparrow, who had previously developed *The Unfinished Swan*, a critically praised and award-winning narrative exploration game that was also co-developed and published by Sony. Annapurna made a deal with Sony to acquire the game after the executives who had been shepherding it left for Annapurna, effectively taking the entire production and development team with them (Giant Sparrow, 2017). Similarly, *Katamari Damacy* creator Keita Takahashi’s surreal puzzle game *Wattam* (2019) and A44’s moody fantasy action-adventure game *Ashen* (2018) were ‘rescued’ by Annapurna several years after Sony and Microsoft, respectively, withdrew their support (Favis, 2018: 24; Seamus, 2018). Annapurna has further taken advantage of their team’s connections to secure the rights to re-issue iconic indie titles to new platforms, including thatgamecompany’s formerly PlayStation exclusive *Flower* and *Journey*, as well as *Kentucky Route Zero* and *Gone Home*.

Annapurna’s second original game, *Gorogoa* (2017), is an enigmatic hand-drawn puzzle game of sliding tiles and optical illusions by developer

Jason Roberts. *Gorogoa* has less obvious pedigree, but enjoyed significant attention and acclaim in the indie scene over its lengthy development process, including multiple high-profile indie game awards (Reichert, 2018). The game also received funding from Indie Fund, a powerful intermediary in its own right made up of a cabal of successful indie developers dedicated to investing in promising projects (Kohler, 2017; Sandovar, 2017). Although *Edith Finch* and *Gorogoa* can be seen as creatively risk-taking, their trajectories and the circumstances of Annapurna's acquisition of both titles reflect a careful, even conservative move to secure their position in a new industry. This strategy has paid off, with both games receiving superlative accolades and awards, firmly establishing Annapurna's reputation as an exciting new publisher.

Subsequent releases have followed a similar pattern of partnering with developers based on proven track records and/or vetting by other indie cultural intermediaries. *Florence* (2018) is a narrative game for mobile devices that uses graphic novel inspired illustrations and simple metaphorical puzzles to tell the story of a relationship. *Florence* was developed by Mountains, an Australian studio founded by Ken Wong, a game industry veteran and lead designer of the celebrated mobile puzzle game *Monument Valley* by Ustwo. Likewise, Ben Esposito, developer of the quirky physics-based puzzle game *Donut County* (2018), is a longtime participant in the L.A. indie/experimental game scene centered on the Glitch City collective and worked on both *The Unfinished Swan* and *Edith Finch* (Greiving, 2019). *Donut County* had been exhibited in different versions at showcases and festivals since 2012, catching the attention of journalists as well as Indie Fund, who invested in an early iteration of the game (Shanley, 2018). Like their predecessors, *Florence* and *Donut County* were warmly received and institutionally recognized with numerous nominations and awards.

Annapurna's most recent game as of this writing is *Outer Wilds*, a time-bending space exploration game featuring a quirky miniature galaxy that exemplifies the company's shrewd approach to developer partnerships. Originally conceived by a student team in 2012, over the years, *Outer Wilds* received considerable attention within the indie scene in its earlier versions, even winning the prestigious Independent Game Festival Grand Prize in 2015. Based on this positive reception, the whole development team was hired by Hollywood actor-producer Masi Oka to develop the game for his newly formed game company Mobius Digital (Shanley, 2019). *Outer Wilds* was also successfully crowdfunded on Fig, a platform that grants backers financial shares in the projects they support, before Annapurna bought the rights to the game in 2018, yielding a significant return on investment for Fig backers (Hall, 2018). The game was finally released in 2019 to significant acclaim, with many critics hailing it as one of the best games of the year (Schreier, 2019; Walker, 2019).

Annapurna's growing roster includes new games from a who's-who of award-winning indie luminaries like Sam Barlow (*Her Story*), Heart

Machine (*Hyper Light Drifer*), and Fullbright (*Gone Home*), among others. Other upcoming releases, such as Giant Enemy Crab's *Due Process* and Beethoven & Dinosaur's *The Artful Escape*, are relative newcomers to the game industry, but both received institutional support and funding from organizations like Indie MEGABOOTH, Indie Fund, and Epic Games' Unreal Dev Grant before being picked up by Annapurna. Annapurna has also aligned strategically with distributors such as the Epic Games Store (Grayson, 2018; Hall, 2019) and the Apple Arcade subscription service (Smith, 2019), lending their cultural cachet in exchange for prominent platform visibility.

A clear pattern emerges in Annapurna's partnerships: virtually without exception, their selections have been well-groomed by other cultural and economic intermediaries in the game industry. In Erik's case, Annapurna's offer was based almost entirely on reputation – he indicates that Annapurna was interested in backing his team even before they had decided what their first game would be. This suggests a high degree of confidence on Annapurna's part in their ability to successfully market any game by a proven developer. As Erik puts it, Annapurna's strength lies in their ability to "spot creatives that are at the right point in their career and figure out how to support them". In other words, they are 'picking winners' by working with developers already poised for creative and commercial success. Annapurna's highly selective approach distinguishes them from other kinds of indie intermediaries like Indie MEGABOOTH or IndieCade that actively seek out and foster untested talent (Parker et al., 2018).

Cultural intermediaries are often overlooked, but play a crucial role in what Liz McFall describes as the "contextual interdependencies between products, producers and consumers" (McFall, 2014: 2). McFall conceptualizes success in cultural industries as a non-linear process of "[f]orging more and more relationships in an expanding network" between a cultural object and the wide variety of human and non-human actors that value and mediate its existence (2014: 5). Annapurna seems to innately understand that "the sheer length of the chain of associations" (McFall, 2014: 5) orbiting a game or developer marks their potential as much as any inherent quality. As in their film division, where they "take 'creative chances' but only with 'filmmakers in the top rank'" (Lyons, 2016: 57–58), Annapurna Interactive is not so much incubating new developers, as 'accelerating' developers that are already well on their way by furnishing them with more substantial budgets and, crucially, marketing and promoting them a much wider potential audience.

Developer support

Above all, what indie developers lack is funding, which publishers grant in exchange for a percentage of the final game's revenues. In Erik's words, "I like being paid a salary [and] I like my teammates to be paid a salary", rather than dealing with the risk and precarity of freelancing or self-funded projects (Ruffino, 2013; Tyni, 2017). Thanks to its cross-media operations, Annapurna is extremely well financed and can offer greater financial support than other actors in the indie space, ultimately enabling more polished and ambitious projects. For Todd's forthcoming game, other indie publishers and sources of funding like Indie Fund or Kickstarter did not seem sufficient: "the amount of money needed was not something that they were at the time able to handle". Other developers similarly leverage publisher resources to enable grander ambitions, larger development teams, and/or higher production values (Flesser, 2018; Shanley, 2019). Taking full advantage of Annapurna's Hollywood connections, *Telling Lies* (2019) features 10 hours of high-quality video featuring professional actors and sets (Favis, 2018: 20), and *Sayonara Wild Hearts* (2019) added narration from rapper-actress Queen Latifah late in development (Cryer, 2019). Annapurna stands out to developers because they strike the right balance between deep pockets, industry experience and connections, and alignment with indie values.

Beyond funding, Annapurna offers the full range of support expected of a publisher, which could be summarized as everything other than making the game. Dealing with digital distributors, porting to different platforms, and quality assurance – all crucial but grueling parts of the development process – are taken on by the publisher. In terms of promotion, Annapurna's carefully crafted brand aesthetic and well-oiled promotional machine are key advantages. Fortyseven Communications, a firm contracted by Annapurna, handles public relations and press on behalf of developers, which Erik describes as a huge weight off his team's shoulders: "I think a lot of indies really underestimate how much energy and time PR takes if you want to do it properly [...] They just entirely took that off our hands". Likewise, exhibiting at conventions like E3, PAX, and GDC is a key part of the commercial game development cycle, but is a huge commitment of time and energy for developers (Parker et al., 2018). Annapurna is able to coordinate the logistics of professional booths, attractive branding, merch, and other necessary resources to ensure a successful showing. All of this encourages developers to focus more of their energy on the game itself, reinscribing traditional notions of creativity unfettered by mundane concerns (Whitson et al., 2018). As Erik succinctly puts it, "I would rather spend my time focusing on the product and letting Annapurna figure out how to deliver it to the world". In other words, financial dependence affords creative independence.

Creative autonomy

In spite of the offloading of a wide range of tasks and 'worry work' to the publisher, this is not seen by developers as a ceding of control. As James Lyons notes, Annapurna Pictures' wider financing, production, and promotional strategy is primarily expressed in terms of the creative autonomy and control that are central to indie cinema's claim to artistic and cultural legitimacy (Lyons, 2016: 58). This is closely related to Annapurna's commitment to auteurism; just as their film division deals primarily with established auteur directors "with recognizable interests and aesthetics" (Reichert, 2018), Annapurna Interactive strongly emphasizes their commitment to supporting the creative vision of their developers (Annapurna Pictures, 2016). *Game Informer* magazine explicitly sets Annapurna apart from industry conventions, saying that they "handle things differently than big publishers by putting faith in the little guy, fostering lasting relationships with developers, and respecting an artist's vision" (Favis, 2018: 16). Developers likewise repeatedly emphasize the company's commitment to indie creators.

According to developers, business and creative decisions are informed by Annapurna's extensive industry experience and knowledge but ultimately rest in the hands of developers: Annapurna provides guidance "without meddling with the work itself" (Favis, 2018: 18). Erik highlights the publisher's flexibility and patience with the artistic process: "They've learned that you can't just make a schedule and follow it exactly. Things will go wrong, and things have to kind of go off the plan. That's how you make amazing games". For Todd's in-development game, "There are probably a handful of features that we think are really important that wouldn't have made it into any sort of release". Annapurna's support afforded Todd's team space and time to take more creative risks, whereas other publishing or funding arrangements might have required them to lock in a more conventional design earlier in the process. This creative freedom and flexibility is possible in part because Annapurna's developer partners are already well-vetted to offset risk, as noted above.

A recurring theme in developer comments is Annapurna as an alternative to the real or imagined evils of traditional publishers, echoing the "emancipatory" origin stories of contemporary indie games (Ruffino, 2013). Todd contrasts Annapurna's "creators first" approach against a more commercially calculating approach; based on his experience:

> they care about games, and they want to make things that matter [...] they're not viewing us as a product to make money out of. It doesn't feel like that at all. It genuinely feels like they care about what we're making.

Numerous other developers share this view of Annapurna as "a breath of fresh air" (Shanley, 2018) and "refreshingly happy to defer to the creatives" compared to the "exhausting formalities" of traditional publishers (Clayton, 2019). *Ashen* creative lead Derek Bradley takes this to its logical conclusion: "it feels like talking to another developer and not a publisher" (Favis, 2018: 17), and Erik similarly recalls that "the word 'publisher' was not on the table for many, many months. It was just like, here's a group of people who are able to help creatives like myself realize their vision". In this discursive framing, familiar from other forms of indie culture, Annapurna is set apart from the stifling or exploitative practices of the traditional mainstream industry as a more authentic, autonomous alternative (Hesmondhalgh, 1999; Newman, 2017).

Aesthetics and indie authenticity

As Lyons argues, the perception of impeccable taste is a key factor in the success of Annapurna Pictures – the ability to 'pick a winner' is closely tied to Megan Ellison's social and cultural capital (2016: 59). In its brief existence, Annapurna Interactive has already cultivated a parallel reputation in the field of games. The enthusiast press has actively contributed to an image of the company as a prestigious seal of quality, dubbing Annapurna "the most exciting publisher around" (Wen, 2018) and a "person of the year for 2018" (Batchelor, 2018). In an effusive preview of the anime-inspired rhythm/racing game *Sayonara Wild Hearts*, *VentureBeat* declares that the game is so creative that "naturally, Annapurna Interactive is the publisher for the game", a telling quote that clearly indicates an emergent brand persona (Takahashi, 2019). Even though Annapurna's partner developers work separately, they are linked through promotion, branding, and exhibition, and journalists and critics further help construct Annapurna as a marker of quality by grouping together different developers together for interviews and other publicity events and identifying connections between their work (gameslice, 2017). Annapurna's body of published work shares a constellation of common features that are articulated across the paratextual promotion and reception of their work. Taken together, these features signify authenticity as a form of distinction.

In Michael Z. Newman's conception, authenticity is a contingent value constructed in specific sociocultural contexts and closely linked to power dynamics and social relations (2017: 30). Jesper Juul builds on Newman's theorization to examine the curation of "indie style" at festivals and awards showcases like IndieCade and the Independent Games Festival, which can also be located in the practices of indie publishers (2014, 2019). Juul conceptualizes the dominant (though by no means universal) style in contemporary indie games as "high-tech low-tech authenticity", using digital technology to emulate or re-work low-tech or low-fidelity representational modes and visual styles. In this context, the hyperrealism associated with AAA production becomes a marker of impersonal inauthenticity, while quirky, bespoke aesthetic choices signal greater immediacy, authenticity, and honesty based on their divergence from the dominant norm (Juul, 2014: 3). This extends beyond the text to the paratextual construction of intimate, deeply personal interconnections between developers, games, and players (Banet-Weiser, 2012: 14; Ruffino, 2018: 54).

The trendy “low poly” 3D aesthetic popularized by games like *Kentucky Route Zero* exemplifies this style and is strongly associated with Annapurna Interactive’s titles, including *Donut County*, *Ashen*, *Wattam*, *Sayonara Wild Hearts*, *The Pathless* (2020), and *Solar Ash Kingdom* (Forthcoming). *Gorogoa* and *Florence* are both explicitly inspired by comics and graphic novels, featuring hand-drawn images that suggest a personal touch. *Edith Finch* and *Outer Wilds*’ quirky, physically impossible spaces are seemingly cobbled together from available materials and densely cluttered with the detritus of past lives. The iconic look and feel of these games visually distinguishes them from mainstream photorealism and positions them as the product of a singular artistic vision (even as, in some cases, they are made by fairly large teams). Music also plays a key role, and many of Annapurna’s releases feature meditative musical scores designed to deepen the player’s emotional experience, such as *Florence*’s use of interwoven classical leitmotifs and *Outer Wilds*’ wistful juxtaposition of sci-fi and folksy acoustic instruments. In the same manner that “hi-tech low-fi” visuals act as markers of authenticity and artistry against hyperrealism and spectacle, these musical textures stand in stark contrast to the standard epic orchestral scores associated with AAA blockbusters.

The curatorial aesthetic described above is reinforced and filtered through Annapurna’s carefully managed brand persona, which extends not only to marketing but also to other official paratextual materials. In keeping with their appeals to indie cultural legitimacy, Annapurna sells vinyl soundtracks, enamel pins, vintage-inspired t-shirts, and other trendy tie-in merchandise for many of their games, as well as deluxe “special edition” physical releases of select titles complete with bonus features (McCall, 2017). This merchandise likely does not generate significant revenue, but instead serves to reinforce the cultural status of the games and foster fan investment in the Annapurna brand. The irreverent and gaming/meme culture-savvy official Twitter account @A_i provides another venue for articulating the overarching brand that connects their body of work, while simultaneously highlighting the publisher’s claim to indie authenticity and hailing a carefully imagined audience.

Hailing a crossover audience

Examining Annapurna Interactive’s operations, partnerships, curation, and branding helps contextualize their unique relationship to established target demographics and cultural categories. By presenting under one banner an eclectic assortment of auteur-driven titles across a wide range of genres, Annapurna challenges the simplistic and frequently deconstructed ‘hardcore’ versus ‘casual’ binary that persists in discursively shaping the industry and culture of games (Chess, 2017; Shaw, 2014). By the same token, its status as a very prominent, well-financed media company that chooses to specialize in smaller, idiosyncratic indie titles that might otherwise be considered niche complicates similarly problematic assumptions about a binary opposition between 'indie' and 'mainstream' (Vanderhoef, 2016). Annapurna Interactive's initial press release makes this central to their mission, emphasizing that the games they publish will "bring memorable experiences to all players, whether experienced or first-time gamers" (Annapurna Pictures, 2016).

I want to tentatively situate Annapurna in the conceptual category of "noncore games" proposed by Shira Chess and Chris Paul. In their words:

> Noncore is a term meant to include casual as well as indie games. By breaking this wall, we hope it will allow for a more expansive treatment of the complex and nuanced styles of gaming that no longer fits into the hardcore/casual paradigm.
>
> (Chess and Paul, 2019: 10)

This hypothetical noncore audience contrasts against other, more exclusive conceptions of indie games. As Vanderhoef argues, Devolver Digital's transgressive "punk" aesthetic is still firmly situated within hegemonic, masculinized gamer culture (Vanderhoef, 2020a), limiting their ability to "instill the importance of the medium on novices and outsiders" (Chess and Paul, 2019: 10). While Annapurna's games have a strong appeal within indie game scenes, their comparatively robust production values and promotional budgets are able to reach the wider gaming audience as well, as evidenced by the positive reception of their titles in the mainstream gaming press and at general industry awards shows.

Beyond these established target audiences, however, Annapurna also appeals to people who "do not fit into the mold the industry has set up" (Chess and Paul, 2019: 10), who play games but do not identify as gamers for a wide variety of reasons (Shaw, 2013), or are not active game players at all. Erik says that the opportunity to make games for an untapped audience of "ordinary people" rather than "gamers" is part of what drew him to Annapurna, and this goal of "bridging the gap between gamers and non-gamers" is shared by other Annapurna developers (Valentine, 2019). This orientation is not completely unique; the co-founders of the IndieCade festival sought to reach beyond gaming culture to "connect with a public we thought had interest in this content" (Sam Roberts quoted in Juul, 2019: 92). However, Annapurna's profile and substantial resources have granted it higher visibility, broader reach, and greater commercial success.

Edith Finch, *Florence*, *Gorogoa*, and *Donut County* in particular are accessible not only in their easy-to-learn controls but also in their peaceful, decidedly un-game-like audiovisual style and user interfaces, and in their exploration of themes and subject matter like romance, mortality, and gentrification not commonly associated with games. Simogo, developers of *Sayonara Wild Hearts*, make this central to their design ethos: "Why are we actively pushing people out of this amazing thing we love? Why can't there be amazing looking 3D games that aren't operated by 12 buttons and 2 sticks simultaneously?" (Flesser, 2018). *Ashen*, *Outer Wilds*, and the forthcoming *Due Process*, while more grounded in conventional genres, control schemes, and modes of play, nevertheless distinguish themselves against expectations via tone, audiovisual style, and theme. As Chess observes, games like these invite a strong affective response from non-gamers, in part because they challenge preconceptions about what games as a cultural form "are capable of being" (Chess, 2020).

By de-emphasizing hegemonic gamer tropes in favor of a broader range of noncore influences and touchstones, the burgeoning Annapurna Interactive oeuvre has achieved popular commercial success and cultural prestige both within and outside of gaming culture. This suggests that, not unlike graphic novels (Beaty and Woo, 2016, 11), unconventional, creative, and/or personal games can, in the right circumstances, connect with a crossover commercial audience.

The rise of mid-budget boutique indie games

At the same time, those right circumstances reflect Annapurna's unique position at the intersection of multiple cultural industries and audiences and their savvy interventions in well-established circuits of cultural and economic value in the game industry, not to mention their substantial economic power. Like Annapurna Pictures, Annapurna Interactive publishes games tailor-made to win prestigious awards and brokers those accolades into economic rewards, a strategy that has already solidified their status in the game industry. Even if their game division is not revenue-generating (Siegel, 2018), they are likely able to convert cultural capital into economic capital in the bigger picture by bolstering their cross-media brand or by trading their prestige to other industry actors as in the case of their partnerships with Epic and Apple. Annapurna's games and the audience they hail may be noncore, but their cultural-economic position in the cultural industries is decidedly central.

As noted above, Annapurna is discursively positioned in opposition to the inauthentic mainstream, but by no means does the company reject the game industry. Rather, Annapurna seems intent on occupying the space between major AAA 'games as a service' titles and craftily monetized mobile apps, profitably co-existing with them in much the same way that award-courting Indiewood pictures attempt to position themselves between franchise monoliths and international art cinema (Tzioumakis, 2012: 14). Erik even distances himself from the term indie and expresses pride that his games have competed for mainstream awards against AAA games rather than indie-specific awards. This is representative of indie games' more general shift from margin to center in the game industry since their initial popularization in the mid-2000s (Vanderhoef, 2016). In the last decade, the industry has witnessed the formalization of a middle tier of polished, moderately budgeted indie and indie-style games that are bigger than traditional low-budget indies but smaller than AAA productions, combining features of both modes of production and finding prominent placement on digital distribution platforms. Indie publishers and funders, whether independent themselves or boutique labels of industry giants, are key actors in the development of new "hybrid configurations and reciprocal dependencies" as Greig de Peuter and Chris Young so evocatively put it (2019: 2).

This growing middle category, sometimes half-jokingly referred to as "single/double-A" or "triple-I" games, takes up considerable cultural space. Even as it expands mainstream conceptions of what is possible in games, it simultaneously pushes less well-financed and/or more radically experimental game-makers and game-making practices that lack the backing of powerful intermediaries further to the margins (Lawhead, 2019; Ruberg, 2019; thecatamites, 2019; Whitson, 2019). In many ways, the phenomenon of commercial indie games has been enmeshed from the beginning in what Brendan Keogh calls the "aggressively formalized" logics of the game industry and has only become more so over the years (2019a); perhaps, the construction of a crossover market for mid-budget indie-style games is the logical outcome of this process.

Conclusion

Among others, I have argued that to understand the cultural-economic ecosystem of indie games, it is crucial to look beyond the activities of game developers alone (Parker et al., 2018) and indeed beyond the game industry proper (Keogh, 2019b). The emergence of Annapurna Interactive as an influential game industry intermediary is emblematic of broader shifts, as cultural industries open up and diversify in significant ways while simultaneously consolidating power and prestige around a gradually shrinking number of corporate and independent actors. Similar concerns resonate in the contemporary film industry, with trade papers debating whether franchise blockbusters, streaming platforms, and media conglomeration are squeezing out indie dramas and comedies completely (Galloway, 2019), or if audiences tired of endless seriality will seek out new kinds of stories (McClintock, 2019). Many midsize indie film studios have not fared well in this uncertain environment, and Annapurna Pictures has weathered a number of underwhelming releases in recent years, as well as financial woes and rumors of behind-the-scenes turmoil (Donnelly, 2019; Siegel, 2019; Sims, 2019). In response to concerns about the future of the company in August 2019, @A_i simply tweeted an animated GIF with the caption "WE GOOD" (@A_i, 2019); nevertheless, it remains to be seen if the cultural-economic middle ground of boutique indie game publishing will hold.

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9 Game Production Studies

Studio Studies theory, method, and practice

Casey O'Donnell

Game Production Studies and its sub-sub-field of Studio Studies compromise a relatively recent addition to the field of Game Studies and the study of game design more broadly. Succinctly, Game Production Studies explore the wide array of processes, practices, texts, technologies, and aspects that take place in and surrounding the game production process. This process is often referred to generally as "game development", which while rooted in the practice of making games actually constitutes a wide variety of tasks, disciplinary perspectives, processes, people, and institutions. While Game Production Studies are not particularly new, as Political Economic Studies of the game industry have long been an important contribution to the field of Game Studies, the actual study of the daily work practices of game design and development is much more recent. These studies, in the form of Studio Studies, have emerged as a much more recent field of study for Game Production Studies specifically and Game Studies more generally. Indeed, much work in the realm of Studio Studies was not viewed, at least initially, as fitting into the broader field of Game Studies, though that perspective is now on the decline.

It is important to differentiate Studio Studies from the broader field of Software Studies, which includes a host of methodologies, conceptual frameworks, and its own set of practices. This isn't to say that Studio Studies is not informed by or deeply connected to Software Studies, but rather that the focal point isn't always the same. Further, I would maintain that it is problematic to circumscribe Studio Studies within the realm of Software Studies, as it is problematic to conflate games, the studios they are produced within and "software" more generally (O'Donnell, 2012). While there are distinct overlaps between software development and the study of software development or simply technical work, I continue to maintain that there are elements of Game Development that make it particularly interesting and distinct from the study of work more generally. There is something simply interesting and important to be found in the perspectives that game developers and the context of their labor.

Perhaps, more significantly, it is important to distinguish Studio Studies from the broader area of study around the Political Economic Study of the game industry. While Political Economic Studies of the game industry were the most initially pervasive that grappled with the "production" side of game studies, their explicit focus is not the "studio" per se and rather more broadly on the game industry and its position within a broader political and social economic context (Couldry, 2004; Dyer-Witheford and de Peuter, 2009; Dyer-Witheford and Sharman, 2005; Kerr, 2011). Studio Studies chooses to engage at a much lower level. Studio Studies begins with the premise that game development is inherently a human endeavor imbricated by culture, technology, politics, economics, and numerous other aspects. Yet, it is that human element that places it solidly in a social scientific tradition. Studio Studies, as it seeks to find the humans making games, draws largely from areas such as the Anthropology and Sociology of Work, Anthropology and Sociology of Science and Technology, and, in some cases, increasingly, the Anthropology and Sociology of Leisure.

This entry explores a variety of the theoretical and methodological foundations of Studio Studies. It examines the increasingly complex practice of research working in the space of Studio Studies and concludes by examining the myriad of issues that can complicate work of this kind and potential pitfalls that the field faces as it moves forward in the studying of game developers making games.

Theories of Studio Studies

In many respects, Studio Studies, conceptually speaking, is rooted in those fields and disciplines that have maintained an interest in the study of human activity in the context of work and technological development. Largely arising from the Anthropology and Sociology of Work and the Anthropology and Sociology of Sciences and Technology was the original fruitful ground from which Studio Studies emerged. In some respect, this places it in a strange position to the rest of game studies, which has largely focused on the product of game development. Studio Studies flips the focus on those creating the object of interest for Game Studies, rather than the object itself. For many who entered this field early, there were few examples of precedents for the kind of work being done (Zackariasson et al., 2006). Some looked to more journalistic accounts of software and technological development for inspiration for the kind of work that slowly began to emerge (Kidder, 1981). Yet, many of those journalistic accounts (Sheff, 1993) failed to really account for the complexity of what game development looked like on the ground. As such, much of the early work in Studio Studies emerged frequently from those working in the game industry that looked around themselves and found a dearth of work examining the daily minutia that frequently goes into the creation of a game (Banks, 2011; O'Donnell, 2014). As such, these early studies found themselves often more at home in their parent disciplines/fields rather than those of Game Studies proper. Yet, recently, Game Studies, as it has increasingly interfaced with

Game Developers, "gamers", and the broader game industry, the desire to better understand the process, activity, and culture of game development has become much more central to the field.

The same turn has been found in numerous other fields, such as Organizational Studies (Barley, 1996), Studies of Work (Orr, 1996), Media Studies (Coleman, 2010), and, of course, Science Studies (Knorr-Cetina, 1983). The same thus proves productive within the realm of Game Studies. The turn toward the human activity that makes up a fields object of study proves a fertile ground for deeper understanding. Separating the human activity from the thing it produces starves our understanding of that object, be it science, film, or games.

As a sub-field-of-a-sub-field, Studio Studies draws heavily on the fields from which it springs. Anthropological and Sociological perspective remain the most influential. This impacts heavily the kinds of theoretical perspectives that come to be deployed in the work of early Studio Studies scholarship. Social Theory, as a broad category, serves to inform the majority of the work of scholars in this field. This also produces the same kind of variety of social theory that is frequently found in these fields as well. Scholars often find productive analytical frames ("theories") from a variety of perspectives. In some ways, this is why early work in the space of Studio Studies found its earliest homes in these more traditional disciplines which were always more comfortable with new empirical domains within which to explore more traditional conceptual frameworks.

Critical cultural theory, as it has dramatically impacted Sociology and Anthropology, also plays an important role in Studio Studies. However, it is a different sort of critical cultural work that often displayed by Political Economic perspectives. It is the continual focus on the developer, and the wide array of social, technological, and political-economic aspects complicates many of these perspectives.

Which is not to say that Political Economy approaches to studying game development are without their own incumbent theoretical frameworks. Indeed, Political Economy work has had a significant influence on work in Studio Studies. Marx and others continue to loom large in these studies, which does not mean that those theoretical frames no longer apply, but rather their dominance in the Political Economy work can often overshadow perspectives that those rooted in other fields may offer in understanding game development work.

What is probably most crucial to understanding early Studio Studies scholarship is to also appreciate the heavy linkages between theory and method for social scientists. In my work, for example, it would be difficult to separate the role that theory in Cultural Anthropology has played separate from the methods that are deployed. Much work in Cultural Anthropology comes from what would be the "interpretive turn" in Anthropology, which often resulted in important shifts in the relationship between researcher and the fieldsite. I have argued elsewhere that the linkage between theory and practice will always be an important touchstone for the kind of work that emerges in game studies (O'Donnell, 2013).

Methodologies of studio studies

Methodologically, with its focus on developers, the bulk of research in Studio Studies has drawn largely on qualitative research methods. The active seeking out of game developers and game studios as sites for "data" collection serves as the most prominent methodological practice, hence the name, Studio Studies. As such, the methods deployed in these studies often relies on actively enrolling developers and studios in the research process.

Interviews of the structured, unstructured, or semi-structured variety serve as one of the most important tools of Studio Studies researchers. Talking to developers and asking them about their work and world-view serve a kind of foundational point for Studio Studies. How can one understand game development or games without some insight into those that create them? While the interpretive reading of games is an important perspective, so too is talking to as many of those people involved in their creation and understanding their perspective on the resulting cultural product. This too reveals an interesting aspect of Studio Studies; in most cases, games are produced by a variety of people. While Creative Directors or Producers or other elements of a team have significant impacts on a game, they are ultimately a collective endeavor and to understand games as something more like a traditional text with one or even a handful of authors are difficult. Thus, speaking with as many developers as possible tends to be the way that most data is gathered for projects of this kind.

Fieldwork, central to Anthropological and Sociological method, has proven to be one of the more in-depth methods deployed in Studio Studies. Placing a researcher within the site of game design and development for an extended period of time creates an opportunity for ongoing conversations between researcher and game developer. Shifts in participants' perspectives over time and the researchers' own observations become data sources for later analysis. The collection of fieldnotes over time and space provide a source of data that goes beyond simply the words of developers. Fieldwork is typically the co-location and participant observation, interviews, and interactions over an extended period of time. While scholars may disagree on what a reasonable length of time for fieldwork would be, the goal is ultimately an extended period of time of interaction, coming to understand the perspectives and stories of those participating. The fieldworker as data collection device is not without its own complications. Indeed, the kind of complex interactions and entanglements between researcher and research participant is a hallmark of the rise of Cultural Anthropology (Faubion and Marcus, 2009; Pratt, 1986; Rosaldo, 1986).

Participant observation is a more general term for what is called fieldwork, though participant observation as a method does not necessarily have a necessary extended time period associated with it. As such, participant observation as a method is more generally used.

Also, critical to understanding game development more broadly is the role that various surveys play in understanding what game development work looks like broadly. Of course, participation in survey work is often difficult to come by, even amongst game developers. Much work has been done on this through important work of scholars in partnership with the International Game Developer's Association (IGDA). This survey work provides important understandings at a broad level of what and who is a game developer. The annual "satisfaction" survey gathers a variety of important information and has been an ongoing project of the IGDA and the scholars it has partnered with (Westar et al., 2018). It is important to note that this research serves as an important tool for even more focused Production Studies work because it often does capture rich data about "the industry" broadly, allowing for all kind of self-identified, hobbyist, professional, independent, or otherwise participation.

This work has supplanted what was once referred to as, the Quality of Life (QoL) survey, which also leveraged the numerous connections of industry networks of the IGDA but through a special interest group or "SIG" within the IGDA. However, these surveys still proved difficult to argue for their generalizability. There is certainly a need to do more work of this kind, though it remains difficult to rally the necessary numbers for more generalizable conclusions. This particular survey was also often criticized for its connection with the SIG, which began after controversies surrounding the "EA_Spouse" controversy within the game industry (ea_spouse, 2004). Of course, this complicated relationship between production studies and the game industry can be felt even more acutely now with the rise of movements such as Game Workers Unite and broader unionization efforts within the broad realm of the global game industry.

Other quantitative survey work can be found deriving from various game developer-focused websites; yet often, the enrollment of subjects in these surveys has proven difficult. If access to fieldsites can be difficult, then so too is simply gaining access to enough developers with the time, desire, and energy to participate. Even commercially funded survey work on game developers labors to gather enough participants to provide statistical validity. The annual "Salary Survey" produced by the industry website Gamasutra has proven successful and informative over its many years, in part due to the role that, historically, Game Developer Magazine and Gamasutra have played in the game industry (Staff, 2014). While these can often inform other work, because less information is known about methods and data collection, they often prove more useful for anecdotal reference rather than definitive references.

Increasingly, the importance of various legacy technologies or even archival documents, such as old notepads, or physically hand-typed design documents have proven to provide a plethora of new information with respect to the game design and development process. Legacy technologies like floppy disks, backup CDs, legacy source-code repositories, and out of date art-asset formats prove simultaneously important and impenetrable for the more historically inclined Studio Studies scholar.

Practicing studio studies

The reality of Studio Studies is that it can be a difficult research endeavor to undertake. Access continues one of the greatest limiting factors for researchers hoping to better understand game development and its place in the field of game studies. As such, in many cases, "indie" (or "indy") or independent game developers have proven to be much more willing to work with and talk with researchers. This too proves to provide its own set of challenges. Indie developers are often distributed broadly and may not have regular "studios" in the traditional sense where all the developers on a team are co-located. Some may work or meet together in person, while others work remotely or some variation on that composition. This places the researcher in the role of trying to tease out how to best gather the kind of data that answers the research questions within their work.

A handful of projects have managed to locate social scientists in the halls of more "traditional" game development studios. Yet, even in traditional studios, walls matter less and in different ways. Increasingly, large portions of game development work depend significantly on contract labor or intensely on tools developed outside any one given organization. Game development work has become much more diffuse, and the methods necessary to studying this craft have become increasingly more difficult. A researcher doing Studio Studies should evaluate in the field what precisely they must pay close analytic attention to.

What can further challenge the role of the researcher in traditional game development studios is not simply access but also the ability to collect data or discuss aspects of what occurs behind the closed doors of game development. Corporate interests may trump the ability of a researcher to discuss or even collect particular kinds of data. In my work with Vicarious Visions, an Activision studio, for example, my work was covered by a Non-Disclosure Agreement (NDA), which meant that my dissertation used pseudonyms for nearly every aspect of the story I was attempting to tell. However, as my NDA expired by the time I was writing *Developer's Dilemma*, I was freer to discuss specific aspects of my research. This can also be seen quite clearly in the first essay I wrote after the expiration of my NDAs, where I could finally discuss the fact that the team had been working on a Spiderman game, rather than some colorful attempt to sidestep specific facts or the names of projects (O'Donnell, 2011).

Regardless, the balance between anonymous developers or specifically identified individuals continues to be an important balance that those working in Studio Studies must maintain. The more work that is done in

Studio Studies that manages to respect both the desires of developers as well as the needs of the researchers working in the field, trust can continue to be developed. There is a balance to be found between the work of the analyst and the perspectives of developers. I have long maintained that my work is strongly governed by the American Anthropological Associations (AAA) Code of Ethics. In particular, one of the clauses that I frequently refer to as critical to Studio Studies are the "Do no Harm" and "Making Results Accessible". Core to many of these tenets is the principle that my work and its results should not make the community I have studied woe to accept another cultural analyst into their midst.

This in particular has struck me as one of the main differences between those working in a Political Economic perspective. In many cases, these scholars have maintained tenuous (at best) connections to those working in the game industry. There are notable exceptions, particularly those scholars who have shifted from Political Economic perspectives to those much more in line with those discussed in this essay (Kerr, 2011). Due to the lack of connection to the "field", where game development work occurs, authors have few qualms with making sweeping claims, which can often alienate and discourage future work that is far more respectful of the rich everyday worlds of game developers.

Perhaps, most interestingly of all is the increasingly connected world of those working in Studio Studies and the developers they study. Scholars and developers encounter one another time and again at conferences, over Twitter or Facebook and even projects that span industry and academy. While Anthropology, and Cultural Anthropology in particular, has become more aware of and sensitive to the complexity of conversations between researcher and those being researched, it is a complex issue that everyone involved must strive to understand and care for as well as not to compromise the ability to take critical stances. One of my fears for the future of Studio Studies is that some sort of roughshod research that leaves developers wary of critical scholarship may mean that the issue of access leaves future work much more difficult to undertake.

The future of Studio Studies

Young scholars continue to enter the field of Studio Studies and find new and interesting ways to explore this broad field. Code, software, development tools, and numerous data sources that have not yet previously been explored will provide new insights into the consistently shifting world of game development. Precisely, what Studio Studies role desires to be with the field of game studies as well as with that of game development have yet to be defined. Each project has been undertaken with its own set of goals and imagination of how it intersects with both. While I do not attempt to summarize all of that work here, I do point to some in the hopes that it shows the innovative attempts to explore the worlds of game developers more broadly.

Production Studies scholars have found themselves working closely with developers in incubators partnered with academic institutions. These collaborations have provided extremely important and productive insights into the relationships between not only academia and industry, but also the role that education and how developers come to learn their craft in-context (Whitson, 2017). The notion of voodoo and the role that tools play in the production process is a critically important one. Other works that spring from this tradition have been expanding the scope of Production Studies by looking at the tasks that publicity and aspects that were once covered by videogame publishing companies have now expanded to developers themselves or more specialized groups often linked closely with successful independent developer groups (Parker et al., 2018).

Special issues of specific journals have featured and shared broadly the expanding work of Production Studies scholarship (Simon, 2013). This has allowed scholars to discuss a variety of issues facing both production studies but game development and game developers alike. From issues of what it “means” for developers to be “indie”, and the role that work practices, culture, and discussing work means for constructing an identity around the term (Lipkin, 2013; Ruffino, 2013). The ways that work and development have had to restructure (or not) as the importance of consoles have shifted as digital distribution platforms have changed over time (Whitson, 2013). Although some argue that these shifts have merely changed the players rather than the underlying structure of the game industry (O’Donnell, 2019). In this context, researchers have explored how independent developers think of their work as “craft” over a more professionalized industrial practice and how that has enabled the emergence of new communities of developers, particularly amongst demographics that have been underserved by the game development community more broadly (Westecott, 2013).

Within this, it is important to view what “counts” as game development as broadly as possible. Looking at professions not often as examined, such as videogame testers within game companies is also important (Bulut, 2015). While this has sometimes been characterized as “above” or “below” the line, I worry that such a distinction ends up pitting game developers against one another. Who or what kind of work counts as above or below the line benefits only those at the pinnacle of the system. Much like unionization efforts at Kickstarter have been embraced by unionization efforts within the game industry. While Kickstarter might be “above” the line with respect to finding or producing funding for game developers, within that company, you have “below” the line workers as well. Unless we are talking about the highest paid individuals at a publishing company, digital distribution platform, or otherwise, there are far more numerous “below” the line people that ought to be explored as part of Production Studies.

By the same impetus, as already referenced, looking at the role that Twitter, publishers, communities of developers, co-housing, co-working, and various forms of shifting labor practices must remain in the realm and visibility of Production Studies. Artists on various crowdsourcing/or freelance websites may be performing game development work and remain out of the realm of visibility. Or the role that press or "press kits" play in how game developers market their material. The use of third-party platforms or freelance work within game development also remains an integral element of what should be characterized as Production Studies. Indeed, while numerous "studios" will continue to exist and produce games, the traditional brick-and-mortar game development studio has become much more diffuse with work happing within the walls of those organizations and without.

Also, critically important have been special issues focused on subcommunities within games and game development, particularly the rising role of queer games and queer game developers (Ruberg and Phillips, 2018). These studies are particularly interesting for a variety of reasons, but perhaps, most importantly, because these communities are often besieged by players of mainstream games. The developers working in this space are often marginalized in/for a variety of ways, making those working in the space particularly precarious (Pozo, 2018). This also often makes the developers much more reflective about the intersection of game development work with other theoretical understandings of work, power, labor, and capital. These special issues allow for new emergent work that also understands tools and the role and power that they wield throughout the development process of a game (Freedman, 2018).

For scholars looking to *do* Production Studies, it is critically important to be in conversation with game developers. I do not mean to say that in order to do Production Studies, a researcher must physically be co-located (though doing participant observation is good) or physically talk (though interviews and fieldnotes from conversations with developers are good) to developers. Much work could be done through a variety of methods. Paying close attention to what Cultural Anthropology has learned about the need for reflexivity and the fact that informants will (and do) talk back (Marcus and Fischer, 1999). The work of researchers in this space is read and reflected on by (some) developers, and this should inform those undertaking this kind of research. Further, because this work does often occur in corporate (or semi-corporate) settings, recognizing that doing research in these spaces is constrained and structured in its own set of ways (Cefkin, 2010; Orr, 1996). Further, there is a deep literature related to ethnographic work in the context of work (Smith, 2001).

As this collective volume indicates, there have also been significant gains in our understanding of the global scope of game development and research studies focused much more closely on regional aspects of game development. This too is of critical importance for Production Studies, as regional, social, cultural, and numerous other aspects of game development matter critically for understanding the broad nature of what many might claim cannot be called any kind of unified game industry, but rather a variety of game industries interfacing through a broader global system.

It also remains important to see the work of Production Studies within Game Studies as connected with a broader literature studying Production and Media Work in other fields (Deuze and Prenger, 2019). The importance of fostering connections between scholarship in these fields enables a kind of connective tissue, but also encourages and enables scholars to publish in a variety of venues. Production Studies within Game Studies views itself too narrowly, rather than as part of a broad community of scholarship focused on media production.

I am certain that there are aspects and communities of Production Studies that I have left out here, but truly, it has been a delight to see this work expand and grow. I too hope that this area of research continues to connect academics and game developers in new ways. The potential that the field offers is one that hopes to study, critique, connect, push, and also empower.

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Part IV

Politics

10 Game workers unite

Unionization among independent developers

Jamie Woodcock

The Game Workers Unite (GWU) movement started early in 2018, following a panel on unionization at the Game Developers Conference. Since then, local chapters of GWU have sprung up across the world in different countries. What is particularly striking about the example of GWU has been the speed with which a network has been established to promote unionization, outside of the workplaces that activists are based in. While this has contributed to the speed of its development, it also means that there remain important unanswered questions about how this will translate into organizing at a workplace level. The aim of GWU is to build a network to unionize the games industry, rather than being a trade union itself. Only in the UK, South Korea, Ireland, and France (although this precedes GWU), so far, this has converted into the formation of a trade union.

There are clear grievances for workers in the videogames industry, for example, crunch and long working hours, lower pay compared to similar roles outside games, lack of diversity, as well as precarity and project-based employment. There are a series of challenges to unionizing the videogames industry: first, the trade union laws and industrial relations systems differ greatly from country to country; second, there is no history of trade unionism in the videogames industry, meaning that the attempts are breaking new ground; and third, the videogames industry involves a range of different kinds of workplaces, with distinctions between AAA and independent developers, as well as the prevalence of contract and freelance work. However, regardless of the organizational form, no videogames are made solely by an individual, but come out of combinations of formal and informal networks of production.

Rather than focusing on how these can be overcome in practice across the entire industry, this chapter seeks to explore these questions in the context of independent developers. There has been a surprising level of interest amongst smaller and independent studios for unionization. Many independent studios have owners or managers who have experienced the pressures and problems with larger studios and therefore attempted to do something different when going independent. However, independent studios still involve a relationship between capital and workers, albeit one that can be closer and more blurred than within AAA.

The chapter draws on an ongoing empirical and collaborative research project with GWU in the UK to explore how these issues are being addressed in practice. It is part of an ongoing workers' inquiry with videogame workers and the GWU branch of IWGB in the UK. This is not a traditional academic research project, but rather a form of co-research that has sought to combine research with organizing (see Woodcock, 2017). I have participated in GWU in the UK since the beginning, from when it only had one member. My involvement has included ethnography, interviewing, as well as actively participating in the project of organizing game workers (see, for example, Woodcock, 2019). This is the context from which this chapter has been written. My own position is that of a researcher and organizer, rather than a worker within the videogames industry. The decision to write this chapter was to reflect on the specifics of trying to organize within indie studios. What follows is intended as a discussion that can inform thinking around the challenges and opportunities with indies, not as a closed project, but rather as part of an ongoing movement to unionize the videogames industry.

Understanding the dynamics of unionizing is important because it has the potential to reshape many of the existing relationships of videogame production. Most attention is paid to AAA studios, so this chapter seeks to correct this by exploring and highlighting the implications for independent studios. This chapter will focus on the challenges and opportunities of organizing videogame workers in small workplaces. It first examines the existing literature on videogame production in indie studios. The next part considers the challenges that trade unions have had in organizing at small workplaces, as well as adapting methods to new forms of work. The contribution of the chapter focuses on drawing out the emerging class composition of workers in indie studios, starting with an illustrative example of Greek delivery drivers, then exploring the technical, social, and political composition. It ends with a brief reflection on the potential directions this could develop in.

Videogame work and independent studios

There has been much recent attention on production and the videogames industry, particularly involving critical accounts (see: Kerr, 2006; Deuze et al., 2007; Dyer-Witheford and de Peuter, 2009; Kirkpatrick, 2013; Parker et al., 2014; Woodcock, 2016, 2019). However, these have tended to focus on AAA studios. One of the key challenges for GWU involves making sense of what will work to organize in different workplaces across the industry. Beyond mainstream AAA studio, there are a range of other forms of workplaces in the industry. These other forms involve "a far more diverse range of creators, audience, and modes of videogame production and consumption" that "has emerged with the rise of digital distribution and a proliferation of platforms" (Keogh, 2015: 152). For example, as Simon (2013: 2) has argued, to discuss indie games is "not to speak only of the games themselves or of the experiences of gameplay but rather of the cultures of game development from whence they came". The problem here is to understand what the common characteristics of indie development might be, in order to think about what sorts of organizing challenges or opportunities they might present for newly organizing workers.

The indie adjective tends to refer to independence from publishers, which may involve – although not always – a smaller workforce that is organizationally separate from other parts of the chains of production and circulation. For example, an indie is unlike to be involved in the production of consoles or other hardware, while it may sell a game through digital distribution channels. This can involve smaller teams (maybe as small as one) producing games without the support or involvement of a publisher. However, as Ruffino (2013: 107) argues, "it is also clear that the phenomenon of independent games cannot be explained solely in terms of a business model". As Ruffino (2013: 109) continues, this would miss the rise of indies rooted within a "complex series of influences coming from contemporary forms of labor that tend to put forward similar arguments in favor of individual emancipation". This individual emancipation is one response to the "poor working conditions" that await many "in the industry, and the indie community serves as an outlet for people to work closer to their own personal limits and time tables" (Lipkin, 2013: 11). Even on the issue of crunch, it becomes "inherently self-imposed in a way that is different from industry practice" (Lipkin, 2013: 11). As Williams (2013) pointed out:

> Many observers pin their hopes on indie studios. But working in a five person shop making small games is no guarantor against the same sorts of abuses the corporate players truck in. The rise of indies must be married to a just and equitable approach to work and employment if they are to serve as a collective counterweight to corporate abuses. Given the size of most indie studios, radical approaches such as cooperatively owned studios and flat hierarchies are within easier reach than most anywhere else in the wider tech sector.

Apart from a few notable examples of worker cooperatives in the videogames industry (for example, Pixel Pushers Union 512), this form of independence does necessarily entail any kind of politics around worker empowerment – let alone ownership. It is also worth noting the potential for self-exploitation of workers engaged in cooperatives (see Mellor et al., 1988). For example, Dovey and Kennedy (2006: 141) have warned that "the notion of independence needs to be interrogated somewhat if it is to have any purchase. As we have seen in the film and music industries, the 'indy' tag may not signify much more than 'wannabe'". Thus, care needs to be taken not to assume that the independence may be read as something similar to what the "literature on independent, alternative, oppositional, radical, or otherwise non mainstream media tends to suggest or advocate" (Martin and Deuze, 2009: 277). Often, this might be caught up in the increasingly complex process of outsourcing game production to "so-called second and third party studios' tied into the corporate system" (Martin and Deuze, 2009: 277). As Pedercini (2012) argues:

> There's no absolute independence because you'll always be constrained by technological platforms, protocols, hardware or infrastructures. Beyond gaming, you'll be entwined in a web of power, privilege, exploitation, and dependency, as long as the current modes of production persist.

This means that indies should not be considered *as independent* from the rest of the industry. Instead, as Keogh (2015: 156) suggests, indie is "a carefully cultivated antagonism against the status quo of the mainstream industry". As Joseph (2013: 100) has rightly pointed out in the context of the indie scene in Toronto, this can involve a "geographic space teeming with activity, filled with artists, organizations, cultures, histories, musicians, and television shows". These are linkages that tie indies together across different workplaces, as well as a "dynamic situation where exposure to the pressures of the commercial mainstream is a constant" (Crogan, 2018: 680). This means that there are important similarities with the industry as a whole: challenges in clearly identifying or delineating between 'labor', 'management', and 'capital' in the workplace; unstable forms of work, which may not map onto traditional notions of a wage-effort bargain; non-traditional workplaces that may blur work and play, both in terms of the product, as well as its production (O'Doherty and Willmott, 2009: 939–943). While there are differences throughout the industry, "what remains constant, for 'AAA' and 'indies' alike, is that game development is seldom an individual venture" (Whitson et al., 2018: 3). This means that there remains the potential for collective organizing. While distinctions between labor and capital might be harder to unpick in an indie context, it also begins from a potentially radicalizing context. As Lipkin (2013: 21) noted, the changes discussed could mean that "the possibility of indie developers clearly presenting anti-establishment perspectives by their mere existence is ending"; however, this also opens up new possibilities with what used to be known as industrial relations, as nascent unions begin organizing in both AAA and indie studios. It is to that possibility that the chapter now turns.

Issues for trade union organizing

There is comparatively little research on trade union organizing in small businesses (Rainnie, 1989), with more attention focused on the higher profile organizing at large organizations. Many workers do, in fact, work in small businesses – which are often family business too (Holten and Crouch, 2014). This means that there are similarities with some of the dynamics between small and large business that can be found in the AAA and indie distinction. Eurofound (1999) reported that there are many reasons given for low trade union membership in small and medium enterprises (SMEs), but that common answers involved "the recruitment difficulties faced by trade unions; hostility to unions among employers; and the closer relations between employers and employees in smaller firms".

Historically, the UK differed from the rest of Europe in that there were no thresholds on the number of employees required for collective representation to be applied (for example, provisions on trade union recognition of workers councils). However, this is not because these were automatically acquired (like, for example, in Norway or Sweden), but rather because there are few regulations that apply within a 'voluntaristic' industrial relations framework (Eurofound, 1999). Much of this was changed through EU legislation on employment rights, a situation that remains uncertain to continue at the time of writing. Despite this, there are clear challenges organizing in smaller workplaces than larger ones for trade unions. SMEs may be seen as too small to engage in company-level bargaining, limiting the potential collective gains or rather meaning that more negotiating is necessary to cover the same number of workers. There have been suggestions of 'territorial' bargaining, bringing together SMEs within a geographical area (Eurofound, 1999). This is an attempt to circumvent this problem, but there has been little evidence of success. There have also been warning that in 'new' – at the time the report was written – "high-skill firms, in information technology, professional services, and media and design, where an entrepreneurial approach predominates. Here, rewards can be high and traditional employer/employee models are inapplicable" (Eurofound, 1999).

There is increasing evidence beyond the videogames industry of successes organizing in new professional industries. The Tech Worker Coalition has had visible successes organizing with tech workers in the USA (see Woodcock, 2019). These have built upon earlier attempts to organize software developers, particularly at Microsoft (Brophy, 2006; Rodino-Colocino, 2012). Similarly, online journalists have seen recent successes too (Cohen, 2016; Salamon, 2016; Cohen and de Peuter, 2018). Coles (2016: 457), writing about the creative industry, has noted that the "prominent features of labour markets and employment relations based on contract, freelance or self-employment, income insecurity, excessive overtime and where risk is both individualized and devolved from the employer to the worker". There are clear similarities with the videogames industry, as well as early examples of how unions can organize within the creative industry more widely (Hesmondhalgh and Baker, 2011; Coles, 2016). However, many of these examples have focused on larger workplaces. The story of Washington Alliance of Technology Workers (WashTech) at Microsoft is far from the experience of a worker in an SME.

Different ways to identify an antagonism

The differences between AAA and indie, particularly when considered in terms of SMEs and large companies, would seem to lead to a conclusion that indies would be hard to organize. The evidence from other forms of tech worker organizing – so far – appears to be focused around large companies. There is a risk of missing some of the opportunities within indies if only examples are drawn from other sectors. The key to understanding this is the changing class composition of workers within the industry. To explore this, the chapter will briefly divert to outline these dynamics within another sector: food delivery. There have been a series of struggles on food delivery platforms (see Woodcock and Waters, 2017; Cant, 2018, 2019). As I have argued with Callum Cant elsewhere (Cant and Woodcock, 2019), the wave of struggles in platform capitalism are beginning to mature:

> What has taken place is a huge attempted decomposition of workers by capital. Taking Uber as an example, drivers who may have previously worked for a local taxi company now find themselves as bogus self-employees engaging with a multinational platform. Along with many other workers drawn onto the platform due to the promise of flexibility - or lack of employment options elsewhere - they are now part of a workforce that is estimated to be around three million. As Uber has sought greater amounts of venture capital, it has also savagely driven down wages and tested and refined new forms of algorithmic management. Unlike the diverse and disconnected taxi companies, these decomposition has also created shared conditions among many workers logging onto the Uber app. These have forged new international connections through which tactics and strategies are being shared, laying the basis for a powerful new recomposition of platform workers.

This would strengthen the argument that union organization is potentially strongest in larger workplaces (whether physical or not) that recompose greater numbers of workers. However, processes of class composition do not follow in linear or stable stages.

In April 2019, Greek delivery drivers held a 24-hour strike (ANA, 2019). However, these were not newly recomposed platform delivery drivers, but rather worked for a large number of restaurants spread across the city. At first glance, these isolated restaurants each employing a small number of delivery drivers would look like a difficult context to organize within. Yet the Driver Workers' Informal Assembly (SVEOD) had managed to organize an effective and disruptive action. My co-writer Callum Cant and I spent time with the delivery workers on May Day in Athens, following their successful action. There are a number of important lessons that can be drawn out from their example.

The first is the antagonism identified by the drivers. A conventional understanding might be that these were workers with a very large number of SME employers, so the antagonism was between small groups of drivers and restaurant owners. While this antagonism was maintained by SVEOD, they also identified further demands for the April action. They demanded that the motorbike and protective equipment should be provided by the SME employer, but also that the government reclassify the drivers' profession as hazardous. This, along with a unified job description, would entitle the drivers to higher pay and lower retirement and so on. This provided a national focus to the campaign, that could then be transferred into local bargaining, strengthened by the collective power at a national level.

The second is the form of organizing. SVEOD is a radical workers assembly, much closer to an anarcho-syndicalist network than a mainstream union. Rather than relying on paid staff and collective bargaining regulation, the drivers drew upon the pre-existing networks of delivery drivers. Athens, like London, has subcultures that have developed around couriers and delivery drivers, who share the streets as a workplace with a common set of experiences and grievances of work. In Athens, there is a long history of this kind of political organizing both within the delivery industry as well as with unions more widely (Kretsos, 2011). However, identification with political forms of organizing like this build upon previous histories, as well as being actively fought for. We witnessed the level of networking when we accompanied the drivers on one of their regular organizing rides around the city. A convoy of drivers sets off multiple times a week to tour the small restaurants and talk to other drivers. In between the meetings, workers riding pillion hoped off at traffic lights to staple posters to lampposts. The conversations involved a mixture of on-the-spot case work and political arguments, ensuring that the network is continuously remade across the city. SVEOD then acts as a focal point to bring these workers together for a biweekly meeting, formulating demands, and discussing strategy.

The third is the relationship between employer and consumer. The videogames industry has a close – albeit quite different to delivery – relationship between workers and consumers. This was potential antagonism that the delivery strike could cause, but it was directly addressed by SVEOD. In addition to the posters about the strike demands, they produced propaganda to build support amongst consumers that featured a simple food recipe. The message itself was clear: the strikers were only stopping the delivery of food for one night, using humor to show that consumers did have another option that evening.

These three examples highlight how different forms of class recomposition can take place amongst workers who are employed by 'independent' companies within a broader network. The activities of SVEOD provide a powerful example of how methods like workers' inquiry can make sense of the challenges and opportunities of organizing in particular sectors. This, as Cleaver (1979: 58) argues, starts from the "actual struggles" of workers, "their content, how they have developed, and where they are headed". Building from Marx's (1880) early attempts at a postal survey, inquiry has been used to look at more traditional, as well as newer, sectors of work. Although the struggles of workers in the videogames industry tend to be organizing networks beyond the workplace for now, the theories of class composition that have developed from workers' inquiry can provide an important starting point for making sense of organizing at indie studios.

Class composition provides a framework to make sense of inquiry, understanding the relationships between the form of work and the struggles that emerge. While initially developed by Italian Workerists as a twofold framework, the editors of *Notes from Below* (2018) have added a third dimension of social composition. This puts forward class composition as:

> a material relation with three parts: the first is the organisation of labour-power into a working class (technical composition); the second is the organisation of the working class into a class society (social composition); the third is the self-organisation of the working class into a force for class struggle (political composition).

This framework will now be used for sketching out the challenges and opportunities for organizing indies, drawing on both the literature discussed before, as well as the example of delivery drivers.

Strategies for organizing at indies

The first aspect that requires attention is the technical composition of workers at indie studios. There are many similarities here with the technical composition of AAA studios, involving similar labor processes for individual workers, albeit on a much smaller scale. This involves a similar difficulty to manage "creative labour process", but rather than in a major studio, taking place in a "marginal position in the global production network, leading to strong pressure on costs and flexibility" (Thompson et al., 2016: 322). This means that there is a different relationship between capital and labor experienced by many workers in indies. There are not the clear lines of managerial control, going first to project leads, then to departments and managers, producers, and so on – and ultimately upward to various agents of capital, shareholders, and corporate investments and profits. Instead, there may be – at a surface level at least – more of a flat structure of authority, with small workplaces having the appearance of either less hierarchy, or that hierarchy being centralized with fewer people. This creates a blurring of the distinctions of labor and capital, as well as work and play – as noted earlier by O'Doherty and Willmott (2009: 939). Thus, there are pressures of the wider videogames industry, both around crunch and overwork, as well as diversity (Woodcock, 2019), that indies cannot escape through formal independence, but rather operate from within the pressures of the existing industry. However, what is important to note is that the identity of indies, as discussed earlier, is “a carefully cultivated antagonism” (Keogh, 2015: 156) against AAAs. This attitude of difference has been one of trying to point out alternatives to mainstream methods of videogame production, which has the potential to change expectations of working conditions.

The smaller workplaces mean that individual workers are more exposed to the risks of organizing. If an individual decides to start unionizing within a workplace of a 1,000 members, they are far less visible than the worker sharing a single office with four people. However, as the Greek delivery workers example showed, the formal boundary of the firm does not need to be considered as the boundary for worker organizing. In the UK, there are clear clusters of videogame production, both for AAA and indie. For example, found in Leamington Spa or Bristol (see: Crogan, 2015), as well as London.

The importance of this potential social composition of workers is particularly sharp within the production of videogames. As a cultural industry, workers are often invested in not only the production but also the consumption of the culture in question. Many videogames workers cite their passion for games as a reason for working within the industry, which comes with shared culture, points of reference, networks, and so on. A powerful example of this can be found with the propaganda produced by GWU, featuring tongue-in-cheek versions of organizing material in the style of videogame walkthrough guides. This social dimension has been an important factor in the rapid growth of GWU as a network. Unlike many forms of greenfield organizing, the international network for unionization preceded the experimentation with organizing in the workplace. This builds a network beyond the small workplace – like that found with the Greek example – that can be then used to strengthen local organizing with the large collective body.

Both the technical and social composition shape – and are then shaped – by the political composition of workers. Similarly, with SVEOD, previous forms of political composition (for example, histories of anarcho-syndicalist organizing) shape future recompositions too. However, with the new organizing in videogames, there are no clear examples of what can succeed with organizing within indie studios. However, there are two potential points of recomposition that are beginning to emerge. The first is the wider movement of GWU. Successes in large studios or different national contexts can be rapidly circulated with other workers. The process of forming GWU has been one of beginning to form a new subjectivity: the game *worker.* This pushes back on the widespread trope that workers in the technology sector, whether in software development in general or in games, are not the kind of people who need union organization. The popularization of the term ‘game *worker*’, particularly with the GWU branch joining the IWGB (Independent Workers Union of Great Britain), brings the culture of videogames together with an antagonistic worker subject.

Second, the Greek example shows that the antagonism in a small business does not have to be limited to just the immediate owner. Given the role of indie studios within the broader ecosystem of the production of videogames under capitalism, there is a range of other points of antagonism that can be organized around. Like the IWGB organizing at Uber, as well as against TFL (Transport for London – the licensing body) and in the courts, there are multiple points of contestation for workers in smaller firms. For example, the issues of crunch and diversity within the industry are both ones that can be fought within an indie, as well as more broadly across a region or country. In the UK, there are widespread tax breaks and incentives for the videogames industry (Woodcock, 2016), which have often operated while companies have ignored labor laws and regulations.

To conclude, no videogames are made solely by a single individual, but rather they come out of formal and informal networks of other workers – within shared disciplines, communities, geographic spaces, and so on. Highlighting this is an important starting point for understanding how the unionization efforts of GWU can be translated from experiments in AAA studios to an indie context. Emphasizing the connections present within indie videogame production shows how new points of antagonism, as well as solidarity, can be found beyond a small (and increasingly individualized) understanding of the indie workplace. Like all new, greenfield forms of organizing, lessons can be learned from other sectors, but they require adapting to the context – including the technical and social composition – for successful and sustained political recomposition of workers within the sector. There is nothing automatic about the process of organizing, but the experiments of GWU show promising potential for how indie videogame development could be transformed by workers within it.

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11 Playing with risk

Political-economy, independent games, and the precarity of development in crowded commercial markets

Nadav Lipkin

> The [Steam] Lunar New Year sale was never a winner but given how poorly things are going I might have to revise my guestimate for when the first wave of indie closures hits to be earlier than 6 months from now
>
> Eniko, Twitter, February 5, 2019

"The situation is dire and mood among devs in back channels is grim" (Eniko, 2019). A statement like this seems unexpected for a sector whose explosion into popular culture in the mid to late-2000s, with games like *Braid* and *Minecraft*, seemed to suggest that independence was taking over the industry. Yet, over the course of around a decade, worker sentiments commonly reflect such despair. How did the thrill of independence become a precarious nightmare for so many?

This chapter explores the political-economic changes in the independent games industry that both contribute to the remarkable growth and rampant production within the independent games markets in the global North as well as the related rise in worker economic precarity. While independent game developers share many similarities to workers not only in the broader games industry but also in other cultural industries, more generally, specific changes to the relationship between workers and markets have accelerated a trend of economic insecurity and risk acceptance in return for more open markets and access to means of production. Where once the industry was restrained by relationships between interrelated firms and industries, changes to development tool access, dropping minimum production costs, and the proliferation of digital distribution have left many independents on their own without stabilizing forces. Put simply, there is a price to pay for political-economic independence, and by exploring its causes and effects through the examination of worker sentiment online, existing scholarship on the political-economy of the industry, and the assessment of shifts in technology and distribution schemes, the industry's future can be better understood.

Precarity and commercial cultural labor

Understanding precarity in contemporary-independent game production requires first a review of this condition across similar industries. Economic precarity is not a unique condition to commercial independent games; rather, it is often described as a common feature of commercial cultural labor broadly across scholarship. Caves (2002) points to the "art for art's sake" property in cultural labor, whereby "artists turn out more creative product than if they valued only the incomes they receive, and on average earn lower pecuniary incomes than their general ability, skill, and education would otherwise warrant" (4). This systematic overproduction in cultural industries creates downward market pressures on product prices and wages. Some, including Caves, suggest that this is because there are more than simply market considerations driving creative production. Menger (1999), for example, suggests that such work provides additional "psychic income" in the form of nonmonetary rewards (554) such as the ability to engage in interesting work, have personal autonomy, and opportunities for self-actualization. Specifically, Ross (2013), Terranova (2013), Florida (2012), Andrejevic (2013), and Caves (2002) point to the particular draw of promised "free promotion". How much the offer of work for exposure lowers the value of labor and product cannot be precisely determined, especially given the extensive backlash against that acceptability of work for exposure (see de Peuter, 2014, or a large number of posts on Reddit's "ChoosingBeggars" subreddit). Some, including Csikszentmihalyi (2013), Florida (2012), Pink (2001), and Leadbeater and Oakley (1999) suggest that the creative impulse is, at least in part, a personality trait that drives workers to suboptimal arrangements. This perspective has been rightly criticized as an attempt to naturalize poor worker conditions and expand neoliberalism (see Kerr, 2017; Lee, 2012; Ross, 2008; Stahl, 2009).

While risk is an inherent quality of cultural labor even for major industry (see Lipkin, 2016: 39–40), there are strategies that attempt to mitigate that risk – especially for major industry firms. Decisions on what products to produce often depend on a cultivated instinct by industry professionals (what Zafirau, 2009, calls the "gut" instinct) based on a record of past successful intuition. Culture industries frequently turn to past successes directly. This could involve backing a previously successful creator's second attempt. As Caves notes, "a successful first book raises readers' interest in the next, just as a successful second book increases demand for the first" (2002: 58). Elberse (2013), Caves (2002), and Becker (1982/2008) note the tendency to bandwagon on a "winning formula" with sequels and franchising (Elberse, 2013: 33). Elberse (ibid) discusses the economic validity of the "blockbuster strategy" whereby large firms place the majority of marketing resources behind a single product that is most likely to succeed. This may not be merely an attempt to keep on top of trends currently in vogue with the audience but also to drive cultural taste in a predictable direction;

Becker (1982/2008) notes, "Innovative dealers thus find that their aesthetic and financial judgments and activities are thoroughly mixed… [and] they cannot, and do not, wait for history to speak" (110).

Still, most new entrants to cultural fields will fail to break through. Next, Big Sound's 2013 report (Next Big Sound, n.d.) concludes that over 90% of music acts studied accounted for merely 3% of total fans across social media platforms, while the 0.2% who represent the top-tier megastars account for around 53% of Facebook page likes, 49% of video views on YouTube and Vevo, and 65% of Twitter followers. The division in the film industry is similarly, though less dramatically, top-heavy: an article by Deadline (Cieply, 2018) points out, "For the last three years, at least a third of domestic tickets sales have gone to the first ten movies". For industries that depend on creators working with producers to finance content creation, there is a natural need for gatekeeping to avoid both financing projects that are likely to fail and avoid wasting limited resources when they could be spent toward projects with higher predicted returns. If the up-front costs to the label, studio, publisher, etc. are low, it is common to see highly disadvantaged initial contracts that cost the financier little upfront in the probable event of failure but huge percentage returns on a surprise hit. Caves (2002) points to these contracts as a response to the music industry's high "stiff ratio", with 80% of albums and 85% of singles not covering costs (61).

Aside from these factors that contribute to production in excess of market capacity decreasing economic stability for workers, there is a cultural dimension in the West as well that normalizes and validates instability for these workers. Longstanding is the "starving artist" motif (see Caves, 2002; Ross, 2008; Whitson et al., 2018), going back at least to the 1800s, most explicitly conveyed in Puccini's opera *La Boheme* and subsequently reinvigorated a century later in Jonathan Larson's musical *RENT*. McRobbie (2002) points to the extensive media coverage of creative workers presenting "the public (especially young people)… with endless accounts of the seemingly inherent rewards of creative labour" (517). Beyond the normalization of overwork exemplified by dominant figures in the tech industry like Elon Musk (see Musk, 2018), this tech culture has also encouraged the blending of work and leisure time into an indistinguishable miasma in which work is ever-present and recast as leisure (see Florida, 2012; Gill and Pratt, 2008; Hardt and Negri, 2004; Lazzarato, 1996; Leadbeater and Oakley, 1999).

This individualization of risk through personal work ethic is part of the broader neoliberalism of creative labor. This includes the acceptance of working for nonmonetary reasons like networking (see Andrejevic, 2013; Caves, 2002; Florida, 2012; Gill and Pratt, 2008; Hracs and Leslie, 2013; McRobbie, 2002; Ross, 2013; Terranova, 2013) and taking pleasure in challenging work (see Florida, 2012; Pink, 2001). In conjunction with these ideological shifts, there has been a shift in expectations for worker tenure demonstrated by the rapid rise in independence: "free agents", as

Pink (2001) describes them, or "freelance and self-employed", or "contract workers" (Florida, 2012). The actual number of such workers in the United States economy is impossible to determine from government statistics (see Lipkin, 2016), but a 2018 poll suggests that 20% of American workers are contract workers (NPR/Marist Poll, 2018). This couples, especially in the American context, with a general diminution of the social safety net, furthering precarity for these workers. McDowell and Christopherson (2009) note that contract work subjects workers to lost access to healthcare, pensions, and other benefits, and while since 2009 the Affordable Care Act has successfully reduced the precarity of self-employment as it relates to employer-provided healthcare, efforts to weaken and challenge the law in the proceeding decade only demonstrate the broader instability of the American healthcare system for independent workers. Around the world, social safety net issues obviously vary, but the overall numbers of independent workers similarly point to the growth in neoliberalism and the challenges that accompany it; according to the McKinsey Global Institute (Manyika et al., 2016), between 20% and 30% of working-age people in Europe and the United States fall under their definition of independent workers.

Risk and the games industry

The digital games industry, as a subset of the cultural sector (Lipkin, 2016: 51), is and has been historically subject to these same conditions, both in regards to political-economic structure and labor ideologies. Like other cultural industry work, probability of financial success remains low, with the vast majority of money spent in the industry going to a small number of high-profile titles (see Kerr, 2017; WePC, 2018). Historically (though, as will be discussed, less so today), the games industry's large firms have relied on incorporating gatekeepers and other strategies to mitigate risk for console game production. Such gatekeeping tactics include strict quality and content controls, expensive development kits and software licenses, and regimented release schedules that limit the number of games released to avoid over-competition (see Dyer-Witheford and de Peuter, 2009; Kerr, 2006, 2017; Montfort and Bogost, 2009).

Ideologically, workers in the games industry find themselves in similar conditions as workers in other cultural industries. As discussed extensively in Dyer-Witheford and de Peuter (2009), Guevara-Villalobos (2011), Ruffino (2013), Bulut (2015), and Kerr (2017), the neoliberalism-influenced, self-sufficient precarity that characterizes other creative laborers is at least as commonly found in workers in the games industry. Challenges to these ideologies are beginning to grow in prominence, particularly in the growing efforts to unionize the industry spearheaded by Game Workers Unite starting in 2018 (see the chapter by Jamie Woodcock in this same volume). In the context of the United States, only voice actors are predominantly represented.

Expansion of access for independent game development

What makes independent game production different, then? A number of factors contribute to precarity and risk-acceptance in independent game development in the global North, and many result from otherwise positive political-economic and technological developments. A major factor is that commercially viable games have never been easier to produce and distribute, expanding the pool of potential producers. For one, there is the matter of production costs. While mainstream game budgets have exploded over time (see Costikyan, 2005), independent game development has seen many financial restraints loosened or removed. While it has fallen out of favor, the emergence of Flash in the late-1990s provided a relatively accessible and easy-to-learn development platform that could export playable (and sellable) titles, though not on console. Since then, development costs for small-scale production have collapsed. High-quality, professional-level development engines, especially Unity and Unreal, have followed to provide (mostly) free professional tools to practically anyone, with reasonable license rates for successful games. Even production for consoles, which once required expensive development kits and licenses (see Kerr, 2006; O'Donnell, 2014), has become significantly more independent-friendly, with those costs dropping considerably. Of course, PC development has been possible without licenses and development kits for some time, but changes to other platforms further help expand the markets for independent games far beyond PC without necessarily incurring massive additional costs. Kerr (2006) notes that even prior to the massive changes caused by digital distribution, PC games would normally experience more competition (twice as much as console games, around 2001) because of the absence of license and dev kit costs increasing the number of games vying for shelf space (82).

In conjunction with diminishing financial demands on the production side, digital distribution and the emergence of retail-distributors have similarly increased access and lowered minimum up-front costs of production. The largest factor for PC development has been Steam, which has offered distribution at no up-front expense and limited pre-approval based on quality or content (intermittently) since 2012 when they introduced Steam Greenlight. Steam Direct, the platform's current submission path in 2020, facilitates distribution for a limited up-front cost of $100 USD – far below the historical costs required for publishing physical products distributed at retail. Likewise, the service's take of 30% of revenue is vastly below the split for physical retail; Kerr (2006) estimates that in the early to mid-2000s, bringing a game to market using the retail model would cost between $3 and $10 million USD, and Swain (Irwin, 2008) estimates that a game sold at retail returns only around 17% of its price to the publisher. Distribution is only becoming increasingly more accessible as competing services rise to claim Steam's territory on the PC. Aside from its long-time competitors offering similarly free/low-cost distribution, notably GOG, Big

Fish, and itch.io, the end of 2018 saw a sudden emergence of serious direct threats to Steam. Both Epic and Discord announced competing distribution platforms offering similar submission paths to developers, though both take a far lower cut of game revenue than Steam's 30% to entice developers away from the dominant force in the market.

Diverse distribution opportunities notwithstanding, similar outcomes can be found even in markets with singular or few distribution channels like on consoles and mobile. Google's Android, Apple's iPhone, Nintendo's Switch, Microsoft's Xbox One, and Sony's PlayStation 4 all include stores that facilitate sales of independent games digitally. Each option varies in accessibility, requirements for prospective developers to seek pre-approval, pay for development hardware and software kits, non-disclosure agreements, and basic checklists for quality controls. That said, even the more restrictive platforms like home consoles are more accessible and independent-friendly than a decade prior, with Sony, Microsoft, and Nintendo deliberately courting independent developers to publish on their digital storefronts.

Not only are there numerous pathways for development at limited, or historically low, cost, but the institutional and development knowledge required to produce marketable products has also never been more accessible. Making games, historically, demanded significant education and effort. Anthropy (2012) explains that this extended to simply using an early computer: "To have access to a computer, then, generally required being connected to an engineering school" but one also needed to know how to code (24). As computing becomes increasingly black-boxed in the 1990s and beyond, the tools needed to develop a commercially viable game became increasingly complex and private as well; Swalwell (2012) notes that earlier microcomputer developers learned mostly by tinkering with open access programs (by necessity as well as for fun), and as soon as both the code and the need to tinker faded, so did the means of accessing development knowledge first-hand. Consequently, as O'Donnell (2014) describes, game production knowledge became hidden from the public in "increasingly secretive inner circles of the videogame monoliths and the acquired knowledge locked into powerful, legally circumscribed spheres" (148). Development tools became increasingly proprietary, developed in-house by studios, and their partners for their use only. So secretive is mainstream development knowledge that, O'Donnell explains, even experienced workers who have made games before have to learn skills again from scratch, because they depended on entirely different tools and processes (147–148).

The biggest change in this dynamic for independents is the proliferation of development tools that do not shun institutional knowledge becoming public. Development tools commonly used in independent and mainstream development like Unity and Unreal have developed large public archives of technical documentation, corporate-produced training modules, sample programs, YouTube tutorials, and community forums where developers can learn from each other at no cost. The public availability of development knowledge presents would-be independent developers greater opportunities to get a foot in the door and develop higher quality work without first acquiring access to a university program or on-the-job training.

That said, formal education has exploded in availability as well to serve an audience of interested potential games industry workers. This includes not only bachelors' programs but also online programs, two-year degrees, master-level programs, and certificates at institutions ranging from prestigious private schools like NYU to trade schools like Full Sail to community colleges and state universities across the United States and worldwide. This is not an entirely recent development, as Kerr points to the proliferation of such options for formal education in 2006, either through more general degrees in art or computer science or game-specific programs (Kerr, 2006: 89). Those game-specific programs have increased in popularity since; an IGDA report released in 2018 claims that 44% of respondents had specialized degrees somewhat related to game development, with 95% having at least some college, university, or post-graduate education (Weststar et al., 2018).

The lower base costs and higher accessibility of development, distribution, and production knowledge all serve to empower potential creators and open multitudinous pathways to entering commercial game development spaces. The fact that all of this is facilitated over the Internet means that geographic constraints are similarly diminished, provided the worker has even meager access to it. This reduces the reliance on outside financing and publishers (at least at the lowest levels of commercial production). GDC's survey results indicate that a significant proportion of respondents take advantage of those opportunities; 38% of respondents claim, "I do marketing myself in addition to helping develop the game" (GDC, 2018: 11), taking ownership of one of the historically primary functions of a publisher (Deuze, 2007: 211; Dyer-Witheford and de Peuter, 2009: 39; Kerr, 2006: 65; Whitson et al., 2018). The report also indicates that a large number of respondents forgo the traditional publisher route, at least at first. Whereas, in 2002, Williams suggests that between half to two-thirds of game production is performed under a publisher (47); by 2018, GDC puts the number at just 23%, and only 34% report being interested in working with a publisher on their next project (11).

The fact that many independents see publishers broadly as antithetical to their ways of making games (see Lipkin, 2013; Whitson et al., 2018) conveniently masks the reality that developments in access to markets render publishers less of an integral part of development anyway. When Team Meat finds a way to circumvent Microsoft's policies of requiring games to feature paid DLC for new content, they note, "It is nice to have the power to totally say 'fuck you' to that system and go our own way" (Goldman, 2010). Such power being where it is almost a decade later, developers no longer need this open hostility to publishers, because there are more plentiful opportunities to go to market without them.

Instability from accessibility

While conditions for independent developers in the global North make the industry prime for expansive production and distribution, this accessibility comes at a cost. Of primary concern is that the relaxation of limitations to production means more games are made and distributed, contributing to exploding competition and concentrating risk on independent workers. In this way, the very conditions that make independent development expansive and accessible also negatively affect economic viability through a number of means. The fact that games can be made and distributed more easily without staff dedicated to producing and publishing has contributed to an environment in which those jobs become necessary, thus adding work on top of the already extensive development labor. The minimized costs of production reduce the dependence on external financing, contributing to an environment in which developers themselves are more likely to shoulder the risks of failure exclusively. Digital distribution has likewise minimized the reliance on external firms (such as manufacturers, distributors, and physical retailers) with stakes in the success of a game at market, leaving that pressure of success firmly on the independent workers in the absence of publishers.

The shunning of publishers and producers for ideological and financial reasons serves to make independent game development more precarious, both as a result of absent expertise that contribute to success at market as well as the lack of a secondary firm with vested interest in that game's success. Whitson et al. (2018) detail extensively the ways in which the absence specifically of producers and other professions more common in mainstream industry, such as "community management, quality assurance, HR, marketing, R&D agent, spokesperson, CEO, corporate relations officer and so on" (10) cause problems for lean independent teams who see this work as associated with "hierarchical, risk-averse and creatively stifling large-scale game production" (4). While making and distributing games in the current structure does not explicitly require workers dedicated to this work, thus contributing to some teams seeing those jobs as unimportant, Whitson et al. note how the absences of producers end up adding work to the other members of the team or causing important work to be ignored, to the eventual detriment of the project's financial viability. Even if workers are qualified and capable of accomplishing these added tasks, the process of "juggling multiple 'hats'" (10) serves to add even more labor on workers already conditioned to work to excess. The authors state specifically that the additional, largely relational, labor that goes into networking, reaching out to press, and building relationships with the game's eventual audience is important to the success of contemporary games at market; yet, it ironically ends up being the work most likely ignored or undervalued.

This trend that Whitson et al. identify exemplifies the tension that typifies these independent workers. A product can be made with less money, fewer interrelated firms, and less oversight, but what remains is an independent development ecosystem that thrives on production without significant regard for quality, thus leaving development teams alone at risk if a game fails at market. The rise in self-financing, facilitated by the drop in price floor for access to tools, skills, and distribution, means that there are more often no external financiers with stakes in a project's success. This naturally provides creative freedom – a long-held prize championed by independent workers in the past decade (see Costikyan, 2005; Lipkin, 2013; Whitson et al., 2018); unfortunately, there can be financial value in someone saying 'no' on a project. As Caves (2002) points out, film executives need to know when "to write off a large sum spent on an out-of-control project", even if they are reluctant to do so (142). Film particularly is full of such cases where no one said "no": studios and filmmakers both have crumbled under the uncontrolled creative impulses that created failures like *Heaven's Gate* (1980), for instance. With games being made without the involvement of these financially interested parties, there is no definite voice within the project dedicated to its financial viability.

The same problem extends to other industry partners. In other industries, especially those dealing in physical goods, success is mutually beneficial – a successful product does not merely benefit the creative team that developed it but also manufacturing companies (who do not have to incur the costs needed to refit factories for new products), distributors (who continue getting contracts to move the goods), and retailers (who continue making reliable money selling those goods to customers). The current independent games landscape in the global North, and the emphasis within independent games for exclusively digital distribution, has sidelined most of these interconnections. Most obviously, digital goods have no manufacturers – thus, no partner with stakes in the product's quality that physical products have. Moreover, the distributor is also the retailer, producing its own differences in incentives. The most definitive benefit to digital products for retailers is that the absence of physical stock means the elimination of many of the elements of scarcity that retailers derive profit from and use to justify limiting what products are available for sale. Digital storefronts have essentially unlimited showroom space for unlimited time. Unlike physical retail where unsold stock is a liability and display space is at a premium, digital retail-distribution does not lose money for products that do not sell to that degree. Generally speaking, a dollar spent on a game that sells 10 copies is worth the same amount to the distributor as a dollar spent on a game that sells millions.[1] So long as total sales across the platform rise, how that is distributed between games is not relevant to them, leaving little incentive to deny low-quality games access to the platform. Kerr (2017) notes, "[t] hese services are less interested in 'gatekeeping' and managing the quality of content and more in content and data flow" (46).

There is a caveat to this idea; distributors that publish anything regardless of quality risk reputation damage that can ultimately turn customers and developers to other platforms. Montfort and Bogost (2009) suggest that this is a possible contribution to the market crash in 1982 (with retailers, frustrated that massive numbers of unsold games were wasting space, turning away from the games industry). They note that "the industry was putting effort into licensing rather than design and programming, inundating the market with less innovative work that had been forced through the development process quickly" (134). The revitalization of home consoles in the West ushered in by Nintendo came with a remarkable shift in priorities, from allowing anything onto the platform without quality or content checks to the strictly controlled licensing approach Nintendo and subsequent console publishers adopted "to recover from the market crash in the West [in 1982] and [ensure] a limited amount of high quality content and premium pricing" (Kerr, 2017).

If this is the lesson the industry learned from the 1980s, it is not always reflected in contemporary digital markets in the West. In January 2019, Steam reached the milestone of having 30,000 games for sale (Patel, 2019), Google Play offered approximately 670,000 games in the first quarter of 2018 (Statista, 2019b), and both Google Play and the Apple App Store have consistently had over 300,000 games on their stores since 2015 (Statista, 2019a, 2019b).

It is possible that the rising number of games, especially the number of low-quality titles brought to market, thanks to the hands-off publishing attitude governing many digital markets, is or soon will affect the platforms' bottom lines. In justifying a change to the search and discovery algorithm in 2018, Google stated, "Over the last year, we've been enhancing our search and discovery algorithm's consideration of app quality and user engagement", so that "apps and games that have high retention rates, low crash rates, low uninstalls, and many other factors, are recommended more often" (Lanier, 2018). This suggests that Google feels that app quality matters, and the platform has an interest in changing discovery to highlight "better" games and apps. That said, the economic conditions and relationships that made overproduction a liability for retailers/distributors in the 1980s until the digital distribution boom emerging after the global recession have changed. Google Play still offers hundreds of thousands of games at any given time and has done so consistently for years – even if some of those games are difficult for a potential buyer to find. Developers may complain about these changes that affect download rates and make it even harder to get a foothold and an audience (as Lanier, 2018, highlights), but so long as quality is adequate enough overall to maintain the user base, the raw number of games available on the platform is of little economic consequence for the platform. It remains to be seen if reputations will worsen to the extent that competitors can steal users from current market leaders, as opposed to the previously mentioned threat of losing opportunities to sell high-profile games as developers seek better revenue splits on emerging platforms.

Responding to the challenges of economic viability

Such is the portrait of the broader political-economy of independent games in the global North, characterized by the minimized need for partnerships to produce and distribute products through platforms whose interests are in aggregated sales across all games and not in any one of them particularly. Developers, for their part, have not necessarily just accepted these conditions. While the discourse around the subject, under the term "indiepocalypse" or not, is fraught and produces endless think-pieces blaming themselves, each other, platforms, or just the inevitable developments of markets (see Coster, 2015; Eniko, 2019; Jeremiah, 2018; Lipkin, 2019; Serviss, 2013; Vogel, 2013; Warner, 2018), developers have taken a number of direct steps intended to bring stability to their lives and bank accounts.

A significant point is that the economic conditions that create risk relate to overproduction and over-flooded markets. As such, developing markets may be more economically viable. With PC and mobile overcrowded, some are turning to virtual reality (VR) as the next frontier (see the work of Paweł Grabarczyk in this same volume). Why games for VR have not yet risen in quantity to place it in the same category as other digital markets come to a few key points. First, a part of overproduction is the wide availability of production knowledge, but while traditional two and three-dimensional games have extensive amounts of user-generated instructional content to draw from, VR is newer, less standardized, and has less information to draw from. To date, each VR headset operates differently, meaning that it is not a simple task to port games from one headset to another. While certainly possible, the difficulty of bringing a single game to multiple VR platforms contributes to fewer total titles as developers tend to balkanize. Thus, producing games for VR mirrors multi-platform development of prior decades, where each port had to be made as its own entirely separate game (see O'Donnell, 2014). A second factor is that VR is a growth industry that many still suspect that it will flop (much like it did in the 1990s; see Moss, 2017, for perspectives). Given the time and money that goes into development, potential developers may shy away for producing content for a still small, though generally wealthy and content-hungry consumer base. What games do end up published for the hardware have less competition than on other digital markets as a result, though how much this translates into financial viability remains to be seen; Jesse Schell (Moss, 2017) estimates 90% of games made for VR are not yet turning a profit.

Another strategy becoming increasingly common is escaping commercial markets for patronage (or, at least, using donations to supplement sales and other revenue). The rise of Patreon in particular is a potentially dramatic one for solo developers making large numbers of small titles. Rather than depending on small amounts of money from a large group of customers who split money and attention between a large number of available games, a developer can make a living wage (or closer to it) supplementing/substituting larger contributions from a smaller number of dedicated fans. Little has been written on this practice amongst game developers or how patronage relationships vary globally, but the platform itself is experiencing a period of continued growth generally; over two years, the number of individual pledges on the site has risen over 230%, and the estimated monthly payouts have risen by 43% (Graphtreon, 2019). Additional research is required to say more about the impact that this and other forms of crowdfunding will have on workers (rather than more generally for projects; see Lipkin, 2016). Patreon, given its monthly payout compared to the all-of-nothing approach of Indiegogo or Kickstarter, may work poorly for conventional game developers, who produce games at a much slower pace than YouTube content creators produce videos or artists produce web comics. Alternatively, it may lead to a rise in the number of (or at least prominence of) shorter, less technically ambitious, more experimental, and more niche games that would never be commercially viable.

Conclusion

It is a complicated but exciting time to be an independent game developer in the global North. Changes to the political-economy of the games industry have opened the floodgates to production, with more opportunities at various budget levels to produce games without the extensive constraints from publishers or technology that limited development in past decades. With this freedom, however, comes upheaval as the rapid influx of new entrants to the market collides with a market environment that has little stake in individual games and more interest in the money spent across the industry as a whole. If in the wake of the crash in 1982, the industry was saved by limitations on production imposed by powerful firms, it seems the response to a crisis caused by the democratization of the industry may, fittingly, come from independent workers themselves. Perhaps, the ideology amongst prospective workers will change, discouraging overproduction, or more workers will start their careers in non-commercial marketplaces. Discussions of unionization in the broader industry suggest that workers across the gaming landscape are rethinking their financial statuses and the amount of precarity they are willing to shoulder. Perhaps, markets will do as Nintendo did in the wake of Atari's collapse, closing off pathways to production and increasing visibility for certain creators while forcing the rest to chart new paths. Or, perhaps conditions will remain; these precarious conditions are the ones under which musicians and authors have struggled for even longer than game developers. What is clear is that the genie cannot go back in the bottle, and the key question left to independent workers collectively is a difficult one: is it preferable for the industry to have fewer workers with more sustainability or more workers with greater precarity?

Note

1 There is some variability on this issue, as the percentage of sales or in-game purchases the distributor takes may differ from platform to platform and over time. Steam announced in late 2018, for instance, that sales after 50 million dollars will split revenue with only 20% going to Steam, unlike the standard 30% the platform otherwise takes. This issue is even more variable for itch.io, which does not take a mandatory cut-off sales of games on its platform, allowing those who upload the game to decide how much of a cut itch.io will get from sales. Still, the broader point is unchanged; Steam earns a cut regardless of what game the customer purchases. If anything, the changes to revenue splitting further incentivize the platform to accept low-selling titles, as the platform earns a larger portion of those dollars than money spent on more successful games.

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Part V
Local indie game studies

12 Playful peripheries

The consolidation of independent game production in Latin America

Orlando Guevara-Villalobos

Since the emergence of the videogame industry, Latin America has been an actor mostly present in videogame cultural representations and media discourses. Although not at all absent, the region has historically played a minor role in the global networks of game development, concentrating productive capacities and consumer markets mostly in Anglo-America, Europe, and main Asian markets (Johns, 2007). As an initially unfamiliar artifact, the videogame label was filled by local media narratives, which would swing between moral panic and an echo of the global industry expansion. It is not until the turn of the last decade that the media, international organizations, and State institutions of the region turned their attention to their local game production. This was made possible by the convergence of specific sociotechnical processes such as soaring videogame digital distribution markets, the local booms in videogame production, as well as the increasing presence of innovation, information, and knowledge economy discourse in governmental policy agendas.

Furthermore, scholarly literature about the region has been rather feeble, especially regarding independent game development. Still, it can be reasserted that amidst the economic, social, political priorities that have oriented social science research in the fields of new media production and digital economies, there have been moments in Latin-American videogame history that have attracted academic attention. These works deal with varied theoretical assumptions, empirical scopes, and practical concerns, but they still provide a solid ground to understand better the presence and conditions of independent videogames and production hubs in Latin-American countries.[1]

It is true that as an expressive cultural medium, videogame production has been subjected to different cultural appropriations from various social actors involved in their consumption. Latin America was also part of this process, and the claims of games as forms of art, political persuasion, and pedagogic value have manifested in a plethora of expressions beyond market commodification. Nonetheless, this chapter delves into a general overview of the history of Latin-American attempts to create a sustainable industry, against the backdrop of the global expansion and deepened intensification of free market economies and neoliberal reforms in national-states during the 1990s (Robinson, 2004). By focusing on the complex social relations of production within which independent studios and development are embedded, I aim to provide a better understanding of the relationship between Latin America and global videogame industry, and the crucial role of the critical political economy on global capitalism and cultural industries to analyze it. These assemblages show the multi-scalar networks that independent studios rely on, as their actions strengthen the denationalized spaces that render global corporate networks arrangements possible (Sassen, 2006; Kerr, 2017). Thus, in this chapter, I argue that the Latin-American videogame industry faces the particular conditions of its marginal position in both consumers' and industry labor markets. The demand for low-cost labor power in the current international division of labor has structurally locked the region as part of the global outsourcing world. Moreover, although digital distribution has provided developers with the chance to dedicate time to videogame projects, it is through their cultural and formal organizations throughout the region that they manage to gain the creative, technical, and business expertise that enable game production and more market awareness. The work of state institutions and private organizations is slowly addressing the game industry as subject of public policy. In spite of this, strategies across governments are not just uneven, but insufficient to leverage their local creative/cultural/media industries at both local and international market scopes.

The argument I unfold goes as follows: although game development can be traced back to the grassroots cultures around playful consumption of videogames since the early 1980s, the neoliberally laden expansion of global game production networks locked the region mainly within the global outsourcing networks, as short-term creative specialists building up brand content for corporate actors from the goods, culture, and services economies. The structural grid changes as digital distribution and mobile markets for independents emerge, revealing both the potential of Latin-American studios, as well as the needs for economic and State institutions to address their national infrastructures for the digital economy.

In the first two sections, I address the first forms of consolidated videogame industry activities in the region, embodied initially in manufacturing capacities and third-party outsourced labor locked-in around corporate global networks and their brands. The third section deals with the reorganization and diversification of videogame markets via digital distribution and mobile technology, while growing cultures of production and institutional organizations of the sector enable the creation of independently made videogames. Last, I briefly address the general directions that private and public institutions take in order to support national videogame hubs, pointing at further local and comparative research, as well as public policy that emphasizes the sustainability, independence, and collaboration within and between local hubs.

Playful beginnings and the off-shore manufacture destination

The beginnings of Latin-American game production can be traced back to the cultures of production that spanned with the early introduction of game consoles and personal computers, featuring occasional local or isolated foreign ventures struggling against the conditions of national market structures. Given the absence of economic, cultural, and political interests converging in the creation of strong cultural and innovation systems in low-income countries, the culture of videogame consumption and production in Latin-American countries had been left to their own processes of domestication. Videogames entered the region first through the circulation of game consoles and personal computers. They were initially acquired through migratory flows and tourist visits mainly to the United States of America, but also significantly potentiated by the distribution of Chinese-cloned FAMICOMs (Penix-Tadsen, 2016: 47). At the same time, personal computers were being introduced, and videogames became commonly used between computer science and engineering students and professionals alike.

Within this early context of computer appropriation, a culture of reverse engineering, software modification, and hacking appeared in some countries. These practices provided the first forms of software and game circulation through informal and solidarity networks of computer and game enthusiasts, amidst the absence of industrial capabilities, and similar to British bedroom coders (Marisca, 2014: 167). Key examples in this early history include accounts of game development in Argentina, Brazil, and Peru. The games developed at the time were based on modified versions of other titles, as well as original productions based on local cultural expressions (Penix-Tadsen, 2017: 108). Other experiences narrate a landscape of small halls and store corridors featuring coin-up machines in the region, many of them traced to a small assembling industry that boomed mainly in Mexico during the 1980s, given its geographical proximity to California (Gonzalez Seguí, 2000).

Nonetheless, the first glimpses of an emerging game industry were found after free market reforms deepened throughout the region during the 1990s, pushed by supranational organizations such as the World Trade Organization (WTO), International Monetary Fund (IMF), and World Bank (WB) along with local elites. These reforms were another step in the expansive wave of global capitalism after the fall of the socialist power bloc, entailing the creation of subnational spaces: institutional and regulatory frameworks within national states that could sustain the logistics of global production networks and financial flows headquartered in the main world economies and the intensification of free market operations worldwide.[2] As a consequence of trade liberalization, deregulation of direct foreign investment, and forced privatizations, Latin-American economies would kick start a process of economic re-integration of the region to global capital financial flows and the new Information and Communications Technology (ICT) markets (Roldán, 2005).

In fact, videogame production first came to academic attention in the context of the expanding global networks of consumption and production deployed by the major corporate conglomerates engaged in the industry. Legal reforms strengthened the power of transnational capital owners to shape their productive embodiments, within unevenly integrated Latin-American export economies. In this context, economic globalization and their most dynamic corporate correlates – such as science, technology, engineering, and mathematics (STEM), financial, and media industries – have entailed the creation of highly uneven relations of production. As a feature of these production relationships, legal frameworks would ensure private informational markets through which transnational corporations could leverage their market position, productive networks, and Internet Protocol (IP) appropriation (Xhardez, 2012). As a result, the integration of the local economies to global markets has come at the expense of a subordinated role in the ICTs and media relations of production, holding the lowest stakes in the complex and highly specialized value chain.

The shape this first industry took was not characterized by the development of local game companies or expressed interest in consumer markets. Given the low articulation of state policies around these newly opened economic trajectories, there were grounds for only a fledging industry shaped solely through foreign interests, to whom the region would be integrated as 'maquila' style economy, using countries like Mexico as one of their global assembly lines for cartridge manufacture, exporting at low cost to the US and Canada (Lugo et al., 2002; Kline et al., 2003). Only US Microsoft has extended this relationship to console manufacturing with a plant that opened in 2013, after console penetration increased in the region (Heim, 2013).

From outsourcing to independent production

The early readings provided by Lugo, Sampson, and Losada of the Latin-American industry were skeptical toward the possibility of an authentic regional videogame production hub, given the interest shown at the time in the region as a location for manufacturing. Nonetheless, by the turn of the decade, further research reports the existence of a small videogame industry powering up in several countries with its epicenters being in Argentina, Brazil, and Chile (Xhardez, 2012).

Still, the recurrent trend evidenced in these works reveals a production creatively subsumed into the logic of decentralization and distribution of global industries, fulfilling a role as a source for short-termed, specialized and cheap high-skilled labor, whose cultural labor is sold to feed through different forms the intellectual property owned and developed by global corporations from the capitalist main centers (Xhardez, 2014: 41).

This tendency corresponds to a growing trend of outsourcing game labor to Asia, East Europe, and Latin America during the mid-2000s that accelerated during the global economic crisis of 2007 (Guevara-Villalobos, 2018).

With low creative and material stakes within these global production networks, game studios would mainly work developing partial or total content for corporate brands, educational projects, or work as specialists in games, media, and related industries. An overview of a main center of production could help exemplify this trend. In a study of 30 Argentinian videogame companies during 2010, Paricio and Seivach (2012) found out that 84% of the income reported was made through licensed projects (such as localization) or work for hire for game studios, advertisement agencies, and other companies; only the 16% of revenues came from owned IP. Even more, the trend was expected to keep stable as the technical expertise of developers was favoring a positive feedback from foreign investments. Likewise, it reports a hierarchy of the demand for developers' work in the areas of advergaming, developing content for game publishers, general services to commerce and retail, thus creating games mostly in low-cost outlets such as web games.

This context suggests an international labor market where videogame studios are locked into a corporate rationale in regards to aesthetic and work processes. However, it is here that independent initiatives emerge as an attempt to bypass it. Exploring these spaces of possibilities, Verónica Xhardez's (2014) case studies in the Argentinian game industry offer an in-depth overview of the inner logic of videogame relations of production at a national level. Her research adopts a stand from critical labor sociology in order to shed light on the capitalist organization of game labor, its relationships with the ICT industrial complex, and their effects on creative input/output. Key findings arise as she explores the organizational culture, business models, creative and technical work processes, as well as developers' subjective well-being of the companies Argames and JugAr, showing us how these elements dramatically change by vertical integration and market demands.

The case of ARgames tells us a story of an independent game company that grew to have 40 employees and operations across Buenos Aires and Los Angeles. Following a trend seen globally, the company whose main product was advergames for companies such as Disney, Lego, Coca-Cola, MTV, and Movistar changes direction in 2008 toward developing their own IP and work toward a commercially successful webgame. The commercial success of their webgame *Gol* (2009) led them to be acquired for almost $7m US dollars by a company that three months later would become a subsidiary of a renowned transnational conglomerate. The acquisition came with new sociotechnical controls on the work process that would clash with traditional communications, followed up by developers' creative degradation and disenchantment, as their work became pigeon-holed and based on repetitive techniques.

Similarly, the case of 80 employee studio JugAr shows the collapse of a company from the conflicting relationship between developers, publishers, and platform holders. After more than seven years of experience in the market, the company's process of collapse started when a Mexican publisher invested in their expansion while managing their latest Mexican wrestling inspired game *Lucha* for the Nintendo Wii. In this process, Xhardez describes the restrictions exerted by the publisher in content, number of employees, work organization, along with arbitrary creative decisions that would meet technical difficulties, and exacerbated work intensity amidst a rigid management of milestones and deliveries. The company collapsed after developers, who had not received their salary for several months, realized that the publisher had stopped paying the company, triggering a wave of resignations.

This early set of local and regional studies presents a general outlook of the conditions observed in the regional industry and their impact not just in the content and ownership of their production, but in their final shutdown. The risks especially for medium-size studios when approaching an unbeknownst enterprise in a complex and global market seemed to make developers more vulnerable to labor abuse and firm instability.

Independent production in the Latin-American new economy

As I have addressed in the first section, the 2000s was a turning point for the region. Both free-market and public policies were implemented in order to fully adjust countries to the demands of global markets and development expectations portrayed by new ICT economies. At the same time, international consolidation of videogames as a culture of consumption had already inspired software engineers and media workers to try out working in the industry, as early research suggests. During this period, there would be attempts of third-party studios, developers, and academics to move toward independent production of game content for commercial or social ends.

Similar to the experience in the main hubs of videogame production, the deepening of ICT infrastructures, with services such as broadband internet, mobile technologies, and digital distribution channels introduced new technical and commercial capabilities oriented to low-cost markets previously impossible. This is how from the mid-2000s to 2019, the frequency of Latin-American game studios emerging and reporting the release of their own titles has certainly increased. For instance, when regional organization Centro de Estudios para America Latina (CEPAL) published its 2016 report on the state of ICTs networks and services for the region, it demonstrated not only just a lower-cost access to mobile technologies and broadband but also a dramatic increase of ICTs services and hardware in recent years. By 2015, 55% of Latin-Americans were regularly using the internet, which was 20% more than in 2010. Within the same time frame, mobile internet had expanded from 7 to 58% of the population. An expansion led by Chile, Costa Rica, and Uruguay and followed up by the much bigger markets of Argentina, Brazil, Colombia, and Mexico.[3] Along this accelerated expansion, the market for web as well as mobile phone apps and services became a new arena of foreign and local investment, strengthening the demand for outsourced and licensed work. Even more important, newly found access to corporate social media and digital distribution channels came with increasing expectations of self-publishing or developing original IP.

Within this process, national videogame hubs intensified their participation in global industry events, while strengthening their organization and relationship with governmental institutions. Although there is still little reliable information providing precise data on their size, we can still identify Brazil with more than 300 studios and Argentina with above 100 studios as the largest centers of game production, followed up by smaller hubs like Chile, Colombia, Peru, Mexico, Uruguay, and Costa Rica, with clusters containing between 20 and 60 game studios, most of them created within the last decade.

As relevant literature points out (CAMTIC, 2015; Penix-Tadsen, 2017; HomoLuddens, 2018: 19), Latin-American developers still work within the national or international demand for licensed content, as well as specialized services within their respective areas of expertise in the general ICT and media industries. But the literature also identifies events that show a slow process of institutionalization of a small to medium-scale entrepreneurial hub of independent production, with their own communities of practitioners, participatory cultures, and institutions. Independent studios try to professionally catch up as they work within a complex labor process, standards, and "game mechanics" of an industry that presents itself as an almost unattainable opportunity/dream.

At the same time, the extent of foreign and corporate investments in regional game development is not very substantial, featuring some offshore subsidiaries of major companies from overseas, such as the mobile game subdivisions for Disney Interactive Studios, Gameloft, Riot Games, and Ubisoft present in Brazil. However, corporate integration comes in the form of outsourced labor such as licensed projects, game testing, and quality assurance in the region (CINDE, 2017). The extractive nature of these businesses as support within internal corporate geo-networks of production has shielded them from dealing with local game studios, therefore not contributing much to the creation of independent content by local developers. Parallel to this, very few initiatives have been created from main videogame actors to actively support independent development. There have been cases such as the Sony incubation program, which has brought titles from studios such as the Chilean IguanaBee's *Monster Bag* (2015) and Costa Rican Green Lava Studio's *Fenix Rage* (2014), games that later on found their way through Xbox One indie outlets.[4] Although Sony seems open to promote the region as they opened their incubation program in Colombia in 2015 (PlayStation Blog, 2015), no further involvement has come from these actors.

As I have shown before, this feeble and rather recent support for the region has created the need to find other sources of income and low-cost market outlets to publish games. Nonetheless, as the literature demonstrates (Marisca, 2014; Penix Tadsen, 2016; Pereira and Bernardes, 2017), the growing hubs around the region have been dependent on a series of regular activities bringing together developers, academia, private, and public sectors and acknowledging the significance of the game industry for their economies. Here, game developers have shown their capacity for self-organization and synergistic collaboration with the industry and other social institutions, triggering processes of social learning (Sorensen, 1994) which have promoted their professionalization and stabilization within the digital economy.

The coalescence of stakeholders and market size has allowed the integration of cultural and institutional intermediation spaces, networking, and learning from main hubs of game production, as well as the emergence of new ones (Parker, 2013; Parker et al., 2017). Industry conferences such as the GDC in San Francisco have historically provided developers with a space to market themselves, as well as building an effective network of contacts for outsourcing work and potential collaboration (Xhardez, 2012). In recent years, public and private organizations have managed to create similar experiences in the region, bringing in experienced independent developers, amongst other corporate representatives (El País, 2015). Likewise, big gaming expos in main Latin-American hubs provide spaces to showcase and reward local talent, as well as bring in publishers and platform representatives. Amongst them, the Brazilian Independent Games Festival (BIG) in Sao Paulo focuses on showcasing, curating, awarding, and connecting local independent studios with global corporations. For instance, in its 2015 edition, 60 international companies such as Sony, Microsoft, Electronic Arts, Bandai Namco, HTC, and Square Enix participated in the manifold events; more than 190 companies scheduled 2040 business meetings, and the expo was attended by almost 20,000 people (Keengamer, 2016).

Furthermore, the digital cultures underpinning game development networks have connected professionals, academics, college students, and hobbyists in game jams and meet-ups, providing the basis for group identification, skill acquisition, and training, and considered fundamental for the local industry (Gómez Urra, 2014: 123; Marisca, 2014: 94; Mastrocola, 2016). Their crucial role in independent production was demonstrated in a presentation at the 2014 edition of Games Developers Conference in San Francisco (2014a). Brazilian developers and lecturers Arthur Mittelbach, Bruno Campagnolo, and Arthur Protasio analyzed the active and vivid community of developers that has been formed around the Global Game Jam. The synergies created through their active coordination are perceived as greatly beneficial in integrating local developers to international business networks, while contributing to cultural identification and social learning (Preston et al., 2012; Guevara-Villalobos, 2014).[5] The network they created assumed the organization of game jams as well as participating in the IGDA Brazil chapter, becoming part of the growing institutional ecosystem behind the Brazilian hubs of production, organizing even a local edition of the Indie Humble Bundle (Garotas Geek, 2014).

Similar experiences are seen in other countries, although with their own trajectories and levels of success. In Costa Rica, the 2011 born IGDA local chapter has carried out the organization of Global Game Jam events, working to bring together dozens of game studios, students, academic institutions, trade organizations, and governmental agencies. Thus, current Latin-American developers producing independent content have thrived through their participation within local and regional clusters, in a global network of independent developers that coalesce in located and dislocated participatory cultural events. Much of this shared culture has informed the organization of these studios. Research on the business and organizational side of many Brazilian-independent studios points out their experimental and discovery-based design practices, both with aesthetic and entrepreneurial concerns (Marisca, 2014; Penix-Tadsen, 2016; Gonzalez, 2017; Pereira and Bernardes, 2017). Certainly, these forms of organization span throughout Latin-American game studios.

Toward a public institutionalization of videogame studios in the region

The practices of networking of independent developers have gone beyond their corporate and participatory cultures, finding articulations with public and private institutions and creating new forms of organization. As in the previous section, these multi-scalar interactions are sites where negotiations of interest are hold between developers and other stakeholders in the industry. In each country, they have been encouraged to pursue more formal channels of industry representation, creating a wave of new national game developers' associations through Latin America. Although the process started early on in Argentina and Brazil, in the last 10 years, national game developers' associations have emerged in at least 13 more countries.

The successful negotiations amongst organized developers have even resulted in the recent foundation of the Latin-American videogames Federation, representing 11 countries from the whole region. The association seems to provide a unionized voice in negotiating with cultural intermediaries, public institutions, and the global industry. It also aims to improve the mechanisms of recognition for regional developers work, as well as to assist in creating spaces that boost the circulation of knowledge, information, networking, business making, and cultural awareness within the region and with global actors. Some developers see the Federation as a bloc negotiating space in game Expos, networking as a region, keeping everyone in touch and informed (Chan, 2018). So far, it has managed to negotiate scholarships for developers at the GDC, promote regional industry events, maintain awareness on awarded titles, and attract foreign investments. The growing visibility and quality of Latin-American videogames have enabled them to strike deals with independent publishers, such as Brazilian studio SwordTales' videogame *Toren* (2015), published by American publisher Versus Evil (Homo Ludens, 2018: 217). Similarly, the Chilean modding group that founded ACE Team had their game *Zenoclash* (2009) adopted by Microsoft XBLA for Xbox 360, after its critically acclaimed release on Steam was followed up by contracts with American publisher Iceberg Interactive, as well as Tripwire Interactive from Netherlands (Fahey, 2009; Smail, 2009).

It is still unclear the direction that these associations will give to the regional industry, and how ideological and economic interests would make independent studios react. The perceived need to grow, reduce investment risk, and consolidate their position could stir the direction of these associations to strengthen ties to big corporate actors and strategies of IP expropriation and outsourcing work. On another level, the need to boost their local markets could lead to the consolidation of a solidarity network with financial and logistical support from governmental institutions, developers' organizations, and ICTs trade associations. They could be oriented to internationalize their products, generate local awareness and recognition for their videogames, but also promote authenticity, creativity, political, and cultural awareness in videogame content. It is significant that in countries like Brazil, the indie style of social organization has provided developers with the opportunity to periodically develop their own intellectual property. Although the markets are mostly restricted to mobile technologies, PC games have become the second market outlet for developers.[6] Still, it would be a mistake to generalize this experience to the region, since most developers are full-time outsourcers, while others struggle to overcome this position and bring their titles to the market.

In this context, the growing presence of institutionally organized independent studios has exerted pressure on public and private national institutions for improved governance and amelioration of market volatile effects, raising the profile of game development within public institutions (Campbell, 2015). At the same time, supranational neoliberal organizations (World Bank and Inter-American Development Bank) and regional think tanks (CEPAL) work as knowledge and financial agencies, circulating public policies oriented to grow national own capacities for transnational operations, while keeping the functioning of global capital flows unfettered. As a result, a range of public institutions oriented to support export goods in most countries have unevenly started to help game studios to market and internationalize their projects (among the most important: ProMexico, PromPeru, Apex-Brazil, Procomer in Costa Rica, and ProExport in Colombia). At the same time, some of them work in coordination with other policy programs aimed toward ICTs and innovation-based services which developers can compete for. For instance, in his exploration of the public institutions supporting game development in Peru, Marisca found developers profiting from the National Science and Technology Innovation Fund (FINCyT) and the Competitiveness Research and Development Fund (FIDECOM). In addition to the above strategies, state-funded accelerators or incubators have been created in many countries precisely to plant seed investments in national companies. Institutions like Start-Up Chile, Start-Up Brazil, Start-Up Peru, Incubar in Argentina, and iNNpulsa in Colombia are potential public channels for financing and commercialization.

Nonetheless, these first steps from Latin-American countries into innovation system, ICT and cultural oriented policies are still targeted to the broad and diverse sectors of the cultural and informational economy, remaining uncertain whether developers would find assistance through some of these institutions. After all, the public policy programs aimed to register and support the new media, cultural, and technology industries were not in the agenda of governmental discussions until 10 years ago (Quartesan et al., 2007).

Thus, developers keep building their position through the creation of organizations that are helping them to market themselves in the industry, improve the quality and organization of their work, and configure networks with private and public institutional programs in order to find work stability and economic performance in the market. This could help them engage under better terms with more restricted corporate channels or at least create a space to discuss the risks and limits of corporate acquisitions and industry contracts, avoiding situations like the ones pointed out by Xhardez. Although the region is increasingly advertized as a captive market for big videogame capital investments, independent studios are yet to be in the scope for regular corporate acquisitions. This could be an important factor to consider with regards to public policy, in terms of both strengthening authentic national production and contribution to the global economy.

Conclusion

As I hope to have shown, Latin America has not been absent within the global networks of game production. Its small contribution is mostly buried underneath the anonymous outsourcing circuit and transmedia approach of corporate brands. In spite of this, the recent history shows a promising growth of independent game releases, as well as original and contextual appropriations of the medium.

The increasingly organized hubs of development, public institutions, as well as private and international organizations are creating the conditions which enable developers to produce independent content and better connect to global markets. However, as scholars warn, not all the peripheries possess the same access to and harnessing of these complex circuits. Unequal budgets and institutional development across the region are the norms. For instance, there seems to be structural differences between the trajectories of the two main videogame markets of the region. Although Mexico seems to start gaining momentum, its developers' base is more than 10 times smaller than Brazil's. Public policy seems to be unevenly developed in Latin-American countries, where main Mercosur game producers and lately Mexico seem to benefit from public institutions and organizations that correlate better with game developers' needs than countries from Central American and the Caribbean. Although historically and culturally bounded, state institutions and social policies are mostly defined at national level. At the same time, besides non-refundable public financing, it is unclear under which criteria the private innovation accelerators' sponsorships are negotiated. The Silicon Valley discourse around these institutions generate doubts about whether their programs are oriented to create stable independent local enterprises or whether they search for suitable candidates to be acquired by corporate interests.

The above indicates new lines of enquiry in relation to videogame markets and independent development. There are still many areas to explore in order to understand the nature and conditions of game labor at national and local scales and in comparative perspective across the region. At the same time, intensified communications and interactions raise questions about the specific sets of technical, aesthetic or industry norms, aspirations, and meanings that are negotiated through them. These areas of enquiry are critical to the understanding of the national infrastructures that promote ICT and media industries, as well as the efforts to support and inform ethical and radical engagements with game design in different cultural arenas.

Notes

1 It is clear that the idea of independence takes specific shape within continuous negotiations and struggles in the cultural, political, and economic arenas of the media industries (Banks, 2010; Lipkin, 2013; Ruffino, 2013), where developers' organizational, creative, and publishing claims about autonomy can be negotiated/overridden as they engage in market relations of production (Garda and Grabarczyk, 2017).

2 This process of global transformation comes to a region with strong history of economic dependency between the capitalist centers and the Latin-American periphery (Gunder Frank, 1969; Cardoso and Faleto, 1977). As Robinson (2004: 94–98) argues, Third-World countries debt became a political tool to impose structural reforms to ensure their reintegration in the global economy that was slowly taking shape.

3 Careful attention should be paid to the relation between ICTs penetration figures and market size. Take Uruguay, Ecuador, and Brazil, by 2018, the first country had around 66% of online users accounting for two million people and a market of around 30 million US dollars. Meanwhile, the 52.9% of Ecuador's population accounting for a market of nine million people provides 73 million US dollars to the global revenues. Still, none of these countries can be compared to Brazil's 139 million-strong online market and its 1,324 million US dollars' worth in game revenues (BBC, 2018).

4 One of the most renowned and recorded case of big publishers investing in the region was Square Enix initiative to find video games with Latin-American authenticity and take their cultural references to a global audience (Tadsen, 2014: 121). Nonetheless, the project shut down and only managed to release Argentinian Okham's *Ultimo Carnaval* (2014) for iOS Appstore, from initially planning 10 titles in co-production (Garst, 2018).

5 Developers' need for information and knowledge goes beyond acquiring work and organization skills, involving business dynamics of the industry. Failure to grasp the intricacies of dealing with industry actors, along with businesses naivety, has resulted in cases of developers losing their IP and right to any royalties. Their lack of knowledge made them easy prey to corporate leverage when decided to deal with publishers without signing any contract (GDC, 2014b).

6 In Brazil, 97.9% of national videogame studios have developed their own videogames and retain their intellectual property rights. Mobile and Tablet platforms are the markets for 59.2% of the titles developed, seconded by PC distribution channels with 42.1% of the games industry (Homo Ludens, 2018: 32, 67, 68).

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13 The Melbourne indie game scenes

Value regimes in localized game development

Brendan Keogh

Every year in November, the Australian city of Melbourne, capital of the state of Victoria, hosts Melbourne International Games Week (MIGW). Curated by the Victorian government body Creative Victoria, MIGW is an umbrella that brings together a number of different events of varying shape and size related to aspects of the videogame field. At the start of the week is Games Connect Asia Pacific (GCAP), an industry-facing conference where professional developers share practical insights. At the other end is the Australian Penny Arcade Expo (PAX Expo), the largest consumer-facing geek culture expo in the Southern Hemisphere, where both Australian developers and international publishers present upcoming games to the public. In the middle of the week, the Freeplay Independent Games Festival presents Parallels, a one-night showcase of alternative and experimental videogames made by Australian students, artists, and hobbyists. Later in the week, for MIGW 2018, the 'arty after party' was held in a warehouse in the suburb of Richmond, open only to those affiliated with the local game development fringe communities who knew it was happening. LCD projectors sat on milkcrates to project bespoke art-games onto the walls of a gutted, graffiti-covered shop. One of the party's organizers told me that the 2018 party is the first that continued to its scheduled conclusion, rather than being shut down early by the police.

MIGW is a bookended period of time in which a heterogeneous range of videogame-related events are held by different parties for different stakeholders with different values. It renders visible over the course of the week the complex and diverse range of actors that now define 'videogames' as a cultural, economic, and technological space. Videogame makers, audiences, and intermediaries do not simply constitute an industry; they constitute a *cultural field* – an internally structured space of positions and dispositions given shape by a distribution of various forms of capital and power (Bourdieu, 1983: 323; Keogh, 2019b). The cultural, economic, and political values of the videogame field are neither homogenous nor universal but instead represent a vast range of practices, communities, aesthetics, and intentions that even the broadest understandings of 'the videogame industry', with its inherently economic and globalist framing, struggle to adequately account for.

This is vividly clear in Melbourne, where large corporate studios, entrepreneurial coworking spaces, artist collectives, student cohorts, bedroom coders, government funding bodies, 'trash games and media art' bars, and slick international consumer expos all interact within and as the local videogame field. Videogame making, in Melbourne, happens at a range of scales, in a variety of contexts, across different networks of people whose relationship with others in the local field may be positive, antagonistic, or non-existent. Crucially, the majority of these differently focused videogame makers almost all fit the bill to be considered 'independent' or 'indie' game-makers. In a regional context, such as Australia, where large multinational studios and publishers are most notable in their absence (McCrea, 2012; Apperley and Golding, 2015; Banks and Cunningham, 2016), 'indie' tells us very little about local videogame production contexts, because nearly *everyone* is indie. As Vanderhoef (2020) details, 'indie' exists primarily in its relationality *to* blockbuster triple-a production. So, what does it mean to consider independent videogame production when 'independence' is the only option? Other frames of differentiation are needed.

In this chapter, I draw from empirical research to consider the Melbourne videogame field as one that is constituted by a number of interlocking local videogame *scenes*. Across interviews with Melbourne-based developers, participants often spoke to me of 'the Melbourne game scene', but they just as often referred to subsets of this scene as themselves scenes: the Arcade scene, the Sharehouse scene, the RMIT scene, the Freeplay scene, the Unity scene, and the Twine scene. Each of these scenes is a loose, vague articulation of a community orbiting the most visible event, location, or software common to that community. Each has ambiguous and porous borders. The word 'scene' is commonly (but not unanimously) used by game-makers I interviewed to describe their local community, albeit often with a cringe or trepidation that the term itself might be pretentious or elitist in its cultural framing. I nonetheless find the term valuable in both its articulations and connotations to describe how local game development is driven by social and affective registers as often as it is by commercial ones. Commercialization, aesthetics, identity, community, and ideologies all intersect in local game development fields and are manifested through the organic emergence of dynamic scenes.

As part of a larger project, I interviewed 60 Melbourne game-makers in mid-2018 about their practices, resources, identities, ambitions, and local communities. In addition to this, from 2012 to 2015, I was an active member within the Melbourne game scenes and have maintained close connections and friendships within its scenes in the years since. In this chapter, I draw from both my formal body of research data and my personal experiences of the videogame field in Melbourne to develop the argument that, ultimately, games research needs more fine-grained focus on what game making looks like in its local context beyond the formal industry. 'Indie', as an alternative resistance to a 'mainstream' industry, only really helps in articulating a countercultural space if that 'mainstream' industry actually exists. In the majority of locations where 'indie' videogame development now happens, the historically mainstream industry and its employment opportunities are absent. While many developers still hold onto an indie identity due to the pervasive hegemony of the 'global' videogame industry's distribution networks, we require more nuanced articulations to account for videogame development in its local context.

The first section draws from existing research on scenes and localized videogame production to better consider what it means to conceive of local game scenes. A key notion here is that a scene emerges when there is an overproduction of value within a local cultural field. The next section, then, details some of the key actors and groups active (or recently defunct) in Melbourne's videogame field, as articulated by my participants, to demonstrate how local game-makers themselves articulate an irresolvable plurality of local activity as not a singular scene, but an intersection of a number of scenes. Finally, the chapter concludes with a consideration of the need for videogame production researchers to account for different, often contradictory, value regimes amongst independent videogame makers.

Toward local game scenes

While different communities or networks of people are commonly, in the vernacular of those in a cultural field, referred to loosely as a scene, the term remains notoriously difficult to pin down for both those within a scene as well as for academics researching them. Scenes have primarily been theorized through sociological research of music subcultures. For Will Straw, a scene "designates particular clusters of social and cultural activity without specifying the nature of the boundaries which circumscribe them" (2004: 412). A music scene "is that cultural space in which a range of musical practices coexist, interacting with each other within a variety of processes of differentiation, and according to widely varying trajectories of change and cross-fertilization" (Straw, 1991: 373). For Holly Kruse, scenes describe "both the geographical sites of local music practice and the economic and social networks in which participants are involved" (2010: 625). But scenes are also always in flux, never stable, deriving "their effervescence from the sense that the 'information' produced within them is forever in excess of the productive ends to which it might be put" (Straw, 2004: 412). Further, and perhaps most crucially for our current consideration of the Melbourne game scene, Straw suggests that a scene emerges "from the excesses of sociability that surround the pursuit of interests, or which fuel ongoing innovation and experimentation within the cultural life of cities" (2004: 412). Straw provides a number of examples where cultural scenes – such as the Montreal disco scene of the 1970s or the Beat generation of poets post

World War II – emerged in part as an *overproduction* of culture unable to be fully captured by the commodifying forces of the cultural industries. Instead, the work of these scenes is mobilized as other forms of capital (social, cultural) that solidify into shared stylistic traits of locally specific scenes.

Considering a scene as 'overproductive' – as spilling out beyond a formalized industry to pervade social spaces and draw new actors and aesthetics into its jurisdiction – goes some way toward explaining why those cities with a history of cultural production but with relatively few formalized game industry opportunities are also those cities whose videogame activity is most readily referred to as a 'scene': Berlin, Austin, New York, Toronto, Utrecht, and, indeed, Melbourne. As the barriers of entry to both videogame production and distribution have lowered over the past decade, a wider range of people have begun creating and sharing a wider range of ostensibly 'independent' (but not necessarily 'indie') videogames. These creators have variously been called 'videogame zinesters' (Anthropy, 2012) or 'everyday game makers' (Young, 2018), and they often represent a range of marginal identities historically excluded from the formal videogame industry (including from commercially feasible indie development) (Harvey, 2014). These new communities of creators are what Ramon Lobato and Julian Thomas (2015) would term 'necessity entrepreneurs' or Kate Oakley (2014) would call 'forced entrepreneurs'. That is, rather than the romanticized entrepreneurial 'go getters' who identify an opportunity to corner a new market and go for it, these new creators are otherwise marginalized and excluded from the means of entering a particular market (in this instance, the videogame industry) and use what available resources they can, by necessity, to get a foot in the door. Thus, there is in one sense in cities such as Melbourne, an *overproduction* of videogame making activity insofar as the number of people able and eager to create videogames has outgrown what the local industry is able to commodify and instead manifests as informal networks and communities of practice: shared spaces, friend groups, meet ups, parties, resource sharing, and game jams.

Such a framing, however, can risk centering the formalized videogame industry as the core that all other game making activity is oriented toward; it potentially implies that these scenes are the overproductive 'leftovers' of the global videogame industry after that industry has taken its fill. To avoid this, the 'global' videogame industry needs to be reconceptualized, as many in the nascent subfield of game production studies have been doing, as emerging *from* the multitude of local videogame development situations that exist in different regions. In *Global Games*, Aphra Kerr provides an extensive political-economy analysis of the global games industry and demonstrates that to comprehend global videogame production is to understand videogame production in specific local contexts (2017: 201). Other researchers have been applying this local lens, most predominately in Canada and the Nordic countries, deploying ethnographic methods to understand the constitution and attributes of different local game making communities (see Jørgensen et al., 2017; Parker and Jenson, 2017; Švelch, 2018 as well as the other chapters in this section for particular examples). In a 2013 article, Daniel Joseph provides a detailed analysis of the Toronto indie games 'community' (Joseph does not use the word scene himself) and its relationship to the broader industry and the local government's cultural policy. Drawing from assemblage theory, Joseph considers how the city of Toronto itself "allowed" its independent videogame developers to create a community of practice: "there is no 'independent' community here without a city-size assemblage capable of fostering close ties between organizations and persons" (2013: 101–102). Canadian cultural policy, an established indie music scene, and urban geographies all contribute to the particular shape of Toronto's scene. Joseph similarly identifies in Toronto the overproduction Straw sees as indicative of a scene:

> The very lack of an 'established' big-studio system in Toronto appears to have led to incubating a small, almost counter-cultural videogame development movement. In the absence of major studios like Electronic Arts or Ubisoft, those interested in videogame development as a career had to strike out on their own, without the institutional support of an established industry.
>
> (2013: 100)

Crucially, while this local, cultural approach to videogame making often exists in the absence of a formalized videogame industry, it is also often the foundation from which a formalized videogame industry emerges. In cities such as Montreal, massive blockbuster studios such as Ubisoft are attracted by not only generous tax breaks but also the pre-existence of active scenes that serve as the skill pool from which the studio can build its employment base. Thus, the overproduced cultural capital of a local scene is, at times, recaptured by a formalized industry increasing its capacity to capture production. Crucially, then, local game scenes are not that which is leftover by a sated global game industry in a top-down sense, but the broader field of informal creative, affective, and social activity that a local formalized industry sometimes emerges *from*.

Perhaps, the biggest challenge to, and greatest value of, considering local videogame production communities as 'scenes' is the historical baggage videogames as a medium still hold as a commodified technological field, rather than a cultural field. Videogames are still dominantly configured in both popular and scholarly discourses as a software industry producing entertainment products, not a cultural field constituted by cultural workers. This configuration is a consequence of the dominance of particular sectors of the videogame field – what I have called elsewhere the videogame industry's 'aggressive formalisation' (Keogh, 2019a) – and, earlier, to the emergence of videogames in male-dominated computer science faculties and computer clubs as opposed to among more traditionally 'cultural' spaces. By empirically focusing on videogame making communities in their local context, however, it becomes evident that those who constitute the videogame field are really no different to those in any other field of cultural production:

> [they] are not stars, nor are they rich or even particularly successful—in fact, the majority of cultural workers toil in relatively anonymous enterprises, either living off the erratic incomes from 'projects' or more conventionally on low or subsistence-level wages.
>
> (Banks, 2007: 10)

To consider local videogame communities as scenes, as many of those within them do, is to capture the broader (often contradicting) range of social, affective, political, and cultural reasons that people produce videogames.

A final note on conceptualizing scenes generally before this chapter shifts its focus to Melbourne specifically: the internet. Kruse warns that in the networked age, "looking at scenes in isolation [now] makes even less sense, as the ability to connect with others across scenes and to disseminate independent music has become easier than ever before" (Kruse, 2010: 629). Indeed, both Kerr (2017) and Parker and Jenson (2017) make similar cases that videogame development is not only local but *trans-local*, with social media platforms and online communities connecting, and sometimes even creating, scenes beyond geographical affordances. For instance, the game making tool Bitsy has an active 'Bitsy scene' that is not located in any one specific city but primarily anchored in a Discord chat channel and distributed across regular game jams held on the itch.io distribution platform (Reed, 2020). The trans-locality of networked communities does not speak to a homogenization or unification of local scenes, mind, but to the importance of local and *trans-local* interactions and networks in bringing different scenes into contact and in mediating the shape and constitution of scenes.

The Melbourne indie game scenes

In a 2013 article, Sebastien Darchen and Diane-Gabrielle Tremblay compare the economic organization of Melbourne's and Montreal's game industries and conclude that while Montreal's game industry can be considered a creative cluster, Melbourne's activity was "very much in silos and there has been limited cross-fertilization with other creative sectors" (2013: 318). While this assessment is understandable within the economic organization framing of the research, it jars with my first-hand experience of the vibrant scenes active in Melbourne at the time – many of which are not themselves 'economic' but which have a direct impact on the local field. What is absent from Darchen and Tremblay's account is an appreciation of game making practices undertaken beyond the commercial 'firms' – the overproduction of the scenes.

In the first half of the 2010s, the Melbourne (and Australian) industry shifted by necessity to an almost exclusively 'indie' industry following the closure of nearly every large studio in the country. Melbourne, more than any other city in Australia at the time, successfully fostered this transition to independence, thanks to a wide range of cultural and political advantages. In addition to the simple fact that many of those let go from the defunct larger studios were already based in Melbourne, Melbourne also had a long history of and existing infrastructure to support creative culture and the arts, including the areas of digital media. Institutions such as the Australian Centre for the Moving Image, and funding bodies such as Film Victoria, have been supporting local videogame production and cultures since the early 2000s. Countercultural videogame makers were part of this cultural field long before the industry collapsed. For instance, the Freeplay Independent Games Festival started in 2004 and is the longest running independent games festival in Australia and possibly the world. Around the same time, Melbourne game developers were collaborating with artists and journalists on political games such as *Escape From Woomera* (see Swalwell, 2007). As of 2018, Melbourne now accounts for over half of the entire Australian videogame development industry (GDAA, 2018) – in part due to the growth of the local industry, but more due to the shrinkage of the industry in the rest of the country, lacking the support provided in Victoria.

Also, in the 2010s, the possibility and opportunities to create and share videogame works far outweighed the opportunities to obtain employment in the videogame industry. While industry veterans pivoted to an indie industry, younger generations of hobbyists as well as students from the growing number of dedicated tertiary game development programs began to emerge in great number. Yet, while a number of Melbourne teams were achieving both commercial and critical success, they were not necessarily looking to dramatically grow the teams and hire extensively. This goes against the default logics of how the videogame industry is imagined as a commercial industry, but is a pattern common among independent videogame production teams in different regional contexts. For instance, Jennifer R. Whitson et al.'s research with Canadian indie developers reveals that despite taking on the language of entrepreneurship and growth when presenting themselves to external actors, the same developers articulate their own ambitions as more modestly wanting to 'keep on keeping on' (2018: 6–7). Consequentially, even when independent teams are commercially successful, they more often than not remain at a relatively small size, rather than rapidly scaling, and don't necessarily create more employment opportunities for other local developers.

For those more experienced developers transitioning from working in now-defunct large teams to perpetually small groups or even individually with a shoestring budget, new forms of mentorship, collaboration, and social support were crucial. As one solo developer who had previously worked in a large team explained:

> Without the community it makes everything very difficult. Because you can't externalise problems. You can't get help. That's just a hugely isolating problem. It's a business where you can act remotely and you can work in your bedroom, but without a whole bunch of people on Skype you can call up with a moment's notice, you just go mad.

Thus, in the early 2010s, in Melbourne emerged a number of initiatives to foster community or, following Straw (2004: 413) what we might call a form of 'horizontal' mentorship between teams. Commonly brought up by participants was the 'IGDAM' (International Game Developers Association (IGDA), Melbourne chapter) monthly social meetup. As in many other cities, IGDA provided a regular event in a bar on a weeknight that, despite the organizers' express ruling against 'networking' at what they perceived to be a strictly social communal gathering, was as much a professional event as it was social, and which became crucial for the fostering of friendships and communal relations that, in turn, become vital for the precarious indie developer to obtain new gigs in a way that is exemplary of what McRobbie (2002) calls a "club-culture sociality".

More formally, this community solidified with the development of the games-centric co-working space, The Arcade. Run by the Game Developers Association of Australia (technically as a different company, but by largely the same team of individuals), The Arcade provided (relatively) affordable office space for small game development teams and allowed teams and individuals to pull together and share resources and knowledge. The overwhelming majority of those residents in The Arcade spoken to for this research highlighted the cross-team collaboration and support that the space afforded as vital to their operations:

> For me [The Arcade is] a lot of the positives of being in an office space and none of the negatives. A chance for me to meet with people and make new connections. … It's helped me prepare for the Film Victoria [funding process], because … I feel like half the people I meet here have been involved in that process either as an applicant or on the selection committee.
>
> That's half of the reason we don't want to leave, because of these relationships. We helped another studio to grow and other studios here have helped us grow. So much of our existence now is tied to people here, and so much of our social culture is tied to people here. It feels like if we ever left, we'd be taking part of our studio away.

Crucially, The Arcade also made an otherwise nebulous and loosely affiliated network of teams more visibly and centrally located, providing a professionalized space that industry advocates could point publishers and government officials toward as *the* Victorian game industry – or where researchers, such as myself, immediately turn to in order to locate research participants.[1] However, this also came with its drawbacks for the broader community not occupying The Arcade. As Tony Reed, the founder of The Arcade explained:

> The thing I didn't see coming was The Arcade becoming a clique. People in The Arcade felt they were special because they were in The Arcade, and people outside of The Arcade felt they weren't part of things because they weren't part of The Arcade. That really bothered me.

This was especially true for those younger teams and individuals not transitioning from the old industry of large studios and who perhaps were not interested in producing the same sort of commercially feasible games that one would have to create in order to justify renting an office or desk in a co-working space. As one game-maker external to The Arcade expressed: "that world has always seemed … too inward looking and also just distastefully commercial … I think because we never kissed the ring, we've never been acknowledged by them".

Thus, also in the early 2010s, parallel game making communities were emerging in Melbourne, buoyed in large part by the large number of students emerging from dedicated tertiary programs in videogame design and development from universities such as RMIT and Swinburne. Loose groups and organizations such as Glitchmark and Hovergarden, alternative coworking environments such as (now defunct) All Day Breakfast, the computer labs and student rooms at RMIT University, and the long-running Freeplay Independent Games Festival became crucial hubs for these alternative collectives and communities who did not necessarily feel catered to by the industry-centric organizations of IGDAM or The Arcade.

Shang Lun Lee, a founder of the group Glitchmark, which met regularly to discuss and learn game design skills and theory, described the origins of that particular group thus:

> the people that would go to something like [Glitchmark] were hungry for a community and for that kind of vocabulary and development of a scene or of a discipline. And so those people continue to create networks and communities that are interested in those same things and you can see the effects of those individuals and those communities have had in fragmenting, fracturing in a very positive way in my opinion, the Melbourne independent game scene.

This sense of a fragmenting or fracturing of the Melbourne scene, of its energies being directed toward different organizations and groupings rather than inward to a singular space or group, was similarly articulated by a range of Melbourne-based game-makers from different backgrounds:

> There are studios that make mobile games that are hugely profitable, but I think their values is less on creating meaningful experiences and more so about maximising profit. And there's also another side of games in Melbourne which is like 'games should just be an expression of yourself' and money is far less valued than the statement. And obviously I'm not trying to say one way is better than the other. I just feel like we're the greedy studio that wants it all.
>
> I'm realizing increasingly that I have this such particular view on what Melbourne is. Like, I'm part of a particular scene. But then I suddenly realize like fuck I don't know like at least half the people in The Arcade or whatever.
>
> In my mind there's two Melbourne game worlds. There's The Arcade and what I think of as Bar SK.

The Bar SK mentioned by this last respondent is a bar located in the trendy entertainment district of Collingwood. Founder Louie Roots had previously been part of a game making collective in Perth, Western Australia, focusing on exhibition and party videogames. Bar SK serves as both a local, public-facing bar, and as a self-defined exhibition space for "trash videogames and media art". Bar SK served as a solidification of the subcultural side of Melbourne game making, providing an alternative space and visibility that counters the dominant visibility of The Arcade.

While this countercultural or subcultural side of videogame production in Melbourne is still overproducing in the sense that much of its activity is not captured entirely by the local industry, this overproduction and its "subcultural capital" (Thornton, 1995) has in turn had a direct impact on the local industry. A founder of a small studio, who had moved to Melbourne with the explicit intent of starting this studio, explained why they chose Melbourne over any other city:

> Access to talent is probably the biggest one, there is a scene here, there's you know there's universities here, there's support here like The Arcade, that is a huge reason.

The broader scenes thus become a source from which the formalized industry draws various forms of capital, even as those scenes remain somewhat marginal and detached from that industry that it affords. In a longer quote worth citing at length, another participant described this interaction between 'the industry' and 'the scene' as more exploitative and extractive:

> I've found that before the industry and the scene decided to be friends the arts people were really happy hanging out making cool shit, talking

> about cool shit, pushing really hard for gender diversity. We were all around Melbourne doing our thing and having a good time. And then only in the last year or so the industry has sort of opened up to us and I think the scene has kind of opened up to the industry. … The wants of that scene were to make good art and to make stuff that was very ethically considerate and inclusive, genre-defining, medium-pushing shit. When that stuff started to become commercially viable, which it has in the last two years, and we found people with money and people with studios coming and saying oh shit this is cool, this has an audience, this is hip, we want to make this stuff, [they broke] up that scene with their money and their opportunities. I feel like that's when it's become political and that's when it's become ugly and that's when it's become really difficult to enjoy it. When it wasn't surrounded by money and we were all still able to get by, it was a delight.

Here, very different values and ideas of what it means to be an independent videogame maker are at play. To say that Melbourne has 'an indie game scene' would be to homogenize what are distinct but overlapping communities among Melbourne game-makers in terms of resources, values, ambitions, and opportunities. A rich tapestry of activity and constituents across Melbourne are all part of the Melbourne videogame field, but reflect a vast range of values, identities, and groupings that deeply intersect but which are just as often conceptualized by their constituents as discrete spheres.

Value regimes of indie development

When I first discussed the idea of there being multiple 'Melbourne indie game scenes' with a colleague, they challenged this idea and said what I had described, with its various tensions and contested values, was quite simply just *a scene*.[2] This is a very valuable challenge and an important one to consider. Yet, when one speaks to Melbourne game-makers, the sense of a multiplicity of scenes, communities, and spaces is felt very deeply and is anecdotally reinforced through my own affective reactions to the Melbourne community during my participation in it. In Melbourne, it remains true that the two sides feel somewhat distinct, even as they are interrelated – each defined by its own internal struggles, relationships, in-and-out crowds, and flows of various forms of capital as much as such tensions exist between the different communities as well. Just as a single city could easily have, for instance, a jazz music scene and a rock music scene and an electronic music scene, Melbourne has a multitude of videogame scenes. The challenge, then, and what I want to conclude this chapter with a discussion of, is to consider how best to articulate the different value regimes that all too often get lumped homogenously under 'indie games'.

While 'indie' and 'independent' are identity markers used by a wide range of videogame makers around the world, the participant remarks in this chapter serve as an important reminder that these labels do not define a singular nor a homogenous way of being a videogame maker. During the 2000s, as the game industry began to shift from a period of aggressive formalization into one of intense in/formalization (Keogh, 2019a), 'indie' as quite simply 'not triple-a' was a strategically useful label for those (still primarily white and male) developers who chose to follow more entrepreneurial models to economic pursuit. Or, at least, this is how it was romanticized by the industry, and movies such as *Indie Game: The Movie* move very much aligned to modern neoliberal reconfigurations of cultural work around the rhetoric of 'creativity' (McRobbie, 2016). Indie remains strategically useful in a triple-a-dominated global industry for independently working game-makers to differentiate their products on the market. However, beyond such commercial considerations, it becomes far less articulate as a critical framing.

'Indies' make videogames for a vast range of reasons, in a vast range of different contexts, with a vast range of different resources. In Melbourne alone, we can see – much as we could see in any city if we were to look close enough – a number of different identities and ambitions and networks among videogame makers, all of whom are ostensibly 'indie'. Considering indie as homogenous (or even as heterogeneous but peripherally circling a hegemonic core 'industry') reinforces the dominance and normalization of a particular conceptualization of what videogame production looks like. Instead, we need to start with the complex local scenes that constitute local industries that in turn constitutes the global industry. This demands a bottom-up approach to understanding the constitution of the videogame field, one that starts with the lived experiences and navigations of game-makers themselves, interacting not only with a videogame industry but their own local (and trans-local) contexts.

As researchers of the videogame field, it is crucial that we reflect as to whether our accounts of what this field looks like are only accounting for those who are 'employed' by a formalized 'industry', or those few indie developers who, through a combination of stubbornness and privilege and luck, 'make it' financially. We also need means to account for those game-makers who are not making money but whose games and networks have a tangible impact on the shape of the field and industry both locally and globally. Ultimately, looking at videogame development in its local context, beyond the formal companies that constitute the industry, demonstrates that videogame development is no more a homogenous activity as is music production, acting, writing, or any other cultural field, and videogame makers are no more a homogenous group any more than 'musicians' or 'actors' or 'writers'. Videogame makers, and the scenes they constitute, are complex, contested, and ever-changing. It is the responsibility of researchers of this field to articulate, and not resolve, this complexity.

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Notes

1 Participants interviewed in other parts of Australia often expressed a frustration that The Arcade, especially due to its close affiliation with the GDAA, diverted attention and resources away from the industry everywhere else in the nation. As one non-Victorian developer put it: "They're meant to be the Game Development Association of Australia, not the Game Development Association of the Arcade!"

2 My thanks to Dr. David Murphy for these critiques and forcing me to better articulate my arguments on this topic.

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14 Modes of independence in the Finnish game development scene

Olli Sotamaa

This chapter sheds light on the different modes of independence in the Finnish game development scene by putting the focus on two fairly recent games: *Nex Machina* (2017) and *Baba is You* (2017/2019). While both of these games can be considered 'independent' in some sense, they differ from each other in many ways. *Nex Machina* is a fast-paced twin-stick shooter game, whereas *Baba is You* is a quirky and thought-provoking puzzle game. While *Nex Machina* is developed and published by Housemarque, one of the oldest active game studios in Finland, *Baba is You* is created by Arvi Teikari, an independent game developer who has so far released most of his game projects openly in his personal website. Together, these two games will help us understand how the conception of independent game-making is situated in time and place and shaped by different cultural and techno-economic trajectories.

In methodological terms, I'm not aiming for a close reading as such but rather a contextualizing reading in which the discussed games are connected to their relevant cultural and economic framings. I'm particularly interested in the funding schemes, production networks, and distribution channels that make these games possible in the first place and shape the forms they take. The study relies on a decade-long study of the Finnish game industry, including observations in local game studios and dozens of in-depth interviews with developers. This chapter utilizes diverse materials from industry reports and recent academic studies to documentaries and online interviews.

With the emergence of popular digital distribution channels and accessible game development tools, we now witness game development happening in a variety of different modes and contexts. As Keogh (2019: 14) points out, the new audiences and the new contexts of production are together "expanding the videogame industry into an ecosystem that is at once broadly global and intensely localized". While studies of digital games may often overlook it, specifically, local aspects of game playing and making are often very revealing (Parker & Jenson, 2017). Accordingly, we need to explore how local actors interface with global ones and how international networks of production and distribution get shaped and territorialized by regional factors (Kerr & Cawley, 2012).

The objective of this chapter is to explore the interplay between global game industry trends and a local game development scene. Although indie games are often seen as primarily North American phenomenon, it has been argued that the game development community within the Nordic countries has played an important role in introducing some of the key ideas and rapid development and prototyping mentalities we often associate with indie game development (O'Donnell, 2012). In this respect, the Finnish game development environment in which massive AAA productions have always been rare provides an intriguing context for exploring the changing conceptions of independent game-making.

Garda and Grabarczyk (2016) make a distinction between financial independence, creative independence, and publishing independence, showing how independent games are a historical phenomenon that changes over time and is determined by the evolution of videogame cultures. This chapter will trace these modes of independence by looking at the two aforementioned games. The chapter begins with a quick overview of the Finnish game development scene that highlights some local aspects relevant to contextualizing the studied games. After this, we take a look at the production of Nex Machina and the history of Housemarque. The second case focuses on *Baba is You* and the regional game jam networks. Finally, the similarities and differences between the studied games are discussed, and the key lessons related to the global and local aspects of independent game development are concluded.

Game-making in Finland

While Finland has for a long time been a small and peripheral node in the global circuits of game production, global hits like Rovio's *Angry Birds* and Supercell's *Clash of Clans* have drawn significant attention to the Finnish game development environment in the past few years. With the annual turnover of around 2.1 billion euros, Finland was placed within the top-three game developer countries in Europe by turnover (Neogames, 2019). While several journalistic accounts have traced the history and nature of the local industry (Niipola, 2012; Kuorikoski, 2014; Lappalainen, 2016), they still often focus on the success stories and have relatively little to say about independent games.

In 2018, Finnish game industry included a little over 200 game studios that employed altogether around 3,200 people (Neogames, 2019). On a global scale, most companies are small, and even the larger ones consciously rely on small independent teams that have a lot of control of their respective projects. Supercell webpages (Supercell, 2019) explain this as follows:

> We've found that the best quality work comes from small teams in which every single member is passionate about what they do. Often times when teams become bigger, processes, bureaucracy and even politics emerge, and the work just isn't fun anymore.

Reliance on autonomous teams does of course not mean that every company follows a recognizable "indie mindset". What it means though is that their processes often include significant aspects of independence, highlighting the blurred boundaries between top-down corporate game production and bottom-up participatory indie culture. Related to financial independence, it was only in 2017 that the first Finnish studios got stock-listed to Nasdaq Helsinki. While many studios have received significant investor funding and others have been acquired by global conglomerates, majority of the Finnish studios are independent in terms of ownership.

The historical development of the Nordic game industry follows a bit different trajectory when compared to the key game industry centers with large-scale home markets (Jørgensen et al., 2017). Still, in the 1980s, game development was largely a hobbyist endeavor that received no serious interest from the existing technological or cultural industries. First, long-lasting game companies were found only in the mid-1990s. Still, one can say that Nordic welfare society has provided a firm foundation for game-making (Sotamaa et al., 2019). High economic level and welfare state regime made it sure that home computers were widely available and obtainable even for the working class. In addition to state investments in computer development, access to free education and cheap student loans have also created a cultural environment in which dabbling with machines has been relatively common.

One of the often-mentioned factors for the advent of the Nordic game industry was the subculture around computer demos – audiovisual showcase works rendered in real-time (Saarikoski & Suominen, 2009). Although demoscene was definitely not only a training ground for game-making, many early game companies recruited at least programmers and artists from the local demo groups. While particular skills like multimedia programing were relatively easy to apply in game-making, the overall transition from hobbyist demo making to professional game development was from straightforward process (Jørgensen et al., 2017). In terms of skills formation, the Finnish scene comes close to the UK game industry that mostly developed bottom-up from the self-taught 'bedroom coders' (Izushi & Aoyama, 2006). The role of the demoscene also participates in creating a particular 'technicity' that encapsulates the centrality of technical virtuosity as a critical aspect of identity construction (Dovey & Kennedy, 2006; Tyni & Sotamaa, 2014).

Hobbyist-based companies had little connections to local industries that could have helped them. With small domestic markets and lack of regional publishers, they quickly targeted the international markets. This also meant that the small companies did not necessarily see each other as direct competitors but instead quite openly shared experiences with each other. In this respect, it is not a coincidence that the Finnish game industry is still often characterized by communal spirit, openness, and willingness to help each other (Sotamaa et al., 2011). On a general level, the hierarchy between employees and their supervisors is relatively low in the Finnish workplaces.

The recent decade of global game industry is often told as a transformation from AAA production-lead model into smaller-scale productions that spawn around digital distribution channels and mobile platforms. While the traditional game publishing system was partly based on locked proprietary systems and benefitted primarily established companies with a certain turnover, the new paradigm has provided more alternatives for small-scale game developers (Sotamaa & Karppi, 2010). Also, in Finland, a clear majority of annual sales and financial success stories come from mobile games sector. It is, however, notable that apart from Remedy, a pioneer studio behind such games as *Max Payne*, *Alan Wake*, *Quantum Break*, and *Control*, Finland has never hosted long-lasting AAA development studios.

Case *Nex Machina*

In August 2017, just two months after the launch of their arcade-style console shooter *Nex Machina*, game studio Housemarque published a blog post in which CEO Ilari Kuittinen proclaimed that "arcade is dead". After expressing his gratitude to the fans and partners, Kuittinen bluntly reported that despite the critical acclaim, Housemarque's recent games had been financial disappointments. The piece also stated that the company's "longstanding commitment to the arcade genre" had now come to an end, and it was "time for Housemarque to move forward with the industry" (Kuittinen, 2017). This was, however, not the first time Housemarque was forced to reinvent itself. In order to understand the various aspects and trajectories of the occasion, we will first take a look at the production of *Nex Machina* and then provide a larger historical context for the developments.

The discussions around the project started in 2014 when the co-founders of Housemarque accidentally met Eugene Jarvis, the creator of legendary early arcade games. Jarvis' games had been a source of inspiration for some of Housemarque's prior games, especially *Resogun* (2013), and they quickly agreed to collaborate on a new game project (Souppouris, 2016). The initial idea was that the funding for the project would come from the studio. Potential crowdfunding campaign was also introduced in the early phases of the project. In a feature-length documentary film about the making of *Nex Machina* (2018), lead designer Harry Krueger describes the project as follows:

> I guess this is the first time, in a long while, that we're doing something on our own. And going the self-funded or crowdfunded route will of course give us a lot more freedom to just make our own decisions without being tied to a specific publisher.
>
> (Name of The Game, 2018)

The excitement related to the freedom and independence is nicely visible at the beginning of the film. The documentary also painfully captures how the details of the project are re-evaluated and revised several times in the following three years. In a relatively early phase, the Kickstarter campaign idea is dropped, and the project is soon close to running out of money. Sony is then introduced to the project as a publisher and co-funder; but soon after this, the favorable Sony contact person leaves the company. Later on, copyright issues ruin the original trailer, and the official announcement of the game is eventually postponed from Paris Games Week 2015 in October 2015 all the way to PlayStation Experience 2016 (PSX 2016) in December 2016.

At the same time, it turns out that the sales of *Alienation* (2016), Housemarque's previous game, do not meet the studio's expectations, resulting in significant losses for the company. This has an immediate effect on the *Nex Machina* team as community manager Tommaso de Benetti is laid off. Furthermore, Sony begins to cut down the budget and its marketing efforts. Once the Housemarque team enters the PSX convention halls, they realize that they are the "indie booth guys now" forced to visit a local printing service to get their marketing materials finished (Name of The Game, 2018).

In June 2017, a week before the launch of *Nex Machina*, Mikael Haveri, the head of marketing at Housemarque, travels to Electronic Entertainment Expo 2017 (E3 2017) to promote the game. As the project has practically no money left for marketing, Haveri sets up his modest roll-up next to a vending machine in one of the corridors of Los Angeles Convention Center. He looks lost and disillusioned (Name of The Game, 2018). Although most things have changed in the quarter-century, this one-man trip bears an ironic resemblance to CEO Ilari Kuittinen's trips to European Computer Trade Show (ECTS) and E3 in the 1990s. The deals agreed in these trade-events often guaranteed continuity in the early days of the company.

The story of Housemarque begins from a merger of two small game companies, Bloodhouse and Terramarque, who had been developing games mostly for Amiga. Programmer Harri Tikkanen had been a member of a Bloodsuckers demo group, and with some of his demoscene friends, he went on to develop *Stardust* (1993) and *Super Stardust* (1994). At the same time, Ilari Kuittinen was collaborating with Stavros Fasulas to create *Elfmania* (1994). In the early years of Housemarque, the company focused on developing PC games, and *Supreme Snowboarding* (1999) was the breakthrough they had been waiting for. With the help of French publisher Infogrames, the company next released *Transworld Snowboarding* (2002) that was the first Xbox game developed by a Finnish studio.

Despite the critical acclaim and relatively good sales, Housemarque was soon in trouble (Karjalainen, 2016: 45–48). Springtoys, a spin-off company they had created to explore the early mobile game market, first collected significant investments but only after two years in business had used most of the funding and had to be offloaded to another company. At the same time, publisher Infogrames ran into financial difficulties and were no more capable of funding Housemarque's future projects. The company spent its funds on developing new concepts but never found a publisher for the projects. The company came up with a survival plan that included, for example, developing games for Nokia's N-gage and working as a subcontractor for Buena Vista film company. Still, the company reported significant annual losses and the small core group that was left agreed to voluntary pay cuts.

The way out from this bleak period was eventually found with the emergence of a new console generation. Microsoft Xbox 360 and Sony's PlayStation 3 came out with integrated online marketplaces that allowed players to download games directly to their machines. This also opened a market for smaller-scale games that did not need large development teams or million-dollar budgets. Housemarque agreed to a publishing deal with Sony and with *Super Stardust HD* (2007), a title paying homage to Housemarque's history, and they were in the game again. The game both reintroduced Housemarque as the master of the twin-stick shooter genre and cemented their close collaboration with Sony, two things that characterized the following century for the studio and came to an end with the "arcade is dead" press release.

Interestingly, *Resogun* was not the first time Housemarque explored the opportunity to self-fund a game project. Reportedly, they tried this already in 2006 and again in 2011 as they collaborated with a local co-financing fund (Lukkari, 2011). Over the years, Housemarque has acquired funding from private investors and but also from several public funds including e.g. Finnish Funding Agency for Technology and Innovation (Tekes), European Union Media program, and Nordic Game Development Fund. Housemarque was one of the very first game companies to receive R&D funding from Tekes already in the late-1990s (Kuorikoski, 2014; Sotamaa et al., 2019). In many ways, the studio – while definitely small in global comparison – has played a big role in the local development scene by providing an example of how to succeed in international markets. CEO Ilari Kuittinen has served as the chairman of both Suomen Pelinkehittäjät (Finnish Game Developers' association) and Neogames, the hub of the Finnish game industry, actively working toward a more sustainable national game industry ecosystem.

With long-lasting collaboration with one of the major platform holders, Housemarque may not appear as an average indie developer. Still, the story of the studio allows us to examine a particular historical trajectory in which different modes of independence fluctuate. Repeated efforts to self-fund and self-publish game projects give a hint that relative independence is still seen as a worthy objective. Housemarque also nicely exemplifies how integrated online marketplaces have provided small developers access to larger audiences. The launch of Sony's PlayStation Network and the quick success of games like *Cloud* (2006) importantly opened opportunities for selected indie developers (Sharp, 2016). While Valve's Steam and mobile app stores have been central in transforming the distribution of digital games, console manufacturers have also been quick to make their digital storefronts more open and accessible to indie games (Keogh, 2019: 25). Housemarque also nicely illustrated how small studios need to be witty, flexible, able to change direction quickly, and willing to explore various different funding opportunities and how even this does not guarantee any long-term sustainability. While downloadable console game market provided a steady revenue stream for Housemarque for several years, at the same time, dozens of game studios worldwide failed in this hyper-competitive market. Finally, Housemarque is celebrating its 25-year anniversary in 2020. While this is a major achievement from any game studio, they are still pretty much one failed project away from insolvency, just like in the beginning.

Case *Baba is You*

Nordic Game Jam, organized annually in Copenhagen since 2006, is one of the oldest and largest annual game jams in the world. In 2017, Helsinki-based psychology student Arvi "Hempuli" Teikari won the jam competition with his game *Baba is You*. Already before its official release for PC and Nintendo Switch in March 2019, Teikari's game won awards for "Best Student Game" and "Excellence In Design" at the 2018 Independent Games Festival (IGF) Awards. *Baba is You* is Teikari's second commercial title, the Metroidvania-style action-adventure game *Environmental Station Alpha* (2015) being the first one. In addition to this, Teikari has over 30 games openly available at his website hempuli.com. Many of these games have been created at different game jams over the years.

In *Baba is You*, the player aims to pass the levels by manipulating blocks that have particular words in them or by modifying the rules by which they behave. By shuffling the blocks around, players can make the level passable with some clever wordplay. The original game jam build had 13 levels, and it is still available via itch.io, a website that allows anyone to host, sell, and download indie video games. The full version with 200 levels is available via Steam and other online distribution services.

With its pixel art graphics and quirky mechanics, it's easy to identify *Baba is You* as an 'indie game'. One important aspect that contributes to the composition of any game is the development tools. Alongside the streamlined online distribution platforms discussed already in connection to *Nex Machina*, more accessible development tools have worked to diversify the modes of game-making (Young, 2017). Instead of a pure technological change, this is at least as much a cultural one: while alternative game engines and editors like GameMaker (Anthropy, 2012) or Twine (Harvey, 2014) have existed for a while, with the growing popularity of recognizable indie games, "these engines themselves have been re-evaluated by developers as something other than second-class software for amateurs" (Keogh, 2019: 26).

In regard to game-making tools, *Baba is You* provides an interesting case study. For over a decade, Teikari has been using the products of Clickteam, a small-scale French software development company (Couture, 2018). While Teikari was aware of GameMaker and other similar software already in the primary school age, lacking skills in English and programing delayed his early projects. Only after finding the script-free programing tools like The Games Factory and Multimedia Fusion, he really got started. In Teikari's words: "During that time there was a shortlived boom of young hobbyist game developers using tools like these, at least in Finland, and I guess the beginning of my gamedev hobby was somewhat a part or a result of that" (Lambie, 2019).

Multimedia Fusion 2, Teikari's main utility, comes with a fairly simple graphical interface and a sprite editor. The program in itself does not support scripting, but, through plug-ins, users can utilize common scripting languages often used in contemporary game projects. For example, most of the game logic of *Baba is You* is created in Lua, a scripting language that has its origins in Brazil. Teikari's tools that are not integrated into Multimedia Fusion 2 include Notepad++ for scripting and different graphics and audio programs (Lambie, 2019). Teikari describes that using Multimedia Fusion 2 for *Baba is You* grew mostly out of habit: once you know the strengths, quirks, and shortcomings of a tool well enough, it is tempting to stick to it (Harris, 2019; Reddit AMA, 2019). Particular tools also end up having an effect on how the game looks and feels. As Nicoll and Keogh (2019: 4) point out, development software and game engines are never just neutral tools, but they always encourage their users to "adopt particular design methodologies and thus possess varying grains—protocols, standards, and affordances—that give shape to creative expression".

Teikari also highlights how he has received a lot of support from friends and colleagues in finding and creating the right plug-ins that have importantly improved the usability of the key tools (Reddit AMA, 2019). In this sense, it is obviously not only access to the right tools but also access to related knowledge that is important to any game developer. As Parker and Jenson (2017: 874) point out, finding the right resources through online communities is a paramount concern for many indie game developers. Communication between developers – be it game design advice, technical troubleshooting, or networking tips – utilizes a diverse range of channels and venues from social media groups to tool-specific forums. Teikari also mentions that he has got some trusted game-maker friends who have playtested his levels and provided other feedback (Lambie, 2019).

Much of the interaction between Teikari and other developers seems to happen online, and Teikari openly questions the importance of a particular location. He even repeats the idea of Finnish indie developers not competing with each other as everyone is selling their games directly to the international market (Reddit AMA, 2019). At the same time, Teikari has been actively collaborating with other well-known Helsinki-based self-identified indie developers like Petri Purho (IGF Seumas McNally Grand Prize winner in 2008 with Crayon Physics Deluxe) and Olli Harjola (Special Recognition Award winner at Indiecade 2011 with The Swapper). In some of the interviews, he also mentions an Indie Beer gathering organized in Helsinki. This monthly event is originated by Petri Purho and provides an alternative to more commercially oriented developer meetings. While local International Game Developers' Association (IGDA) meetings have typically been key venues for developers to meet and socialize, need for less-business-minded smaller gatherings has been identified in different sites as for example Parker and Jenson's study (2017: 877) on Canadian indie developers shows.

For Teikari, attending game jams is one of the most effective ways of building a collegial support network (GOFIG_news, 2019; Harris, 2019). Game jams are events that bring together a diverse collection of professionals, students, and others interested in game development to create games in a relatively short timeframe. They provide an accessible forum to learn more about game-making and a chance to network with like-minded individuals. Teikari highlights the importance of game jams as a venue for prototyping ideas and evaluating how much potential they have (Harris, 2019). The short duration forces people to concentrate on the core design issues and frees them from the slow process of polishing the game (Chan, 2018). At the same time, Teikari is aware of how game jams can familiarize the future game developers with a crunch-oriented working environment and therefore warns indie developers from "overdoing" jams: "A potential danger is, though, that an overly competitive/intensive approach to jamming can lead to overworking and crunch, neither of which healthy, and can reinforce these kinds of behaviors in other game development work" (Harris, 2019).

In the past few years, local game jam scene has grown in popularity: jam goers can select from several annual jams with different foci and scale. Finnish sites have participated in Global Game Jam since 2010, and, in 2013, the key collaborators registered a non-profit organization Finnish Game Jam Association to coordinate the operations. In 2016, Global Game Jam attracted over 900 participants in Finland, and according to Kultima et al. (2016), Finland had the highest number of created games per capita in a global comparison. Teikari also pays attention to the change:

> I'd say that the indie community in Finland has been gathering momentum over the past few years; however, when I started participating in jams, the Finnish indie scene was pretty quiet [--] It's just that there hasn't been an active 'scene' for indie devs here in the same way as, say, in Sweden or Denmark.
>
> (Chan, 2018)

He also mentions that personally he got to know many indie developers by visiting events and jams in other Nordic countries, once again highlighting the wider regional aspect of the indie. Here, one can see a connection to prior hobbyist cultures like the demoscene that also relied on active Nordic networks (Jørgensen et al., 2017).

Discussion and conclusion

As discussed already in the beginning of the chapter, studying different modes of game development requires combining global and local perspectives (Kerr & Cawley, 2012; Keogh, 2019). As the case studies illustrate, both distribution platforms, development tools, and individual's networks are highly international. Housemarque's small core team has members representing several different nationalities, and Teikari's game jam network includes indie developers worldwide. At the same time, Ilari Kuittinen has been actively contributing to the local development scene for example by consulting public authorities about customized support structures, lobbying for advanced game education, and speaking in various local events. Similarly, Arvi Teikari's personal achievements and his Nolla Games collaboration with the other well-known Finnish indie developers have provided inspiration and much-needed example to many local developers.

Parker and Jenson's (2017) work on Canadian indies suggests that having both active local communities and extensive transnational developer networks reduces the need for strong national identification. The Finnish case may be a different here. A small nation located far away from the traditional game industry centers has in the past decade witnessed mainstream success stories like Rovio's Angry Birds and Supercell's Clash of Clans that have raised global awareness of the Finnish game industry. Journalists have been eager to reveal the reasons behind the success and often referred to as national particularities (Niipola, 2012; Karjalainen, 2016). Neogames, a non-profit game industry organization that receives support from the Ministry of Education and Culture, has been actively promoting the idea of a particular Finnish game development culture. At the same time, IGDA Finland gatherings are often mentioned as remarkably open and collaborative environment, contributing to the welcoming public image of the national industry.

While self-identified indie developers surely differentiate themselves from the most extravagant national branding, they still sometimes end up repeating the same marketing arguments as we saw in Teikari's case when talking about local developers not competing with each other in the same market. There are also more indirect consequences from the local industry success: talented hobbyists and game school graduates have found it relatively easy to find satisfying and safe employment within the local game studios contributing to the slow growth of the local indie scene Teikari also mentioned.

One interesting observation is that the "Finnish" found in branding materials in most cases refers to the capital Helsinki and the region around it. In 2018, whopping 97% of the Finnish game industry's turnover was generated by companies located in the capital region, and around 75% of all employees also work in this area (Neogames, 2019). In addition, many central IT and media companies are also located in the capital area, making cross-sectoral collaboration much easier than in other parts of the country. While the figures talk about the "mainstream" game development, independent developers also witness this bias to a certain degree. As previous research (Parker & Jenson, 2017) shows, independent developers often see the larger size and diversity of particular cities as a major advantage as they can provide more room for different identities and perspectives. In the case of Helsinki, this means that you easily access different communities and modes of game-making and also find collaborators from other creative fields. In other towns, a monthly IGDA hub meeting may be the only public opportunity to meet likeminded individuals (Komulainen & Sotamaa, 2020) and in more isolated locations you mostly rely on your online networks.

Another interesting aspect discussed already in the case studies is the Nordic dimension. Teikari designed the first version of the game in a game jam in Denmark and openly identified with a Nordic community of indie developers. Housemarque folks have also been involved in the Nordic collaborations, for example, by sponsoring and giving talks at the Nordic Game convention organized annually in Malmö, Sweden. They have also received grants from the Nordic Game Development Fund and been benefitting from the visibility provided for example by the shared Nordic booths and parties in key international game industry events. This regional aspect – active Nordic networks – has been there from the early days of the local industry (Jørgensen et al., 2017). While the importance of studying the role of Nordic welfare society as a foundation for game-making has been addressed (Sotamaa et al., 2019), it is clear that we need more detailed studies looking into the regional indie developer networks.

As already suggested by some previous work on indie games (Lipkin, 2013; Ruffino, 2013), instead of a single entity, independent game-making needs to be seen as an umbrella term for a variety of activities. Features like digital distribution, small teams, and game jams, at some point, considered central markers of independent production and have become crucial features of almost all game development. The two games discussed in the article actively negotiate the different dimensions of independence (Garda & Grabarczyk, 2016). After almost a quarter of century in action, Housemarque founders, Kuittinen and Tikkanen, are still the biggest shareholders of the company. Between 2006 and 2017, the studio benefitted significantly from close collaboration with the gaming giant Sony. Sub-contraction work for a major publisher provided a steady stream of revenue and allowed the studio to create a line of easily recognizable games. While the publishing schedule for the contracted projects was closely commanded from the outside, the company still had some room for maneuver as they simultaneously explored the opportunities for more independent publishing models.

In Teikari's case, it took two years to develop the project from a quick game jam build into a proper game released through Steam for PC and ported for Nintendo Switch. As this process has obviously required a lot of polishing, negotiating the deals with publishers and, for example, a more organized attitude toward testing the game (Lambie, 2019), even an individual developer becomes dependent on a variety of actors. While the two cases discussed in the article present relatively different modes of game-making, the discussed projects face very similar challenges. Networking with other industry actors and interacting with players has become increasingly central and time-consuming. This 'relational labor' (Baym, 2018; Whitson et al., 2018) is highlighted in Teikari's observations about the importance of the support he receives from game jam communities and online networks. Housemarque team also faced this challenge in a very concrete manner when the support from publisher decreased, and they had to lay off the community manager from the project.

One of the key reasons for Housemarque's survival over the decades has been their ability to quickly adjust to ever-changing industry environment (Lappalainen, 2016). In order to create sustainability and continuity, they have utilized a few different strategies. While contract work for Sony has been used to create stability, both service-based model relying on constant updates and reuse of existing IP has been actively used to boost the sales of individual titles. Super Stardust HD (2007), Housemarque's first game to Sony, was a remake of Harri Tikkanen's 1994 original, and several updates and downloadable content packs were released after the release to increase the "shelf-life" of the game. The following years also witnessed the releases of Super Stardust Portable (PlayStation Portable, 2008), Super Stardust Delta (PlayStation Vita, 2012), and Super Stardust Ultra (PlayStation 4, 2015). Many independent companies and developers rely on similar strategies, highlighting the importance of prior games as a portfolio and a way of attracting a wider player base. At the same time, as Housemarque's post-2017 shift away from small arcade shoot 'em ups toward larger AAA productions shows that no strategy will last forever in the fast-moving game development environment.

At the same time, Arvi Teikari's down-to-earth views underlining the importance of community and creative freedom nicely illustrate how developers also consciously move away from the marketing discourses focusing on venture capital, business models, and windfall profits. When asked about the benefits of having released a widely recognized title, he openly expresses some mixed feelings about being in the spotlight, at the same time capturing something essential about the pressures associated with being an independent developer today:

> It's actually somewhat scary to suddenly be in a situation where I have this very successful game under my belt, because I don't really know what exactly that entails. [--] What kind of extra requirement does it

add for me as a business owner? How does it affect my future game design process?

(Schilling, 2019)

The examples discussed in this article show how every new project forces developers to rethink their objectives and constantly evaluate the risks associated with game-making. Long-term sustainability is almost impossible to guarantee. It is also important to remember that both case studies discuss critically acclaimed games. At the same time, thousands of indie game projects were never finished or published, not to mention globally distributed. For the study of videogames and associated production cultures, these failed, marginalized, and forgotten games should be at least as important as the successful ones. The case studies also remind us about the internal biases of the indie game development space. The most recent industry report indicates that every fifth employee in the Finnish game industry is female (Neogames, 2019), and there are no reasons to believe that the gender division of the indie scene would be any more equal. In this respect, when evaluating the results of this study and other game industry narratives, we need to consider if all developers fit to these stories and who are potentially left out of the picture.

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15 The rebels across the street

IndiE3 and the strategic geography of indie game promotion

John Vanderhoef

Over the last decade, indie developers and publishers have cultivated an identity defined by outsider authenticity in order to differentiate themselves aesthetically and discursively from the comparatively slick and corporate-dominant game industry. Scholars have discussed this differentiation in terms of aesthetics (Juul, 2019), popular discourses, and production dynamics (Ruffino, 2013). Yet, the strategic geography of indie promotion has not been sufficiently highlighted or examined. Studies of space in games have tended toward spaces of play (Bryce and Tutter, 2003; Flynn 2001; McNamee, 1998; Nitsche, 2008), the representation and use of virtual space, and, more recently, the spatial geography of production (Crogan, 2015; Huntemann and Aslinger, 2013; Johns, 2005). Studies of the spaces of videogame promotion have been more limited, with notable exceptions including those attending to gender dynamics in the use of "Booth Babes" at promotional events (Huntemann, 2013) and recent work on indie intermediaries in the context of the Indie Megabooth and other aspects of indie game production (Parker et al., 2018). This chapter contributes to this growing body of work by attending to a trend of promoting and marketing indie games at ancillary events and adjacent spaces to the industry's largest gatherings, including the Game Developer's Conference (GDC) and the Electronic Entertainment Expo (E3). Each of these events occurs annually in San Francisco and Los Angeles, respectively, and focus on industry trade discussions and networking in the former's case and marketing and promotion in the latter. Some of the ancillary indie promotional spaces within these larger venues include events organized by The MIX (Media Indie Exchange) at E3, the Lost Levels Unconference which coincides with GDC, and various meeting places organized by indie publishers and partners at GDC and E3. Drawing from site-visits, marketing materials, social media discourse, and press articles, this chapter specifically deploys a cultural industry and cultural geography framework to examine the circuits of strategic spatial promotion of indie games in the Los Angeles area during E3 week.

In existence since at least 2014, the clever portmanteau "IndiE3" is a series of online and real-world indie game events that coincide with the annual E3 in Los Angeles. The original organizers of IndiE3 attempted to offer a corrective to the large budgets, flashy prerendered game trailers, and cynical marketing of the E3 convention, largely focusing on glitz and gunfire, which has been the industry's largest marketing event since 1995, after splitting off from the annual Consumer Electronics Show. Leading up to and during E3 week in early June each year, giant posters plaster the LA Convention Center, advertising high-profile game releases such as the latest iterations in the *Call of Duty* or *Mortal Kombat* franchises. Burly men, scantily clad women, firearms, and the threat of latent violence have become the icons of the videogame industry's gigantic promotional event, which features massive press conferences from the industry's most powerful companies, hundreds of booths and game demos, and thousands of meetings between journalists, developers, publishers, distributors, and other industry professionals from around the world.

Starting with a tweet from developer TJ Thomas using the hashtag #indie3, a now-defunct account called @projectindiE3 emerged to act as a rallying cry to other disaffected small developers who felt that a corrective to the E3 machine was needed (Kunzelman, 2014). In 2014, IndiE3 first manifested in an online-only "un-conference" featuring panels with topics ranging from game design and promotion to more critical topics including the influence of racism and capitalism on videogames. The online videos of the panels, still available via YouTube, contrast drastically with the marketing and promotional rhetoric of E3 proper and illustrate the critical and socially conscious goals of the initial IndiE3 idea. The counter-conference not only focused on sharing the knowledge base necessary to successfully develop and launch an indie game, but more importantly, it highlighted how developers can address other industry problems like the lack of a diverse workforce and the often-unremarked influence of neoliberal capitalism on the gaming industry and game design itself. Furthermore, the conference came together thanks largely to the efforts of queer and black game developers, critics, and other intermediaries, groups which historically have been marginalized at the larger E3 conference, particularly when it comes to who gets to take the stage at the massive corporate press conferences.

Unfortunately, not everybody involved with IndiE3 shared the same level of social consciousness, as contributors ranged from developers to academics and critics to streamers, all of whom came to the event with different politics and perspectives. As a result, the conference was initially plagued by a few racist and transphobic remarks on the stream and in the associated community chat that led to key organizers leaving and the remaining organizers writing a mission statement that outlined expected community behavior and language (Kunzelman and Walker, 2014). While the discourse created around the initial IndiE3 was productive in many ways, game critics Cameron Kunzelman and Austin Walker worried over the event's failures and considered whether the event was just recreating E3 on a smaller scale. This concern would prove adroit several years later when the moniker of

IndiE3 resurfaced in reference to ancillary indie game promotional spaces dominated by boutique indie game publishers Devolver Digital and its sibling company Good Shepherd. This reemergence of IndiE3 embraced the contra attitudes of the initial IndiE3 but also leaned heavily into an aesthetic of ironic masculinity endemic to many indie cultures.

Framed as a computer technology in popular culture texts and emerging from the same outsider male geek subculture as early computer communities, videogames were shaped early by patriarchal language, rituals, and customs that systematically excluded women and girls (Thornham, 2008) or framed their inclusion in decidedly gendered ways. Despite efforts to cultivate and design for the sizable but neglected female gaming population in the mid-1990s (Cassell and Jenkins, 1998), women were systematically victimized, erased, or marginalized in the games, marketing, and discourse surrounding the most popular videogames of that era. As casual games became the dominant force in the economics and industry of videogames after the introduction of smart phones, the enmity hardcore gamers felt for any genre, aesthetics, mechanics, or audiences that deviated from the hegemonically masculinized norm continued to grow (Kubik, 2012; Vanderhoef, 2013). The toxicity of hegemonic masculinity in gaming culture reached a fever pitch in August 2014 with the formation of the #gamergate movement across social media, which waged a campaign that targeted feminist and progressive critics, developers, and scholars engaged in critiques of the gaming industry and culture (Chess and Shaw, 2015). However, the emergence of a robust amateur and indie game development sector has offered challenges to the hegemonic and patriarchal formation of masculinity historically linked to videogames. Unfortunately, the indie embrace of outsider masculinities has often manifested in a gender performance characterized by a post-feminist irony, an ideological position that accounts for the successes of feminism while embracing elements of hypermasculinity or toxic masculinity for comedic effect. The joke derives from the fact that since we're all aware of critiques of exaggerated forms of masculinity, focusing on hyperviolence, phallic imagery, or female victimhood, it is okay to enact them. It's just a joke, after all. It is this form of ironic, self-reflective gender performance, exercised particularly by indie publisher Devolver Digital, which colors the gender dynamics of IndiE3 marketing spaces.

The IndiE3 of recent years purposefully juxtaposes itself against the frantic schedule and crowds of the LA Convention Center, positioned across the street, by providing a space of reprieve, indie game counter-programing, and a more relaxed atmosphere, replete with informal meetings over beers and bar food. In this chapter, I suggest that boutique indie publishers and indie intermediaries associated with IndiE3 use proximity to the dominant industry event in order to perform a deliberate outsider authenticity by establishing their anti-E3 exhibition spaces in eyesight of the event they disavow. By placing alternative indie promotional spaces in proximity to a dominant industry event, boutique indie publishers construct their indie opposition not only discursively and aesthetically but also geographically, establishing a distinction by contrasting the often busy and cacophonous halls of E3 to the much more low-key and casual spaces of restaurant bars and trailer park cookouts positioned just across the street. This allows for the continued construction of a discourse of rebellion while also being strategically timed and placed to garner the media attention and consumer interest already gathered for a large event like E3.

Cultural geography, authenticity, and indie game promotion

Although hardly absent (Lin, 2008), the field of cultural geography has been under-represented in the study of videogames, especially compared to other media (Ash and Gallacher, 2011). In general, cultural geographers have tended to address the ways videogames represent spaces virtually and the consequences of such representation (Leonard, 2003; Longan, 2008). James Ash and Lesley Anne Gallacher (2011) have argued that "experiences of videogaming are filtered and emerge through a variety of cultural, spatial and political processes" and therefore suggest attending to the multiple geographies of videogames' distributed production, embodied play, digital worlds, and social fields (1). To this list, I would add that scholars and critics should also attend to the promotional spaces of video games since they act as significant constituents of the medium's social and cultural reception.

Ash and Gallacher specifically argue for the use of a (techno)cultural practice when studying the geographies in and of videogames and videogame production. This involves attending to the concepts of materiality, practice, and embodiment when attempting to understand the cultural spaces of videogames. Nigel Thirft (1998) suggests that we treat all spaces as constitutive of the practices and performances they host and therefore as never truly settled or stable. Despite their cyclical nature, spaces of videogame promotion at festivals, conventions, and expos are temporary installations that reinvent and reproduce themselves with each new event. In his examination of arts festivals, Stanley Waterman (1998) suggests that routinized art festivals "transform places from being everyday settings into temporary environments that contribute to the production, processing and consumption of culture" (1). Festivals in general have become areas of interest to cultural geographers who often highlight the spatial construction of the festivals through the lens of cultural, social, political, theoretical, and historical analysis (Cudny, 2014).

Cultural geography provides a productive tool to discuss indie game culture because it highlights the ways people and cultures shape and are shaped by the use of and development of spaces and how these spaces change over time. For instance, Tyni and Sotamaa (2014) examine the scene and space of Assembly, an informal local area network (LAN) and demo party found in 1992. They explain how Assembly developed from an amateur event into a formalized convention with links to the commercial videogame industry, an incorporation process that threatens every indie and counter-cultural initiative. The indie game industry and culture have undergone myriad transformations over the last decade or so, including an "aggressive formalization" (Keogh, 2019) process involving an incorporation into the dynamics of the dominant videogame industry. One central way to track this development is through their promotion and marketing at festivals and conventions.

In their study of the Indie Megabooth roaming promotional space, which has a presence at events like the Penny Arcade Exo (PAX) and E3, Parker et al. (2018) argue that the Megabooth and other curatorial promotional spaces, like The MIX, IndieCade, or Good Shepherd's Indie Heaven, act as "'cultural intermediaries' that are constitutive of the field of indie games and represent a key area of investigation and intervention for the critical study of digital games" (1955). Drawing from the work of Pierre Bourdieu (1984) and later scholars, Parker et al. (2018) define cultural intermediaries as "workers involved in the production and circulation of symbolic goods" that contribute to the way those goods are valued and legitimized within culture (1956). Part of the role of cultural intermediaries and the promotional spaces they cyclically establish at festivals and conventions, particularly when it comes to indie media cultures, is to (re)produce and perform a kind of indie authenticity.

Michael Z. Newman (2009) suggests that indie media cultures tend to share a social construction as "authentic, autonomous alternative[s] to mainstream media" (16). This construction hinges on both the formation of a culture of opposition to the supposed mainstream *and* a culture of distinction, which gives greater cultural value to those works and practices that deviate from mainstream trends. Although indie culture constructs its authenticity in many ways, including discursive, industrial, and aesthetic considerations, another way it constructs authenticity is through carefully crafted and strategically located promotional spaces. Following Newman above, as well as Richard A. Peterson (2005), I treat authenticity as a social and performative construct deployed to give people and companies credibility in their respective fields or cultures. Authenticity in the case of indie game promotional spaces is achieved, I suggest, through the development of a specific performance of outsider authenticity that doubles as a market brand. Adam Arvidsson (2007) and others have argued that brands act as constitutive pillars within our current era of information capitalism that provide additional value to companies and consumers beyond the actual products being sold. In this case, boutique indie game publishers like Devolver Digital and Good Shepherd craft a brand based on performative authenticity by strategically locating their E3 promotional spaces adjacent to but not within the official expo.

This differs from other indie promotional spaces that exist within the walls of E3 and share association with the large event. For instance, The MIX (or Media Indie Exchange) is a promotional event held during E3 that is designed to connect videogame journalists with standout indie game developers in order to garner media attention the small games might not otherwise receive during the chaos of these large industry events. Additionally, while often cloistered from the chaos of the rest of the convention floor, the art-focused exhibition of IndieCade has had a presence at E3 for over 15 years. Usually on the periphery of the convention floor, the IndieCade space features indie developers personally answering questions and demoing their small games, which are positioned unpretentiously on foldout tables, topped with PCs, game consoles, or mobile devices. The IndieCade E3 space is lowkey and acts as a welcome break from the intense crowds who fill up the other areas of the convention floor, often shoulder-to-shoulder, and in long lines at the various game demos from publishers like Ubisoft or Activision. Yet, despite the intimate nature of The MIX or IndieCade, neither can be disentangled from their official relationship with E3. While their claim to authenticity rests on the face-to-face contact the press receives with indie developers and the humble demo setups, both organizations use E3 as a platform and rely on the convergence of press within the LA Convention Center that E3 cultivates in order to promote their respective missions and the small games they curate. In contrast, the indie publisher sister-companies Devolver Digital and Good Shepherd purposely reject official association with E3, instead choosing to establish their performative promotional spaces across the street from the convention center, at once visibly declaring their separation from the event while also strategically attracting the same throngs of enthusiast press that gather to cover the larger games and announcements of E3. Even though The MIX and IndieCade obviously contribute to the promotion of indie games at E3, it is Devolver's makeshift indie trailer park and Good Shepherd's nightclub spaces that have appropriated the moniker of IndiE3, drawing on the history of the concept as representative of counter-programing and resistance.

Constructing an "indie heaven"

At E3 2018, Good Shepard hosted an exhibition, social, and business space they called Indie Heaven for three days in the now permanently closed Don Chente restaurant and nightclub. The venue was next door to Devolver Digital's annual, infamous parking lot trailer park exhibition and press space, which has been staged across the street from the LA Convention Center for years at E3. As the two dominant physical venues for IndiE3 2018, the Devolver and Good Shepard spaces illustrate how the current deployment of "IndiE3" represents marketing strategies that simultaneously distance indie games from the dominant industry even as indie intermediaries position their events and associated games strategically adjacent to the E3 main event both temporally and geographically. These geographic and time-based promotional strategies contribute to the complex performance by industry cultural intermediaries of "indie-ness" and "independence" that

Parker et al. (2018) have discussed in relation to the Indie MegaBooth's role as a curator, promoter, and purveyor of indie games. In these cases, Devolver's indie trailer park and Good Shepherd's 2018 nightclub-based Indie Heaven perform 'indie-ness' by renouncing the corporate and compromised promotional spaces of E3 and establishing sites of 'resistance' within eyeshot of expo attendees.

As a prominent publisher of indie titles, Devolver Digital attends most large industry trade shows and conventions, but they strategically position themselves geographically and figuratively on the margins of these events. For example, one year at GDC, Devolver rented out a bus and drove past the Moscone Convention Center in San Francisco to invite listless indie developers to take a ride, drink a beer, and discuss their latest projects. The publisher called this stunt the "indie bang bus", riffing on a pornographic video series filmed with amateur aesthetics inside a bus. Elsewhere, I have argued that such performative marketing stunts contribute to the construction of an overwhelmingly white masculine identity for the boutique publisher, one heavily invested in ironic, self-conscious, post-feminist ideology, a retrograde ideology which reflects the problematic and contradictory identity politics of the commercial indie game sector more broadly (Vanderhoef, 2020). Of course, realizing that they were breaking the law by offering open alcoholic beverages in their "indie bang bus", Devolver moved its outreach operations to a "basement bunker" at GDC 2014, being sure to remind journalists during interviews, in between tequila shots, that they were not officially part of GDC. Devolver has since moved their GDC presence yet again to the Galvanize speakeasy room in San Francisco and rebranded this space Devolver Underground (2018). This move perpetuates Devolver's connection to alcohol and illicit locations linked to illegal bars during prohibition while also providing those who visit Devolver during these events with a certain subcultural capital gleaned from 'being in the know'. In these examples, a roving bus, a basement bunker, and a bar that evokes surreptitious drinking establishments under prohibition, Devolver constructs its indie authenticity and brand explicitly through promotional spaces that orbit but do not inhabit the gaming industry's largest networking event.

Devolver's strategy of adjacent proximity can also be seen at E3 where Devolver has rented out a parking lot for the past several years across from the LA Convention Center where the rest of the expo is held. Describing the strategy of adjacent proximity, Devolver Co-Founder Nigel Lowrie says, "Indie publishers, we're smart folks, despite how we look, and we can come up with some clever ways to be a part of E3 without paying exorbitant prices to be inside a convention center" (Conditt, 2018). As part of the economic reality of the indie games business, Devolver and Good Shepherd save a substantial amount of money by not renting out space in the official LA Convention Center. In their parking lot headquarters, Devolver sets up a barbecue with tents, picnic tables, and mobile homes to shelter their indie partners and their games from the grueling LA sun – a space they are quick to remind us that had been next to a Hooter's, the American restaurant chain famous for barbeque wings and scantily clad waitresses – for several years.

While I attended E3 in 2015 and 2017, Devolver representatives offered me something hot off the grill and a cold drink as I checked in for my meetings with several indie developers in Devolver's makeshift trailer park. In between appointments, industry folk like publisher reps, journalists, and developers chatted in the shade of trailer overhangs. Indie developers invited me into their (sometimes) air-conditioned trailers to discuss and demo their games to me personally, not through the traditional public relations intermediaries that pepper the official E3 show floor. Inside the trailers, I sat alongside indie developers shoulder-to-shoulder on small couches and played their in-progress games.

Supporting its status as industry rebel, Devolver for years accused the Entertainment Software Association, the industry's lobbying group that organizes E3, of conspiring with the city of LA to block Devolver's trailer park on Figueroa Street, a charge the ESA has repeatedly denied. For instance, after the ESA allegedly parked semi-trucks to block the view of the indie publisher's trailer park from E3 attendees, Devolver decided to erect a giant white balloon tagged with the Devolver logo the next year (Conditt, 2018). Whether or not the ESA has indeed schemed to thwart Devolver's promotional event is beside the point. Devolver uses the accusation to bolster its status as industry outsider and construct a discourse that positions them and their promotional space as antithetical to the corporate practices of the industry's largest publishers across the street at the LA Convention Center. By highlighting its dispute with the city of LA and the ESA publicly via Twitter, Devolver constructs itself as an authentic industry rebel. The idea that the industry's central lobbying group, the ESA, would try to shut down an ancillary promotional event not officially connected to E3 proper contributes to Devolver's authenticity as a small maverick publisher standing up to the video game industry's corporate overlords. It also contributes to the mystique of Devolver's trailer park village, a mystique that has been adopted and cultivated by video game-focused media outlets.

The enthusiast press has repeatedly embraced Devolver's yearly publicity stunt and covered the event using headlines like, "Cold Beer, BBQ, and Indie Brilliance: How Devolver Digital Won E3, Again", (Murphy, 2015) and "A Tour of Devolver's E3 Anti-Booth" (Engadget Staff, 2016) with the use of "Anti-Booth" positioning Devolver's manufactured indie trailer park in stark contrast to the strobing lights, loud music, and massive crowds emblematic of the booths in the official E3 convention. Games journalist Brendan Caldwell has also adopted the strategy of configuring coverage of indie and alternative games alongside traditional E3 coverage at the website *Rock, Papers, Shotgun*, using the hashtag 'anti-E3' to designate a series of articles and previews for games that cannot be found at E3.

In addition to its trailer park across the street from E3, starting in 2017, Devolver began holding its own digitally streamed E3 press conferences, at once mocking the established format embraced by large platform holders like Sony and publishers like Ubisoft but also benefiting from the exposure of being named alongside the other, much-larger press conferences. Even while considered one of the least important press conferences, Devolver embraces this position with a winking pride. Devolver has self-consciously called its short, 15-minute events "The Devolver Digital Big Fancy Press Conference: Tomorrow's Unethical Business Practices Today!" leaning into the irony that defines much of its marketing persona. Like its E3 trailer park, Devolver's digital press conference, more than three years running now, benefits from proximity to the larger press conferences by Microsoft and Electronic Arts, even as it saves an enormous amount of money by opting for a digital-only presentation, a strategy Nintendo has adopted since 2013.

In 2018, Good Shepherd followed Devolver's adjacent proximity strategy when they hosted an exhibition, social, and business space they called Indie Heaven for three days in Don Chente's margarita bar and rooftop lounge next to Devolver's trailer park. The exhibition space featured food, drinks, indie game kiosks, VR demos, art installations, and a video dome on the roof. The similarity between the two company's adjacent proximity strategies likely stems from the fact that Devolver purchased Good Shepherd (formerly Gamebitious) in 2012, and Devolver's co-founder, Mike Wilson, later went on to head Good Shepherd. Good Shepherd shares Devolver's interest in edgy indie games and, in the case of E3, the same aversion to paying the ESA for official space within the convention center.

Leading up to Indie Heaven in 2018, Good Shepherd's (2018) Twitter account shared a flier for the event reminiscent of a flier for an indie rock band venue, complete with a list of developers meant to resemble a list of bands. The flier advertises Indie Heaven as "directly across from the LACC", and a Facebook post similarly describes the promotional space as "just steps away" from the convention center. The proximity of Indie Heaven to the official E3 proceedings became a refrain when people mentioned the space across social media. For instance, the Twitter account for development studio Nyamakop (2018) tweeted:

> Journalist, influencer, and friends in general at #E3: If you're looking for some rad games and a chilled place to hang out, eat and drink, then make sure to drop by @GoodShepherdEnt et al's 'Indie Heaven' across the street from the convention center!

Reportage of the Indie Heaven space highlights the largely retro, pixel-art games, social dynamics, cash bar, ambient lighting, and nightly musical acts available at the venue.

Significantly, Indie Heaven is in broad view of all E3 attendees, positioned as it is directly across the street from the convention center. When crossing

South Figueroa Street from the convention center side, the reduction in foot traffic is immediately palpable. The lack of shoulder-to-shoulder crowds at Good Shepherd's events contributes to the 'calm' atmosphere Indie Heaven advertises itself as promoting.

Indie Heaven trades on its barroom atmosphere to contrast itself conceptually, in addition to visibly, with the tumult of the LA convention floor. A large Good Shepherd logo adorns the front of the building hosting Indie Heaven while a poster advertising the games featured within hangs on the building's side. Upon entering the front doors, attendees confront a red-carpeted reception area and staircase with a sign above reading, "Good Shepard Presents". Attendees' eyes are drawn to the room beyond the top of the stairs, which beckons with neon blue lights and a large sign highlighting what exactly is being presented: Indie Heaven. Another carpeted staircase on this landing invites attendees to the exhibition space on the building's second floor, giving physical shape to the implied metaphor of ascending to a kind of utopian indie game afterlife. Unlike the postmodern digital hellscape of the E3 convention floor, the design of Indie Heaven instead evokes mid-century art deco aesthetic, despite also trading in bright game screens and its own blend of neon lights. In this way, Good Shepherd's Indie Heaven is the classy, though still informal, cousin to Devolver's lowkey, trash aesthetic that it cultivates with its indie trailer park. Besides an area for musical performances and a few stocked bars, this converted restaurant and nightclub space hosts a stage for live streaming interviews with developers and a suite of computers hosting indie games and VR experiences.

At night, the rooftop space of Indie Heaven acts as an extension of the party occurring indoors, sporting neon laser light shows, smoke machines, live music, and plenty of drinks and networking opportunities. The rooftop exhibition and party area provide a bird's eye view of the LA Convention Center. Attendees are literally positioned above the masses of journalists, developers, industry reps, streamers, players, and even executives that wander in and out of the convention center all day. Besides supporting, again, the Indie Heaven metaphor, the rooftop location also contributes to the idea that the space, as well as the indie games it hosts, holds an elevated place in the cultural and social hierarchy of E3. Indie Heaven offers intimacy rather than the LACC's massive crowds, one-and-one discussions rather than the LACC's traffic jam of unruly attendees, and unique, personal games rather than the iterative, blockbuster, high-profile titles whose trailers repeat on wall-sized screens on the LACC's exhibition floor.

Despite their tactical locations on the periphery of E3, the promotional spaces of IndiE3, centered on Devolver's trailer park and Good Shepherd's nightclub, do not offer a significant rebuke to either E3 or the general promotional and professional posturing of the corporate gaming industry. Devolver and Good Shepherd offer alcohol to press and developers to lubricate social interactions and trade talk, but there are many companies at E3 who do the same. Developers like CD Projekt have given out beer to the press before small-room showings of their blockbuster games like *The Witcher 3* or *Cyberpunk 2077*, and even smaller companies, like Oddworld Inhabitants and Just Add Water, have offered me drinks while chatting with their PR representatives casually near a standing videogame art exhibit in the convention center's main building. As an industry built on conversations over dinner and drinks, especially at large conferences and conventions, embracing and constructing spaces that promote drinking and casual hangouts are not antithetical to the industry's operations.

Instead, such 'alternative' promotional spaces merely reproduce the same strategies the industry has long used in order to downplay its business goals in favor of an illusion of leisure and casual conversation. The dominant industry's embrace of the corporate cool (Liu, 2004), which has profoundly gendered implications, can be found at most large developers and publishers, such as Electronic Arts, where recreation rooms, gyms, and other amenities are provided to create an atmosphere of perpetual play and mask the well-known grueling working conditions many developers face as part of the job (de Peuter and Dyer-Witheford, 2005). The anti-corporate rhetoric and performance within the videogame industry extend back to the early days of Atari where developers and other employees, influenced by counter-cultural politics and a rejection of professional decorum, infamously traded in dirty jokes and drug use in between programing sessions (D'Anastasio, 2018). The culture of these more contemporary indie 'outsider' spaces, defined by post-feminist humor and an ironic embrace of masculinized aesthetics, becomes profoundly gendered and acts, in many ways, to uphold the gendered gatekeeping practices the industry writ large has been guilty of for years. Such practices discursively delineate who is and, more importantly, who *is not* included in these promotional spaces. Although perhaps not purposeful, such geographically based and aesthetic marketing performances, along with their casual alcohol consumption, contributes to the dominance of a particular white outsider masculine identity in indie videogame culture (Fisher and Harvey, 2013) that has been a concern of many women, people of color, and non-binary people working in the indie games space for years.

Conclusion

After the first IndiE3 in 2014, Austin Walker and Cameron Kunzelman worried that the counter-programing event, though based on enmity for the corporate nature of E3, in some ways mirrored, rather than critiqued, the problems of the larger expo. Several years later, the IndiE3 moniker has seemingly lost the potential for critical change it initially offered, instead becoming the namesake of two of the most prominent and performative indie game publishers hosting ancillary promotional spaces across the street from the LA Convention Center. Although these spaces do put human faces to the games on display, highlight small games that would probably otherwise go unnoticed, and construct casual, semi-anti-corporate atmospheres replete with beers and bar food, they also do not offer a critical discourse that might highlight issues like the lack of industry diversity or the influence of neoliberal capitalism on videogames – topics that were featured on the panels during the first online-only IndiE3. Instead, the critique Devolver and Good Shepherd's indie promotional spaces offer is largely surface and performative rather than substantial and impactful.

Do indie game promotional spaces have a responsibility to engage in focused critiques of the dominant industry while also being self-reflective enough to avoid reproducing the same systemic issues? Many would argue that they do not. It is, perhaps, enough to offer parody and alternative spaces within eyeshot of the ESA's marketing behemoth of E3. Yet, if these adjacent promotional spaces want to gain subcultural currency by performing a rebellious, outsider status, they open themselves up to such criticism. Poking fun at dominant industry marketing tropes and the corporate nature of events like E3 provides little tangible challenge to ongoing issues in the industry and culture of games. A more radical approach might involve a true distancing, both geographically and temporally, from the E3 event, a carving out of something that seeks to define itself by what it is rather than what – across the street – it is not. It might involve more honest self-reflection on the culture these publishers produce through their marketing practices. It might involve a move away from irony and toward more honest conversations about what precisely these companies are rebelling against and what values they believe are necessary antidotes. It might involve critical discussions rather than performative aesthetics and affect.

The deployment of promotional events adjacent to the industry's largest gatherings by indie intermediaries is part of a larger trend of holding major press events outside of the spatial and temporal white noise of GDC and E3. Not only have Electronic Arts, Activision, and Microsoft all moved their E3 marketing events away from the LA Convention Center to other parts of Los Angeles, but 2019 marks the first time that Sony and its powerful PlayStation brand have opted not to participate in E3 in any way. The spatial strategy of staging indie promotional spaces adjacent to E3 performs a highly visible, performative critique of the industry while also relying on the promotional momentum and enthusiasm these large-scale events produce. Even as E3 slides toward a slow death, as more and more large companies are choosing to not take part, the event still matters a great deal, at least discursively, to the industry and culture of games. How indie games are incorporated within the marketing machine of E3, either within or outside of the LA Convention Center, tells an ongoing story about their cultural, social, economic, and political trajectories.

The indie strategy of piggybacking off events like E3 is part of a larger tension between the performance of 'indie-ness' by indie intermediaries and their more business-minded global operations. Put another way, through such promotional performances, indie developers and publishers stick a proverbial middle finger up to the dominant industry while simultaneously trading in the same spatial and temporal promotional gimmicks, exclusive industry parties, and rhetoric that masks labor as a perpetual playground in order to sell digital game experiences. If indie games and the promotional spaces of their publishing partners are really purporting to be an authentic alternative to some mainstream or corporatized game industry, we have to ask hard questions about what exactly they are alternative to, and whether or not their alternative posturing represents a political and structural alternative to the dominant industry or simply a re-dressing of established marketing techniques that other giants in the industry have already embraced, including cost-cutting relocations and using the nucleus of E3 in order to stage ancillary promotional events necessary to secure coverage and eyeballs in a space that is increasingly diluted by the static of peak video game content and services.

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16 Freedom from the industry standard

Student working imaginaries and independence in games higher education

Alison Harvey

In July 2019, an American teen made international headlines when he won the *Fortnite* World Cup and its $3 million prize pot. Media interviews with Kyle Giersdorf emphasized his intensive training schedule and his plans to make a 'career' in games, comparing his earnings to other high-profile professionalized players (Bryson and Chokshi, 2019). Beyond competitive gaming, the visibility of those making a lucrative living from the expanding range of entrepreneurial activities related to games is on the rise, with *Forbes* listing among its top 10 highest-paid YouTubers in 2019 five videogame streamers (Berg, 2019). These outlier success stories, as with others who have channeled their energies on social media platforms to create brands, monetize engagement, and stake out a livelihood, are part of a cultural production landscape where independence is framed as a possible route to exciting and rewarding working lives (despite the many structural barriers to actually getting paid for this passion work as demonstrated by Duffy, 2017).

The contemporary landscape of platform-based entrepreneurialism broadens potential engagements with the concept of 'independence' in forms of games labor. This adds complexity to an already tangled field of production studies examining the games industries and self-enterprising work therein. As previous research and the chapters in this book demonstrate, what is 'indie' in videogames is a moving target, conceptually burdened with expectations and values in relation to its aesthetics, economics, politics, and role apropos the 'mainstream' (see also Jahn-Sundmann, 2008). But what media reporting on young amateurs-turned-millionaires in games indicates is that for some, 'going indie' also represents an aspirational dream – a future-looking vision that promises specific kinds of opportunities for greater freedoms and potentially better rewards than traditional work narratives suggest. The emphasis on young people, micro-celebrity, and fantasies of success in these kinds of mediatized narratives have been shown to resonate with 14–18-year-old students articulating their hopes and dreams for future work (Mendick et al., 2018). It is worth then considering the role of independence in how students in formal games educational contexts articulate possible working realities in the short and long-term. This chapter, therefore, focuses on what independence in games signifies for 'gamesworkers[1] in becoming', and how in turn these subjects discursively engage with politicized topics including limits on creative freedom in professional game-making, diversity, and inclusivity in games, and the UK's departure from the European Union, through reference to the notion of 'independence'.

Before exploring the contours of the dream of 'going indie', we must review the significance of the higher educational context and the role students as future working subjects play within not only the games workforce but the creative industries more broadly. As subjects within games discourse, students in higher education are symbolically and discursively important, representing to policy-oriented stakeholders the promise of the national industry's growth, productivity, and sustainability in the form of 'talent' (Livingstone and Hope, 2011). Here, higher education institutions (HEI) are seen as important sites for addressing 'skills shortages' and acting as 'feeders' of human capital to the sector. National industries become 'talent hubs', while educators must concern themselves with watertight 'talent pipelines' enabling large 'talent pools'. The legibility of this approach to gamesworkers before their entry into the industry can be seen in the ongoing dominance of the talent discourse across the mission statements and activities of major trade bodies such as UKIE, advocacy groups including European Women in Games and BAME (Black, Asian, and Minority Ethnic) in Games, and charity organizations like the British Games Institute, which has to date planned two summits focused on games education. The latter agencies all feature inclusion and diversity within their purviews, and therefore the affixing of 'diverse' before talent comes as no surprise. But this occurs in other documents referencing talent in games, communicating awareness of distinctly homogenous workforces while placing responsibility on educational institutions to address pernicious exclusion. Overall, the variable metaphors that become attached to 'talent' indicate that this perspective on the education and training of students enables initiatives and recommendations that presume fluid movement (via pipelines) and tidy containment (within pools). It is a set of images and discourses productive of flows of capital in the British context as demonstrated by the Transfuzer and UK Games Fund programs. Simultaneously, it is a perspective that simplifies and flattens the experiences, contexts, and lived realities of students and other educational stakeholders.

Talent-focused policy maneuvers are common within the creative industries and serve to frame work in these fields as highly individualized while positioning their labor markets as inherently meritocratic (Banks and Hesmondhalgh, 2009). Focusing on talent simultaneously enables quantified measurements of creativity (e.g. Florida, 2003) and essentialization of those who flourish in creative work as innately deserving of their successes based on inborn talent, often aligned with very limited embodied subject-positions stratified by class, gender, and race (for more see Banks, 2007). The implications of the talent discourse are therefore broad, but there are also unique elements to examine when considering talent in games as Daniel Ashton's (2011a) research demonstrates. Student subjects are simultaneously hailed as both passionate consumers and creators, drawing on the norms and values of games to harness and channel their energies toward the creative economy. The working futures articulated for and by game students as creative talent then reify normalized sets of practices and habits within this sector of cultural production, including imperatives of 'perpetual upgrade' of the self (Ashton, 2011b), 'grinding' through skills training in educational working environments with contradictory aims (Livermore, 2013), and adopting masculinized bravado as a mode to deal with normalized exploitation and exclusion in the games industry (Harvey, 2019). This research, examining formal games education across diverse national contexts, has therefore linked HEI to the formation of professional identities that are conservative, uncritical, and risk-averse, which is significant, given that qualifying for degrees in higher education contexts has become an increasingly mandatory criterion in job postings since it was identified as an entry point into gameswork (Deuze et al., 2007). Ongoing engagement with education in games is therefore important for understanding how the future of this creative sector is shaped and structured for its future workers.

The normalization of the above-mentioned professional identities is in some part based on how students in games higher education are exposed to ideas about what working in games looks like. A picture of gameswork is formed from various sources across these programs, including instruction from those working in or previously employed by well-known firms, internships, and placements within game companies, and extra-curricular activities such as game jams and networking events enabling meetings with professional game workers. As a complement to these professionalizing or 'employability' activities, students also describe learning about the less favorable elements of gameswork such as crunch, gender pay gaps, curtailed career lengths, and harassment from news media and online discussions. Beyond these increasingly mainstream narratives, instruction within games programs often overtly refers to the punishing realities of both 'getting into' the industry given the highly competitive nature of most open positions and the brutal expectations normalized for workers once they are in the role (Harvey, 2019).

In sum, while students have variable degrees of hands-on experience of the day-to-day functioning of professional gameswork (with some as Keogh (2019) finds expressing complete uncertainty of the skills they will require for it), all of them have a unique investment in 'work' – a fraught affective relationship that makes conversations about how they envision their next steps especially generative of insight into how independence figures in these imaginaries. To understand these aspirations and how the young people that constitute 'talent' envision working futures, in this chapter, I engage with materials derived from qualitative research in five UK HEI offering undergraduate and graduate games programs. Participants in this study were involved in a range of courses with diverse foci including, among others, business, critical theory, art, programming, and design. My study included 49 semi-structured interviews with students, instructors, and administrators in these courses and observations of the spaces dedicated to teaching, coursework, and exhibition. I also engaged in critical discourse analysis of the plethora of policy, industry, lobby, and trade documents referencing games education, skills training, and talent as well as promotional materials generated by the dozens of UK universities offering games programs.

In this analysis of narratives of independence, I focus primarily on my interviews with students, who came from an eclectic range of backgrounds. While a number of participants in this study did reflect the still-homogenous constitution of the games industry – young, White, male (Weststar et al., 2019) – I also had the opportunity to interview a substantial proportion of students who made reference to how their gender, sexuality, age, and race differed from the expected gamer subject-position and how this impacted on their educational experiences and future working plans. In the below discussion of working imaginaries and independence, I highlight this diversity in student backgrounds and make no claim to generalizability, focusing instead on commonalities in the discourse deployed despite the heterogeneity of the sample. The shared narratives across very different groups of students highlight the potency of some aspirations and the power of barriers associated with them, confirming the significance of how future work is constituted in games education.

Though I did not ask specifically about indie games in my interviews, the topic arose consistently with respect to two lines of questioning – diversity and innovation in games and imagined working futures. It is on the basis of these recurring narratives that the remainder of the chapter is structured around my critical thematic analysis of student responses, focusing on how independence was linked to various kinds of freedoms in games – in creative production, from the constraints that hinder diversity, and in crossing through and occupying spaces and institutions. While many of these aspirational narratives recognize the structural and systemic problems with the contemporary shape of gameswork, they largely still adhere to individualized approaches focused on self-disciplining, leaving unchallenged the norms they seek to sidestep. Indeed, as the below discussions will show, in many articulations, these dreams of independence actually reinforce the standard operations of the industry. Therefore, to conclude this chapter, I reflect on what opportunities there are for short-circuiting the relationship between 'indie' and the individual through collectivist movements, such as campaigns aimed at organizing labor in games and initiatives seeking to diversify the games workforce.

Promises and perils of creative freedom in indie games

Across the five HEI in this study, industry connections were highly valued by both students and instructors alike. Well-known and highly successful local game companies were frequently referenced as providing support of various kinds to teaching and learning materials. They were also named as likely destinations for jobs, frequently with reference to the successes of former students there. In addition to this focus on work as mediated by larger firms in the mainstream industry, when I asked students about the projects they were working on, they often described the genre, mechanics, characters, and aesthetics of their prototypes by drawing parallels with successful Triple-A games and franchises. Furthermore, infrastructure in each program, from games hardware to engines to studios and labs, was typically branded with the names of institutional partners in mainstream companies. In sum, the education and skills training provided by these courses for future work were heavily structured through the universally referenced but rarely explicated 'industry standard', with the industry in this phrasing hinged on the norms of work in mainstream, Triple-A game development firms. In other words, non-independent work in games was the presumed end goal of most of those studying on these courses. Working as an indie (self-employed or freelance) therefore seems by default to be set in opposition to the gamesworker-in-becoming's apparent career path.

For instance, a core component of how games education presumes futures in established, mainstream games firms was through a universal emphasis on team-based work. Students were assigned roles in these groups, typically based on the focus of their chosen courses, and expected through this to develop a range of technical and 'soft' skills. The latter included those supporting collaborative development, such as communication techniques, as well as team working styles, particularly hyper-specialization in specific techniques. These configurations of assessment contribute to the formation therefore of a number of specific traits of the ideal gamesworker, including acculturation to 'agile' development methodologies and relinquishment of sole ownership over one's creative production. But while students accepted these realities as part of working to the 'industry standard', a fair number of them also engaged in game development independently within their courses or outside their studies at game jams or in their spare time. Some, therefore, had these individual and group experiences of game-making on which to draw when they articulated a critical perspective on how standardized teamwork impacted on their specific creative practice as well as more broadly on possibilities for autonomy in gameswork.

To illustrate the typical discourses mobilized when discussing independent versus team-based game development, here, I focus on the perspectives of a mature White trans* student specializing in game design. In our interview, he described to me the pleasures of both the collaborative and independent projects undertaken in his degree program. He reported enjoyment in "the individual project because it gives you freedom to do a thing you want to invest some time in and gives you free reign on that", while describing the group development project as "really a cool experience". Because he emphasized wanting to be a "jack-of-all-trades" able to master each element of producing a game, I asked whether indie work was his preferred work trajectory. His response, again characterizing independent work as enabling "creative freedom", also indicated awareness of the challenges of working outside the mainstream:

> It can be a little bit harder because you've got to facilitate all the things around it, rather than just being a cog in an otherwise larger machine. You've got to be the machine itself.

I have included this student's reference to the metaphor of the cog in the machine as this was an image evoked across multiple student interviews. It exemplifies how indie is configured as a site of creative freedom in opposition to the standard division of labor in the mainstream industry where students see their individual contributions as functionally important but of otherwise negligible significance in the process. This interview perfectly encapsulated how this sits in tension with the second sentiment expressed in relation to indie work in games, as he also noted that but for a lucky few very talented freelancers, "the work isn't always guaranteed and pay isn't always great either".

As these qualifiers highlighting labor insecurity and precarity indicate, within these programs, indie development is not framed as a necessarily viable career route. Indeed, it is described as rife with risks and barriers to financial reward and success, a portrayal embedded in their teaching (another student quoted his instructor's characterization of indie game development as "suicide, financially"). Thus, while indie is evoked as promising creative autonomy and freedom to innovate, it is usually then renounced as a likely working future. Instead, as a third student noted, indie game development is an aspirational dream for those who have put in their dues in the industry, built up financial stability, and can afford to "make games that don't make money". Indeed, creative freedom was seen as the privilege of indies unconcerned with profit or large markets.

Beyond these barriers, what these discussions reveal is that rather than being an alternative working pathway, for many students, indie is actually a stepping stone to work in the mainstream.[2] The employability techniques in games higher education solidify this relationship, framing independent development and attempts to bring individual projects to market via the Apple and Android Stores as a way to "get yourself out there", and a key component of the absolutely imperative process of portfolio-building. The portfolio is as pervasive a requirement in games education as teamwork, and its importance cannot be understated. At the 2019 Games Education Summit, portfolio creation was emphasized by industry representatives as the most important asset required to secure a job in a competitive field. Of significance to the educators at the event, the portfolio was often set in opposition to the university degree, which was denigrated as 'a piece of paper'. This was a discourse also deployed at times by students in the study who argued that many of the technical skills they developed in their degrees could be learned through independent trial-and-error.

While collaborative development in games degrees indicates a normalized mainstream work path, the emphasis on each individual student's portfolio in turn indicates expectations they also engage in indie development. Across my interviews with students, the portfolio was seen as not only necessitating the creation of assets, mock-ups, and other materials aligning with the industry standard, but in many cases full, marketable games, preferably tailored to the company being applied to. One cisgender White male game design student noted that "a lot of students do make their projects into full-fledged games afterwards", but on the main these independently created games are intended to demonstrate capability to future employers rather than marking the start of an indie career.[3] This was most clearly articulated by a cisgender White British male in a postgraduate games degree who noted that indie game development has become a requirement in a competitive job requiring both degrees and experience:

> What I'm trying to do at the moment is get experience through making my own games… everyone wants lots of experience for you to get a job… If that gets green lit or published or something, I can then say, 'Look, I did this. Is this enough for me to get a job?'

In this way, it became clear that for students in games programs, indie became an affectively charged narrative of hope enabling them to negotiate anxieties and uncertainty about their job-readiness and working futures in games, an option to imagine should there be no job "right out of the gate" of degree completion. Such a configuration of the relationship between indie work and industry jobs therefore supports discourses of self-reliance and acceptance of risk in this creative industry, continuing to normalize precarity at the same time as creative constraint in gameswork. As we will see in the next two sections, narratives of the impossibility of indie (aside from necessary unpaid labor as part of training) have serious implications for innovation, diversity, and opportunity in games.

Indie as the route to diversity and innovation

In addition to their exposure to mainstream critiques about exploitative practices in the games industry, all students in this study were well-versed in debates about a lack of diversity in games and aware of the shifting nature of player demographics. Some maintained that meritocracy would enable diverse workforces where others argued that there was a cultural impetus based on the ubiquity of gameplay that would lead to greater inclusion in games. As the study's call for participants overtly noted my interest in diversity in games, student's strong opinions on the topic were not surprising. However, while the interview questions asked directly about diversity and its relationship to innovation, the prevalent connection students made between these concepts and indie games was unexpected. This was expressed through two sets of ideas: first, a consistent identification of innovation through examples of independently created games, and second, that diversity is more likely to be found in indie development contexts. Both of these conceptualizations of indie as the locus of practices and titles promising inclusion and opportunity serve to demonstrate how the 'industry standard' is framed in educational contexts and provides nuance to the idea that students are uncritical consumers.

Perhaps, unsurprisingly, given the association with visions of creative freedom, indie titles were consistently cited by students as examples of innovation in games in terms of mechanics, aesthetics, and the kinds of emotional experiences they might afford. Students cited well-known examples like *The Stanley Parable*, *Journey*, and *Undertale* as well as less publicized titles such as *Anthill* and *Superhot*. Indie titles were set in opposition to Triple-A production that was characterized as, to paraphrase several different students, formulaic, "stale and generic", built on endless clones, and generative of "yet another male power fantasy" and infantile "shooty bang bang" games. In contrast, indies were described as "unique" and "creative", the product of passionate people creating a labor of love, and the promise of what can happen when games are made outside of the constraints of the mainstream industry.

However, rather than seeing indie and Triple-A as distinct, students also envisioned indie innovation as part of a larger ecosystem of game production. As much as indie was framed as free to experiment and the mainstream as risk-averse, innovation was conceptualized as circulating symbiotically between these sites of production. Indie according to students acts as a testbed for ideas that the mainstream industry then adopts once they are tried and deemed successful, often in a smaller venture followed by a blockbuster title. One cisgender White male international student in game design explained this as follows:

> The way the industry seems to be created is if you want innovative story or innovative game mechanics, you look at the indie industry. That seems to be where most of that is coming from. Then, at a big company you'll see one of the smaller games that has a really interesting mechanic and put that into one of their big games. That's how it gets spread out.

This discourse framing indie as the generator of innovation in games more broadly is tightly tied to another association made between independent game development and increased diversity and inclusion in games production. Explicitly, this was seen as an issue with the industry standard – many students described those working in the games industry as less exposed to diverse demographics and outdated in their perspectives on contemporary player markets. One cisgender White male student on a game programming degree characterized the dynamic as "the same men making the same games is always going to end up in the same games", based on a market-driven impulse that enables sequels, franchises, and standardized and even "stagnated" gameplay rather than innovation. This student characterized the contemporary games industry as a holdover of the "wild west days" that is largely not "reflective of the people who play them and people in general".

Indie, in contrast, was characterized by this student as a source of "inclusivity". One of his young White male peers in game design agreed that the openness and accessibility of indie enable greater opportunity for inclusion – "you see more diversity in indie because it's easier to get in to entry level". For many students in the study, accessible game-making in indie quarters was directly linked to the creation of innovative games from the ground up – "indie is good because you can… just have someone who is very liberal and come up with their own ideas and get out there with ease". In addition to suggesting a structural barrier to both diversity and innovation in the mainstream industry, students also argued that those who were underrepresented in the creative sector may be in a unique position to introduce innovation through indie development, generating new ideas "because of their struggles".

However, as with the recognition of the systemic challenges posed for those seeking to make a living in indie development, many of the students who identified innovation with diversity in independent game-making were wary of the challenges posed by relying on this kind of game-making to address the ills of the industry standard. The programming student quoted above qualified his optimism about a culture shift represented by a new generation of game-makers by noting "what makes me a bit more pessimistic is that really the driving of diversity of the industry is going to be mainly passion-based and we've not a lot in the way of funding for that kind of stuff". Additionally, there was a recognition that indie successes are hard-fought and that there are also many less impressive indie titles to list alongside the "diamonds in the rough", including multiple references to the indiepocalypse of an over-saturated indie games marketplace (Pedercini, 2017). Deeply practical concerns about the economic realities of working in indie therefore informed student perspectives on the limitations for greater inclusion in games based on this area of production.

Through these narratives of indie's promises of diversity and innovation, the participants created a connection between the industry standard, homogeneity of the workforce, and inflexibility about inclusion in the existing industry. As with the above-noted observations about creative freedom and autonomy in production, this was framed as the by-product of large production teams, hierarchical decision-making, and "old-fashioned" approaches to game design out of step with contemporary player bases. Therefore, if indie is meant to be the key route to diversity and innovation in games, the challenge is how students already think of it as a closed pathway for their working futures.

Indie games and freedom of movement

The affective resonance of the working futures articulated by students in this study has so far been inflected with anxiety about job prospects and grim financial pragmatism. The freedoms offered by independence in games were quickly qualified with awareness of the structural industrial forces foreclosing creativity, diversity, and innovation. In this final section, I will highlight another element of the student working imaginary that, while no less uncertain, retains an aspirational optimism for the freedoms afforded by indie work – in this case, movement across spatial contexts.

As this study was conducted within UK HEI post-2016, it is perhaps to be expected that participating students referenced the departure of the country from the European Union (Brexit). But rather than endorsing sentiments about gaining independence from the European Commission or burgundy passports, the students in this study reflected on the repercussions of Brexit in relation to their possible working futures, albeit in a contradictory fashion. On the one hand, there was a sense that leaving the EU would have unknown implications for games companies and working futures in the UK, and yet, on the other hand, students still largely envisioned futures that could include work across national contexts. It was indie work in particular – both as freelancers and self-employed game designers – that was often linked to these working imaginaries characterized by flexibility and freedom of movement. But as with the above-noted emphasis on freedom associated with creativity and opportunity via indie games, this was often in response to the foreclosures associated with the industry standard in the specifically British context.

When I asked about aspirations for future work, students were split about whether they wanted to stay local, move around the country, or go abroad for their careers. Those with dreams of movement outside of the UK with indie aspirations were largely focused on 'Europe' (with the United States a frequent target for those wanted to work for specific Triple-A companies). The attraction of moving to the European context was not however ideological but based on a keen awareness of the implications of Brexit for indie game development opportunities in the UK. For one cisgender White male game design student, working freelance and then indie was the dream career path, and he was insistent that this was something he'd prefer to do in Europe. When I asked why, the answer was simply "Brexit". The rationale for going outside the newly closed borders of the UK was largely related to opportunities afforded to independent workers in the EU, as "Europe's got really good bursaries for games designers". Another student, a White trans* student on an indie games course subscribed to the narrative that greater opportunities and diversity can be found in independent gameswork. But she noted that this was conditional on remaining in the European trade bloc as "there's a lot of EU funding for game dev companies", funding that supported many of the indie titles she and her peers most respected. She summarized these concerns about indie working futures by noting “Brexit will pose a huge issue”.

Awareness of the constraints Brexit would impose for independent game development funding did not however translate to recognition of how this would limit the ability of British students to work in the EU after graduation. Leaving the UK – in stark contrast to getting a job in the games industry – was framed as frictionless and imbued with a largely positive affect (aside from the student who answered my question about working locally with “Hopefully not. Fuck the UK, man”). “Working and travelling”, “working anywhere”, “France or Canada”, “seeing new places”, and other visions of cosmopolitan freelancing and indie development were the only part of discussing student futures that were largely hopeful and unqualified with practical concerns. One White British female student at the end of her degree in games writing explained to me that these visions of movement were informed by generational subjectivities (“it’s a millennial thing”) and visions of the creative life (“your inspiration won’t change if you see the same stuff”). But it was also shaped by the economic realities faced by these young people – “I’m never going to own a house. I can’t afford it. I probably never will afford it”.

It becomes clear then that the hopeful narratives affixed to indie work and transnational mobility become a way of managing political and economic uncertainty through recourse to individual agency. The flexibilization of labor in transnational flows of production becomes an opportunity linked to generational discourses – “we’ve got that freedom to move because we’re digital people”. Less optimistically, some interviews made it clear that freedom of movement was actually hinged on deep insecurity about working futures. As one cisgender White British male student in a postgraduate game degree asserted, “If you’re happy to move about then it should be okay”. This indicates that an indie mentality and a willingness to relocate may be more of a necessity than a choice.

While for the mainly White British students under the age of 30 who participated in this study, freedom of movement was something to take for granted until June 2016, the international students provided a different perspective. For example, one mature female student of color studying games production said she wanted to return to her home country. However, she described the situation as challenging: “all the people I met there, they’re very creative. The thing is, since it is a Third World country, we can’t afford to be indie”. While her aspiration involved connecting investors to indies there and mentoring them, in the short-term, her vision for the future of work was that she would go with “whoever gives me a visa to stay. I’m in a beggars can’t be choosers state”. Aside from those with experience of immigration, however, most student sentiments about freedom of movement demonstrated relatively little awareness of the shape of global gameswork. Despite their recognition of some challenges working in the industry such as crunch and short career lengths, none of the students in my study referred to the consequences for labor created by the respatialization of games through globalization such as outsourcing, subcontracting, and automation (for more on this, see Kerr (2016) and Fung (2016)). Instead, their articulations of freedom of movement for work were hinged on a connection being made between digital labor, flexible working arrangements, and dreams of these superseding the frictions of visas, borders, and immigration controls. Local contexts and the realities they create for forms of cultural production and labor therein were not part of student visions; these young people instead see independent gameswork as enabling an open terrain of opportunities and frictionless movement across spaces and locales. This allows a distancing from the constraints of gameswork and the barriers created by Brexit for these young people, another affectively charged working imaginary that highlights concerns about finding stability in their working futures.

Conclusions: challenging the industry standard

As we have seen, independence in videogame development is evoked by students in formal education as an alternative work futurity that illuminates their frequently critical perspectives on the professionalized corporate industry and the shape of the economy underpinning it. While the conclusions from previous research on formal sites of training are supported here in terms of the ways this education is shaped by and reinforces the norms of exclusion and exploitation through adherence to the 'industry standard', as we have seen the frequent discussions of the crisis of gameswork in academic and news media research is not lost on students in these contexts, who do reflect in nuanced ways on structural problems in their chosen field of work. What this chapter has shown is that independence, via freelance work, entrepreneurialism, or other forms of labor flexibility, provides a set of possibilities for imagining more inclusive, agentic, and sustainable ways of working games, though these are at times premised on unspoken racial and economic privilege in particular. Importantly, even in dreams about future work indie opportunities are ultimately disavowed, leaving untouched the very problems that provoked these aspirations.

It might seem that these conclusions would solely support a negative analysis on the possibilities these future gamesworkers suggest. However, it is important to put these imaginaries in context, particularly in the UK, where tuition fees are high, job opportunities are highly competitive, and, as Brexit indicates, the political and economic future is deeply uncertain, particularly in the regions of the country where I undertook this research. Dream narratives are emotive and affectively charged, particularly with anxiety about job-readiness, and it is not only students that can feel powerless when confronted with pervasive structural inequalities. But on the main, students in this study took some critical stance on gameswork – be it in terms of creative stagnation, manipulative monetization strategies, exclusion, or precarity and crunch. It is this political awareness that might be seen as a possible point of intervention. If 'going indie' provides a legible set of discourses and narratives about problems with the industry standard, allowing the possibility of doing something differently in gameswork, then raising the profile of other kinds of pathways may introduce additional viable imaginaries on which to draw. I am thinking here in particular of the collectivist actions represented by labor organizing and diversity initiatives, both of which can productively channel the dissatisfactions articulated by students into less individualized actions.

A number of such organizations already interface with educational stakeholders – Game Workers Unite includes in its literature on labor rights and unionization a student-focused flier, the European Women in Games group interfaces with students regularly at its events and through its Ambassadors mentorship program, and the inclusion-focused Refiguring Innovation in Games network has organized a number of jams and activities open to students. Still, a minority of students in this study were aware of these projects, indicating a ripe terrain for engagement. The critical perspectives articulated by those who constitute the future of the workforce suggest that the task of raising awareness has reaped rewards. However, as this analysis of independence and freedom in working imaginaries indicates, there is still the danger these critiques may be channeled into venture labor subjectivities (Neff, 2012) embracing self-reliance and mandatory flexibility, accepting imposed precarity and risk, and in turn bolstering the norms of the mainstream games industry. The next phase for those interested in addressing structural challenges in gameswork is therefore converting these individualized responses into involvement in global confrontations against the constraints that limit creative, inclusive, sustainable, and open participation in games for all.

Notes

1 Following Deuze et al. (2007), I use 'gamesworker' to indicate the complexity of the professional identities implied in work in this field in addition to the broadness of roles and employment statuses implied by 'games worker'.

2 While independent games programs were not the focus of this study, a minority of participants did study on these kinds of courses, and it is therefore instructive to see that indie development is also framed in some of these promotional materials as a stepping stone to the mainstream industry or in a symbiotic relationship with it (see, for example, courses offered at Teeside and Grimsby, and for an exceptional approach, the MA Independent Games and Playable Experience Design at Goldsmiths).

3 In fact, educational licenses and institutional policies related to copyright and intellectual property in UK HEI actually often make taking student games to market impossible.

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17 The cultural conditions of being indie

Bart Simon

I was around when the field of game studies began to take shape in the late-1990s, and there was very little discussion of what we have now come to call independent (indie) game development and culture. In this chapter, I offer a reflection on my own path toward developing a distinctly 'indie game studies' following on the diverse empirical range and depth of studies presented in this volume. I start with the observation of the empirical anomaly of the videogame indie scene in the context of prevailing ideas about the global media industry. This is followed by an account of my own attempt to make sense of the phenomenon of 'going indie' in Montreal in the mid-late-2000s from a cultural economic perspective. I end with some thoughts about how we might begin to reconcile our empirical work with broader concerns in cultural theory with the hope that indie game studies of the kind collected here might offer new insights to scholars interested in the future of cultural production, creative, and cultural economy and cultural justice.

Game studies and the videogame 'culture industry'

At the turn of the century, digital games were defined by the rapid expansion of large multinational triple-a game studio-publishers like Ubisoft and Electronic Arts and the nascent field of game studies was more or less preoccupied with the legitimation of videogames as an object of academic reflection rather than just a cause for social concern. Our focus as scholars then, which is still the predominant focus now, was on the games – digital games as media, as systems, as literature, as art, as education, or as something else entirely.

This singular focus on game objects, albeit in relation to players or cultures or other media, belied a deeper investigation into the modes of production of those objects. Commercial game development as an economic activity and game design as an organizational practice remained opaque to most game studies scholars for nearly a decade with one of the most important exceptions in the work of Aphra Kerr (2006). And while there were many game designers who reflected more or less openly on commercial game design processes, patterns, and assumptions, there were no formal or codified methods for doing game development, let alone much-shared knowledge about how game studios worked in the larger political-economic context of global media and software industries. More importantly, there was not enough nuanced analysis of game production to be able to articulate how different contexts for making games might lead to different game objects which might lead to different cultural experiences and meanings. This is why the present edited volume is such an important contribution. It underscores the growing collective concern with the textures and nuances of the cultural production and not just the consumption of videogames. To be sure, game studies came about as the early scholar-players of videogames sought to understand their cultural value, but we took the provenance of those games for granted.

Well not quite. There was, in the context of media studies and political economy of media industries, already a prevailing neo-Marxian analytical frame operating behind the scene. Videogames were part of an expansive hegemonic and increasingly digitized global culture industry in which large multinational corporations owned both the means of production and distribution of multiple media forms. Film, music, television, and publishing (magazines and books) were all dominated by large media companies and media studies already had a long history of both ideological and political-economic critique of these culture industries and their products. The fact that the global game industry grew from a handful of software and entertainment companies to become multinational giants was completely consistent with what was occurring across all cultural forms.

If anything, games were the epitome of Theodor Adorno and Max Horkheimer's original 1944 conception of a culture industry producing banal mass culture that effectively masks the operations of capitalism in maintaining social control and inequality (Adorno and Horkheimer, 1992). Equally relevant, the adaptation of Pierre Bourdieu's idea of cultural reproduction (Jenks, 2002) illustrates how large game studios, much like Hollywood film studios, become locked into a logic of profit maximization and risk minimization through the reproduction of labor processes, the form and context of games as commodities, and the capture of markets. Coming out of a Canadian tradition of media studies, these arguments were perhaps the most robust in the work of Kline et al. (2003) and later in Dyer-Witheford and De Peuter (2009).

Following these works and others, with models borrowed from the analysis of film and music industries, a picture of the global game industry as a mass entertainment complex emerged that remains largely unchanged to this day. Game studies scholarship has mostly come to terms with this, either by incorporating elements of media industry critique or looking at forms of resistance in specific cultural contexts of videogame play and interpretation. In this way, the field of game studies has mostly become an extension of media studies in general.

The intriguing wrinkle in this story is that while game studies has been absorbing the critical apparatus of media industry scholarship, the actual shape of the games industry has shifted dramatically. Over the last 20 years, we have seen an explosion in small and mid-size independent game studios which represent an increasing rather than decreasing portion of the global market share for games. Where once large studios would be growing at the expense of smaller studios (through acquisitions or market domination), now there is a proliferation of studios along with increasing differentiation and variety in the forms of games. Big multinational studio conglomerates and logics of cultural reproduction still dominate the market, but the relative success of the indies (as they have come to be called) marks a significant empirical moment in the history of the global cultural economy generally and for game studies more specifically. Game studies must make sense of the indies. Media studies must do so as well.

To be sure, the relative success of indie studios may be short-lived as predictions of 'indieapocalypse' fatally project the collapse of the indie market for games and a resurgence of corporate hegemony, but this does not discount the phenomenon of the indie boom in spite of the powerful economic forces that proceeded it. For myself, indie game studies are primarily concerned with this phenomenon and its impact on game cultures and their modes of production and consumption as well as providing important insights for understanding cultural economy beyond just videogames. I am less concerned with the question of who or what is truly indie than I am with "indie" as a cultural economic space of hybridity, difference, and indeed possibility. Indie in this sense has never represented a radical outside of culture industry production but rather a radical inside as the indie studio is always in an unsettled and precarious position relative to the largest corporate studios and yet is also fundamentally concerned with economic production and market share (if only for their own economic sustainability). Indeed, perhaps, the most pressing argument in my indie game studies work has been to try and show how indies necessarily experiment with forms of economic, social, and cultural sustainability.

Montreal indies: an origin story

Let me back up a bit though so I can better situate the indie game studies work of the present volume as well as begin to sketch a program for future research. My own research on indies began with a co-authored report I helped to prepare for the Canadian government (Gouglas et al., 2010). Working with Jason Della Rocca, who was the former head of the International Game Developer's Association and the current founder and director of the largest indie co-working space in Canada (GamePlay Space), we developed a model of the game development cycle based on a case study of the game studio ecosystem in Montreal. At the time, Montreal had the largest cluster of multinational triple-a game studios in Canada and one of the largest clusters in the world. We wondered why this was and articulated a notion of clustering as a solution to what Della Rocca (2011) called, "the peak overhead" problem.

Peak overhead occurs in the game development process as a project moves from conceptualization and pre-production stage to production stage. In pre-production, project teams are often one-third (or less) the size of the teams at production (and in large studios these teams can consist of hundreds of workers). What this means pragmatically is that for each project, studios must hire on hundreds of workers for production but once the game moves to post-production, they are left with the problem of what to do with all that extra labor. In Montreal, in 2010, that labor was routinely laid-off, and much of the excess labor pool would then be absorbed by other game studios in the city, thus, the importance of clustering (coupled with a relatively low cost of living and generous provincial tax incentives which incentivized hiring).

We had noticed, like others had before, that the peak overhead problem also made studios increasingly risk-averse, because each project represented a kind of cultural bet on market success, and as the overhead increases, the need to hedge the bet increases. Games that had been previously successful became templates for future game projects and effectively established most of the game franchises that still exist today. This also allowed studios to reduce the time between peak overhead phases of their game projects which meant less time (and cost) to ramp up and ramp down between projects. The casualty in this equation is what we described as game developers' sense of cultural or creative autonomy. The prevailing solution to peak overhead means that fewer and fewer creative decisions are made as game developers are increasingly forced to reproduce previously successful games for audiences that are dependable.

This problem of peak overhead and risk mitigation combined with one other observation salient to the situation in Canada. The decisions for how to deal with peak overhead, risk, and labor were not being made in Canadian studios but by the executive producers of multinational companies with headquarters in the United States, Japan, or Europe. We surmised at the time that this led to (a) an excess labor pool in Montreal that was not effectively absorbed in the cluster because studio executives were not sensitive enough to their own local labor pool, and (b) increasing disaffection amongst experienced and senior game developers with creative decisions made elsewhere. Thus, it was in 2010 that we advised the government to develop cultural policies that would encourage what we called, "The Independent Game Development Sector. The term "independent game", we suggested, is:

> used by developers mainly to describe a game that is released or published by the company that made it. In other words, it is a game that is not published through a third-party producer. Independence is a

> relative measure of a game developer's degree of creative and financial autonomy and control over the IP of their game. Independent game development is something of a precarious business for many of the reasons addressed in our discussion of the development cycle but recently there has been new enthusiasm for addressing the sustainability of small-scale independent studios running no more than one to three projects at a time.
>
> (Gouglas et al., 2010, p. 17)

Here, we coupled an economic argument, that indie studios might solve the peak overhead problem by basically having the same labor pool for pre-production, production, and post-production phases of a project (and this helped explain the structure of Quebec's largest indie studio at the time, Behaviour) with a cultural argument that experienced game designers desired more creative autonomy, and this was linked to the emergence of a notion of game design as a career rather than just job. What we wondered at the time was why or how anyone would leave the relatively secure salaries of triple-a studio jobs in favor of starting indie studios of their own.

It was true enough that Montreal was full of developers who were effectively "between jobs", and this created the cultural space for talk about 'going indie' to flourish. Organizations like the Mont-Royal Game Society began at this time (in 2010), and Della Rocca started working on the plan for what became Execution Labs which was founded in 2012 as an indie start-up accelerator. We had also conducted a series of interviews with senior game developers in 2008–2009 who almost all had concerns about their own creative autonomy and a frustrated desire to make "interesting" and/or "different" kinds of games.

It was also at this time (in the late-2000s) that the bottom nearly fell out of the triple-a market through the rise of mobile gaming and the platformization of distribution (iOS and android app stores and Steam). It is generally seen that this shift in the material conditions for making and playing games created an economic environment which, for a time, leveled the playing field for prospective independent game developers. It was no longer the case that one needed a huge capital investment to compete and in principle almost anyone could make a two-dollar mobile game for an app store with minimal overhead. This prospect drove an explosion of commercial game making and indeed most indie games were, and still are, mobile platform games.

I want to be careful here though, while a great many new games were made in this period, it was more difficult than ever to make a living from those games. This was in part driven by the counter-intuitive pricing policies of the distribution platforms that enabled a "race to the bottom" in terms of pricing which meant that only those studios producing a high volume of games and large marketing budgets (including market analytics) tended to survive, let alone thrive. These conditions favored new business models in large studios which were often the very opposite of desires for cultural autonomy.

From my perspective, what constitutes the indie boom was not the proliferation of independently made games on mobile so much as the increasing number of individual developers and small studios that were able to exploit the interstices of shifting platforms, markets, and production tools in order to achieve a degree of financial stability, if not success. It is for this reason that I think that the cultural conditions in Montreal, an excess skilled labor pool coupled with an increasing desire for creative autonomy, formed the basis of an expansion of indie studios, not just the shifts in the material conditions of production as such. In Montreal, at least, most of the indie studios surviving today were started by ex-triple-a studio developers, and while the same cannot be said of indie scenes elsewhere in Canada and the world, it is still important to look to cultural preconditions to help explain the continued development of the indie sector beyond mobile platformization.

Indie cultural autonomy beyond art and commerce

In the last part of this chapter, I want to turn directly to the concept of cultural autonomy as the heart of any definition of "going indie" in the games industry. All the articles in the present volume are premised on some notion of cultural autonomy in terms of how to make sense of indies as being any different (or not) from other kinds of game development practice or organization. Collectively, the articles in this volume are wrestling with the stakes of "being indie" and why that may matter beyond a neoliberal fascination with all things artisanal. It is not the categories that are important here – what is or isn't indie – but rather the processes, relations, and tensions in the cultural economy of game making that these categories reflect.

In a beautiful 2010 essay, Mark Banks quotes Slater and Tonkiss' 2001 Market Society on cultural autonomy in way that describes the situation with indie games quite well.

> Autonomy of culture means at least two things; first, autonomy from economic values, the creation of art in relation to its own inner gods rather than the idols of the marketplace; and second, autonomy from the false and inauthentic 'culture' that arises in and through the marketplace, the seedy demon born when the ignorant tastes of the people mate with the fiscal lust of the capitalist.
>
> (253–254)

This double demand is a kind of ideal type of the artist as the source of authentic culture. It is a kind of aspiration that drives not just artistic self-realization but indeed the idea of creativity itself. It is the demand which marks the contradiction at the heart of what is now called the creative economy. How can culture be creative and economic at the same time when the very definition of creativity requires autonomy from the marketplace? It's an old problem, perhaps the oldest cultural problem, that expresses and re-expresses itself in the tension between art and commerce, the idea of "selling out", of compromising creativity, and so on.

The art–commerce relation here is drawn from Bill Ryan's (1992) 'Making Capital from Culture' and is nicely summarized by Banks:

> as a specific contradiction at the heart of the culture industry – namely, that while cultural production is increasingly organized in accordance with capitalist demands, it is never reducible to those demands since, by necessity, it must encompass other antithetical forms of value derived from the practices and procedures of art.
>
> (Banks, 2010, p. 259)

This is not idealism so much as way of accounting for the continued actual trouble, resistance, feistiness, and vanguard subjectivity of cultural workers of all sorts all over the world. This is why the stakes around the idea of the creative economy are so great – the last battle may indeed be fought on this front, and it may well be game-makers that finally unwind capitalism.

But I would argue that the game-makers are not important here because they are artists. Quite the opposite. Games as cultural objects have as problematic a relationship to art as they have to commerce. Arguably, the origins of game studies lie in the art question with respect to videogames and much ink has been spilled on a somewhat misconceived legitimation project to have games taken seriously as a cultural form on the level of art (Simon, 2017a). In the long history of debates over the cultural autonomy of artists in the context of the cultural and creative economy, game-makers have mattered very little. In the battle between art and commerce, games have always fallen on the side of commerce. Historically, videogames have not been art; they have been, in line with Frankfurt school presumptions, mere entertainment.

This makes the desire for cultural autonomy on the part of game developers even more interesting. The desire for cultural autonomy amongst musicians, writers, and filmmakers is obvious – their formation is defined by the art–commerce relation. But game-makers, at least those who do not begin their careers as artists, must generate their own forms of non-artistic cultural autonomy. While the art game movement aspires to artistic ideas of cultural autonomy, indie game developers typically have no such recourse. And so, it seems indies evolve a notion of cultural autonomy which is not dependent on the idea of art as resistance to the "idols of the marketplace". Indies inhabit the marketplace, and their claims on autonomy are fraught and problematic. How they wrestle with this as individuals, practices, organizations, and scenes is why they are so important. The question of cultural autonomy without recourse to the ideology of art may be the most pressing cultural question of our time.

In our most recent work with the indie interfaces project, myself, Jen Whitson, and Felan Parker, along with several students (see https://www.indieinterfaces.com/), have started a range of investigations in the discourses and practices of independence in an attempt to characterize indie game cultural autonomy. Very early on, we realized a fundamental problem in that commercial forms of cultural autonomy most often take shape in the form of entrepreneurship (Whitson et al., 2018). Entrepreneurs are, in the current celebratory mode, almost like the artists of commerce. They are inspired, creative, industrious, and courageous, and there is much work that can be done to look at how traditional discourses of cultural autonomy have been appropriated in the culture of entrepreneurship, social economy, and venture capital.

If there is a predominant discourse in indie game cultural economy, it is this. Between art and commerce, there is entrepreneurship, and game-makers have been quick to conflate financial and cultural autonomy, such that economic decisions and design decisions become one in the same exercise in meaningful and fulfilling creativity. That is, it is just as meaningful to survive against the odds and succeed as a business as it is to create a beautiful, compelling, or meaningful game. In entrepreneurship, art and commerce are conflated not opposed, but this is a false route to cultural emancipation or social change. As Gina Neff (2012) among others has argued, entrepreneurship reproduces individualistic and competitive social relations that reinforce the status quo rather than undermine it. There is no creative justice in entrepreneurialism.

And yet, beyond this dominant discourse which sets the tone for most of the cultural economic activity of the current indie sector, there are other routes that don't conform so easily (Simon, 2017b; Perks et al., 2019). In our work, we describe the evolving local patterns of interdependence and peer support in indie scenes, the rise of cultural intermediaries like festivals, co-working spaces, alternative and collective models of finance, and the development of new discourses of labor sustainability coupled with social justice. It's not that these things are entirely new in arenas of cultural production, but in games, they are developed without recourse to artistic legitimacy.

Perhaps, the most intriguing observation we have made in our many interviews with indie developers is kind of almost nihilist realization that success as an indie might simply just mean being able to make another game. Such sentiments are missing both the call of art to make a meaningful game, or the call of commerce to make a profitable one. The call of indie is sustainability to make another game. And in that process, practices, relations, organizations, self-conceptions, and even culture itself may change. Indie game studies will have made its mark when we manage to tie this robust and diverse empirical research on indies to the deeper and more far-reaching processes that drive culture and cultural production beyond art and commerce.

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